



Integrated Cloud Applications & Platform Services

Oracle Application Express Workshop I

Activity Guide – Volume II
D79653GC30 | D106874

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Authors

Sayantani Ghosh
Moitreyee Hazarika

**Technical Contributors
and Reviewers**

Suresh Rajan
Anjani Pothula
David Peake
Chaitanya Koratamaddi
Joe Greenwald

Editors

Adrita Biswas
Aju Kumar
Moushmi Mukherjee

Graphic Editors

Prakash Dharmalingam
Anne Elizabeth
Pushparaj Kundar

Publishers

Pavithran Adka
Veena Narasimhan
Giri Venugopal

3009092019

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Course Practice Environment: Security Credentials

For OS usernames and passwords, see the following:

- If you are attending a classroom-based or live virtual class, ask your instructor or LVC producer for OS credential information.
- If you are using a self-study format, refer to the communication that you received from Oracle University for this course.

For product-specific credentials used in this course, see the following table:

Product-Specific Credentials		
Product/Application	Username	Password
Workspace Administrator User	apex_admin	<u>Initial login:</u> apex <u>Changed To:</u> welcome (will be using this all through the course)
Workspace Developer User	apex_dev	apex
Workspace Developer User	brad.knight	apex
Workspace Administrator User	susie.parker	apex
Workspace End User	john.bell	apex

Note: The credentials mentioned above are for the users of the **APEX** Workspace. The first user (that is, `apex_admin`) is already created. You will be creating all the other users during the practices.

**Practices for Lesson 13:
Validating and Debugging Your
Application**

Practices for Lesson 13: Overview

Overview

In these practices, you use the Oracle Application Express Advisor (Advisor) to validate your application; update the Attribute Dictionary based on items and report columns on your page; and examine how to use, debug, and resolve some common issues during application development.

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Practice 13-1: Using the Advisor

Overview

In this practice, you use the Advisor to validate the GlobalMart Management Tool application.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Run the Advisor and correct an Accessibility warning on the **202 - Product Information** page.
2. Run the Advisor for the **202 - Product Information** page.
3. Run the Advisor again for the entire application.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 13-1**.

If you haven't completed **Practice 12-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_12_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Practice 13-2: Modifying the Attribute Dictionary

Overview

In this practice, you update the Attribute Dictionary based on the items and report columns on your page.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Change the widths of items on the Customer Details page of the GlobalMart Management Tool application.
2. Edit `P9_ACCOUNT_MGR_ID`. Change the item label and add help text.
3. Add the items on the Customer Details page to the Attribute Dictionary.
4. Review the UI defaults in SQL Workshop. Change the help text for `CUST_FIRST_NAME` to `Customer First Name` and add a Synonym called `FIRST_NAME`.
5. Check whether `CUSTOMERS` is in Table Dictionary.
6. Update the Attribute Dictionary for the items on the Customer Details page.
7. Update the Customers page to use the Attribute Dictionary defaults. Verify by showing the help text for the First Name column in the report.
8. Edit the Update Customers page to use the Attribute Dictionary defaults. Verify by showing the help text for the First Name column in the report.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 13-2**.

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If you haven't completed **Practice 13-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_13_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

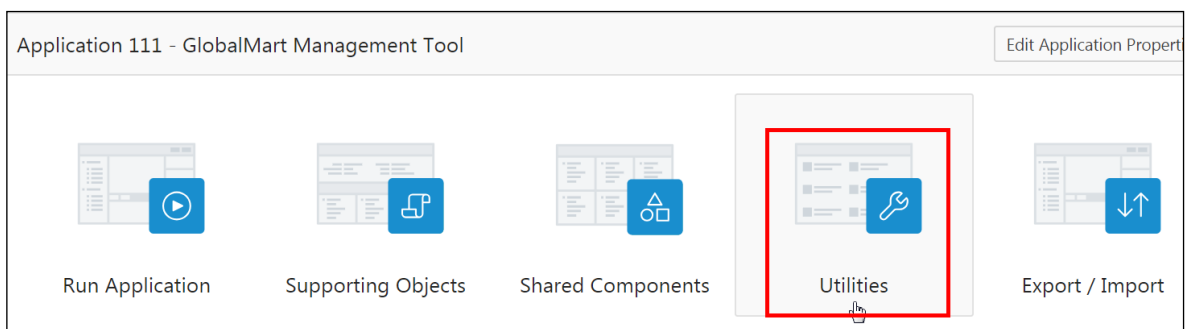
Solution 13-1: Using the Advisor

Overview

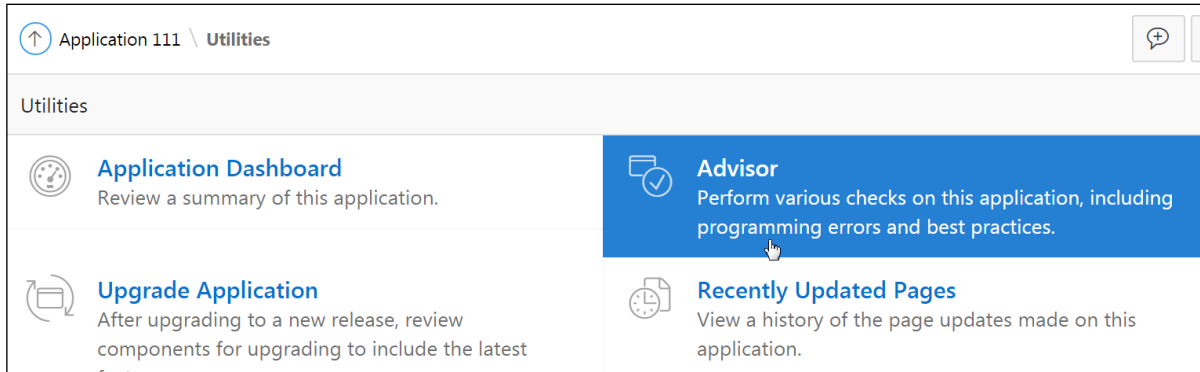
In this practice solution, steps are provided for using the advisor to validate the GlobalMart Management Tool application.

Steps

1. Run the Advisor on the application and correct the warning on the **202 - Product Information** page.
 - a. From the Workspace home page, click **App Builder** and then select the GlobalMart Management Tool application.
 - b. On the Application home page, click **Utilities**.



- c. On the Utilities page, click **Advisor**.



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- d. You can review the pages and checks to be performed by expanding **Pages last edited with the last 7 days** and **Checks to Perform**. For this exercise, accept the defaults. Click **Perform Check**.

▼ Check Page(s)

Perform Check

The Advisor performs various checks on your application, including programming errors, security issues, quality assurance, and other best practices. For the list of checks, see *Checks to Perform*. Once executed, your previous settings will be recalled for the next use.

Specify a comma separated list of pages to check, or leave blank to check all pages.

Page(s)

?

▶ Pages last edited within the last 7 days

▶ Checks to Perform

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- e. The list of violations appears under **Filter Result**. Filter the results by deselecting **Error** and **Security** and clicking **Apply Filter**.

▶ Check Page(s)

Perform Check

Filter Result

Apply Filter

The results are displayed for the checks that you performed. You can filter the result set by unchecking any category or specific checks under Filter Results and clicking the Apply Filter button.

To address the issues identified, click the View link within each result. Once issues have been fixed, you can use the *Perform Check* button to re-run the report or view *Checks to Perform* to select different checks to be performed.

☐

Error (9)

☐ References with Substitution Syntax (1)☐ References with Bind Variable Syntax (4)☐ Declarative References of Application Items, Page Items, Columns or Interactive Report Filters (4)

☐

Security (21)

☐ Authorization (21)

☒

Warning (1)

☒ Protected items in Ajax calls (1)

☒

Accessibility (1)

☒ Region has Row Header (1)

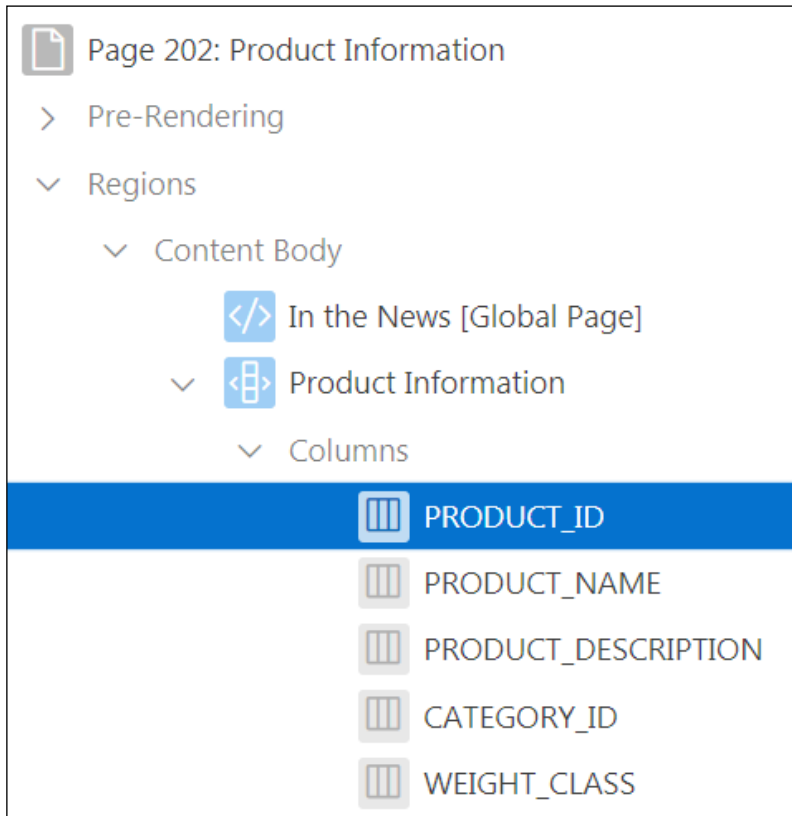
- f. The Accessibility warning Message “**Region has Row Header**” appears. To address it, you will edit a report column and enable the **Use As Row Header** attribute. Click the **View** button.

Applications > 111 - GlobalMart Management Tool > Pages > 202 - Product Information > Regions > Product Information

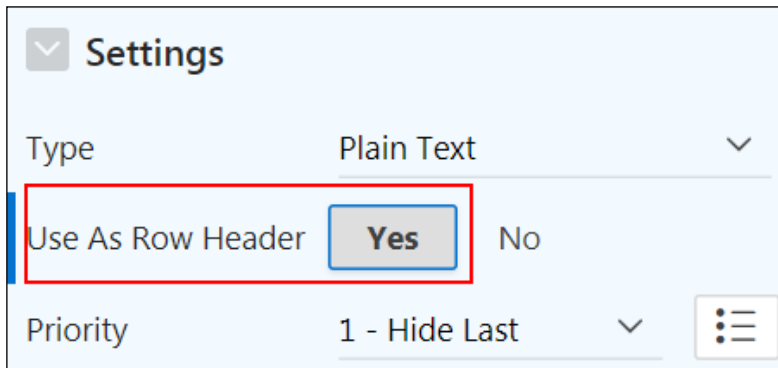
Attribute	Source Type (Identifies how APEX will interpret the Region Source)
Check	Region has Row Header
Category	Accessibility
Message	This Column Toggle region does not have a column with 'Use as Row Header' set to 'Yes'. The Row Header column should be the column that best identifies the current row (for example a customer name in a report of customer information).
Value	Column Toggle Report
	View

Note: Close the Advisor window, which is open in your browser.

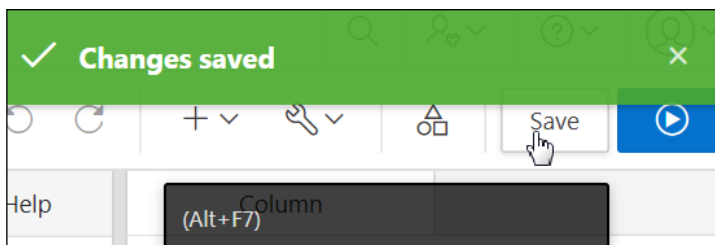
- g. The Product Information page opens in Page Designer view. On the Rendering tab (in the right pane), expand **Regions > Content Body > Product Information > Columns** and select `PRODUCT_ID`.



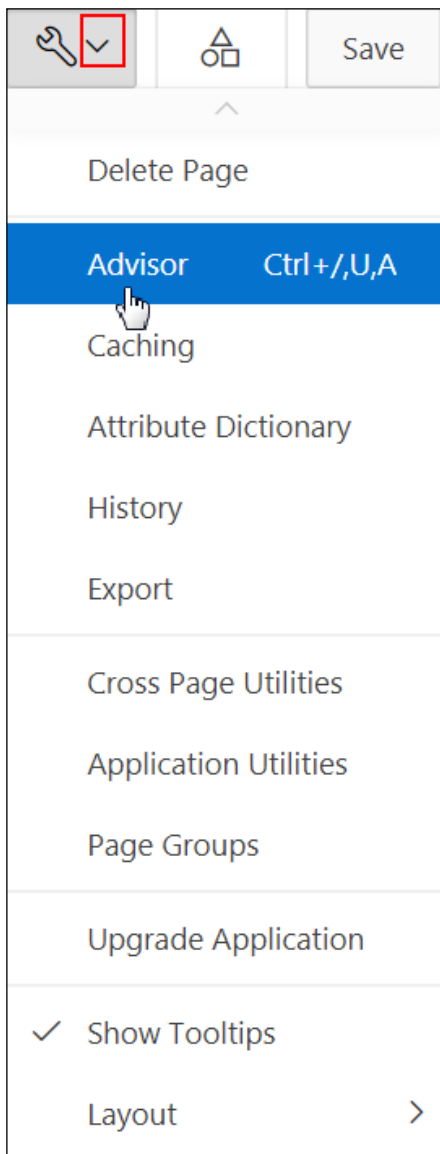
- h. In the Property Editor (left pane), under **Settings > Use as Row Header**, select **Yes**.



- i. Click **Save**.



2. Run the Advisor for the **Product Information** page.
 - a. On the Page Designer toolbar, select the **Utilities** menu and then select **Advisor**.



- b. Advisor automatically populates the Page(s) field. Click the **Perform Check** button.

▼ Check Page(s)

Perform Check

The Advisor performs various checks on your application, including programming errors, security issues, quality assurance, and other best practices. For the list of checks, see *Checks to Perform*. Once executed, your previous settings will be recalled for the next use.

Specify a comma separated list of pages to check, or leave blank to check all pages.

Page(s)

?

▶ Pages last edited within the last 7 days

▶ Checks to Perform

- c. Note that violation has been resolved. Now only a Security, Authorization issue appears. You will be learning about authorizations in a later chapter.

☒ Security (1)
☒ Authorization (1)

Applications > 111 - GlobalMart Management Tool > Pages > 202 - Product Information

Attribute	Authorization Scheme (An authorization scheme must evaluate to TRUE in order for this page to be displayed)
Check	Authorization
Category	Security
Message	Page is not protected by an authorization scheme.
Value	View

3. Run Advisor again for the entire application.
 - a. In the Advisor window, expand the **Check Page(s)** region and delete the number in the Page(s) field. Click **Perform Check** again.

▼ Check Page(s) **Perform Check**

The Advisor performs various checks on your application, including programming errors, security issues, quality assurance, and other best practices. For the list of checks, see *Checks to Perform*. Once executed, your previous settings will be recalled for the next use.

Specify a comma separated list of pages to check, or leave blank to check all pages.

Page(s)

▶ Pages last edited within the last 7 days

▶ Checks to Perform

- b. Close the Advisor window.

Advisor - Google Chrome

Not secure | 10.237.16.240:8080/apex/?p=4000:8101:17345853713259:NO::

▶ Check Page(s) **Perform Check**

Filter Result **Apply Filter**

The results are displayed for the checks that you performed. You can filter the result set by unchecking any category or specific checks under Filter Results and clicking the Apply Filter button.

To address the issues identified, click the View link within each result. Once issues have been fixed, you can use the *Perform Check* button to re-run the report or view *Checks to Perform* to select different checks to be performed.

☒ **Error (10)**

- ☒ References with Substitution Syntax (1)
- ☒ References with Bind Variable Syntax (4)
- ☒ Declarative References of Application Items,

- c. Return to the Application home page. Click the **Application ID** breadcrumb at the top of the page in the Page Designer view.

↑ [Application 111](#) \ Page Designer

Solution 13-2: Modifying the Attribute Dictionary

Overview

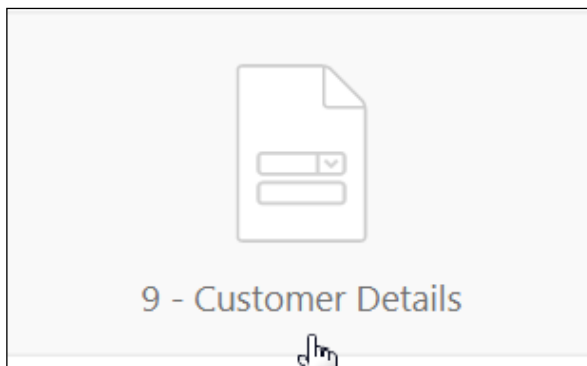
In this practice solution, steps are provided for updating the Attribute Dictionary based on the items and report columns on your page.

Steps

1. Change the widths of items on the Customer Details page of the GlobalMart Management Tool application to the following:

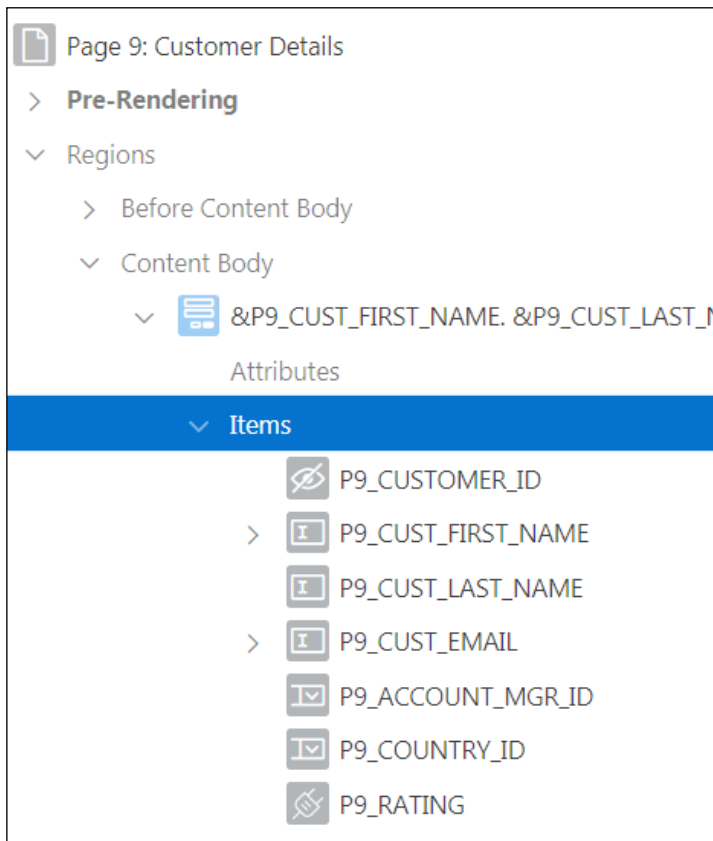
Item	Attribute	Value
P9_CUST_FIRST_NAME	Width	18
P9_CUST_LAST_NAME	Width	18
P9_CUST_EMAIL	Width	18

- a. On the GlobalMart Management Tool Application home page, select the **Customer Details** page.

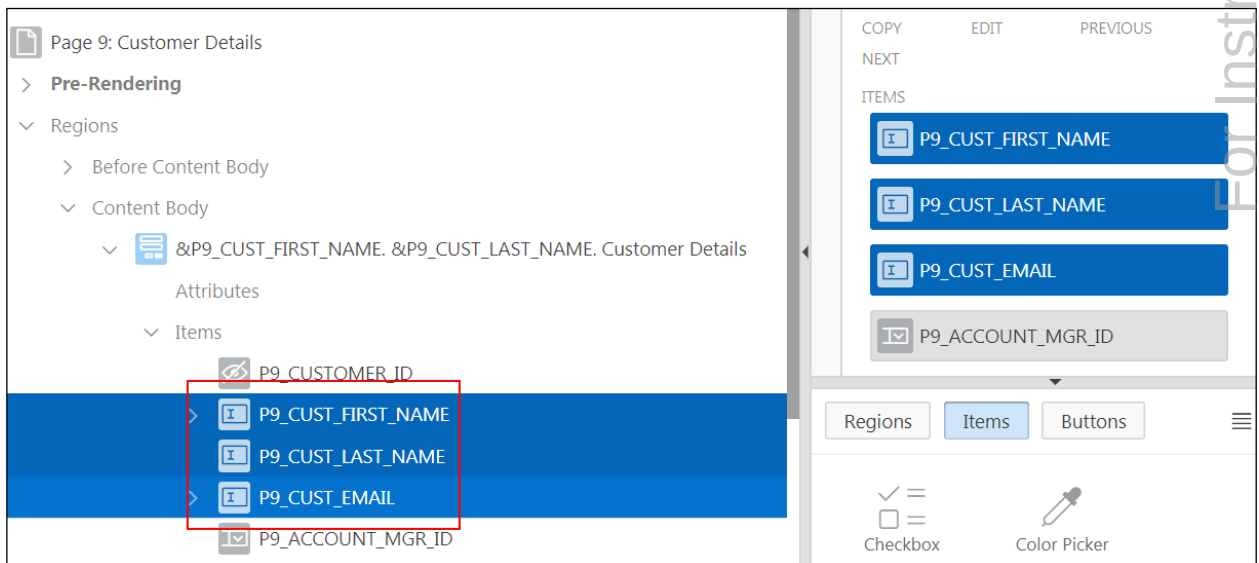


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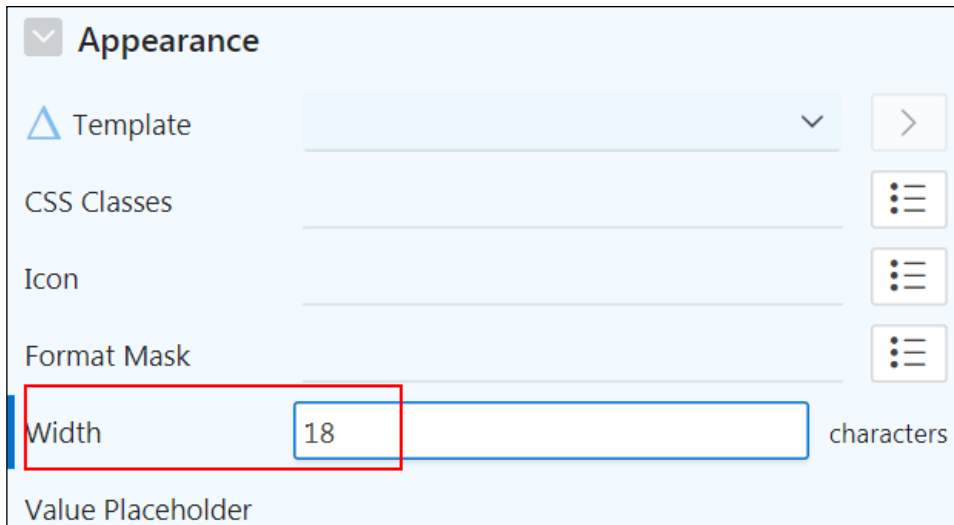
- b. Page Designer appears. On the Rendering tab, expand **Regions > Content Body > Items**.



- c. Under items, press and hold Ctrl + Shift and select P9_CUST_FIRST_NAME, P9_CUST_LAST_NAME, and P9_CUST_EMAIL.

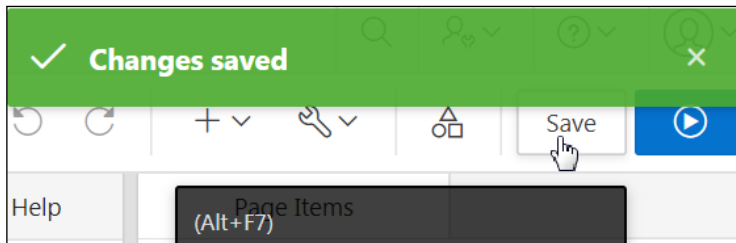


- d. In the Property Editor, under **Appearance** > **Width**, enter 18.

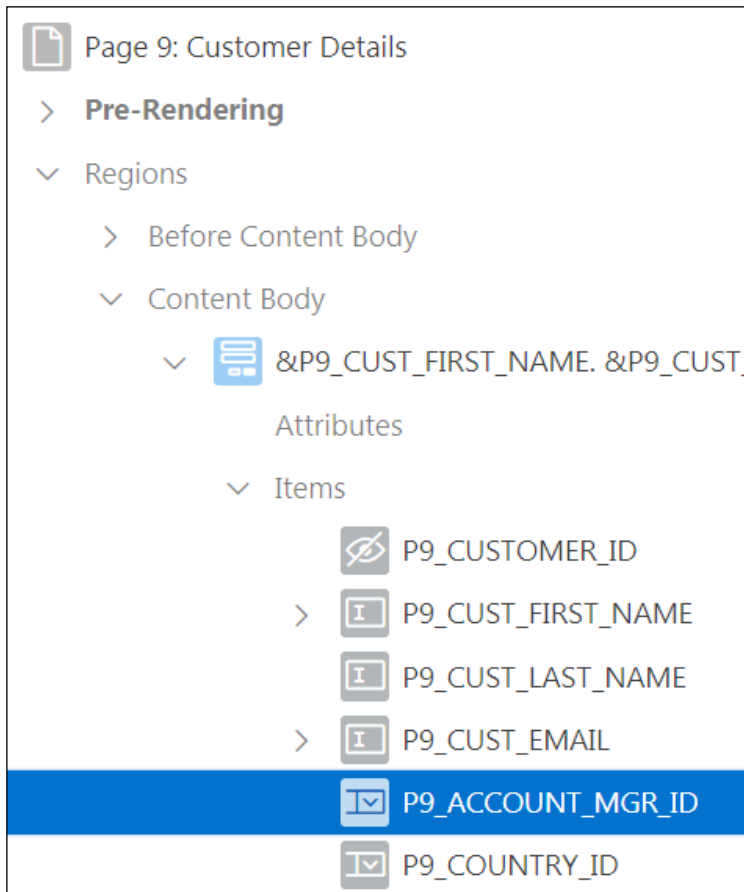


The screenshot shows the 'Appearance' section of a property editor. It includes fields for 'Template', 'CSS Classes', 'Icon', and 'Format Mask'. The 'Width' field is highlighted with a red rectangle and contains the value '18', with the unit 'characters' indicated to its right. Below the 'Width' field is a 'Value Placeholder' field.

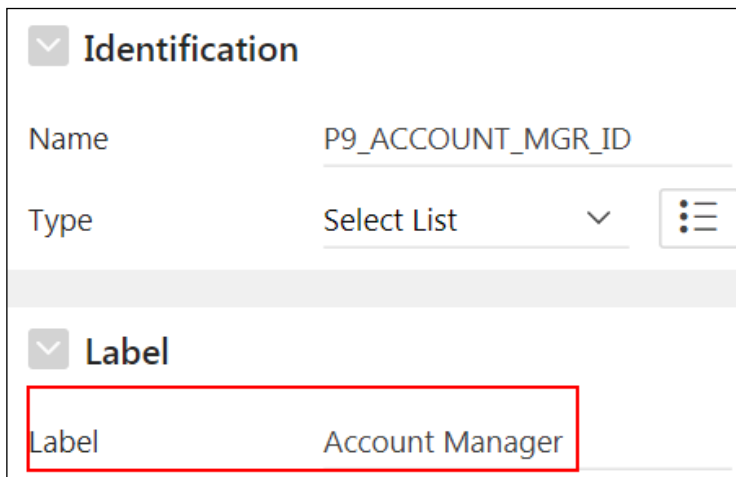
- e. Click **Save**.



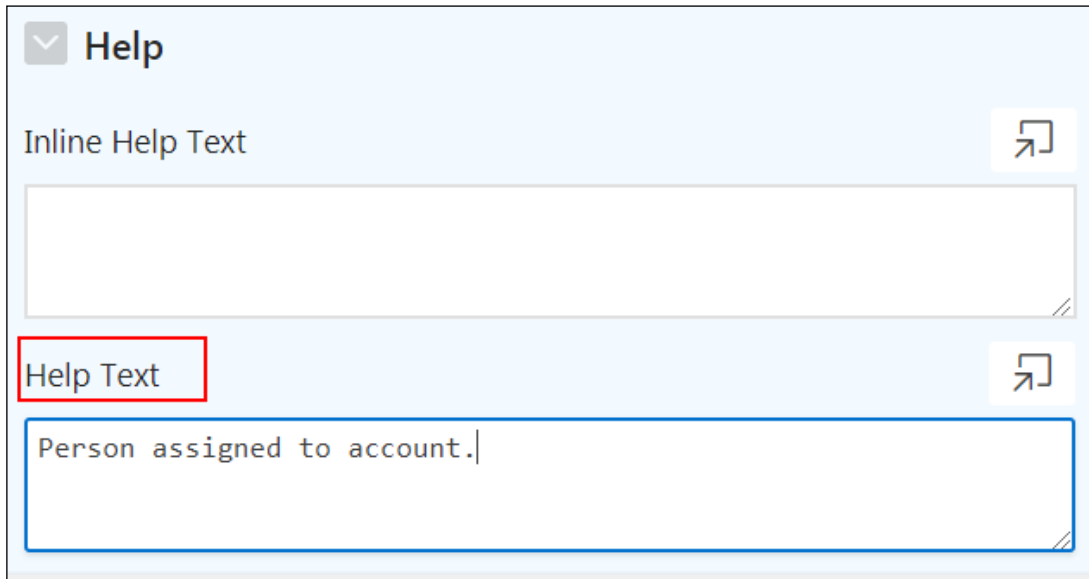
2. Edit P9_ACCOUNT_MGR_ID. Change the item label and add help text.
 - a. On the Rendering tab (left pane), under **Regions > Content Body > Items**, select P9_ACCOUNT_MGR_ID.



- b. By default, the Label Name appears as **Account Manager** in the Property Editor (right pane), under **Label > Label**.

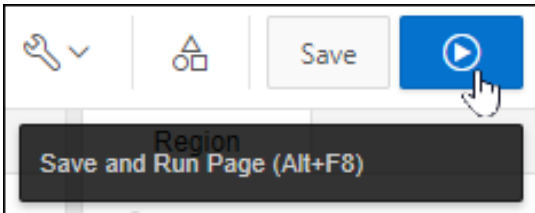


- c. Add Help Text. In the Property Editor, under **Help > Help Text**, enter `Person assigned to account.`

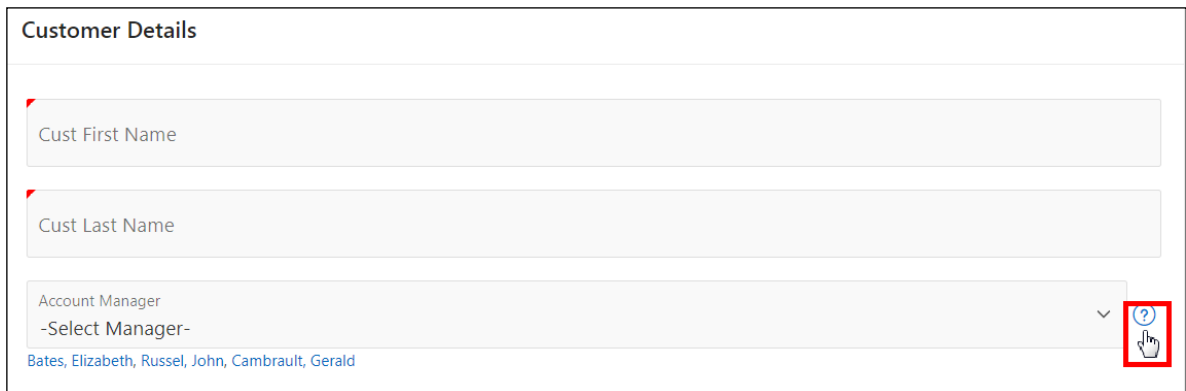


The screenshot shows a 'Help' section in a property editor. It has a dropdown arrow and the word 'Help'. Below it is an 'Inline Help Text' field. Underneath that is a 'Help Text' field, which is highlighted with a red rectangular box. This field contains the text 'Person assigned to account.' and has a blue border. To the right of the 'Help Text' field is a small icon of a document with an arrow pointing out.

- d. Click **Save** and then **Save and Run Page**.

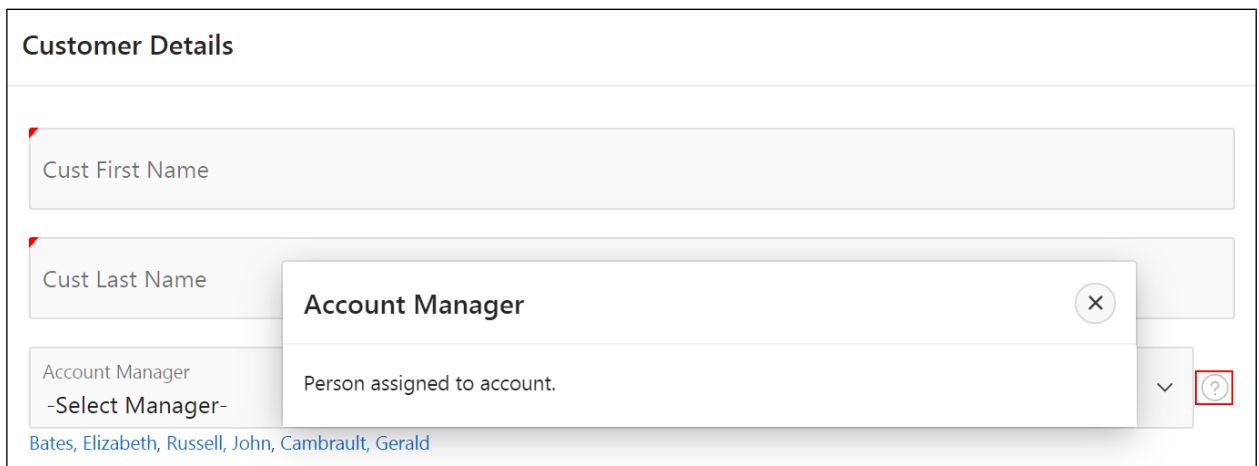


- e. The Customer Details page appears. Click the **Question Mark** icon  next to the **Account Manger** field.



The screenshot shows a 'Customer Details' form. It has three input fields: 'Cust First Name', 'Cust Last Name', and 'Account Manager'. The 'Account Manager' field is highlighted with a red rectangular box. It contains the text '-Select Manager-' and has a dropdown arrow on the right. Below the field, a list of names is visible: 'Bates, Elizabeth, Russel, John, Cambrault, Gerald'. To the right of the 'Account Manager' field, there is a question mark icon in a circle, which is also highlighted with a red rectangular box. A mouse cursor is pointing at this icon.

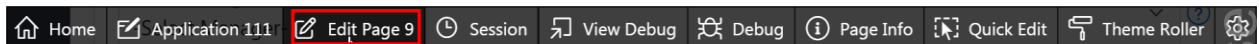
The **Help Text** window appears.



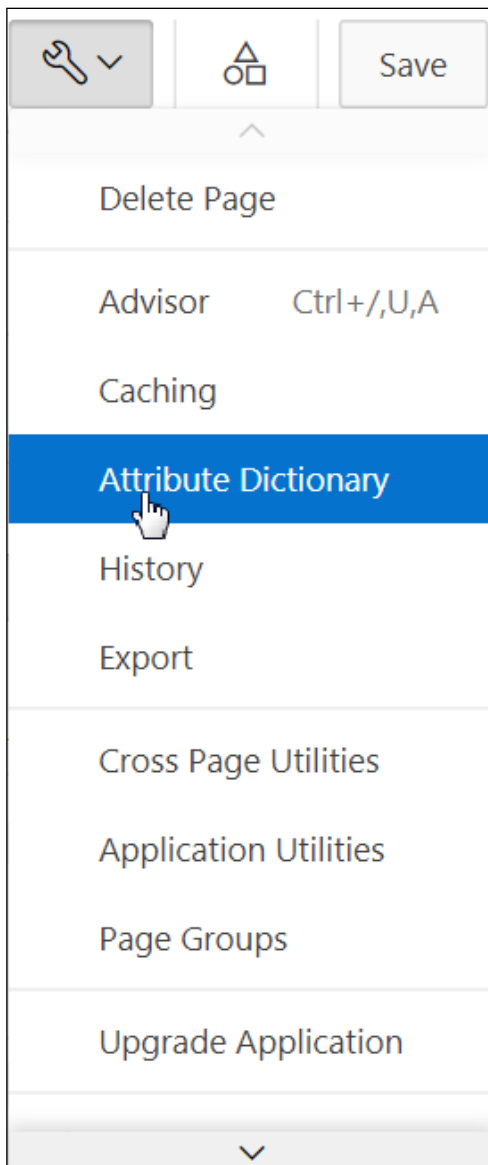
- f. Click the **Close** icon.



- g. Click the **Edit Page 9** link on the Runtime Developer toolbar.



3. Add the items on the Customer Details page to the Attribute Dictionary.
 - a. **Page 9 – Customer Details** opens in Page Designer view.
 - b. On the Page Designer toolbar, select the **Utilities** menu and then select **Attribute Dictionary**.



- c. You need to review the items that you want to add to the Attribute Dictionary. Click the **Review 6 items for insert into the Attribute Dictionary** link.

Update Page
0 Items for update

Update Attribute Dictionary

Review 6 Items for insert into the Attribute Dictionary

0 Items for update of the Attribute Dictionary

- d. The items that you have on the page consist of all the changes that you made previously. Select the check box in the header to the left of the Region header to select all the entries in the list and click **Update Attribute Dictionary**.

Page: 9 - Customer Details ?
Include in Update: ?
☒ Label ☒ Help Text ☒ General Format Mask ☒ Default
☒ Form Format Mask ☒ Width ☒ Height ☒ Data Type

Cancel **Update Attribute Dictionary**

☒	Region ↑	Item	Will Become	Label	Format Mask	Help Text	Default	Width	Height	Data Type
☒	&P9_CUST_FIRST_NAME. &P9_CUST_LAST_NAME. Customer Details	P9_ACCOUNT_MGR_ID	ACCOUNT_MGR_ID	Account Manager	-	Person assigned to a...	-	-	1	VARCHAR
☒	&P9_CUST_FIRST_NAME. &P9_CUST_LAST_NAME. Customer Details	P9_COUNTRY_ID	COUNTRY_ID	Country	-	-	-	-	1	VARCHAR
☒	&P9_CUST_FIRST_NAME. &P9_CUST_LAST_NAME. Customer Details	P9_CUST_EMAIL	CUST_EMAIL	Cust Email	-	-	-	18	-	VARCHAR

Note: The screenshot is truncated in size and is not the entire list.

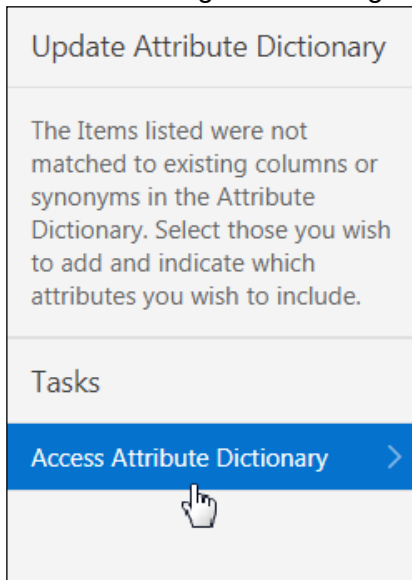
- e. The message **Attribute Dictionary Updated** appears at the top of the page.

Application 111 \ Page 9 \ Attribute Dictionary \ **Update Attribute Dictionary**

☒ **Attribute Dictionary Updated.**

Page: 9 - Customer Details ?
Include in Update: ?
☒ Label ☒ Help Text ☒ General Format Mask ☒ Default
☒ Form Format Mask ☒ Width ☒ Height ☒ Data Type

4. Review the UI defaults in SQL Workshop. Add Help Texts for `CUST_FIRST_NAME` and `CUST_LAST_NAME`.
 - a. In the Tasks region on the right side of the page, click **Access Attribute Dictionary**.



- b. You see the attributes that you added to the dictionary. You can modify them here. Click `CUST_FIRST_NAME`.

Dashboard Table Dictionary Attribute Dictionary			
Go Actions			
Column Name ↑	Synonym of	Label	Help Text
ACCOUNT_MGR_ID	-	Account Manager	Person assigned to account.
COUNTRY_ID	-	Country	-
CUST_EMAIL	-	Cust Email	-
CUST_FIRST_NAME	-	Cust First Name	-
CUST_LAST_NAME	-	Cust Last Name	-
RATING	-	Rating	-

Q v		Go
Column Name ↑≡	Synonym of	Label
ACCOUNT_MGR_ID	-	Account Manager
COUNTRY_ID	-	Country
CUST_EMAIL	-	Cust Email
CUST_FIRST_NAME	-	Cust First Name
CUST_LAST_NAME	-	Cust Last Name
RATING	-	Rating

- c. Change **Label** to First Name, and in **Help Text**, enter Customer First Name. Click **Apply Changes**.

Column: 4 of 6 ?

Cancel Delete **Apply Changes**

Details Synonyms Form Specific Attributes Report Specific Attribu...

* Column Name CUST_FIRST_NAME ?

Label First Name ?

Format Mask ?

Default Value ?

Help Text Customer First Name

The changes appear on the Attribute Dictionary page.

✓ **Action Processed.**

Dashboard Table Dictionary **Attribute Dictionary**

Q ▾ **Go** Actions ▾ **Reset**

Column Name ↑≡	Synonym of	Label	Help Text
ACCOUNT_MGR_ID	-	Account Manager	Person assigned to account.
COUNTRY_ID	-	Country	-
CUST_EMAIL	-	Cust Email	-
CUST_FIRST_NAME	-	First Name	Customer First Name
CUST_LAST_NAME	-	Cust Last Name	-

- d. You also want to change the CUST_LAST_NAME. Click CUST_LAST_NAME.

Dashboard Table Dictionary **Attribute Dictionary**

Q ▾ **Go** Actions ▾ **Reset**

Column Name ↑≡	Synonym of	Label	Help Text
ACCOUNT_MGR_ID	-	Account Manager	Person assigned to account.
COUNTRY_ID	-	Country	-
CUST_EMAIL	-	Cust Email	-
CUST_FIRST_NAME	-	First Name	Customer First Name
CUST_LAST_NAME	-	Cust Last Name	-
RATING	-	Rating	-

- e. Change **Label** to Last Name, and in **Help Text**, enter Customer Last Name. Click **Apply Changes**.

Column: 5 of 6 ?

Cancel Delete **Apply Changes**

Details Synonyms Form Specific Attributes Report Specific Attribu...

* Column Name CUST_LAST_NAME ?

Label Last Name ?

Format Mask ?

Default Value ?

Help Text Customer Last Name

The changes appear on the Attribute Dictionary page.

✓ **Action Processed.**

Dashboard Table Dictionary **Attribute Dictionary**

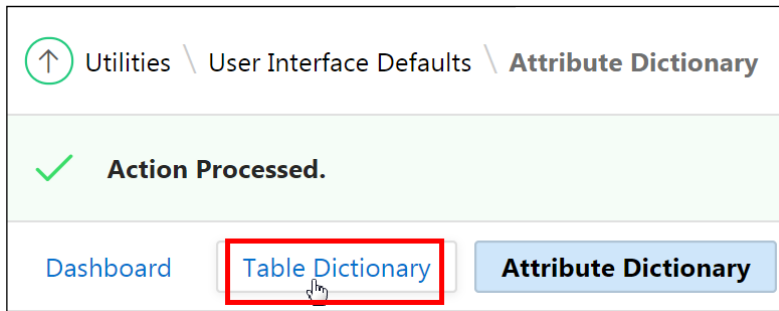
Q ▾ Go Actions ▾ Res

Column Name ↑	Synonym of	Label	Help Text
ACCOUNT_MGR_ID	-	Account Manager	Person assigned to account.
COUNTRY_ID	-	Country	-
CUST_EMAIL	-	Cust Email	-
CUST_FIRST_NAME	-	First Name	Customer First Name
CUST_LAST_NAME	-	Last Name	Customer Last Name
RATING	-	Rating	-

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5. Check whether OEHR_CUSTOMERS is in the Table Dictionary.

a. Click the **Table Dictionary** tab.

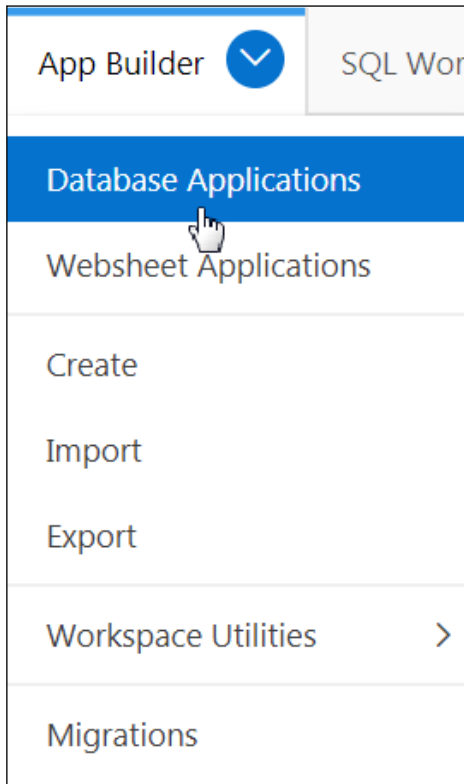


b. Find OEHR_CUSTOMERS. Note that the **Default Exists** column displays **No** for this table, which means that user interface defaults are not configured and, therefore, do not exist in the Table Dictionary.

The screenshot shows the 'Table Dictionary' page in Oracle APEX. It includes a search bar, 'Go' button, view toggles, 'Actions', 'Reset', and 'Synchronize' buttons. A table lists database objects with columns 'Object Name', 'Type', and 'Defaults Exist'. The row for 'OEHR_CUSTOMERS' is highlighted with a red rectangle, and its 'No' value in the 'Defaults Exist' column is also highlighted with a red rectangle.

Object Name ↑	Type	Defaults Exist
DEMO_ORDER_TYPE	TABLE	No
OEHR_ACCOUNT_MANAGERS	VIEW	No
OEHR_BOMBAY_INVENTORY	VIEW	No
OEHR_COUNTRIES	TABLE	No
OEHR_CUSTOMERS	TABLE	No
OEHR_CUSTOMERS_VIEW	VIEW	No

- c. Select the **App Builder > Database Application** menu.

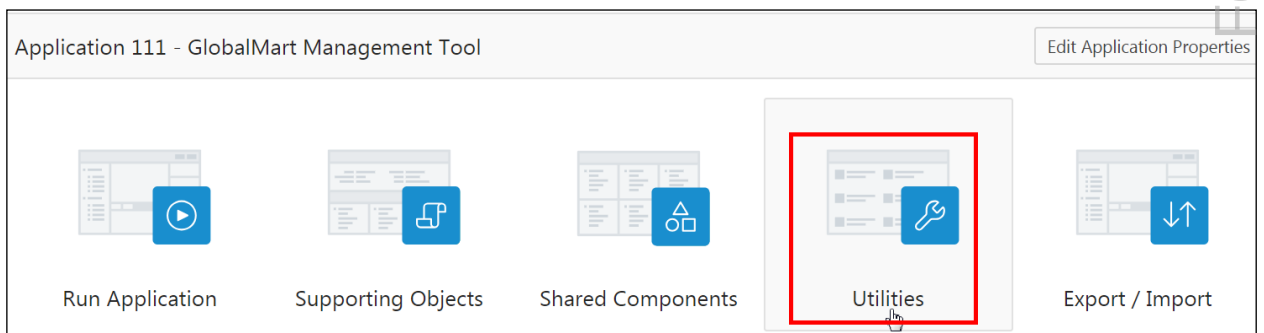


- d. On the App Builder home page, select the **GlobalMart Management Tool** application.



6. To update the Attribute Dictionary for the items on the Customer Details:

- a. Click **Utilities**.



- b. Click **Attribute Dictionary**.

Utilities



Application Dashboard
Review a summary of this application.



Upgrade Application
After upgrading to a new release, review components for upgrading to include the latest features.



Attribute Dictionary
Manage item / column user interface defaults for a selected page.

- c. Click the link for the **9 - Customer Details** page.

Application 111 \ Utilities \ Attribute Dictionary		
Search Go Actions		
Page ↑	Name	Page Type
0	Global Page - Desktop	Global Page
1	Home	Home
2	Top Tier Salary	Report
3	Customer Address List	Report
4	Customers	Interactive Report
5	Employee Commission	Interactive Report
6	Product Information Details	Interactive Grid
7	Products	Report
8	Update Customers	DML Form
9	Customer Details	DML Form

- d. Notice that you have two items to review for update to the Attribute Dictionary. Click the **Review 2 Items for update of the Attribute Dictionary** link.

↑ Application 111 \ Page 9 \ **Attribute Dictionary**

Page Items

Update Page

[Review 2 Items for update](#)

Update Attribute Dictionary

0 Items for insert into the Attribute Dictionary

[Review 2 Items for update of the Attribute Dictionary](#)

- e. Notice that the column in the Attribute Dictionary is the one that you want, rather than the help text for the item. You will leave it as is.

↑ Application 111 \ Page 9 \ Attribute Dictionary \ **Update using Items**

Page: 9 - Customer Details ?

Include in Update: ?

☒ Label ☒ Help Text ☒ General Format Mask ☒ Form Format Mask
☒ Default ☒ Width ☒ Height ☒ Data Type

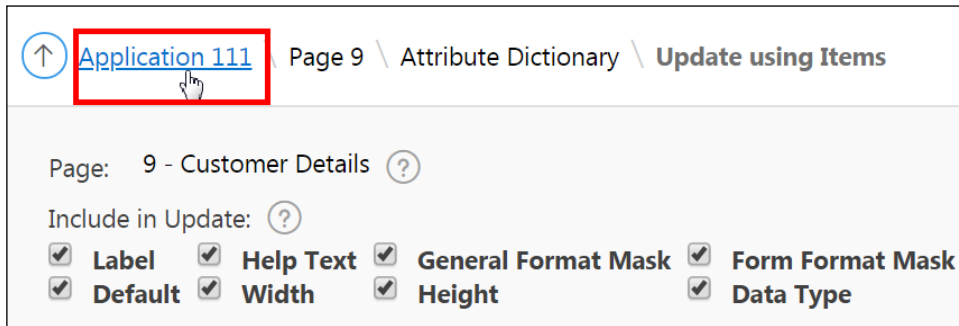
Cancel

Update Attribute Dictionary

Source ↑	Name	Label	Help Text	General Format Mask	Form Format Mask	Default	Width	Height	Data Type
Region: &P9_CUST_FIRST_NAME. &P9_CUST_LAST_NAME. Customer Details	P9_CUST_FIRST_NAME	Cust First Name		-	-	-	18	-	VARCHAR
from Attribute Dictionary	CUST_FIRST_NAME	First Name	Customer First Name				18	1	VARCHAR
Region: &P9_CUST_FIRST_NAME. &P9_CUST_LAST_NAME. Customer Details	P9_CUST_LAST_NAME	Cust Last Name		-	-	-	18	-	VARCHAR
from Attribute Dictionary	CUST_LAST_NAME	Last Name	Customer Last Name				18	1	VARCHAR

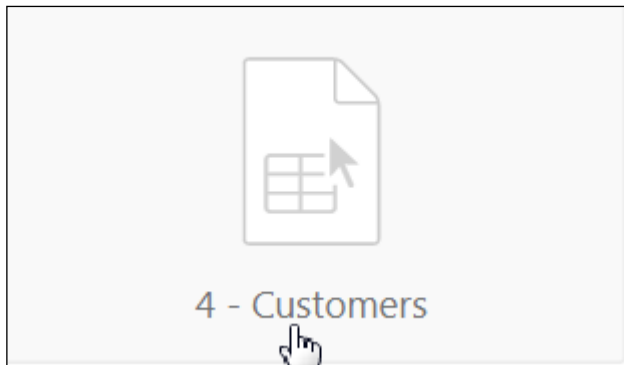
row(s) 1 - 2 of 2

- f. Click the **Application ID** breadcrumb to go back to the application.

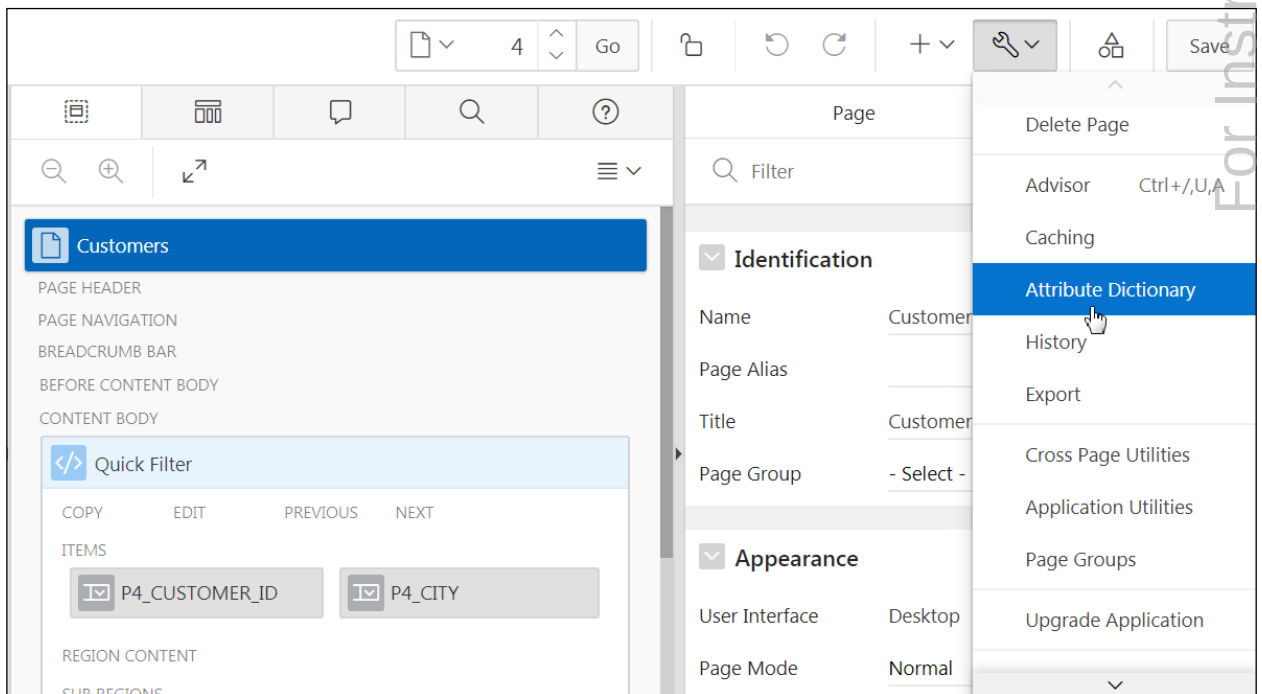


7. Update the Customers page to use the Attribute Dictionary defaults. Verify by showing the help text for the **First Name** and **Last Name** column in the report.

- a. Select **4 – Customers**.



- b. **Page: 4 Customers** opens in Page Designer view. On the Page Designer toolbar, select **Utilities**  and **Attribute Dictionary**.



- c. Because you want to update the page, you want to review the columns for update. Under **Update Page**, click the **Review 5 Report Columns for update** link.

Report Columns

Update Page

Review 5 Report Columns for update

Update Attribute Dictionary

Review 9 Report Columns for insert into the Attribute Dictionary

- d. Review the list. Select the check box in front of `CUST_FIRST_NAME` item and click **Update Report Column**.

Include in Update: (?)
 ☒ Label
 ☒ Help Text
 ☒ Format Mask
 ☒ Report Column Alignment

Cancel
 Update Report Columns

	Source	Name	Label	Help Text	Format Mask	Report Column Alignment
Region:						
☐	Customer Report from Attribute Dictionary	ACCOUNT_MGR_ID	Account Mgr Id	Person assigned to a View	-	Right
☐	Customer Report from Attribute Dictionary	COUNTRY_ID	Country Id	Country	-	Left
☐	Customer Report from Attribute Dictionary	CUST_EMAIL	Cust Email	Cust Email	-	Left
☒	Customer Report	CUST_FIRST_NAME	First Name		-	Left

You get the message Report Columns Updated.

Application 111 \ Page 4 \ Attribute Dictionary \ **Update Report Columns**

Report Columns Updated.

Page: 4 - Customers ?

Include in Update: ?

☒ **Label**
☒ **Help Text**

☒ **Format Mask**
☒ **Report Column Alignment**

- e. Run the page to see the results. Click the **Run Page 4** icon.



- f. To see the help text, select the **First Name** header in the report and click the **Question Mark** icon.

First Name	Cust Last Name	Street Address	Postal Code
Filter... Ajay Alain Alan Albert Alec			15214
			18503
			19103
			19401

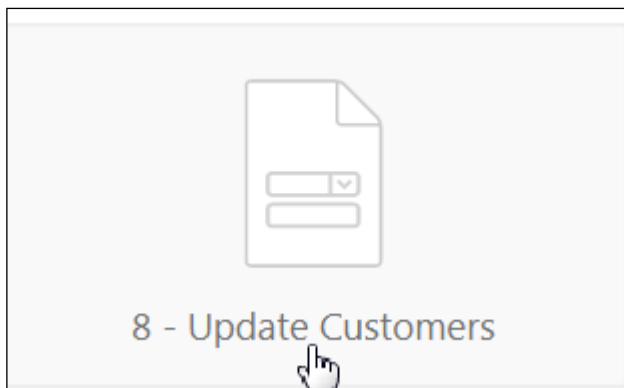
The help text from the Attribute Dictionary appears.

First Name	Cust Last Name	Street Address	Postal Code
Bryan	Dvine	Ave	15214
Customer First Name			
Filter...	Sen	220 Penn Ave # 300	18503
Ajay			
Alain	Jordan	135 S 187 St # 1	19103
Alan			
Albert	Bradford	522 Swede St	19401
Alec			

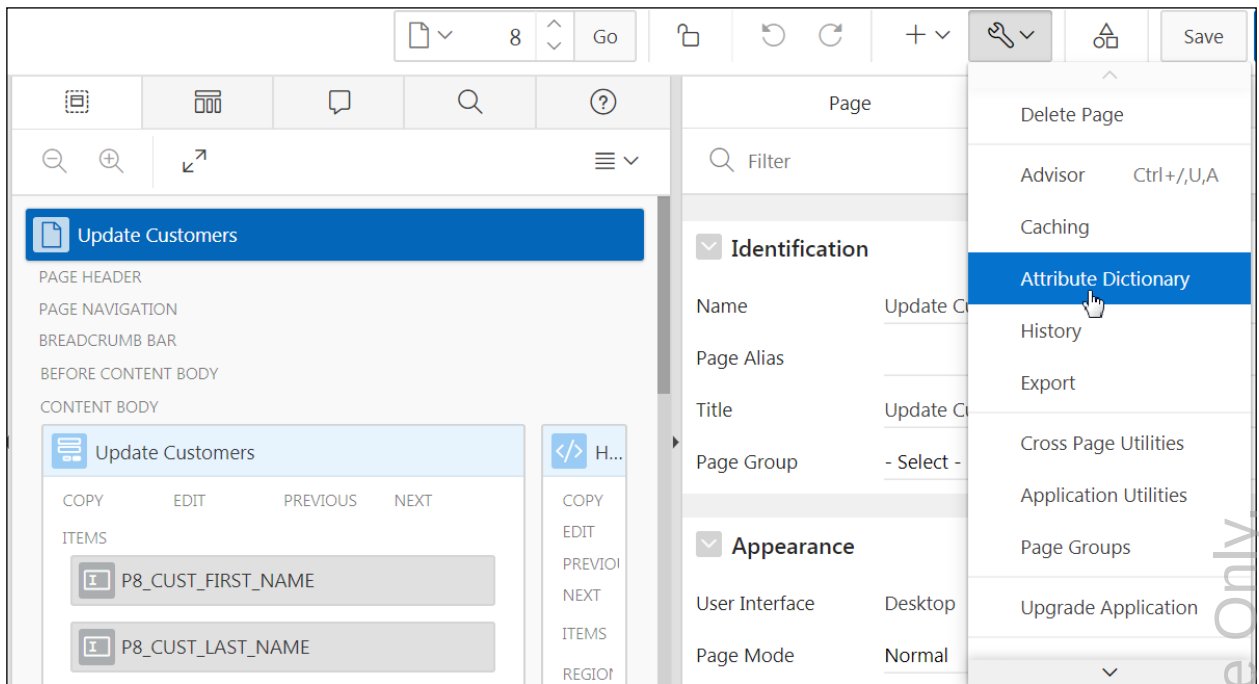
- g. Click the **Application ID** link on the Developer toolbar.



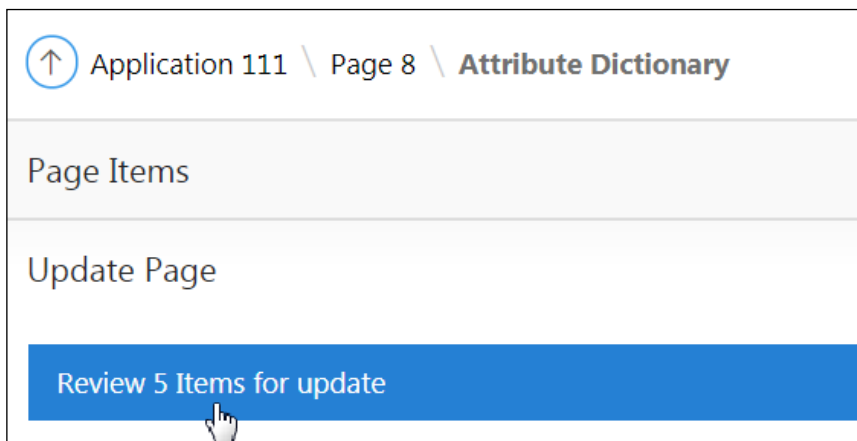
8. Edit the Update Customers page to use the Attribute Dictionary defaults. Verify by showing the help text (that you entered in the Attribute Dictionary) for the **First Name** column in the report.
- a. Go to the GMT application home page and click **8 – Update Customer**.



- b. **Page 8: Update Customer** opens in Page Designer view. On the Page Designer toolbar, select **Utilities** and **Attribute Dictionary**.



- c. Review the items to update. Under **Update Page**, click the **Review 5 Items for update** link.



- d. Review the list. Select the check box next to the `CUST_FIRST_NAME` item. Click **Update Items**.

Include in Update: ?

☒ Label ☒ Help Text ☒ Format Mask ☒ Default
☒ Width ☒ Height ☒ Data Type

Cancel **Update Items**

Source	Name	Label	Help Text	Format Mask	Default	Width	Height	Data Type
Region: Update P8_ACCOUNT_MGR_ID Customers	P8_ACCOUNT_MGR_ID	Account Mgr Id		-	-	32	1	VARCHAR
Attribute ACCOUNT_MGR_ID Dictionary	ACCOUNT_MGR_ID	Account Manager	Person assigned to a<a href="javascript:popupURL('f?p=4000:908:4782780937356:::908:P908_DISPLAY,P908_ID:ADH,5369221862024237')"				1	VARCHAR
Region: Update P8_COUNTRY_ID Customers	P8_COUNTRY_ID	Country Id		-	-	32	-	VARCHAR
Attribute COUNTRY_ID Dictionary	COUNTRY_ID	Country					1	

Region: Update P8_CUST_EMAIL Customers	P8_CUST_EMAIL	Cust Email		-	-	32	-	VARCHAR
Attribute CUST_EMAIL Dictionary	CUST_EMAIL	Cust Email					18	VARCHAR
☒ Region: Update P8_CUST_FIRST_NAME Customers	P8_CUST_FIRST_NAME	First Name		-	-	32	-	VARCHAR
Attribute CUST_FIRST_NAME Dictionary	CUST_FIRST_NAME	First Name	Customer First Name				18	1
Region: Update P8_CUST_LAST_NAME Customers	P8_CUST_LAST_NAME	Last Name		-	-	32	-	VARCHAR
Attribute CUST_LAST_NAME Dictionary	CUST_LAST_NAME	Last Name	Customer Last Name				18	1

You get the message Page Items Updated.

Application 111 \ Page 8 \ Attribute Dictionary \ **Update Items**

Page Items Updated.

Page: 8 - Update Customers ?

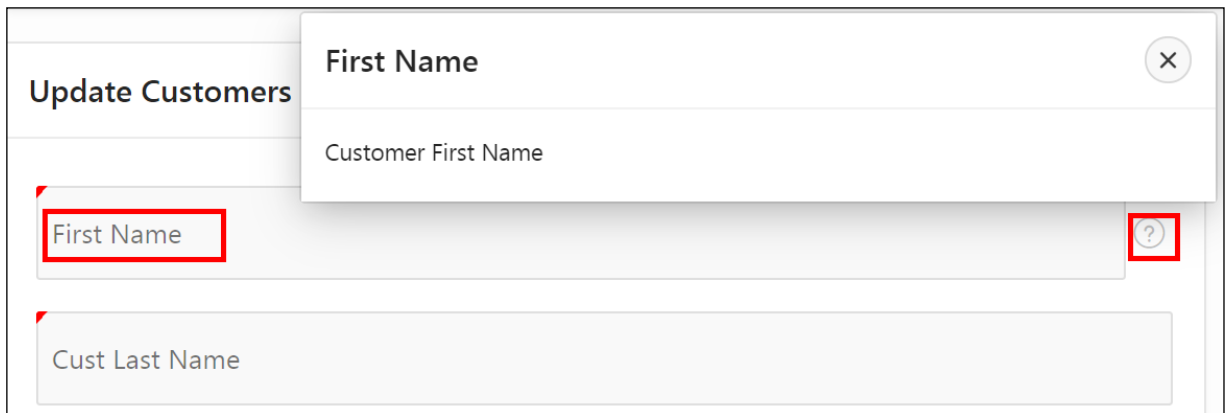
Include in Update: ?

☒ Label ☒ Help Text ☒ Format Mask ☒ Default
☒ Width ☒ Height ☒ Data Type

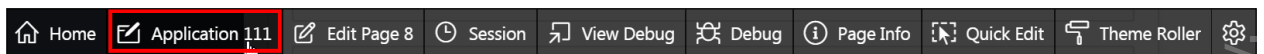
- e. Run the page to see the results. Click the **Run Page 8** icon.

Icons:

- f. To see the help text, select the **First Name** header in the report and click the **Question Mark** icon. The help text from the Attribute Dictionary appears.



- g. Click the **Application ID** link on the Developer toolbar to go back to your application.



**Practices for Lesson 14:
Adding Shared Components
That Aid Navigation**

Practices for Lesson 14: Overview

Overview

In this practice, you create a list and list region, create and edit navigation menu, add navigation bar entries, and add breadcrumbs to an existing page of the GMT application.

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Practice 14-1: Creating Lists and List Regions

Overview

In this practice, you create a list of links that aid navigation. Using these links, a user will be able to navigate to the Customers, Products, and Orders pages.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. In the GMT application, create a list called **Home Page List** with the following list entries:
 - View Customers (linked to the Customers page)
 - View Products (linked to the Products page)
 - View Orders (linked to the Orders page)
2. Create a list region called **Tasks** in the Home region.
3. Create a Dynamic List showing all the Products on the Products page. Create the List Region on the Products page by using the same Create List Wizard.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 14-1**.

If you haven't completed **Practice 13-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_13_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.

- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Practice 14-2: Creating and Editing a Navigation Menu

Overview

In this practice, you create a navigation menu for the application and assign pages to the menu. You also create a Help page and a Help tab to add to the existing tab set.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a navigation menu called **Orders** for the GMT application. It should be the current tab for the List of Orders page. Sequence the Orders entry before the Products navigation entry.
2. Assign the Master Detail forms page to the Orders navigation entry. Run the application and confirm that the navigation is created successfully.
3. Create a page called **Help**, and then add a new navigation menu entry called **Help** to the existing navigation menu. The new Help entry will link to the new Help page.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 14-2**.

If you haven't completed **Practice 14-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_14_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.

- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Practice 14-3: Adding Navigation Bar Entries

Overview

In this practice, you add a navigation bar entry that, when clicked, will show the page-level help. You also modify the name of an existing navigation bar entry and add another navigation bar entry that is displayed on all pages except the page that it navigates to.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a navigation bar entry called **Help** that navigates to the Help page. Run the page and click the Help navigation bar entry.
2. Change the name of the navigation bar entry for Sign Out to **Sign Off**.
3. Add a navigation bar entry called **Home** such that it appears between the **Help** and **Sign Off** button. Display the navigation bar entry on all pages except the Home page.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 14-3**.

If you haven't completed **Practice 14-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_14_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.

- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Practice 14-4: Adding Breadcrumbs to an Existing Page

Overview

In this practice, you create breadcrumbs and add it to the existing pages of the GMT application.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Edit the existing breadcrumb for the GMT application and create a breadcrumb entry for the Customers page. Run the customers page. Is the breadcrumb displayed on the page?
2. Create a breadcrumb region on the Customers page and run the page.
3. On the Customer Details page, create a breadcrumb called **Customer Details** whose parent entry is the Customers page.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 14-4**.

If you haven't completed **Practice 14-3**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_14_03.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).

- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.

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- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

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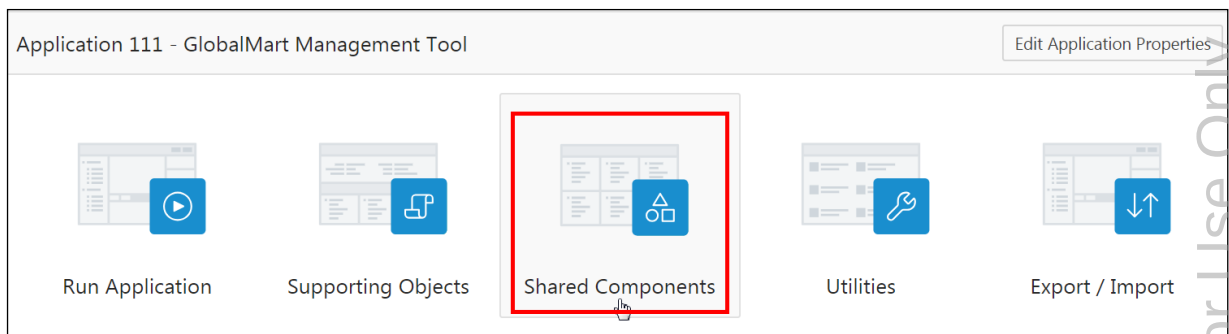
Solution 14-1: Creating Lists and List Regions

Overview

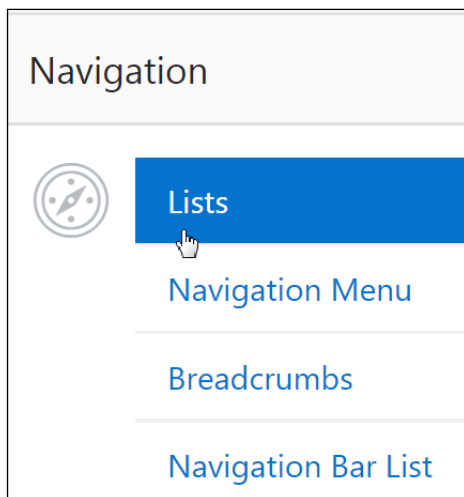
In this practice solution, steps are provided for creating lists and list regions for the GMT application.

Steps

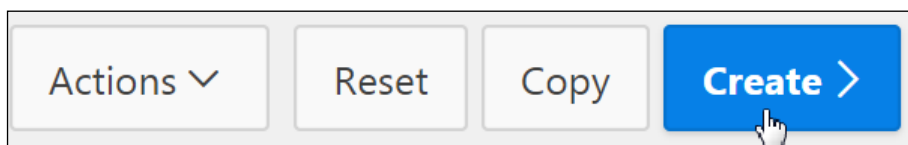
1. In the GMT application, create a list called **Home Page List** with the following list entries:
 - View Customers (linked to the Customers page)
 - View Products (linked to the Products page)
 - View Orders (linked to the Orders page)
- a. On the GlobalMart Management Tool application home page, select **Shared Components**.



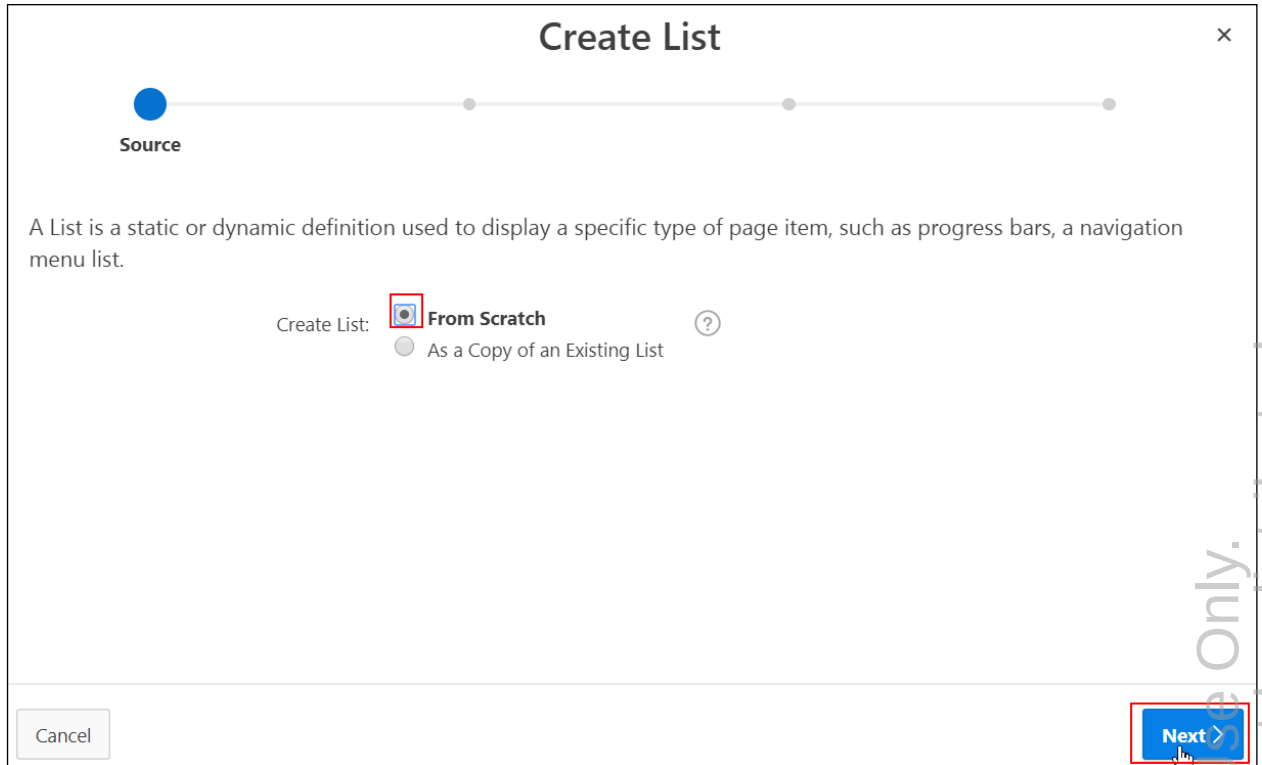
- b. In the **Navigation** section, click **Lists**.



- c. Click **Create >**.

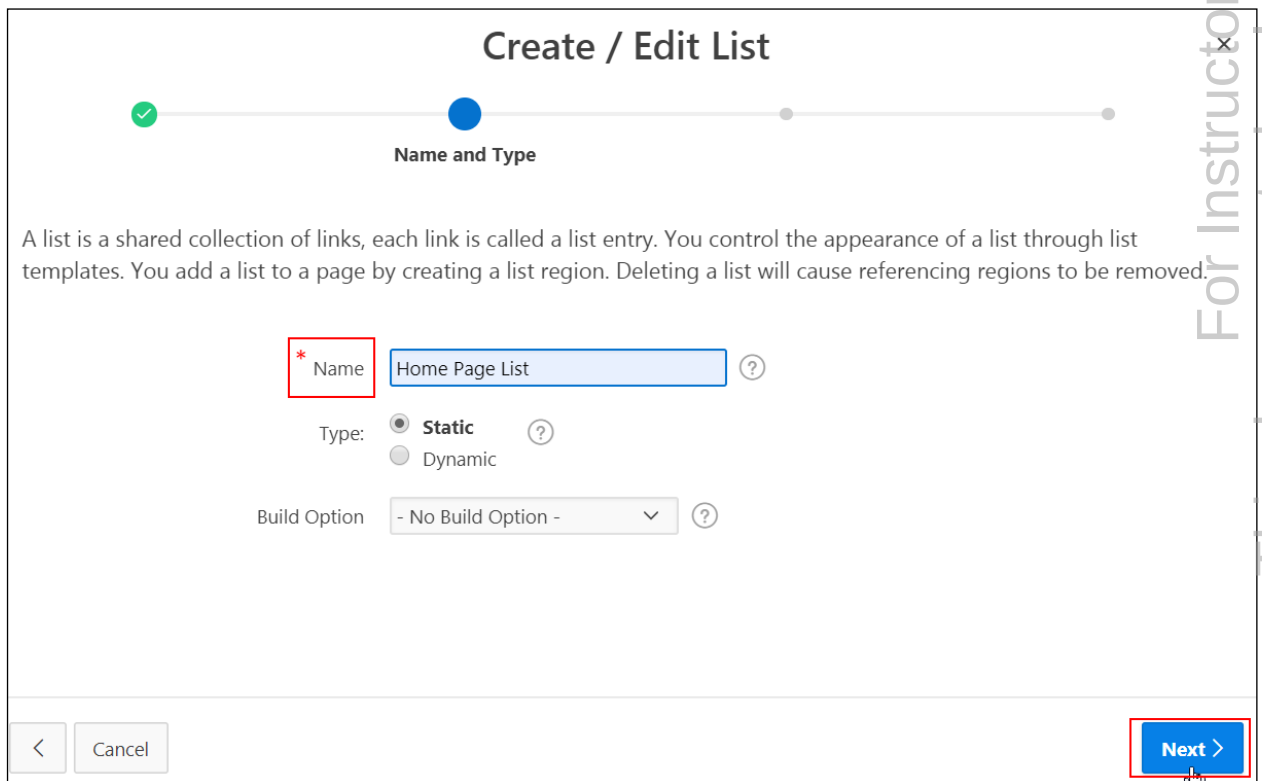


- d. Accept the default (that is **From Scratch**) and click **Next >**.



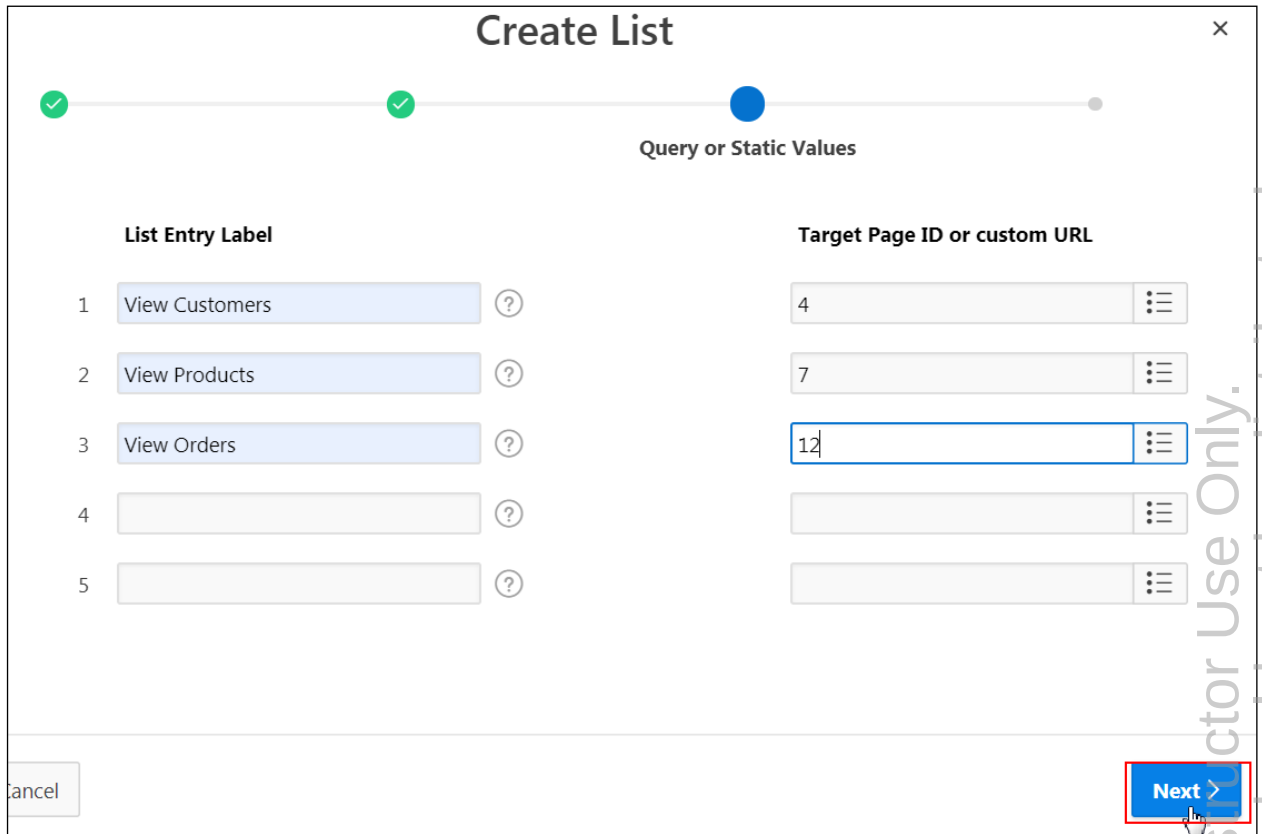
The 'Create List' dialog box features a progress bar at the top with four dots; the first dot is blue and labeled 'Source'. Below the progress bar, a text description states: 'A List is a static or dynamic definition used to display a specific type of page item, such as progress bars, a navigation menu list.' Under the heading 'Create List:', there are two radio button options: 'From Scratch' (which is selected and highlighted with a red box) and 'As a Copy of an Existing List'. A help icon (?) is located to the right of these options. At the bottom left is a 'Cancel' button, and at the bottom right is a 'Next >' button, which is highlighted with a red box.

- e. For **Name**, enter Home Page List and click **Next >**.



The 'Create / Edit List' dialog box has a progress bar at the top with four dots; the second dot is blue and labeled 'Name and Type', while the first dot is green. Below the progress bar, a text description states: 'A list is a shared collection of links, each link is called a list entry. You control the appearance of a list through list templates. You add a list to a page by creating a list region. Deleting a list will cause referencing regions to be removed.' The 'Name' field is marked with a red asterisk and contains the text 'Home Page List'; this field and its label are highlighted with a red box. Below the name field, the 'Type' section has two radio button options: 'Static' (selected) and 'Dynamic'. A help icon (?) is to the right of the 'Static' option. Below the type section, the 'Build Option' is set to '- No Build Option -' in a dropdown menu, with a help icon (?) to its right. At the bottom left are '<' and 'Cancel' buttons, and at the bottom right is a 'Next >' button, which is highlighted with a red box.

- f. Enter `View Customers` in the first **List Entry Label** field. Then click the corresponding up button and select the **Customers** page (Page 4) for **Target Page ID or custom URL**. Similarly, enter `View Products` and `View Orders` in the second and third List Entry Label fields. Then select the **Products** (Page 7) and **List of Orders** (Page 12) pages, respectively, by clicking the  icon. Click **Next >**.



The **Create List** dialog box is shown with a progress bar at the top. The first two steps are completed (green checkmarks), and the third step, **Query or Static Values**, is currently active (blue circle). The dialog is divided into two main columns: **List Entry Label** and **Target Page ID or custom URL**. The **List Entry Label** column has five rows. The first three rows contain the text `View Customers`, `View Products`, and `View Orders` respectively, each followed by a question mark icon. The last two rows are empty. The **Target Page ID or custom URL** column has five rows. The first three rows contain the values `4`, `7`, and `12` respectively, each followed by a list icon (three horizontal lines with a dot). The last two rows are empty. At the bottom left is a **Cancel** button, and at the bottom right is a **Next >** button, which is highlighted with a red rectangle.

	List Entry Label	Target Page ID or custom URL
1	View Customers	4
2	View Products	7
3	View Orders	12
4		
5		

- g. Accept the default and click **Create List**.

Create List

×

✓ ✓ ✓ ●

Confirm

Create List Regions? Do not create list region(s) ▼ ?

	List Entry Label	Target Page ID or custom URL
1	View Customers	4
2	View Products	7
3	View Orders	12

Cancel
Create List

The **Home Page List** is created.

↑
Application 111 \ Shared Components \ Lists

Lists
List Details
Unused
Conditional Entries

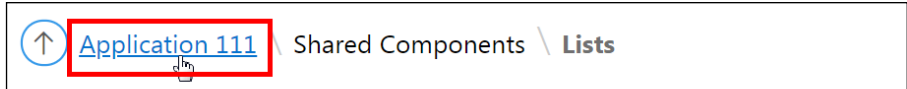
Q
▼

Go

☐☐☐☐

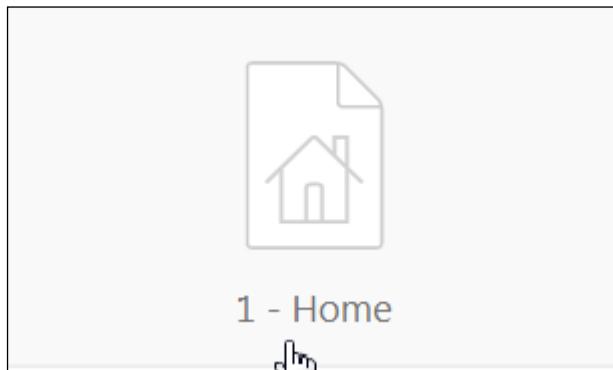
Name ↑≡	Type	Entries
Desktop Navigation Bar	Static	3
Desktop Navigation Menu	Static	4
Home Page List	Static	3

- h. Click the **Application 111** breadcrumb.

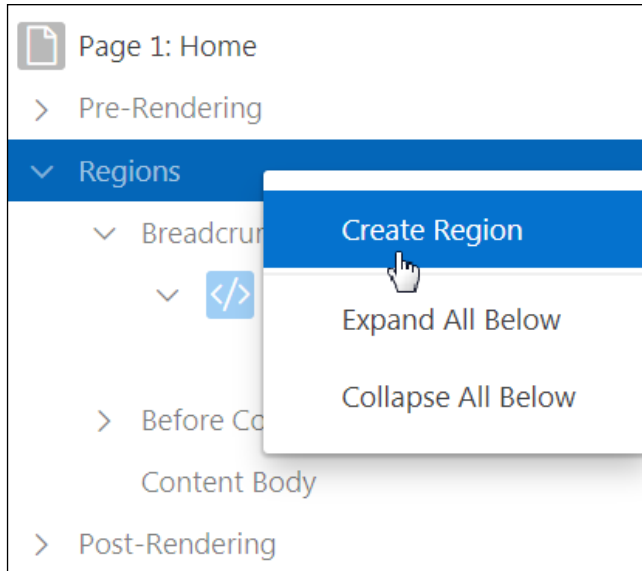


Application 111 \ Shared Components \ Lists				
Lists	List Details	Unused	Conditional Entries	Utilization
Q v Go   Acti				
Name ↑	Type	Entries	References	Entries Updated
Desktop Navigation Bar	Static	3	0	-
Desktop Navigation Menu	Static	4	0	-
Home Page List	Static	3	0	1 seconds ago

2. Create a list region called **Tasks** in the Home region.
- a. On the GMT application home page, click **1 - Home**.

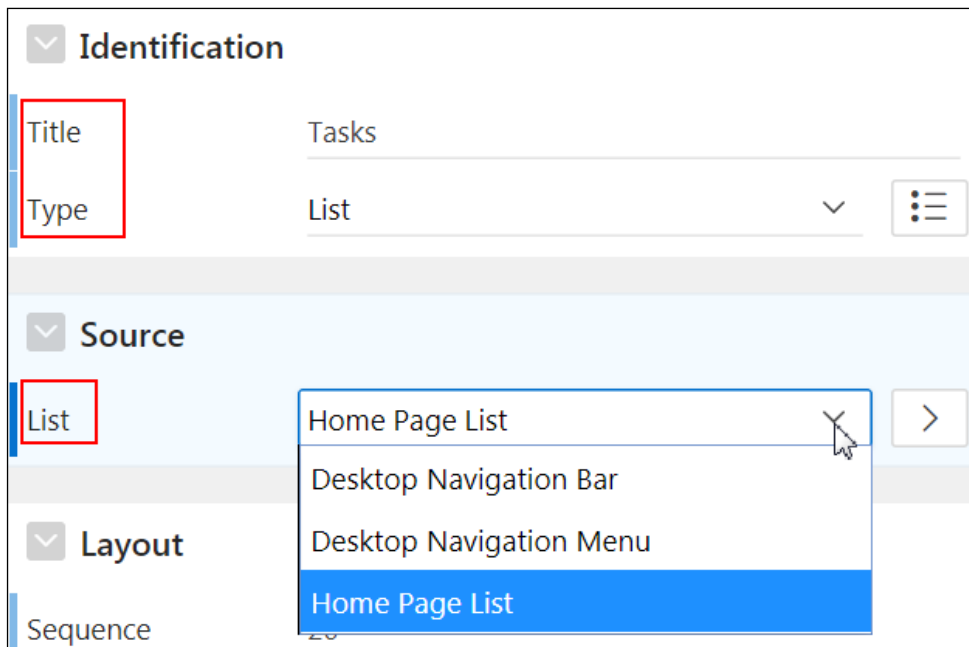


- b. On the Rendering tab, right-click **Regions** and select **Create Region**.

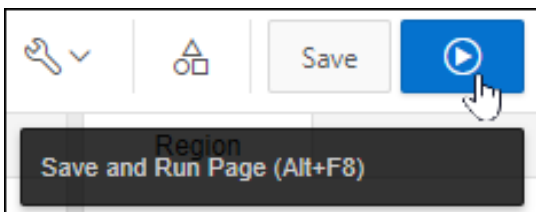


- c. On the Region tab, enter the following value for the respective fields:

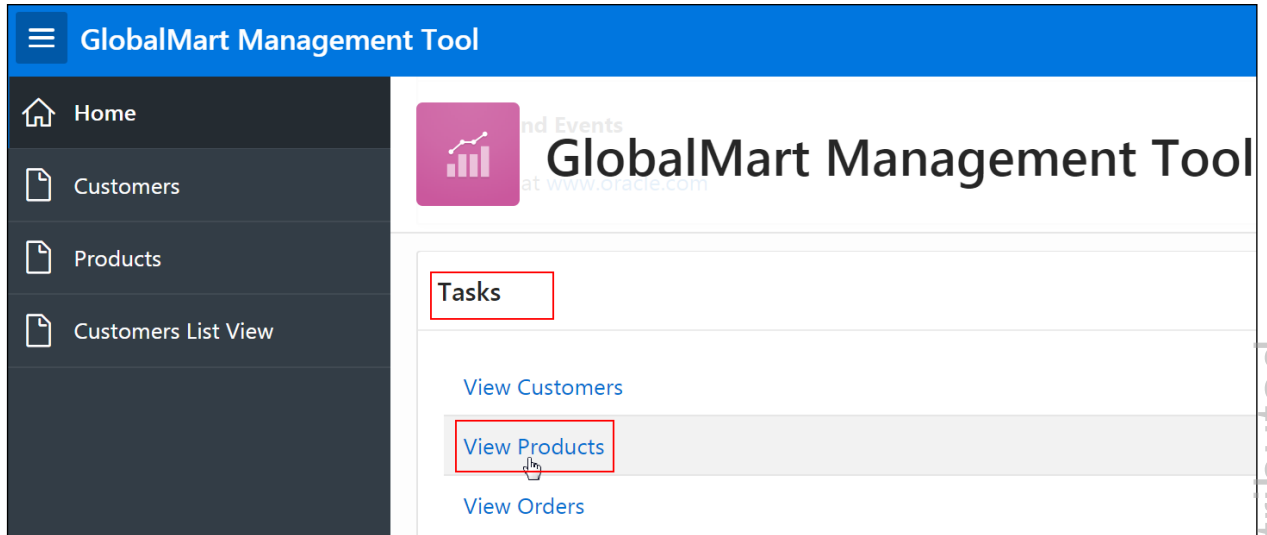
- **Identification > Title:** Enter **Tasks**
- **Identification > Type:** Select **List**
- **Source > List:** Select **Home Page List**



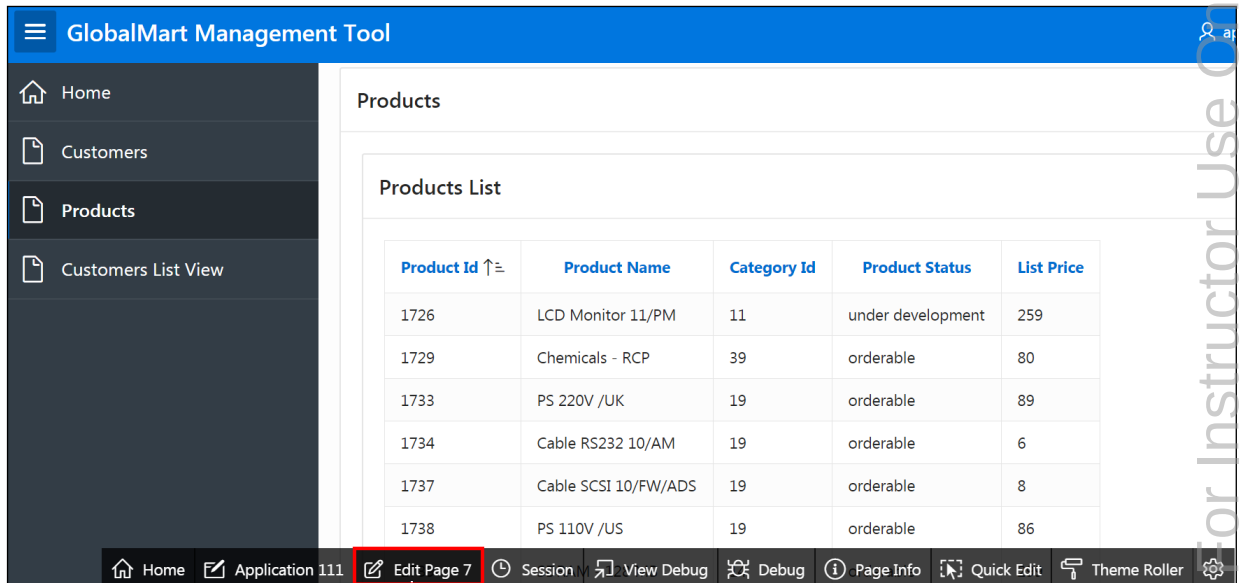
- d. To view the list, run the page. Click the **Save and Run Page** icon.



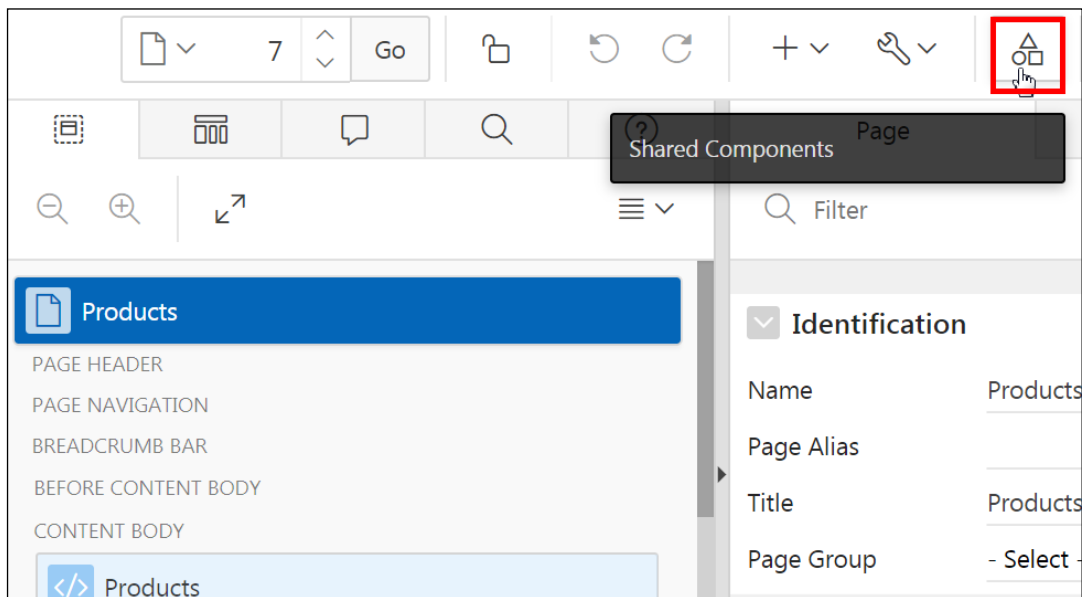
- e. The list is displayed. Click the **View Products** link.



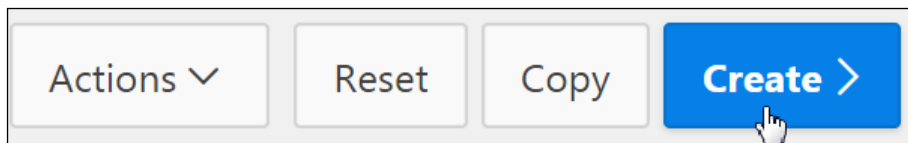
- f. The Products page is displayed. Click the **Edit Page 7** link on the Developer toolbar.



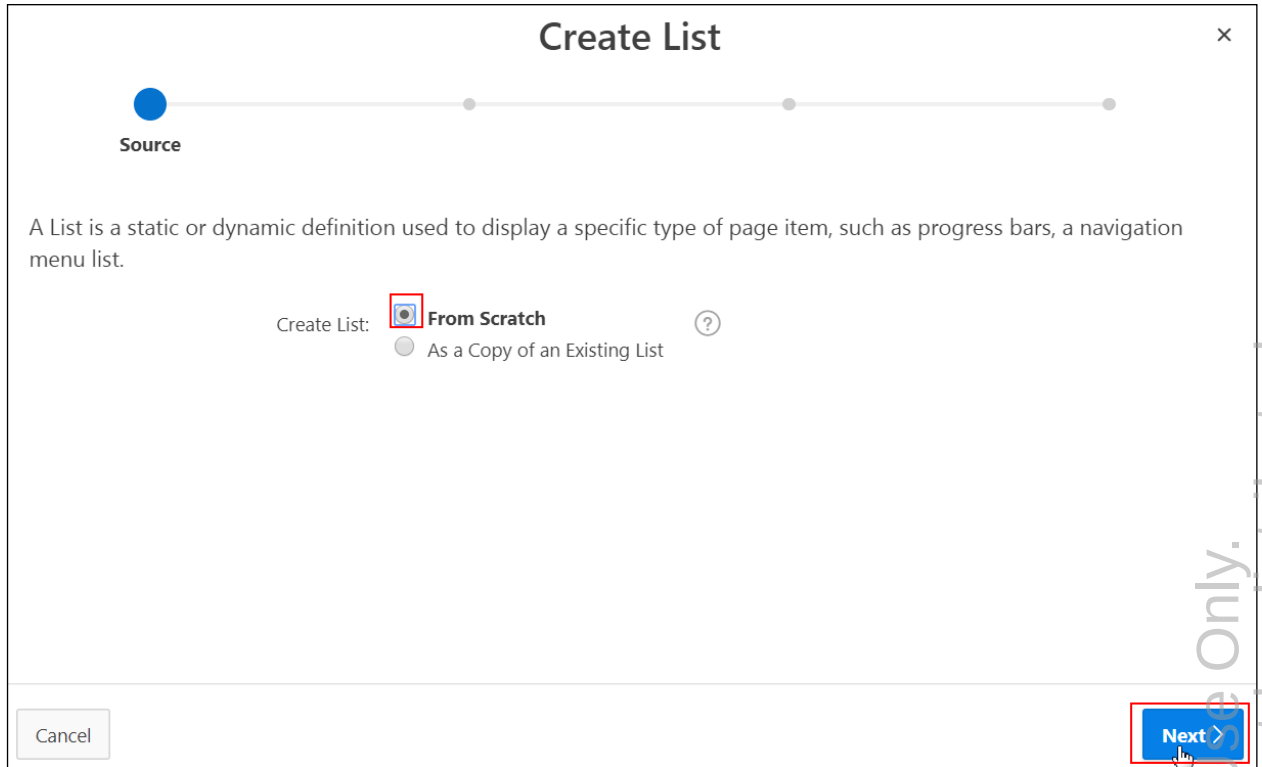
3. Create a Dynamic List showing all the Products on the Products page. Create the List Region on the Products page by using the same Create List Wizard.
 - a. **Page 7: Products** open in Page Designer view. On the **Products** page, click the **Shared Components** icon.



- b. In the **Navigation** section, click **Lists**.
 - c. Click **Create >**.

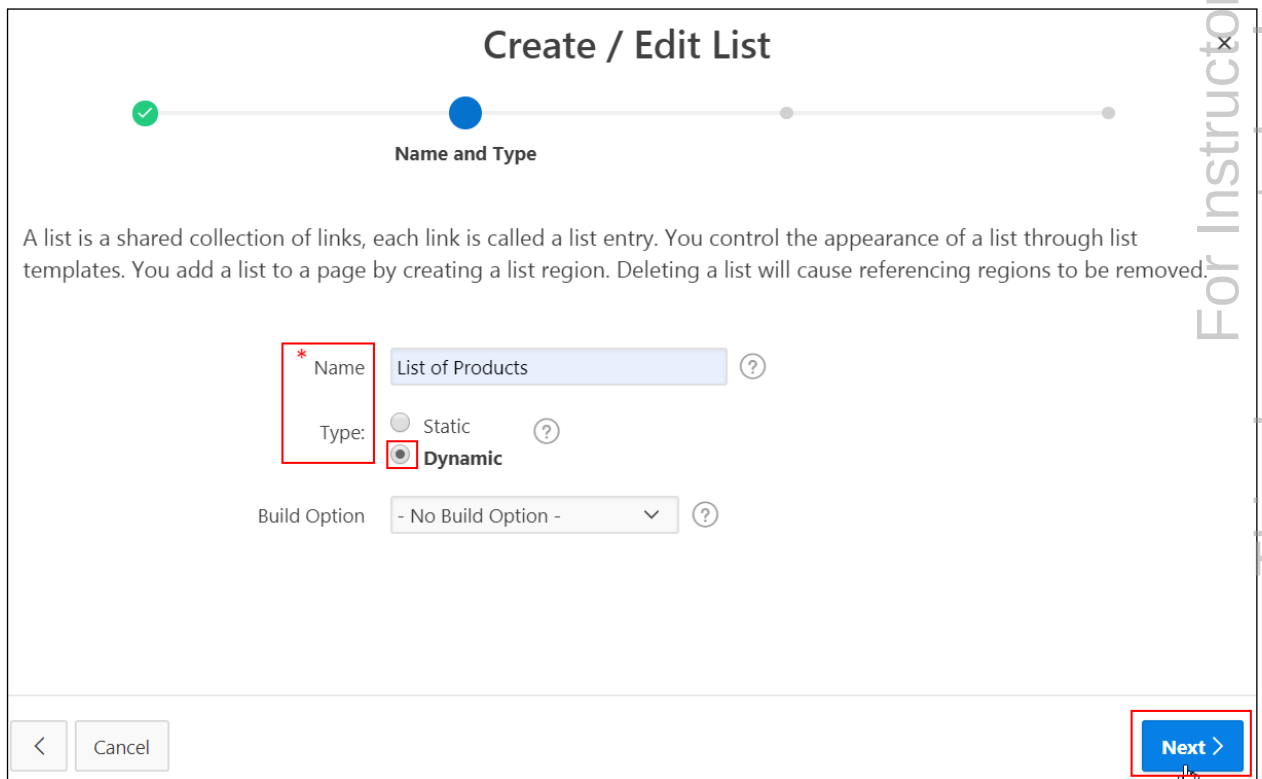


- d. Accept the default and click **Next >**.



The 'Create List' dialog box features a progress bar at the top with four dots; the first dot is blue and labeled 'Source'. Below the progress bar, a text description states: 'A List is a static or dynamic definition used to display a specific type of page item, such as progress bars, a navigation menu list.' Under the heading 'Create List:', there are two radio button options: 'From Scratch' (which is selected and highlighted with a red box) and 'As a Copy of an Existing List'. At the bottom of the dialog, there is a 'Cancel' button on the left and a 'Next >' button on the right, which is also highlighted with a red box.

- e. Enter **List of Products** for **Name** and select **Dynamic** for **Type**. Click **Next >**.



The 'Create / Edit List' dialog box shows a progress bar with four dots; the second dot is blue and labeled 'Name and Type', while the first dot is green. The text description reads: 'A list is a shared collection of links, each link is called a list entry. You control the appearance of a list through list templates. You add a list to a page by creating a list region. Deleting a list will cause referencing regions to be removed.' In the form area, the 'Name' field (marked with a red asterisk and highlighted with a red box) contains the text 'List of Products'. Below it, the 'Type' section has two radio button options: 'Static' and 'Dynamic' (which is selected and highlighted with a red box). At the bottom, there is a 'Build Option' dropdown menu currently set to '- No Build Option -'. The bottom of the dialog features a '< Cancel' button on the left and a 'Next >' button on the right, which is highlighted with a red box.

- f. Select **Build Query**.

Create List

Query or Static Values

Query Source Type SQL Query ?

* SQL Query: ?

Build Query

> Examples

< Cancel Next >

- g. In the Create Dynamic List window, select OEHR_PRODUCTS for **Table or View**. Click **Next >**.

Create Dynamic List

* Table / View Owner OEHR ?

* Table or View OEHR_PRODUCTS ?

Next >

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- h. Select `PRODUCT_NAME (Nvarchar2)` for **Label Column** and `CATALOG_URL (Varchar2)` for **Target Column**. Click **Next >**.

Create Dynamic List

Owner: **OEHR** (?)

Table: **OEHR_PRODUCTS** (?)

* Label Column (?)

Target Column (?)

URL (?)

Is Current Column (?)

Image Column (?)

- i. Click **Finish**.

Create Dynamic List

```
select null as level_value
      , "PRODUCT_NAME" as label_value
      , "CATALOG_URL" as target_value
      , null as is_current
      , null as image_value
      , null as image_attr_value
      , null as image_alt_value
from "OEHR_PRODUCTS"
order by 1
```

Finish

- j. Click **Next >**.

Create List

Query or Static Values

Query Source Type: ?

* SQL Query: ?

```
select null as level_value
      , "PRODUCT_NAME" as label_value
      , "CATALOG_URL" as target_value
      , null as is_current
      , null as image_value
      , null as image_attr_value
      , null as image_alt_value
from "OEHR_PRODUCTS"
order by 1
```

> **Examples**

- k. Select **Create list region on current page** for **Create List Regions?** and **Page Template Body (3)** for **Region Position**. Click **Create**.

Create List

List Name **List of Products** ?

Create List Regions? Create list region on current page ?

* Region Position Page Template Body (3) ?

Region Template No Template ?

List Template: Links List ?

List Query **select null as level_value
 , "PRODUCT_NAME" as label_value
 , "CATALOG_URL" as target_value
 , null as is_current
 , null as image_value
 , null as image_attr value** ?

Cancel **Create**

- l. The **List of Products** Dynamic list is created. Click the **Run Page 7** icon.

Application 111 \ Shared Components \ **Lists**

Lists List Details Unused Conditional Entries

Q | Go

Name ↑	Type	Entries
Desktop Navigation Bar	Static	3
Desktop Navigation Menu	Static	4
Home Page List	Static	3
List of Products	Dynamic	0



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- m. Scroll to the bottom of the page. The list appears. You will now make the list collapsible and **Collapsed** by default. Click the **Edit Page 7** link on the Developer toolbar.

Products

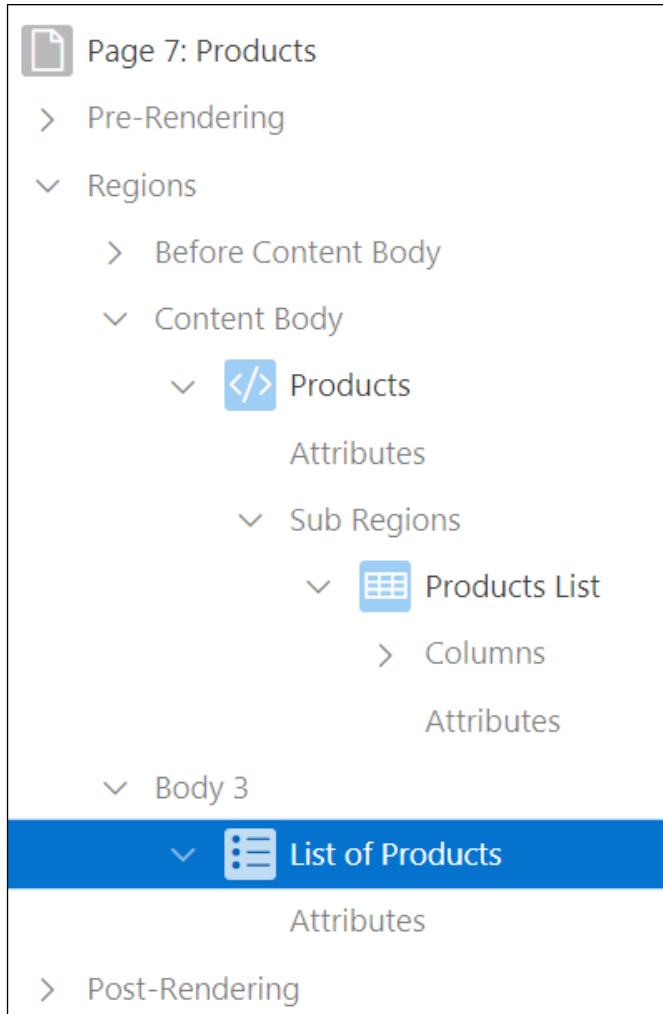
Product List

Product Id ↑	Product Name	Category Id	Product Status	List Price
1726	LCD Monitor 11/PM	11	under development	259
1729	Chemicals - RCP	39	orderable	80
1733	PS 220V /UK	19	orderable	89
1734	Cable RS232 10/AM	19	orderable	6
1737	Cable SCSI 10/FW/ADS	19	orderable	8
1738	PS 110V /US	19	orderable	86
1739	SDRAM - 128 MB	14	orderable	299
1740	TD 12GB/DAT	17	orderable	134
1742	CD-ROM 500/16x	17	orderable	101
1743	HD 18.2GB @10000 /E	13	planned	800

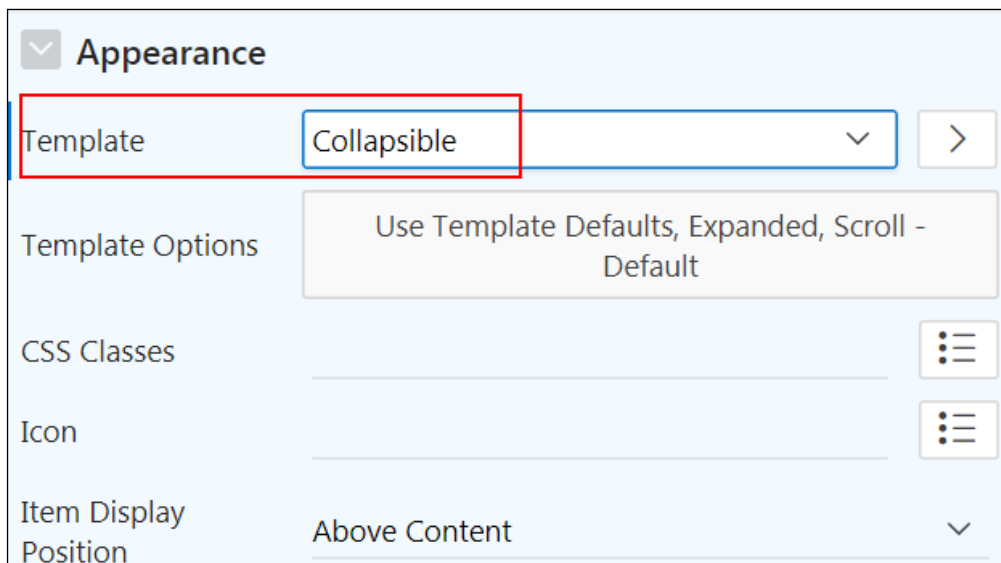
Mouse Pad /CL
Wrist Pad
Wrist Pad /CL
CD-ROM 500/16x
CD-ROM 600/E/24x
CD-ROM 600/I/24x
CD-ROM 600/I/32x
CD-ROM 8x
CDW 12/24
CDW 20/48/E
CDW 20/48/I
DFD 1.44/3.5
DVD 12x

Home Application 111 Edit Page 7 Session View Debug Debug Page Info Quick Edit Theme Roller

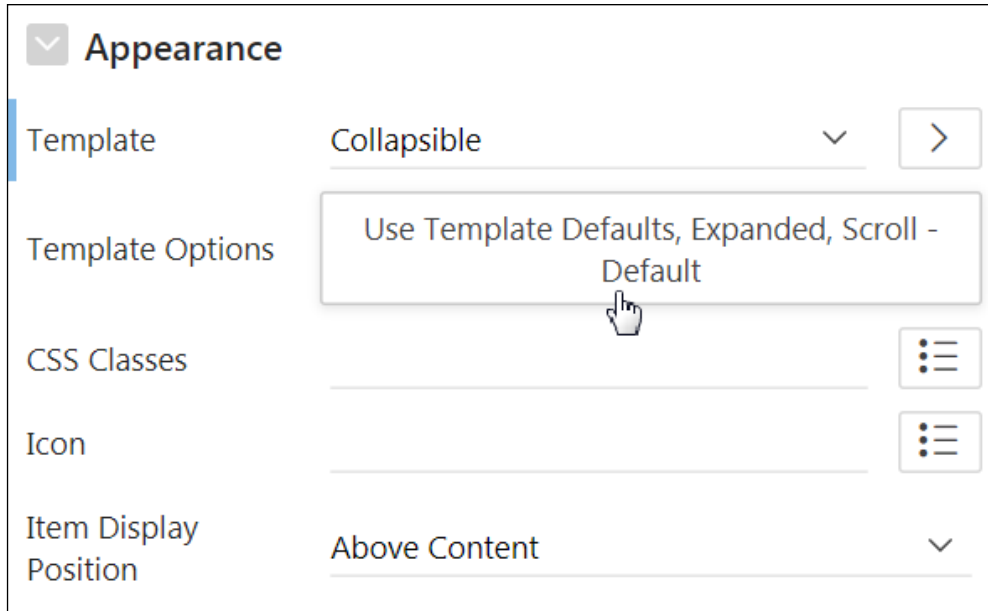
- n. Select **Regions > Body 3 > List of Products** on the Rendering tab.



- o. In the Property Editor, select **Appearance > Template > Collapsible**.

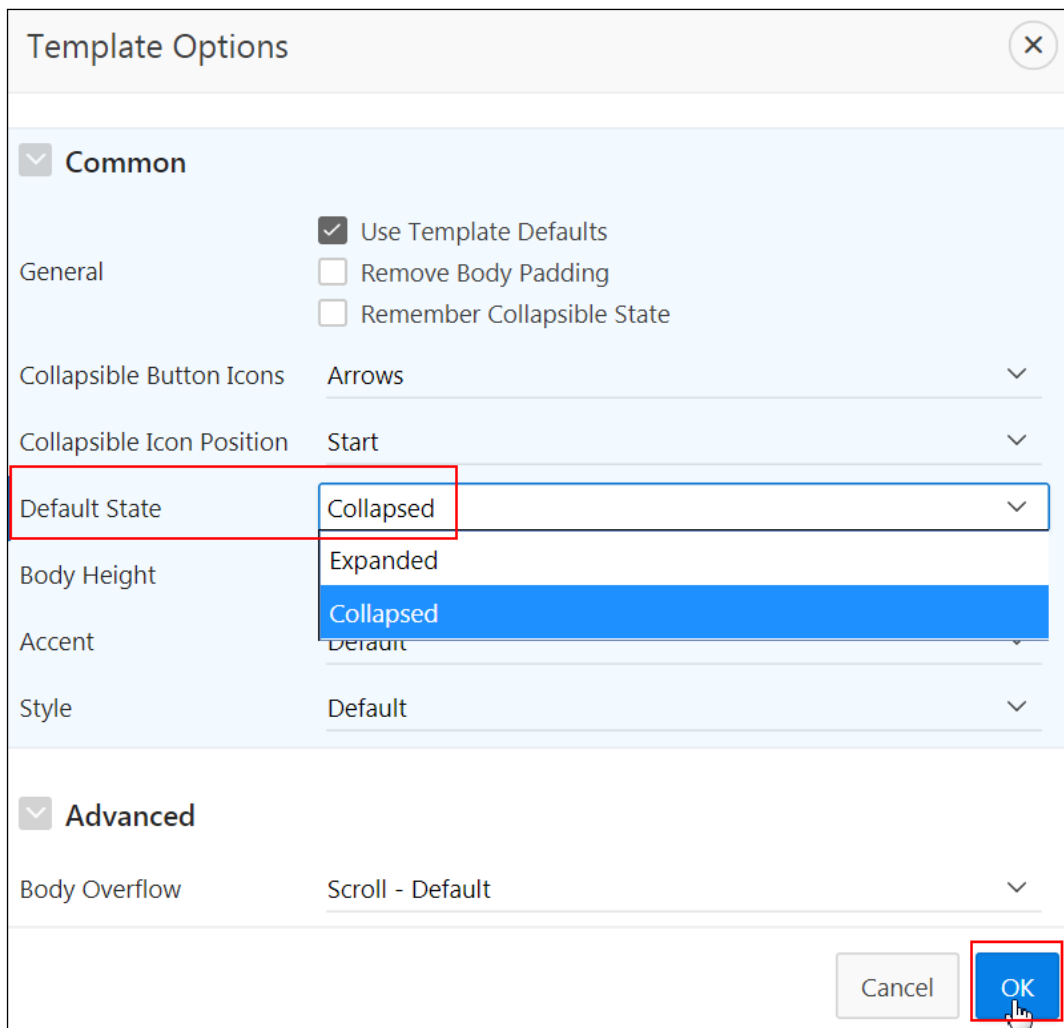


- p. Click the button in the **Appearance > Template Options** field.



The screenshot shows a settings panel with a dropdown menu open for the 'Template Options' field. The dropdown menu displays the text 'Use Template Defaults, Expanded, Scroll - Default'. A mouse cursor is pointing at this text. Other fields in the panel include 'Template' (set to 'Collapsible'), 'CSS Classes', 'Icon', and 'Item Display Position' (set to 'Above Content').

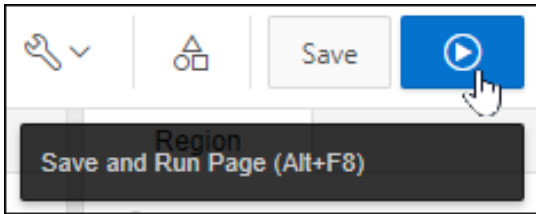
- q. Change the **Common > Default State** to **Collapsed** and click **OK**.



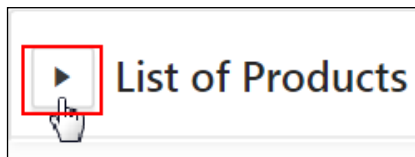
The screenshot shows the 'Template Options' dialog box. The 'Common' section is expanded, and the 'Default State' dropdown menu is open, showing 'Collapsed' as the selected option. The 'OK' button at the bottom right is highlighted with a red box. Other options in the 'Common' section include 'Use Template Defaults' (checked), 'Remove Body Padding' (unchecked), 'Remember Collapsible State' (unchecked), 'Collapsible Button Icons' (set to 'Arrows'), and 'Collapsible Icon Position' (set to 'Start'). The 'Advanced' section is also visible, showing 'Body Overflow' set to 'Scroll - Default'.

Note: By default, the collapsible region is expanded.

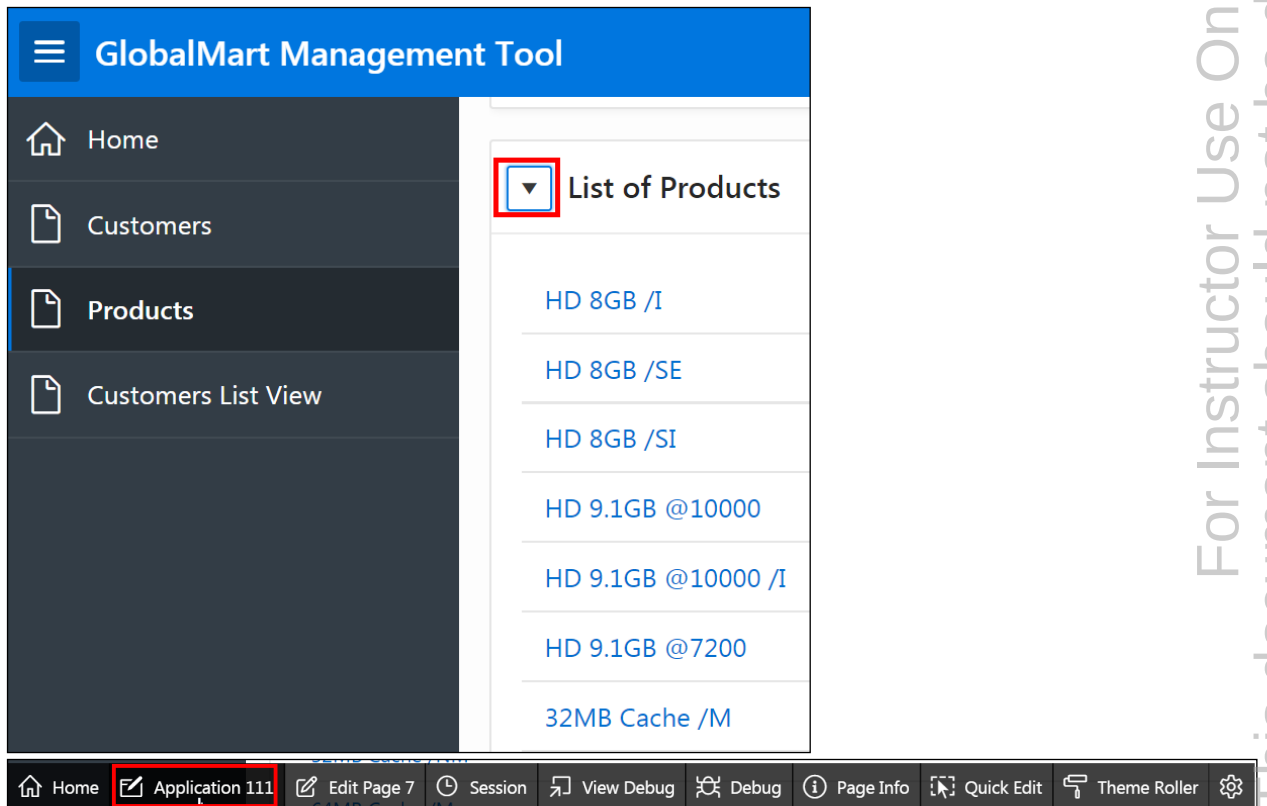
- r. Click the **Save and Run Page** icon.



- s. Notice that the List of Products is now displayed in a collapsible region and the region is collapsed by default. Expand the List of Products region by clicking the arrow next to it.



- t. The list is now expanded. Click the **Application 111** link on the Developer toolbar.



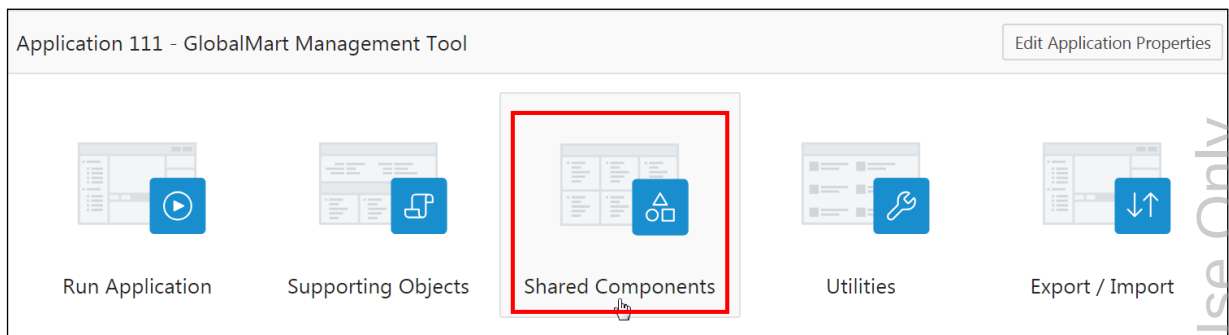
Solution 14-2: Creating and Editing a Navigation Menu

Overview

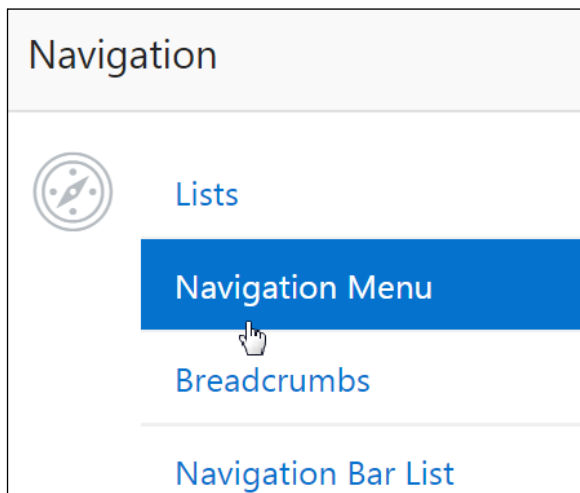
In this practice solution, steps are provided for creating and editing a navigation menu for the GMT application.

Steps

1. Create a navigation menu called **Orders** for the GMT application. It should be the current tab for the List of Orders page. Sequence the Orders entry before the Products navigation entry.
 - a. Click the **Shared Components** icon.



- b. In the **Navigation** section, click **Navigation Menu**.



- c. Click the **Desktop Navigation Menu** link.

Lists List Details Unused Conditional Entries Utilization					
Go Grid Table Actions					
☐ ☒ Navigation Menu					
Name ↑	Type	Entries	References	Entries Updated	
Desktop Navigation Menu	Static	4	0	-	

Click the **Create Entry >** link to add a new navigation entry. Note the **Sequence** number of the Products entry.

Desktop Navigation Menu Grid Edit Edit List Create Entry >									
Go Rows 50 Actions									
Sequence ↑	Name	Parent Entry	Target	Conditional	Updated	Level	Authorization Scheme	Copy	
10	Home	-	f?p=&APP_ID.:1:&APP_SESSION.::&DEBUG.::	-	-	1	-		
20	Customers	-	f?p=&APP_ID.:4:&SESSION.::&DEBUG.::	-	-	1	-		
30	Products	-	f?p=&APP_ID.:7:&SESSION.::&DEBUG.::	-	-	1	-		
40	Customers List View	-	f?p=&APP_ID.:10:&SESSION.::&DEBUG.::	-	-	1	-		

- d. Enter the following values for the respective fields:
- Enter 25 for sequence. (**Note:** The sequence value should be between the Customers and Products sequence value.)
 - Enter **Orders** for **List Entry Label**.
 - Select **Page in the Application** for **Target Type**.
 - Select the **List of Orders** page number (Page 12) for **Page**.
 - Select **Comma Delimited Page List** for **List Entry Current for Page Type**.
 - Enter the **List of Orders** page number (Page 12) for **List Entry Current for Condition**.

[Show All](#)
[Entry](#)
[Target](#)
[Current ...](#)
[Conditio...](#)
[Authoriz...](#)
[Configur...](#)
[Click Co...](#)

Parent List Entry - No Parent List Item - ?

Sequence 25 ?

Image/Class ?

Attributes ?

Alt Attribute ?

* List Entry Label Orders

Target

Target type Page in this Application ?

* Page 12 ?

☐ reset pagination for this page ?

☐ Printer Friendly ?

List Entry Current for Pages Type Comma Delimited Page List ?

List Entry Current for Condition 12 ?

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- e. Click the **Create List Entry** button.

The screenshot shows the 'List Entry' form. At the top, there are navigation buttons: '<', '>', 'Cancel', 'Create and Create Another', and 'Create List Entry' (highlighted with a red box). Below these are tabs: 'Show All', 'Entry', 'Target', 'Current ...', 'Conditio...', 'Authoriz...', 'Configur...', 'Click Co...', 'User Def...', and 'Develop...'. The form fields include: 'Parent List Entry' (dropdown menu with '- No Parent List Item -'), 'Sequence' (text input with '25'), and 'Image/Class' (text input with a menu icon). A red box highlights the 'Create List Entry' button.

2. Assign the Master Detail page to the Orders navigation entry. Run the application and confirm that the navigation is created successfully.

- a. Click the **Grid Edit** button.

The screenshot shows the 'List Details' view for 'Desktop Navigation Menu'. At the top, there is a green banner with 'Action processed.' and a close button. Below are tabs: 'Lists', 'List Details' (selected), 'Unused', 'Conditional Entries', 'Utilization', and 'History'. The 'List' dropdown is set to 'Desktop Navigation Menu'. At the bottom right, there are buttons: 'Grid Edit' (highlighted with a red box), 'Edit List', and 'Create Entry >'.

- b. Select Orders row and click the **Edit** button. Add the Master Detail form **Oehr Orders** page number for **Orders** (Page 13) in the **Pages** field of the **Orders** row. Delimit the page numbers by using a comma.

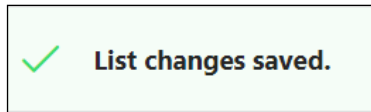
List: Desktop Navigation Menu

Search: All Text Columns					
Go					
Actions					
Edit					
Save					
Add Row					
Reset					
		Sequence ↑	Text	Target	Pages
☐	≡	10	Home	f?p=&APP_ID.:1:&APP_SESSION.::&DEBUG.:	-
☐	≡	20	Customers	f?p=&APP_ID.:4:&SESSION.::&DEBUG.	4
☒	≡	25	Orders	f?p=&APP_ID.:12:&SESSION.::&DEBUG.:	12,13
☐	≡	30	Products	f?p=&APP_ID.:7:&SESSION.::&DEBUG.	7
☐	≡	40	Customers List View	f?p=&APP_ID.:10:&SESSION.::&DEBUG.	10,11

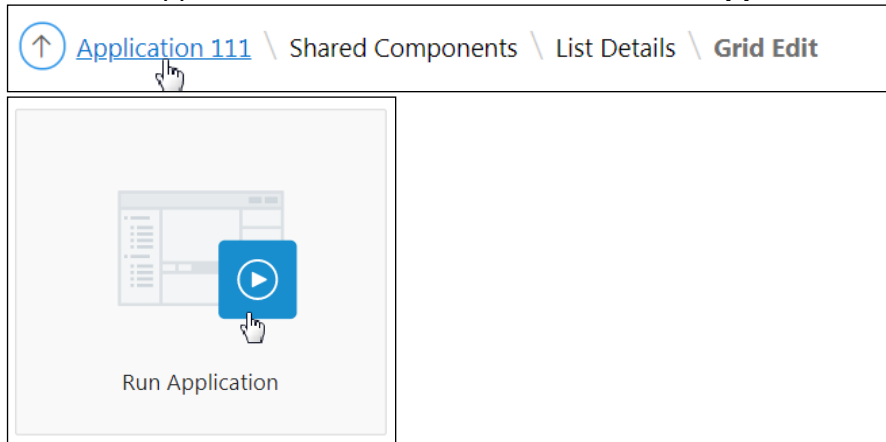
- c. Click **Save**.

List: Desktop Navigation Menu					
Search: All Text Columns Go Actions Edit Save Add Row Reset					
☐		Sequence ↑	Text	Target	Pages
☐		10	Home	f?p=&APP_ID.:1:&APP_SESSION.:::&DEBUG.:	-
☐		20	Customers	f?p=&APP_ID.:4:&SESSION.:::&DEBUG.	4
☒		25	Orders	f?p=&APP_ID.:12:&SESSION.:::&DEBUG.:	12,13
☐		30	Products	f?p=&APP_ID.:7:&SESSION.:::&DEBUG.	7
☐		40	Customers List View	f?p=&APP_ID.:10:&SESSION.:::&DEBUG.	10,11

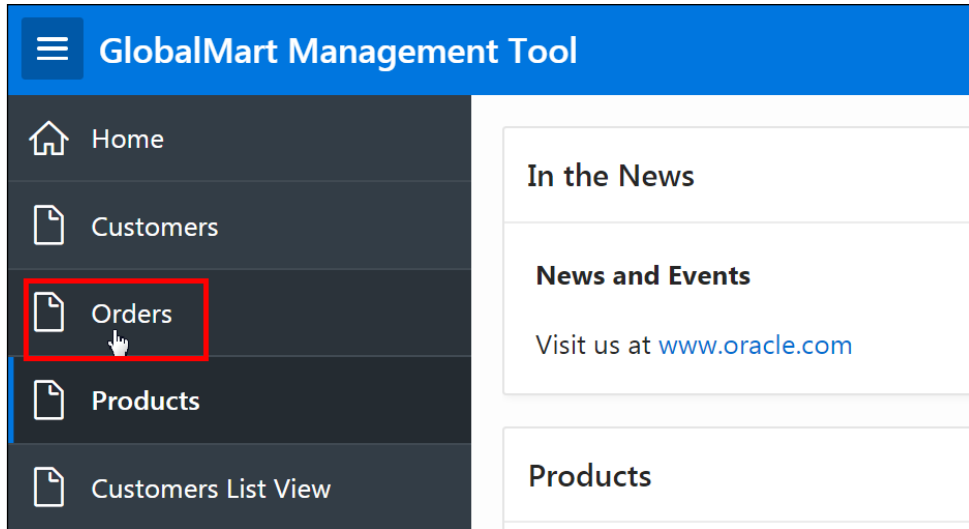
You get the message “List changes saved”.



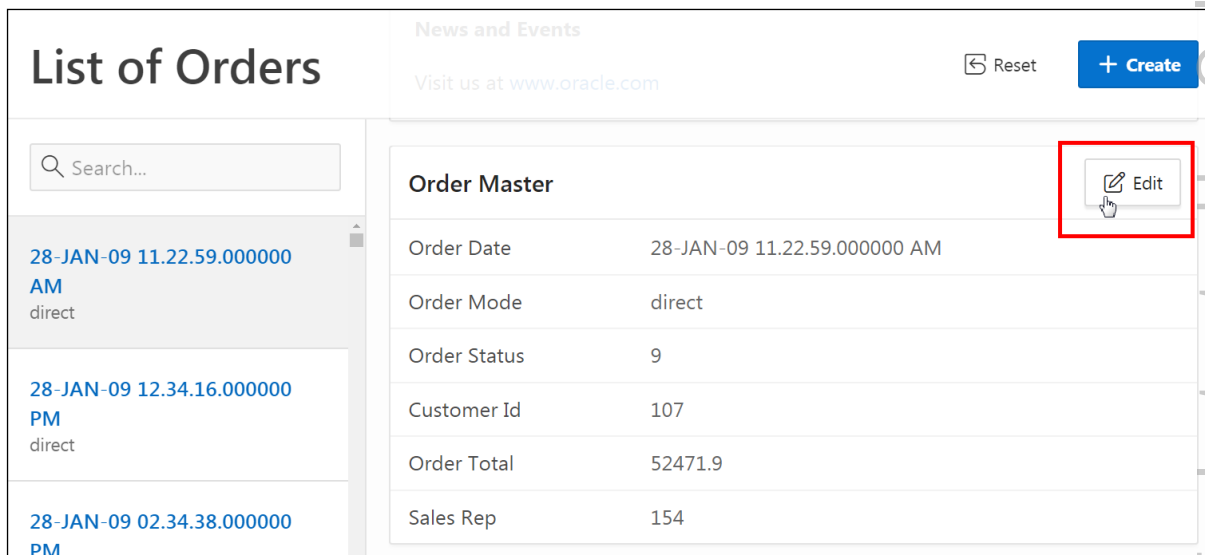
- d. Select the Application 111 breadcrumb and click **Run Application**.



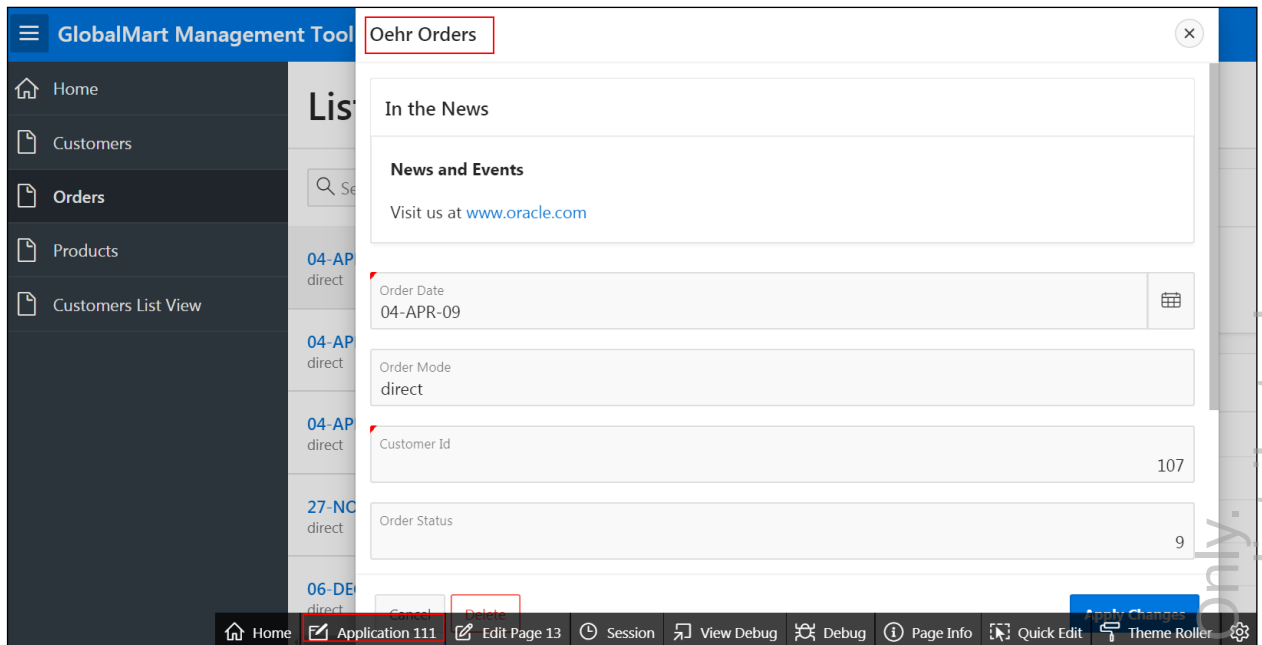
- e. The application is displayed. Click the **Orders** tab.



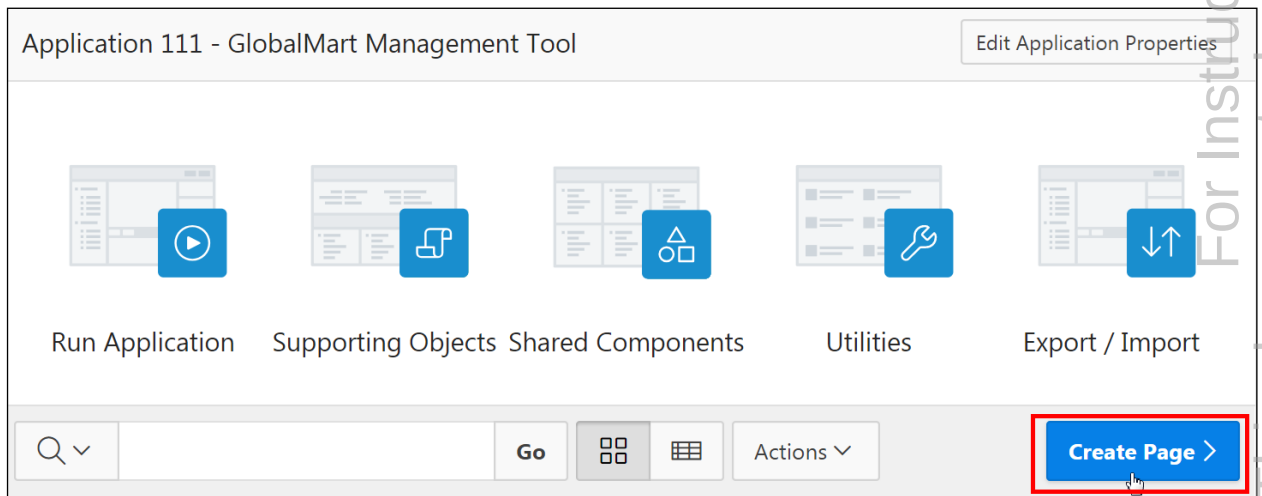
- f. The List of Orders page is displayed. Click the **Edit** icon for the first entry.



- g. The Master Detail form page **Oehr Orders** (13) is displayed. Notice that the **Orders** tab is still active. Click the **Application 111** link on the Developer toolbar.



3. Create a page called **Help**, and then add a new navigation menu entry called **Help** to the existing navigation menu. The new Help entry will link to the new Help page.
- a. You want to create a new page that will contain the help information. Click **Create Page >**.



- b. Select **Blank Page**.

Create a Page

Page Type

Component Feature ?


Blank Page


Report


Form


Master Detail


Dashboard


Calendar


Tree


Wizard

- c. For **Page Number**, enter 20. For **Name**, enter `Help`, and click **Next >**.

Create a Blank Page

Page Attributes

* Page Number

20 ?

* Name

Help ?

* Page Mode

Normal Modal Dialog Non-Modal Dialog ?

Page Group

- Select Page Group - ?

Breadcrumb

- don't use breadcrumbs on page - ?

> **Optional Static Content Regions**

- d. Select the **Create a new navigation menu entry** option for **Navigation Preference**. Ensure **Help** is selected for **New Navigation Menu Entry** and then click **Next >**.

Create a Blank Page

Navigation Menu

Navigation Preference ☐ Do not associate this page with a navigation menu entry ?

☒ **Create a new navigation menu entry**

☐ Identify an existing navigation menu entry for this page

* New Navigation Menu Entry ?

Parent Navigation Menu Entry

- No parent selected -

Home

Customers

Orders

Products

Customers List View

?

- e. Click **Finish**.

Create a Blank Page

✓

✓

Confirm

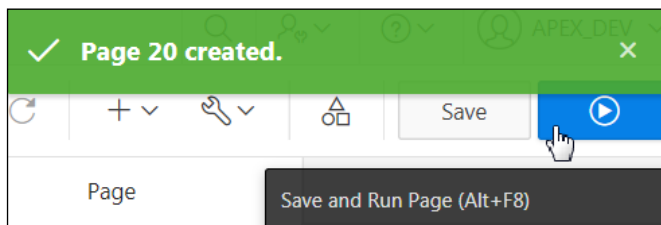
You have requested to create a page with the following attributes. Please confirm your selections.

Application	111
Page	20
Page Name	Help
Page Title	Help

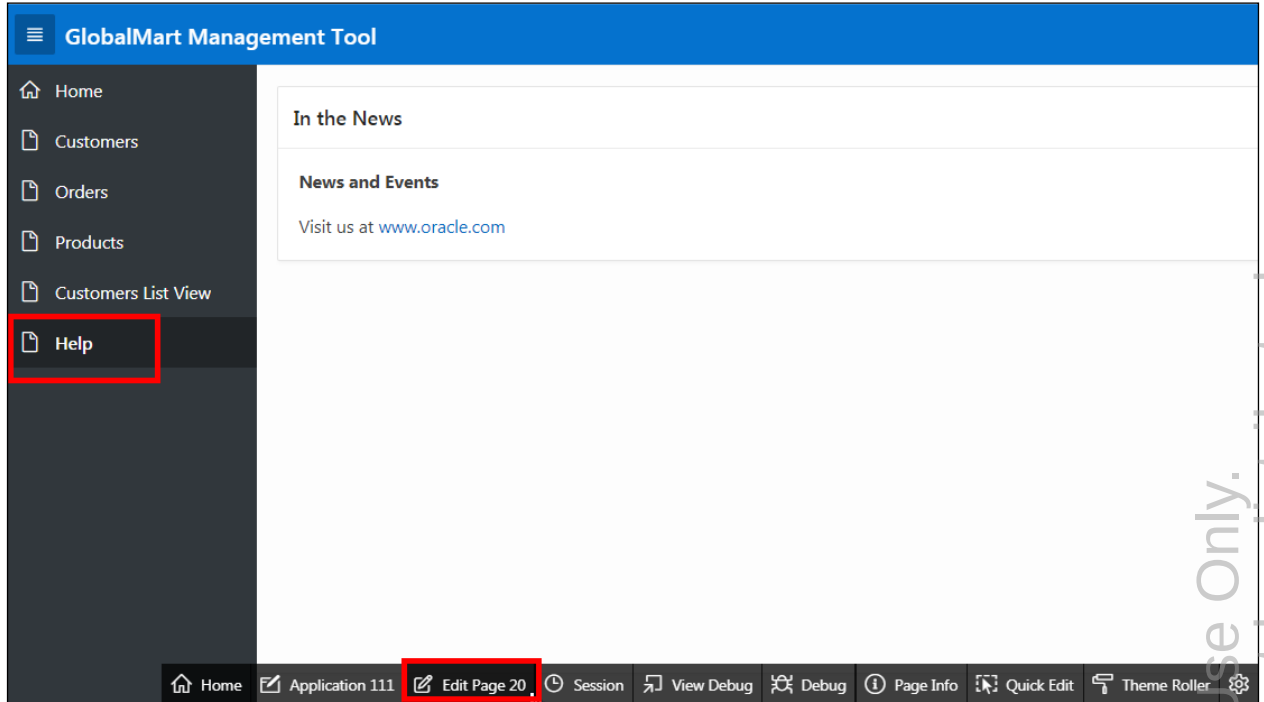
< Cancel

Finish

- f. Run the page to see the new navigation menu entry. Click the **Save and Run Page** icon.



- g. The navigation menu entry is displayed. In the next practice, you add a region to the Help page and add it as a navigation bar entry. Click the **Edit Page 20** link on the Developer toolbar.



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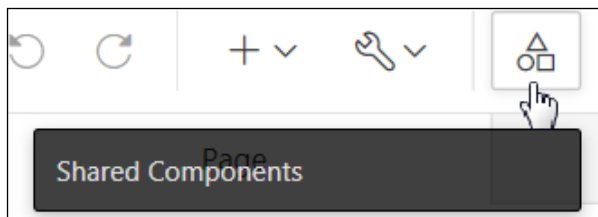
Solution 14-3: Adding Navigation Bar Entries

Overview

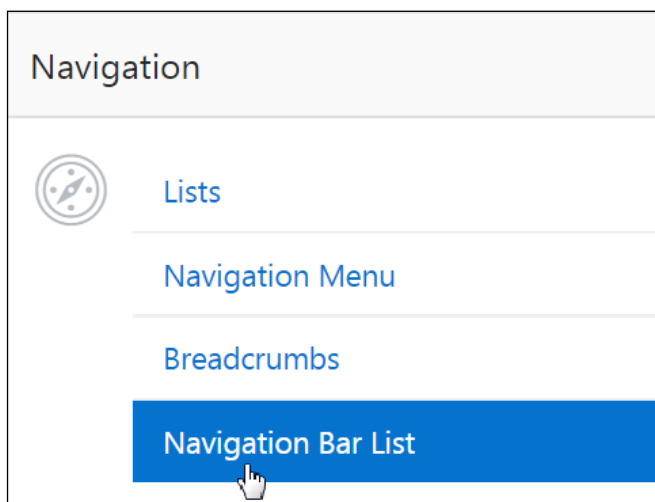
In this practice solution, steps are provided for adding navigation bar entries to the GMT application.

Steps

1. Create a navigation bar entry called **Help** that navigates to the Help page. Run the page and click the Help navigation bar entry.
 - a. Click the **Shared Components** icon.



- b. Under **Navigation**, click **Navigation Bar List**.



- c. Click the **Desktop Navigation Bar** link.

Name	Type	Entries	References	Entries Updated
Desktop Navigation Bar	Static	3	0	-

- d. Click **Create Entry >**.

Sequence	Name	Parent Entry	Target	Conditional	Updated	Level	Authorization Scheme	Copy
10	&APP_USER.	-	#	-	-	1	-	
20	---	&APP_USER.	separator	-	-	2	-	
30	Sign Out	&APP_USER.	&LOGOUT_URL.	-	-	2	-	

- e. Enter the following values for the respective fields:

- **Entry > List Entry Label:** Enter `Help`
- **Target Type:** Select `Page in this Application`
- **Target > Page:** Select `20`
- **Target > Request:** Enter `&APP_PAGE_ID.`

Note: By specifying the `&APP_PAGE_ID.` bind variable as the request, you are instructing the Oracle Application Express engine to display the Help text for the current page when the user clicks the navigation bar entry link.

Click **Create List Entry**.

< > List Entry Attributes
Cancel Create and Create Another Create List Entry

Show All **Entry** Attributes Target Current List... Conditions Authorizati... Configurati... Click Count... User Define... Developer...

List Entry Label

Target

Target type Page in this Application ?

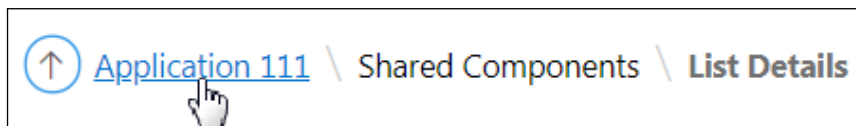
* Page ?

☐ reset pagination for this page ?

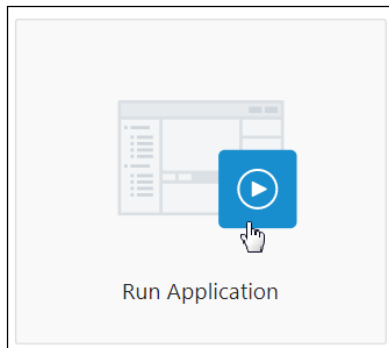
☐ Printer Friendly ?

Request ?

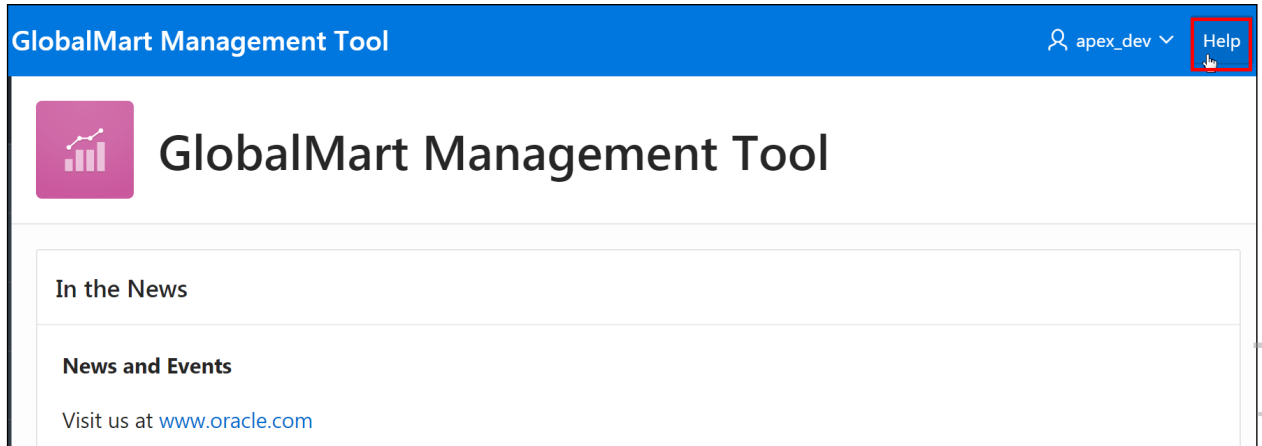
- f. Click the **Application 111** breadcrumb.



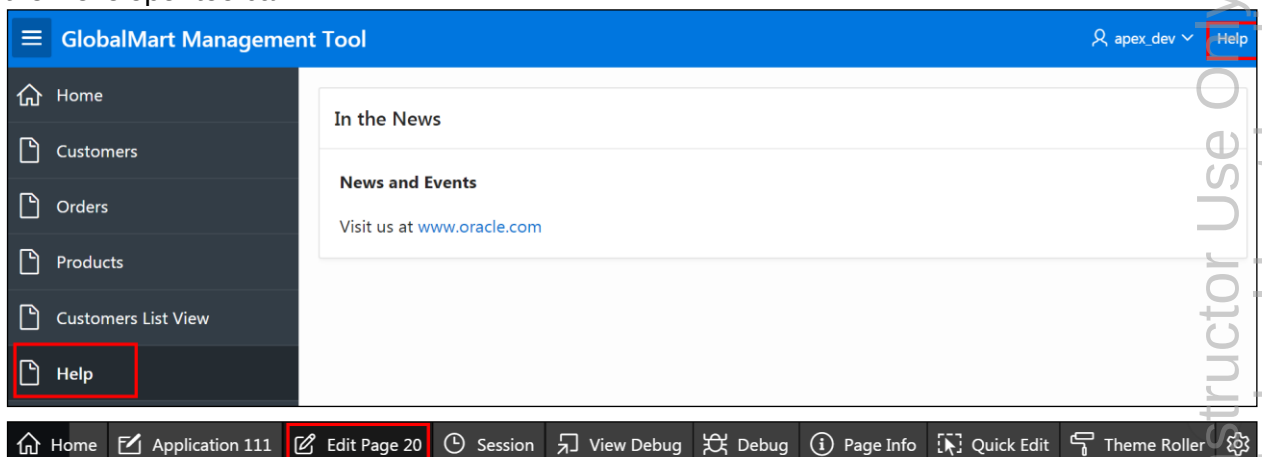
- g. Click **Run Application**.



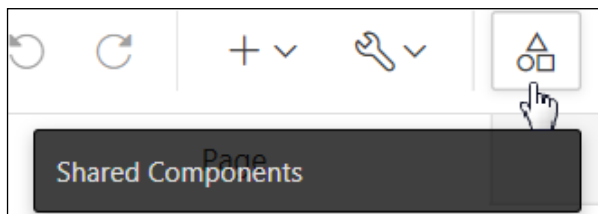
- h. Click the **Help** navigation bar entry.



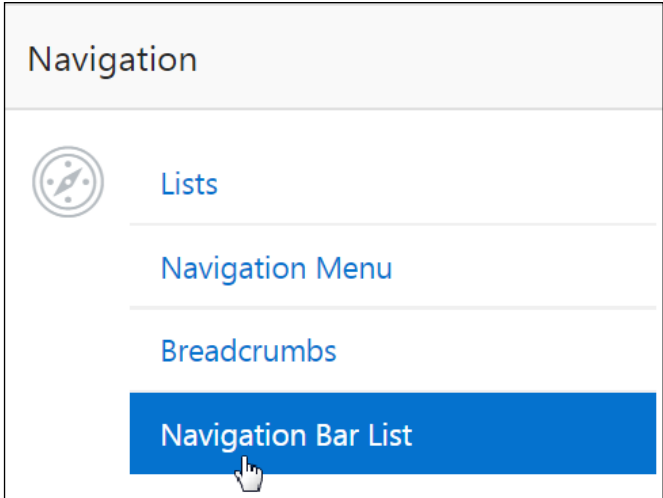
- i. The Help page is displayed. You want to modify the Navigation Bar such that the **Help** button is displayed before **apex_dev** (&APP_USER.). Click the **Edit Page 20** link on the Developer toolbar.



- j. Click the **Shared Components** icon.



- k. Under **Navigation**, click **Navigation Bar List**.



- l. Click the **List Details** tab.

Application 111 \ Shared Components \ Lists

Lists List Details Unused Conditional Entries Utilization History

Search: [] Go [Grid Icon] [Table Icon] Actions [v]

Navigation Bar List

Name ↑	Type	Entries	References	Entries Updated	List Updated
Desktop Navigation Bar	Static	4	0	3 minutes ago	3 minutes ago

- n. Select the check box next to Help and click **Edit**.

List: Desktop Navigation Bar

Search: All Text Columns Go Actions ▾ Edit Save

☐		Sequence ↑	Text	
☒		10	&APP_USER.	#
☐		20	---	separator
☐		30	Sign Out	&LOGOUT_URL.
☒		40	Help	f?p=&APP_ID.:20:&SESSION

- o. Change the **Sequence** of the Help entry to 5 and click **Save**.

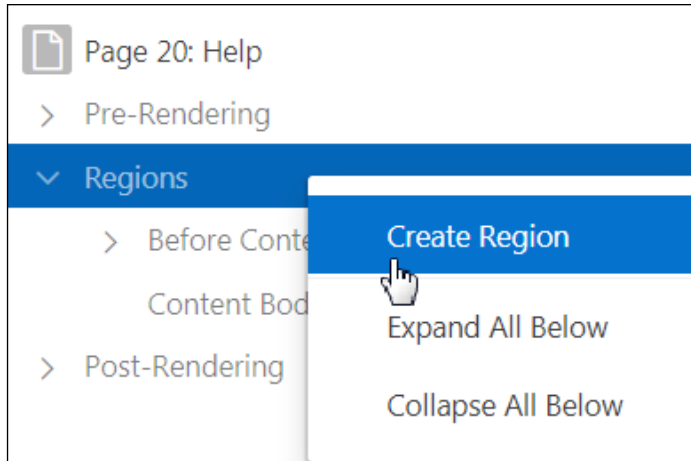
List: Desktop Navigation Bar

Search: All Text Columns Go Actions ▾ Edit Save Add Row

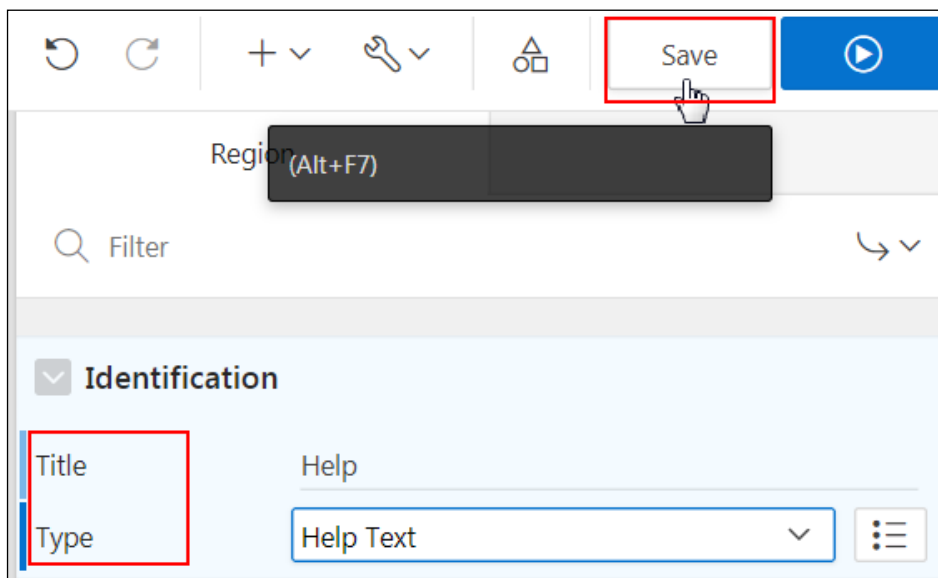
☐		Sequence ↑	Text	Target
☒		10	&APP_USER.	#
☐		20	---	separator
☐		30	Sign Out	&LOGOUT_URL.
☒		5	Help	f?p=&APP_ID.:20:&SESSION.:&APP_PAGE_ID

You get the message List changes saved.

- r. Right-click **Regions** and select **Create Region**.



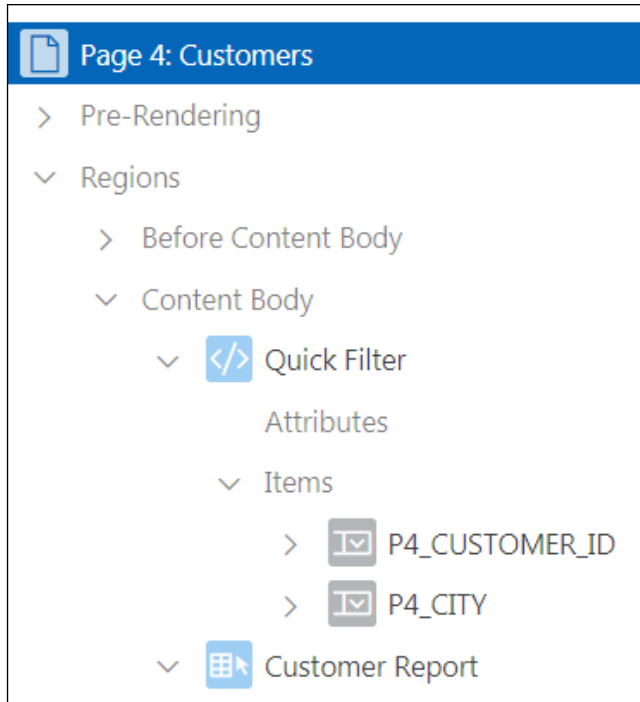
- s. In the Region panel, enter the following values for the respective fields:
- **Identification > Title:** Enter **Help**
 - **Identification > Type:** Select **Help Text**
- t. Click the **Save** button.



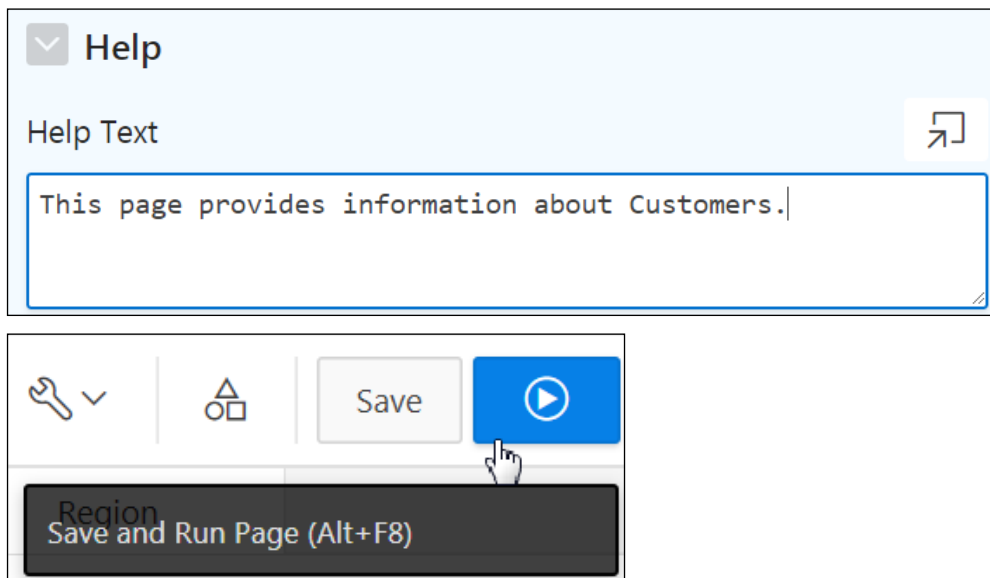
- u. You need to enter some help text for a page. Enter the **Customers** page number (Page 4) in the **Page Finder** field and click **Go**.



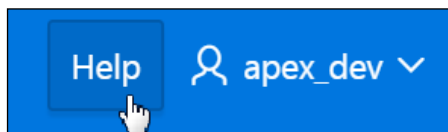
- v. In the Rendering panel, select the **Customers** page entry.



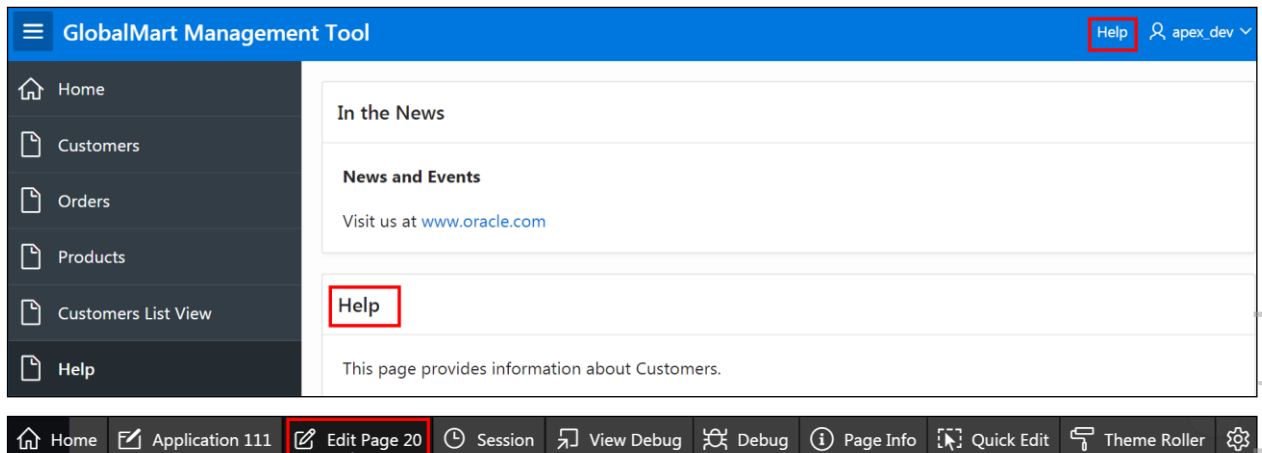
- w. On the Page tab, scroll down to the **Help** subtab. Enter This page provides information about Customers in the Help Text area and click the **Save and Run Page** icon.



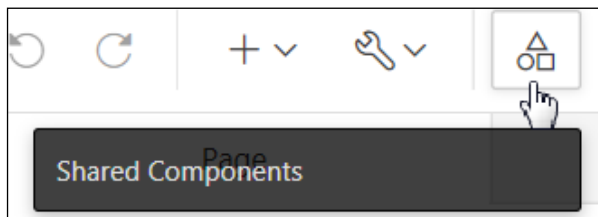
- x. Click the **Customers** page and then select the **Help** Navigation Bar entry.



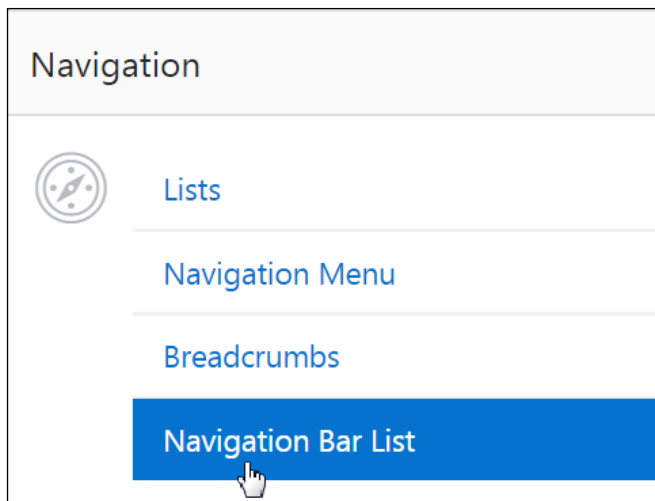
- y. Notice that the Help text from the page definition is displayed. Click the **Edit Page 20** link on the Developer toolbar.



- z. Click the **Shared Components** icon.



2. Change the name of the navigation bar entry for Sign Out to **Sign Off**.
a. On the Navigation tab, click the **Navigation Bar List** link.



- b. Click the **Desktop Navigation Bar** link.

Lists List Details Unused Conditional Entries Utilization				
Q Go Acti				
▼ ✓ Navigation Bar List				
Name ↑≡	Type	Entries	References	Entries Updated
Desktop Navigation Bar	Static	3	0	-

- c. Click the **Sign Out** link.

List Desktop Navigation Bar ?			
Q Go Rows 50 ▼ Actions ▼			
Sequence ↑≡	Name	Parent Entry	Target
5	Help	-	f?p=&APP_ID.:20:&SESSION.:&AP
10	&APP_USER.	-	#
20	---	&APP_USER.	separator
30	Sign Out	&APP_USER.	&LOGOUT_URL.

- d. Change the **List Entry Label** to Sign Off and click **Apply Changes**.

The screenshot shows the 'List Entry' configuration window. At the top, there are navigation buttons: '<', '>', 'List Entry', 'Cancel', 'Delete', and 'Apply Changes' (highlighted with a red box). Below these are tabs: 'Show All', 'Entry' (selected), 'Target', 'Current ...', 'Conditio...', 'Authoriz...', 'Configur...', 'Click Co...', 'User Def...', and 'Develop...'. The main form fields include: 'Parent List Entry' (set to '&APP_USER.'), 'Sequence' (set to '30'), 'Image/Class' (set to 'fa-sign-out'), 'Attributes' (empty), and 'Alt Attribute' (empty). At the bottom, the 'List Entry Label' field (highlighted with a red box) contains the text 'Sign Off'.

3. Add a navigation bar entry called **Home** such that it appears between the **Help** and **Sign Off** button. Display the navigation bar entry on all pages except the Home page.

- a. Click **Create Entry >**.

The screenshot shows a window with three buttons: 'Grid Edit', 'Edit List', and 'Create Entry >'. A mouse cursor is pointing at the 'Create Entry >' button.

- b. Enter the following values for the respective fields:

- **Entry > Sequence:** Enter 7
- **Entry > List Entry Label:** Enter Home
- **Target Type:** Select Page in this Application
- **Target > Page:** Select **Home** (Page 1)

- **Conditions > Condition Type:** Select **Current page != Expression 1**
- **Conditions > Expression 1:** Enter 1

Entry

List: Desktop Navigation Bar ?

Parent List Entry

- No Parent List Item - ?

Sequence

7 ?

Image/Class

⋮

?

Show All

Entry

Target

Current ...

Conditio...

Authoriz...

Configur...

* List Entry Label

Home

?

Target

Target type

Page in this Application ?

* Page

1

⋮

?

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Conditions

Condition Type **Current page != Expression 1**

PL/SQL item / column=value item / column not null item / column
null request=e1 exists never none

Expression 1 **1**

- c. Click the **Create List Entry** button.

< > **List Entry** Cancel Create and Create Another **Create List Entry**

Show All Entry Target Current ... Conditio... Authoriz... Configur... Click Co... User Def... Develop...

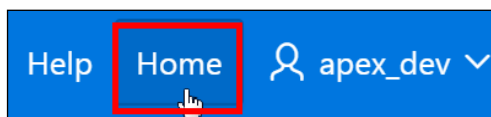
Entry

List: **Desktop Navigation Bar** ?

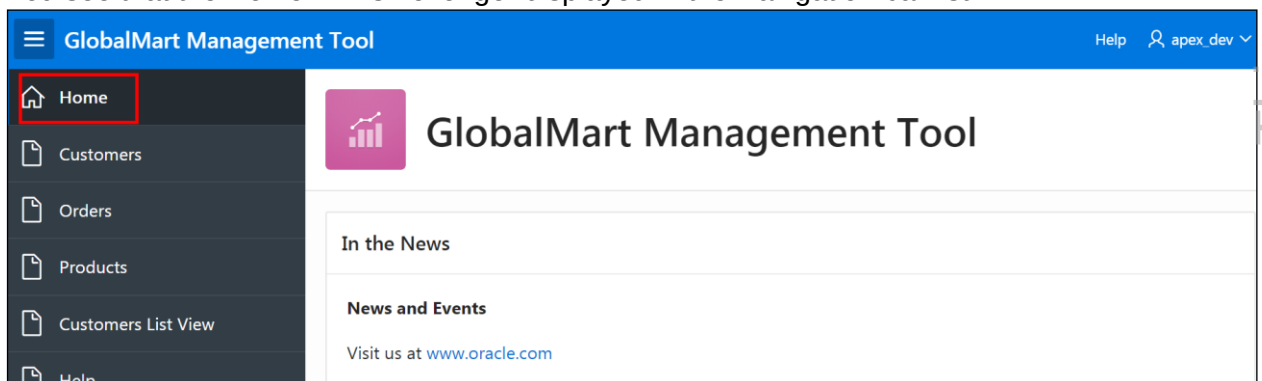
- d. Click the **Run Page 20** icon.



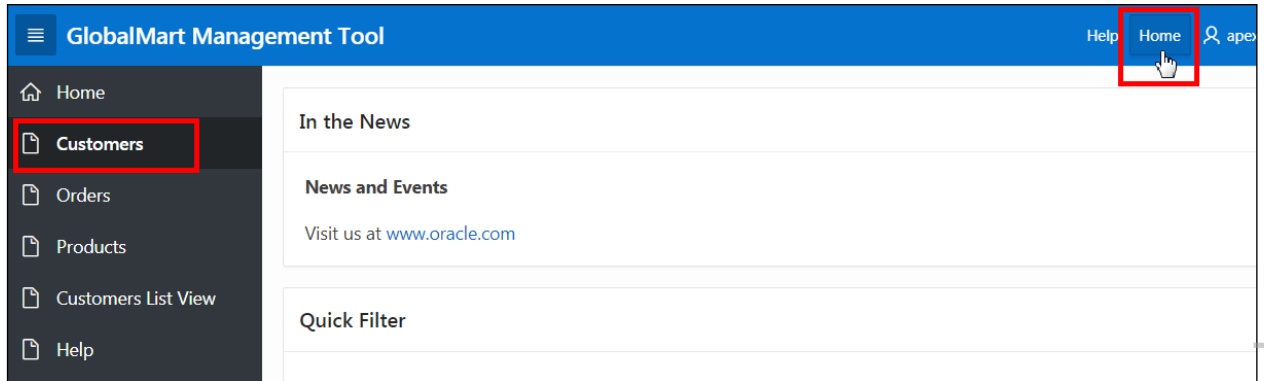
- e. Note that the Home link is displayed at the top of the page. Click the **Home** tab.



- f. You see that the Home link is no longer displayed in the Navigation bar list.



- g. Click the **Customers** tab. Notice that the Home navigation bar entry is displayed again.



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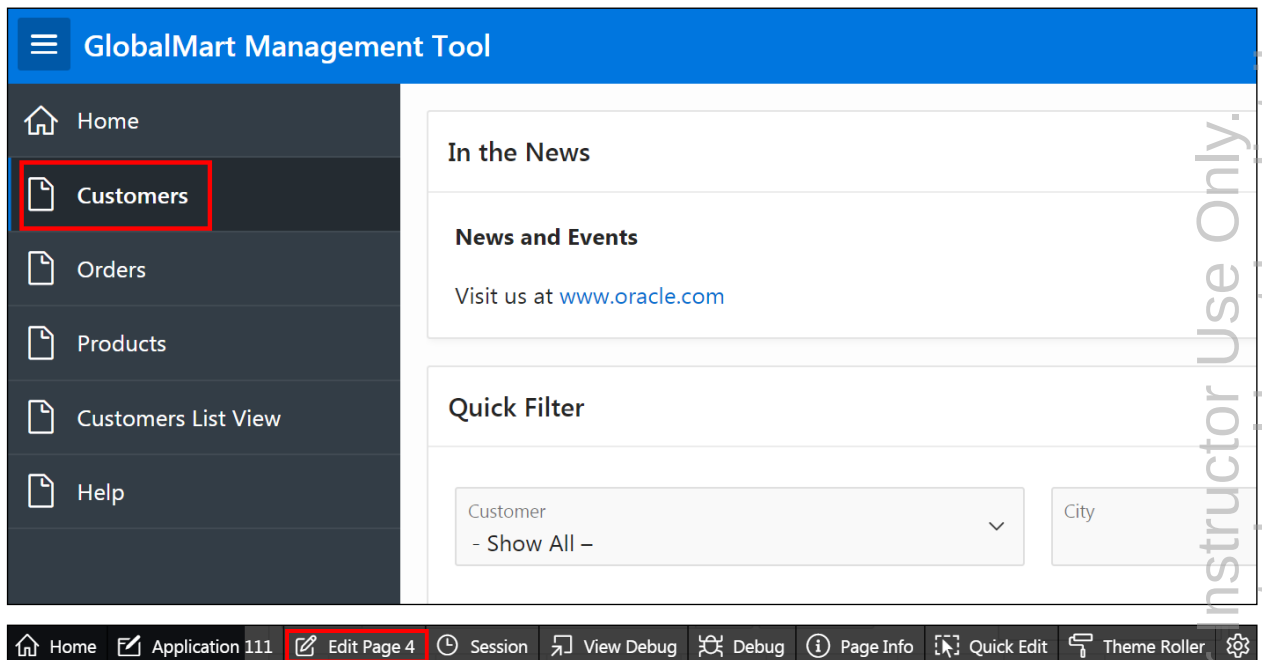
Solution 14-4: Adding Breadcrumbs to an Existing Page

Overview

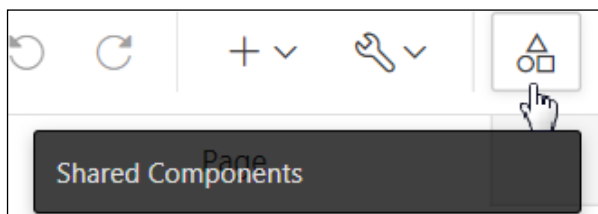
In this practice solution, steps are provided for creating breadcrumbs and adding them to the existing pages of the GMT application.

Steps

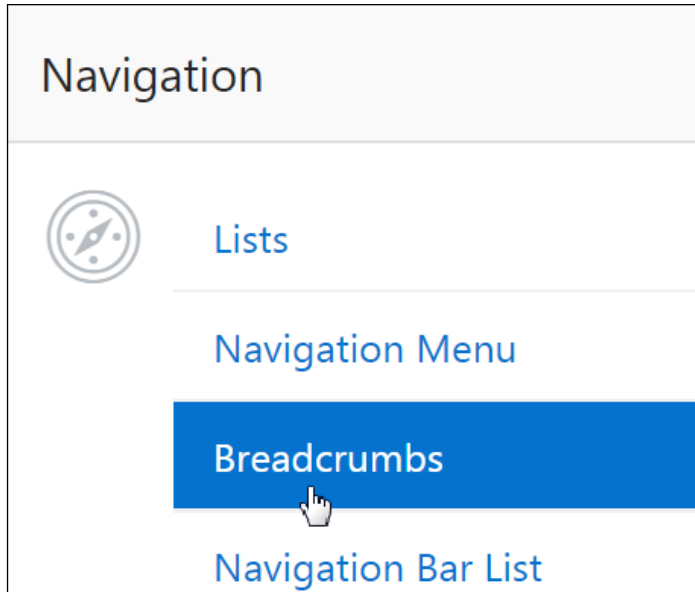
1. Edit the existing breadcrumb for the GMT application and create a breadcrumb entry for the Customers page. Run the customers page. Is the breadcrumb displayed on the page?
 - a. You are currently on the Customers page. Notice that there is no Breadcrumb on this page. Click the **Edit Page 4** link on the Developer toolbar.



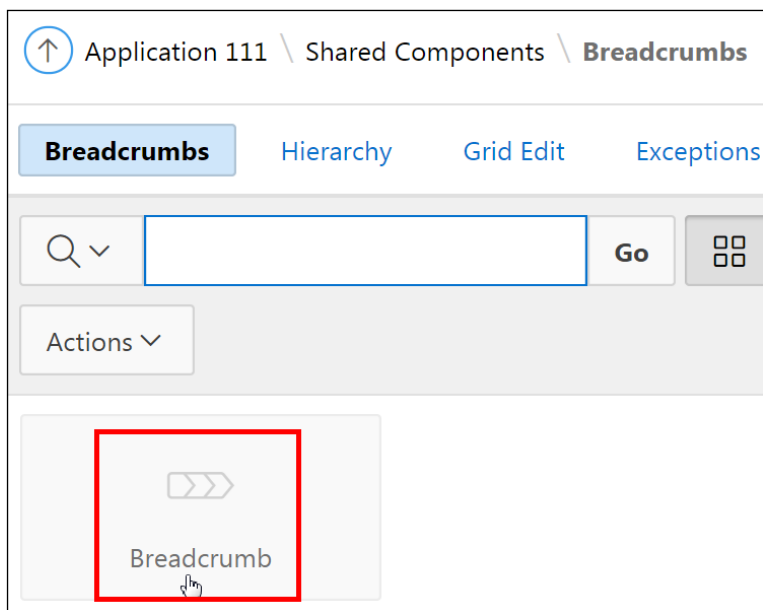
- b. Click the **Shared Components** icon.



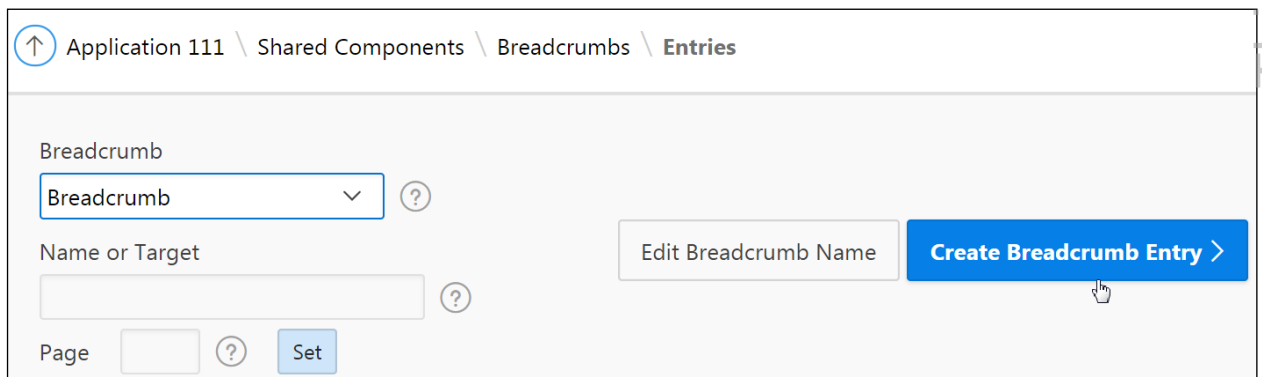
- c. Under Navigation, click the **Breadcrumbs** link.



- d. Click the **Breadcrumb** icon.



- e. Click **Create Breadcrumb Entry >**.



- f. Click the **[4]** quick pick for Page on the Breadcrumb tab. Select **Home (Page 1)** for **Parent Entry** and enter **Customers** for **Short Name** under Entry. Select **4** (**Customers**) for **Page** under Target and click **Create Breadcrumb Entry**.

The screenshot displays the 'Breadcrumb' configuration window. The 'Entry' tab is active, showing fields for 'Sequence' (10), 'Parent Entry' (Home (Page 1)), 'Short Name' (Customers), and 'Long Name'. The 'Target' tab is also visible, showing 'Target is a' (Page in this Application), 'Page' (4), and a 'reset pagination for this page' checkbox. The 'Create Breadcrumb Entry' button is highlighted in red.

Breadcrumb

Breadcrumb Breadcrumb ?

* Page 4 ?

[4] ?

Sequence 10 ?

Parent Entry Home (Page 1) ?

* Short Name Customers ?

Long Name ?

Breadcrumb Entry:

Cancel Create Breadcrumb Entry

Show All Breadcrumb Entry Target Conditions Authorization Configuration

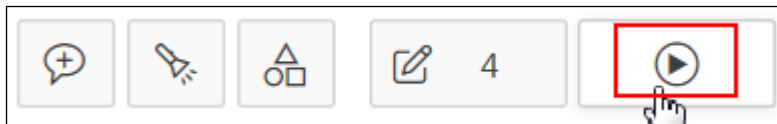
Target

Target is a Page in this Application ?

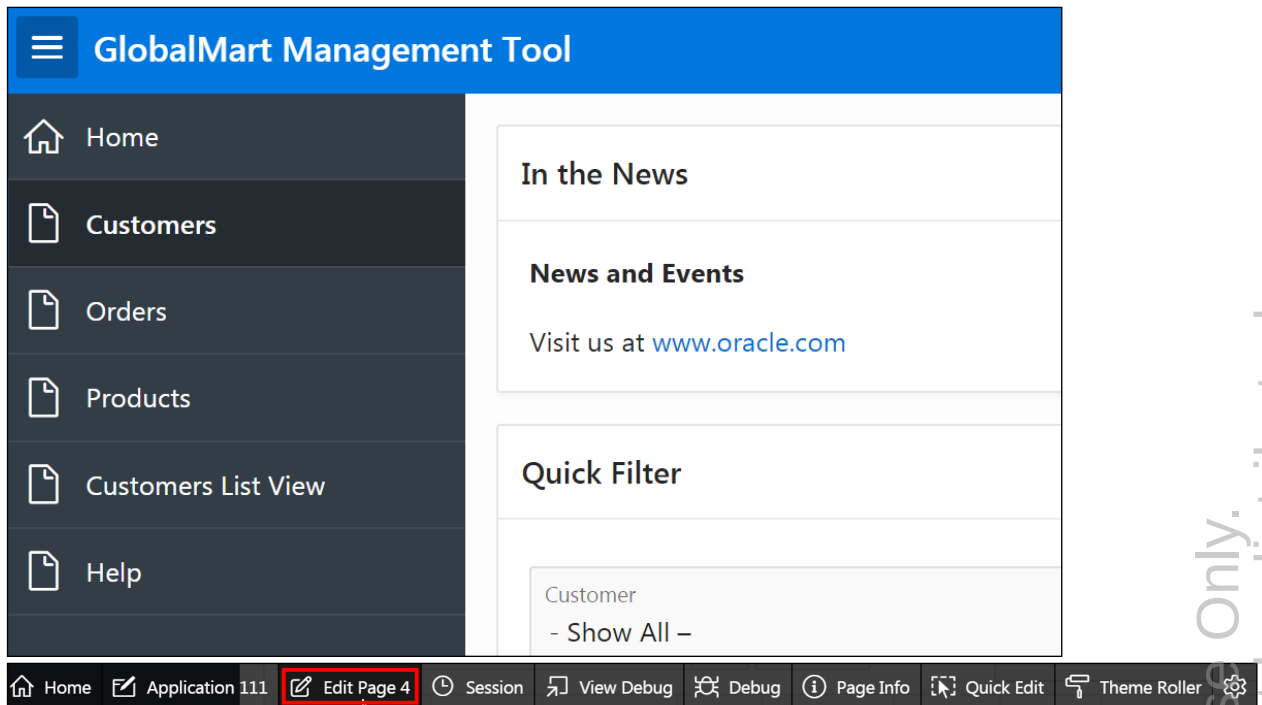
Page 4 ?

☐ reset pagination for this page ?

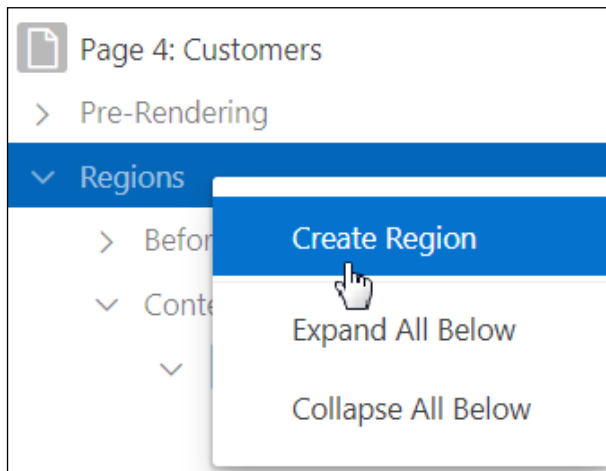
- g. Click the **Run Page 4** icon.



- h. Note that the breadcrumb you created is not reflected. This is because there is no breadcrumb region on this page. Click the **Edit Page 4** link on the Developer toolbar.



2. Create a breadcrumb region on the Customers page and run the page.
- a. In the Rendering panel, right-click **Regions** and select **Create Region**.



- b. Ensure that the newly created Region is selected. On the Region tab, enter the following values for the respective fields:
- **Identification > Title:** Enter `Breadcrumb`
 - **Identification > Type:** Select `Breadcrumb`
 - **Source > Breadcrumb:** Select `Breadcrumb`
 - **Layout > Position:** Select `Breadcrumb Bar`

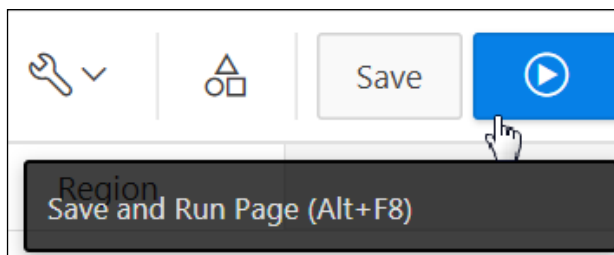
The screenshot shows the 'Region' configuration interface. It is divided into three main sections: Identification, Source, and Layout. In the Identification section, the 'Title' field is set to 'Breadcrumb' and the 'Type' dropdown is also set to 'Breadcrumb'. In the Source section, the 'Breadcrumb' dropdown is set to 'Breadcrumb'. In the Layout section, the 'Sequence' is set to '30', the 'Parent Region' dropdown is set to '- Select -', and the 'Position' dropdown is set to 'Breadcrumb Bar'. Red boxes highlight the 'Title', 'Type', 'Breadcrumb' (in Source), and 'Position' fields.

Identification	
Title	Breadcrumb
Type	Breadcrumb

Source	
Breadcrumb	Breadcrumb

Layout	
Sequence	30
Parent Region	- Select -
Position	Breadcrumb Bar

- c. Click the **Save and Run Page** icon.



- d. Note that now the breadcrumb is seen. Click the **Edit** icon next to any row in the report.

≡

GlobalMart Management Tool

Home

Customers

Orders

Products

Customers List View

Help

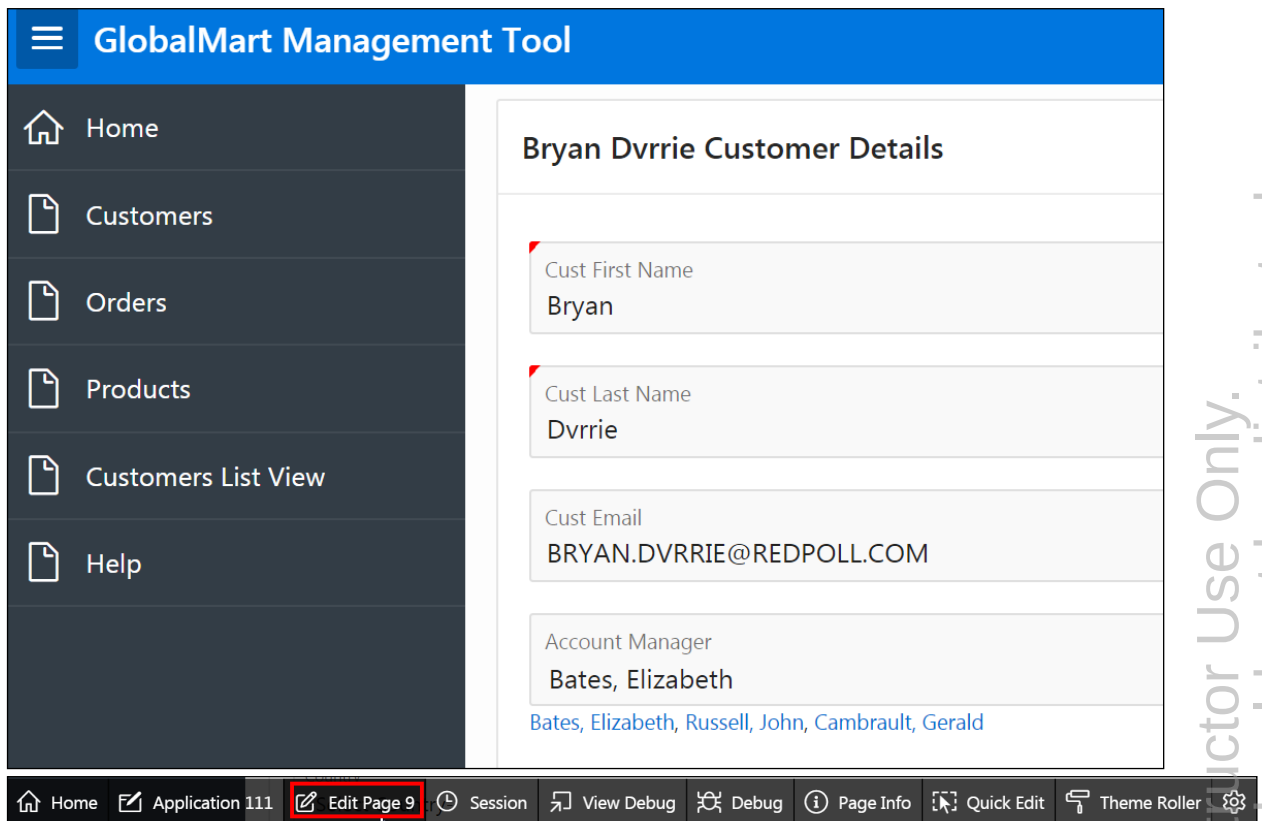
Breadcrumb

Home \ Customers \

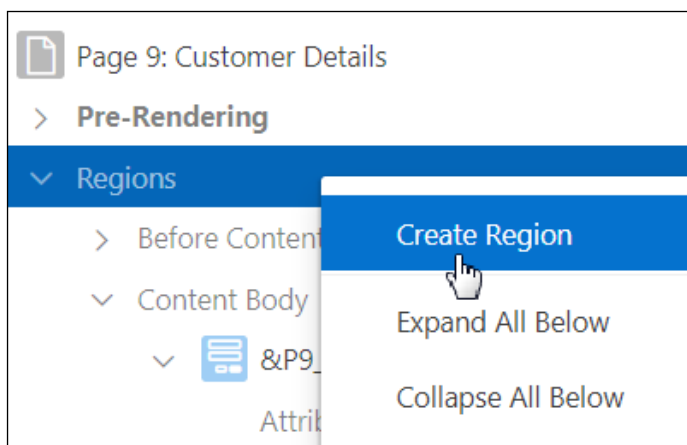
	First Name	Cust Last Name	Street Address	Postal Code	City
	Bryan	Dvrrie	3386 Perrysville Ave	15214	Pittsburgh
	Ajay	Sen	220 Penn Ave # 300	18503	Scranton
	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia

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3. On the Customer Details page, create a breadcrumb called **Customer Details** whose parent entry is the Customers page.
 - a. The Customer Details page does not have any breadcrumbs. You want to add one that has a parent entry to the Customers report on page 4. Click the **Edit Page 9** link on the Developer toolbar.



- b. Right-click **Regions** and select **Create Region**.



- c. Ensure that the newly created Region is selected. On the Region tab, enter the following values for the respective fields:
- **Identification > Title:** Enter `Breadcrumb`
 - **Identification > Type:** Select `Breadcrumb`
 - **Source > Breadcrumb:** Select `Breadcrumb`
 - **Layout > Position:** Select `Breadcrumb Bar`

Identification

Title: Breadcrumb

Type: Breadcrumb

Source

Breadcrumb: Breadcrumb

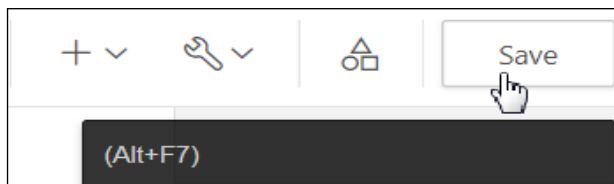
Layout

Sequence: 20

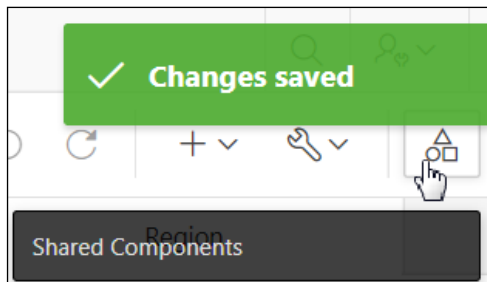
Parent Region: - Select -

Position: Breadcrumb Bar

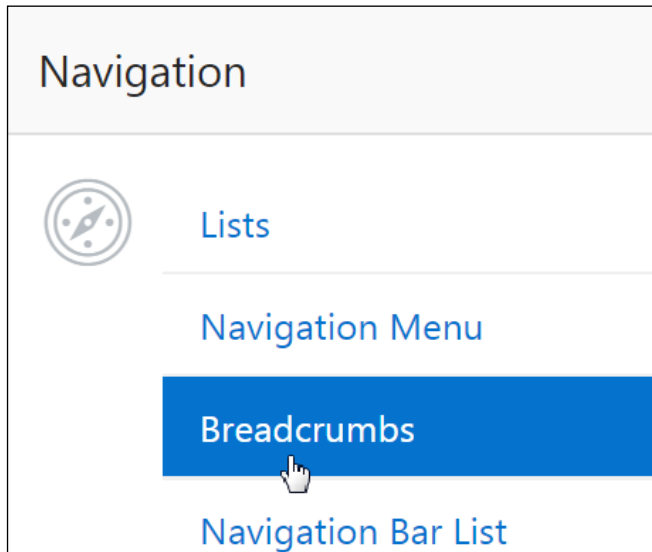
- d. Click the **Save** button.



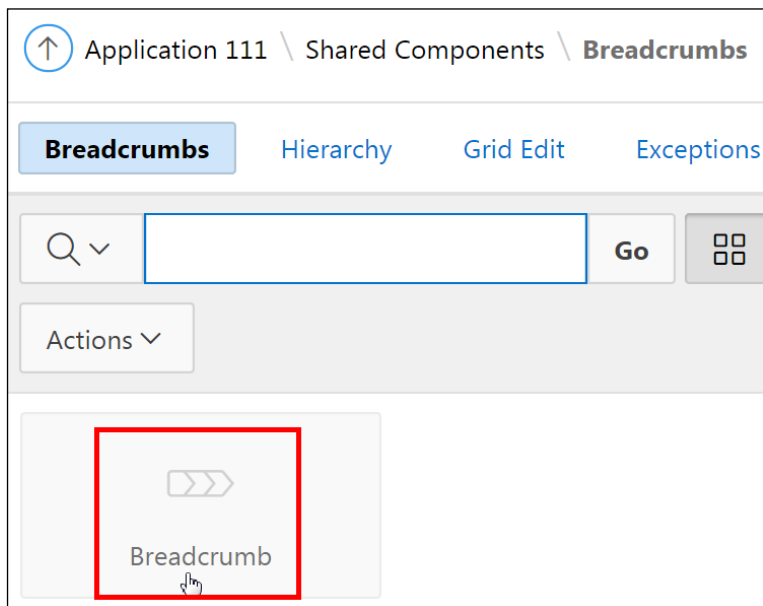
- e. You have created a breadcrumb region. Now you will create a breadcrumb. Click the **Shared Components** icon.



- f. Under Navigation, click **Breadcrumbs**.



- g. Click the **Breadcrumb** icon.



- h. Click **Create Breadcrumb Entry >**.

Application 111 \ Shared Components \ Breadcrumbs \ Entries

Breadcrumb
Breadcrumb

Name or Target
Edit Breadcrumb Name

Page
Set

Create Breadcrumb Entry >

- i. Click the **[9]** quick pick for Page on the Breadcrumb tab. Select **Customers (Page 4)** for **Parent Entry** and enter **Customer Details** for **Short Name** under Entry. Select **9 (Customer Details)** for **Page** under Target and click **Create Breadcrumb Entry**.

Breadcrumb

Breadcrumb Breadcrumb

* Page 9

[9]

Sequence 10

Parent Entry Customers (Page 4)

* Short Name Customer Details

Long Name

Breadcrumb Entry:

Cancel **Create Breadcrumb Entry**

Show All Breadcrumb Entry Target Conditions Authorization Configuration

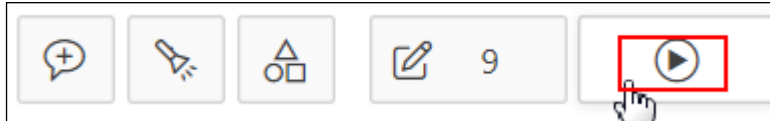
Target

Target is a Page in this Application

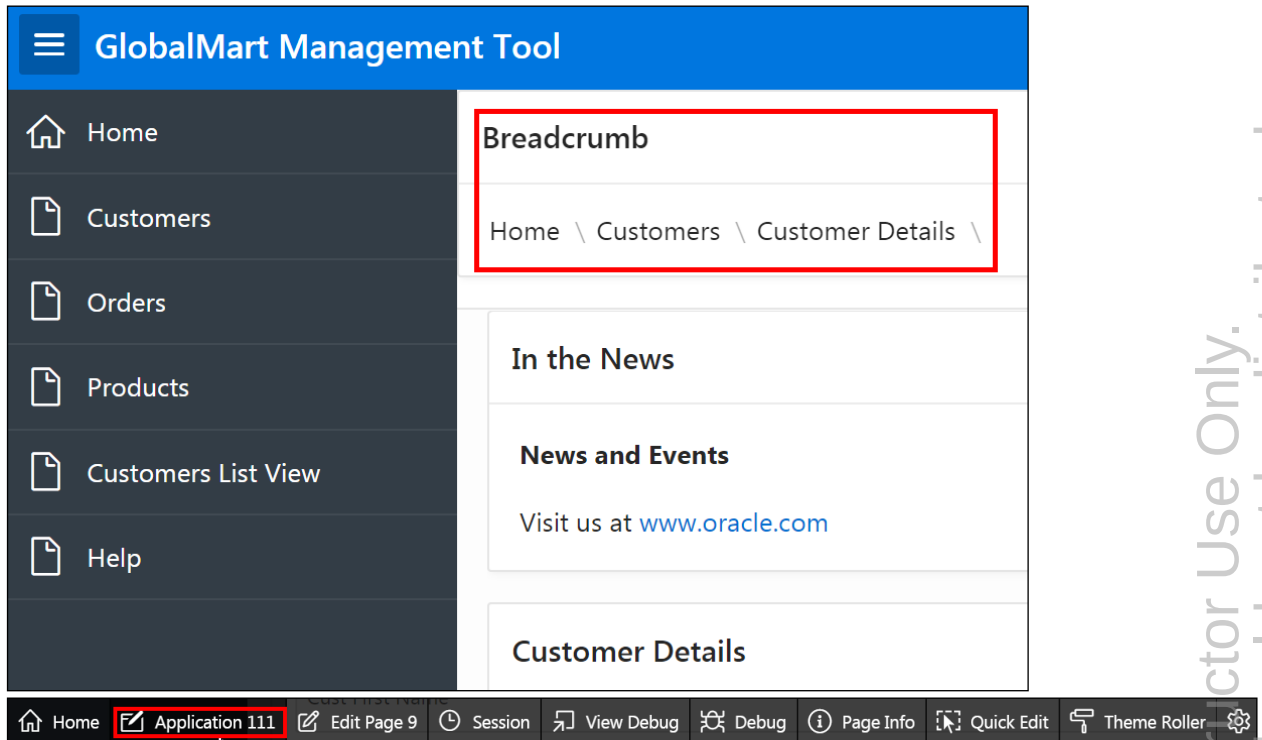
Page 9

☐ reset pagination for this page

- j. The Breadcrumb entry is created. Click the **Run Page 9** icon.



- k. Notice that the breadcrumb is now displayed. Click the **Application 111** link on the Developer toolbar.



**Practices for Lesson 15:
Working with Themes,
Templates, and Files**

Practices for Lesson 15: Overview

Overview

In these practices, you create a theme from scratch; copy, edit, and switch a theme; edit the theme templates; and upload and apply a CSS in the GlobalMart Management Tool application.

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Practice 15-1: Creating a Theme from Scratch

Overview

In this practice, you create a theme from scratch for the GlobalMart Management Tool application.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a new theme from scratch.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 15-1**.

If you haven't completed **Practice 14-4**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_14_04.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.

- j. Verify that the application and the supporting objects are successfully installed.

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Practice 15-2: Copying, Editing, and Switching a Theme

Overview

In this practice, you copy an existing Universal Theme and then edit the theme. You will later switch the current theme to reflect the newly modified theme.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Copy an existing theme and then edit it.
2. Switch the theme of your application to the newly created theme. What do you observe?

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 15-2**.

If you haven't completed **Practice 15-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_15_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.

- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

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Practice 15-3: Editing Templates

Overview

In this practice, you customize the template according to the requirements of the GlobalMart Management Tool.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the catch up section of this practice.

Tasks

1. Upload the `logo.png` image from your `/home/oracle/labs/labs/` directory and add it as a logo for your application.
2. On your Home page (Page 1), change the Tasks list to appear on the right side of the page, vertically.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 15-3**.

If you haven't completed **Practice 15-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_15_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).

- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

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Practice 15-4: Uploading and Applying a CSS

Overview

In this practice, you upload a CSS file and apply it to the pages of the GMT application to enhance its look and feel.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Upload the `apexstyle.css` file from the `/home/oracle/labs/labs/` directory.
2. Add the cascading style sheet to the Help region on the Help page.
3. Change the Help text to use the new style, **bigblue**, by using the following script (located in `/home/oracle/labs/labs/lab_15_04.txt`):

```
<span class="bigblue">No help is available for this page.</span>
```

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 15-4**.

If you haven't completed **Practice 15-3**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_15_03.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.

- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

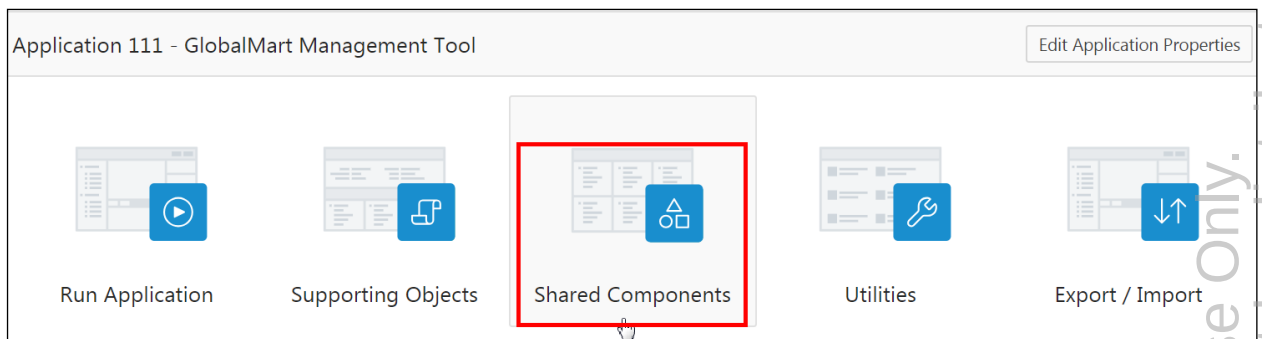
Solution 15-1: Creating a Theme from Scratch

Overview

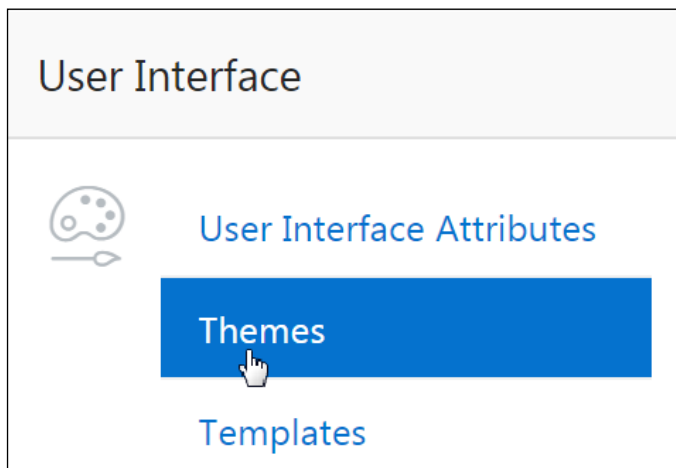
In this practice solution, steps are provided for creating a theme from the beginning for the GlobalMart Management Tool application.

Steps

1. Create a new theme by using the Theme repository.
 - a. On the GlobalMart Management Tool application home page, click **Shared Components**.



- b. Under **User Interface**, click **Themes**.



- c. Notice that there is one existing theme currently in the list. To create a new theme, click the **Create >** button.

Application 111 \ Shared Components \ Themes

Themes Reports History

Search Go Actions

Number ↑	Name	User Interface	Is Universal Theme	Is Current	Subscribed From
42	Universal Theme - 42 *	Desktop	✓	✓	Theme Repository

Reset Switch Theme Create >

- d. Ensure that the **From Scratch** method is selected and click **Next >**.

Create Theme

Method

- ☐ From the Repository
- ☐ As a copy from another application
- ☒ From Scratch
- ☐ From Export

- e. Enter the following values and click **Create**:
- **Theme Number:** Enter 101.
 - **Name:** Enter My New Theme.
 - **Identifier:** Enter BLUE_101. (The identifier is an arbitrary number and must be unique within your application.)
 - **Navigation Type:** Select List.
 - **Navigation Bar Implementation:** Select List.

- **Description:** Enter This is a new theme created from scratch.

Cancel

Create

Application: 111 ?

* Theme Number

101 ?

* Name

My New Theme ?

* Identifier

BLUE_101 ?

Navigation Type

List ?

Navigation Bar Implementation

List ?

Description

This is a new theme created from scratch.

The new theme is created.

✓ **Action Processed.**

Themes

Reports

History

Q v

Go

Actions v

Number	Name	User Interface	Is Universal Theme	Is Current	Subscribed From
42	Universal Theme - 42 *	Desktop	✓	✓	Theme Repository
101	My New Theme - 101	Desktop			

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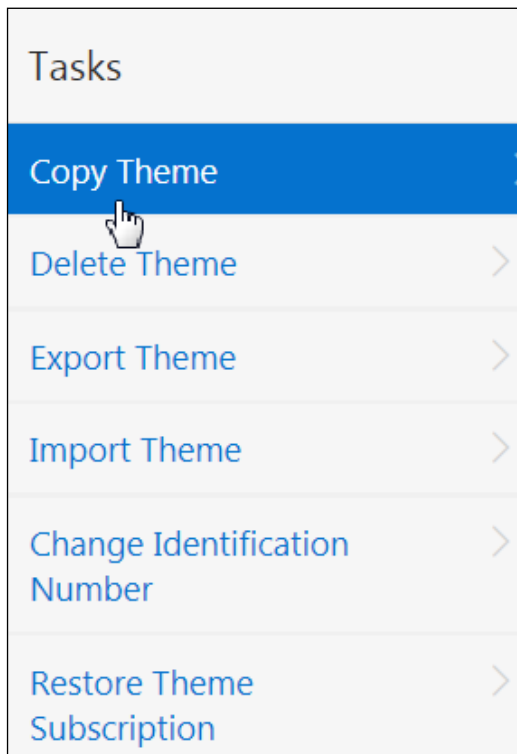
Solution 15-2: Copying, Editing, and Switching a Theme

Overview

In this practice solution, steps are provided for copying an existing theme and then modifying it according to the requirements of the GlobalMart Management Tool application. Later, steps are provided to switch the current theme.

Steps

1. Copy an existing theme and then edit it.
 - a. To copy an existing theme, go to the Themes page. Select **Copy Theme** on the right.



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- b. Enter 102 for **Copy to this Theme ID**. Click **Next >**.

Copy Theme

Each theme is identified by a numeric identification number (ID). Use this to make a copy of an existing theme and specify a new theme ID. Copying a theme is useful if you wish to export a theme with a different ID.

Application: 111 GlobalMart Management Tool

Copy from Theme 42. Universal Theme

* Copy to this Theme ID 102

Subscribe Theme Yes No

Next >

- c. The Confirm page appears. Evaluate the Application (Name/ID), Copy Theme From ID, and Copy Theme ID to options and click **Copy Theme**.

Copy Theme

Application: 111 - GlobalMart Management Tool

Copy Theme ID from: 42

Copy Theme ID to: 102

Copy Theme

d. You can see the copied theme listed on the Themes page.

✓ Theme copied.

Themes

Reports

History

Q

Go

Actions

Number	Name	User Interface	Is Universal Theme	Is Current	Subscribed From
42	Universal Theme - 42 *	Desktop	✓	✓	Theme Repository
101	My New Theme - 101	Desktop			
102	Universal Theme - 102	Desktop	✓		111

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2. Edit and modify the theme that you just copied from the Universal Theme – 42.
 - a. On the Themes page, click **Universal Theme – 102**.

Theme copied.

Themes

Reports

History

Go

Actions ▾

Number ↑≡	Name	User Interface	Is Universal Theme	Is Current	Subscribed From
42	Universal Theme - 42 *	Desktop	✓	✓	Theme Repository
101	My New Theme - 101	Desktop			
102	<u>Universal Theme - 102</u>	Desktop	✓		111

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- b. The Create/Edit Theme page appears. You can make the necessary changes you want for your Theme on this page.

↑ Application 111 \ Shared Components \ Themes

Create / Edit Theme

Theme

Show All

Name

Theme S...

JavaScri...

Compon...

Region ...

Dialog D

Name

Application: 111 ?

* Theme Number 102 ?

* Name Universal Theme ?

* Identifier UNIVERSAL_THEME ?

Navigation Type List ?

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- c. Click the **Component Defaults** tab. Change **Page** from **Standard** to **Marquee**.

Component Defaults

Specify component defaults by component type.

Page	Standard	?
Navigation Bar List	- Select Template -	?
Navigation Menu List Position	Left Side Column	?
Navigation Menu List (Top)	Left and Right Side Columns	?
Navigation Menu List (Side)	Login	?
Login Page	Marquee	?
Error Page	Minimal (No Navigation)	?
Printer Friendly Page	Modal Dialog	?
	Right Side Column	?
	Standard	?
	Wizard Modal Dialog	?
	Standard	?

- d. Click **Apply Changes**.

Theme

Cancel Apply Changes

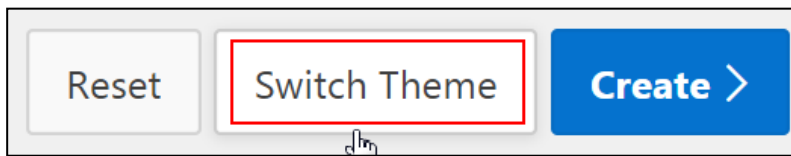
Component Defaults

Specify component defaults by component type.

Page	Marquee	?
------	---------	---

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3. Switch the theme of your application to the newly copied and edited theme. What do you observe?
- a. To use the theme that you just created in your application, you must switch to that theme. Click **Switch Theme**.



- b. Select **Currently Active Theme** from the drop-down list, in this case, **42. Universal Theme**. Make sure **102. Universal Theme** is selected for **Switch to Theme** and click **Next >**.

Switch Theme

Identify Theme

A theme is a collection of templates. When you switch a theme, all templates assigned to components within one theme are assigned to templates in another theme. Application Express accomplishes template mapping through the assignment of template class identifiers.

Application: **111 - GlobalMart Management Tool** ?

* Currently Active Theme: 42. Universal Theme ?

* Switch to Theme: 102. Universal Theme ?

cel

Create Theme

Next >

- c. The Verify compatibility page appears. The Status columns display the check marks. Click **Next** >.

Switch Theme

Verify Compatibility

When you switch to a new theme, App Builder maps all currently used templates to a template in the new theme using the template class. This report displays these template mappings and allow for selecting alternate templates when no templates with a matching template class exist.

Application: **111 - GlobalMart Management Tool** ?

Currently Active Theme: **42. Universal Theme** ?

Switch to Theme: **102. Universal Theme** ?

Template Type ↑	From Template	To Template	Status
Breadcrumb	Breadcrumb	Breadcrumb ▾	✓
Button	Text with Icon	Text with Icon ▾	✓
	Text	Text ▾	✓
	Icon	Icon ▾	✓
Label	Optional - Floating	Optional - Floating ▾	Multiple matches
	Hidden	Hidden ▾	✓
	Required - Floating	Optional - Floating ▾	Multiple matches
List	Links List	Links List ▾	✓

< Cancel
Next >

- d. Click **Switch Theme**.

Switch Theme

Confirm Switch

Switching your theme may result in template issues if multiple templates are defined with the same template class identifier. Before switching a template, it is recommended you create a back up of your application before continuing.

Application: **111 - GlobalMart Management Tool** ?

Currently Active Theme: **42. Universal Theme** ?

Switch To Theme: **102. Universal Theme** ?

< Cancel
Switch Theme

You get the message Active Theme Switched.

✓ Active Theme Switched.

Themes

Reports

History

Q

Go

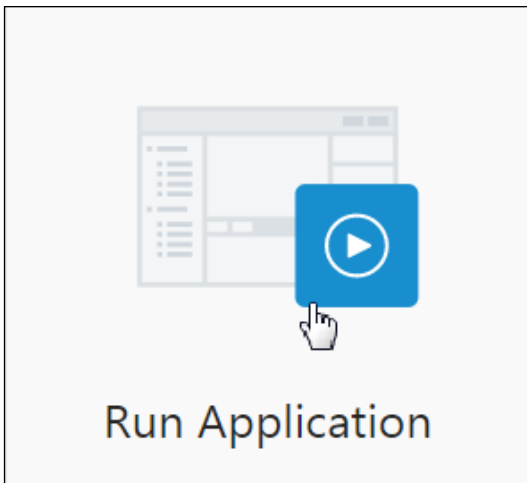
Actions

Number	Name	User Interface	Is Universal Theme	Is Current	Subscribed From
42	Universal Theme - 42	Desktop	✓		Theme Repository
101	My New Theme - 101	Desktop			
102	Universal Theme - 102 *	Desktop	✓	✓	111

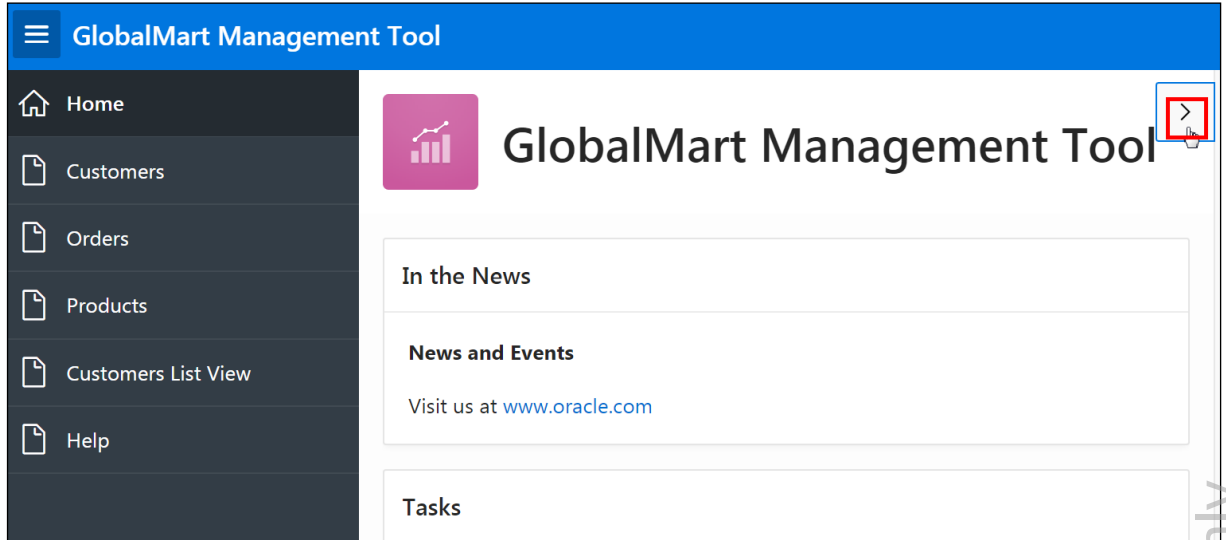
Click **Application 111** breadcrumb and then run the application.

↑ [Application 111](#) \ Shared Components \ **Themes**

✓ **Active Theme Switched.**



You will see that all the pages have changed from the Standard page to Marquee (Expand / Collapse Side Column). This is because you have changed the current theme from **Universal Theme – 42** to the newly edited theme **Universal Theme – 102**.



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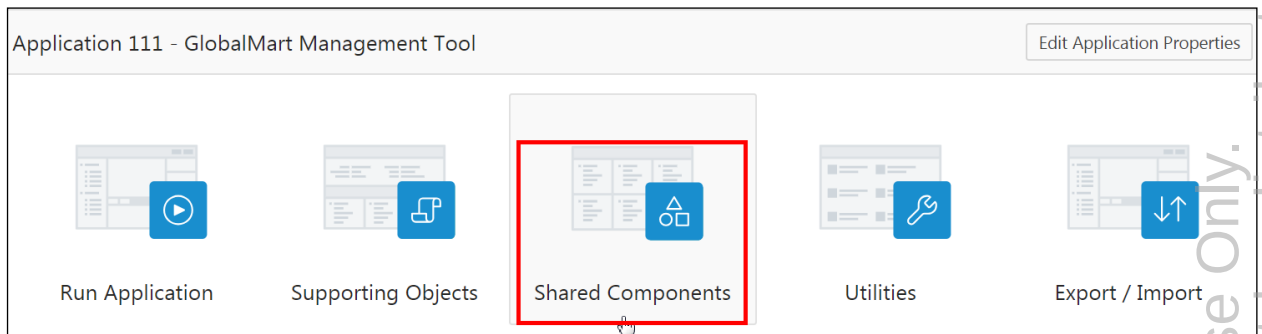
Solution 15-3: Editing Templates

Overview

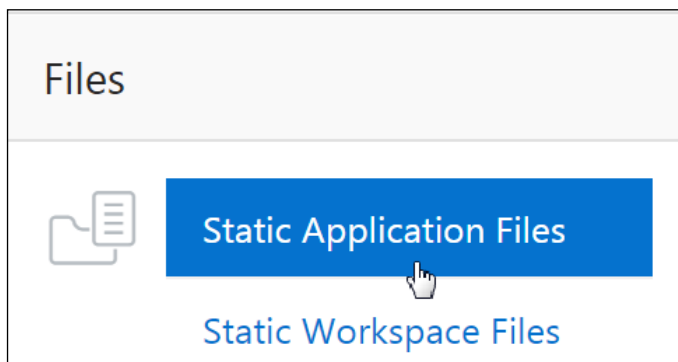
In this practice solution, steps are provided for customizing the template according to the requirements of the GlobalMart Management Tool.

Steps

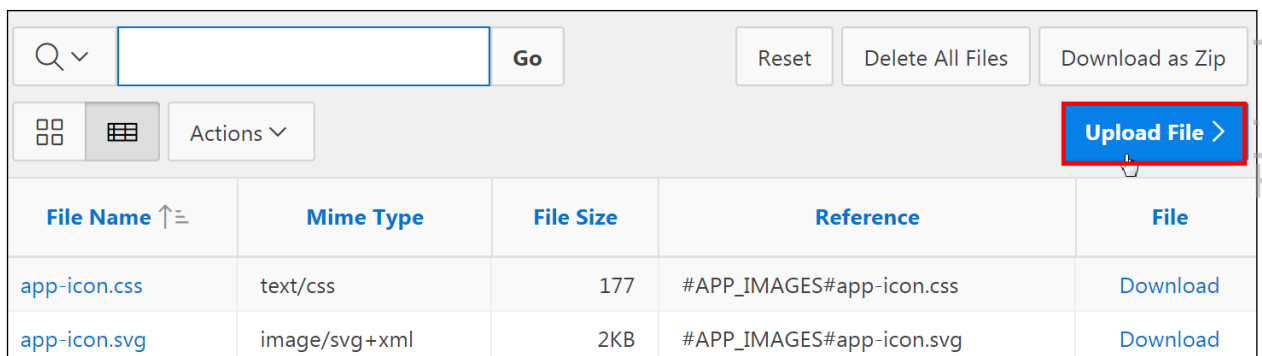
1. Upload the `logo.png` image from your `/home/oracle/labs/labs/` directory and add it as a logo for your application.
 - a. Click **Shared Components**.



- b. Under **Files**, click the **Static Application Files** link.



- c. Click **Upload File >**.



- d. Select the `logo.png` file in your `/home/oracle/labs/labs/` folder for **File** and click **Upload**.

Upload Static Application File(s)

Use this page to associate files like images, CSS or Javascript files with your application. To associate a file with your application, select the file(s), and click Upload.

Directory

* File(s) Choose Files `logo.png`

* File Character Set Unicode UTF-8

* Unzip File Yes No

Cancel Upload and Upload Another Upload

- e. Notice that the file is uploaded. Copy the reference of the file.

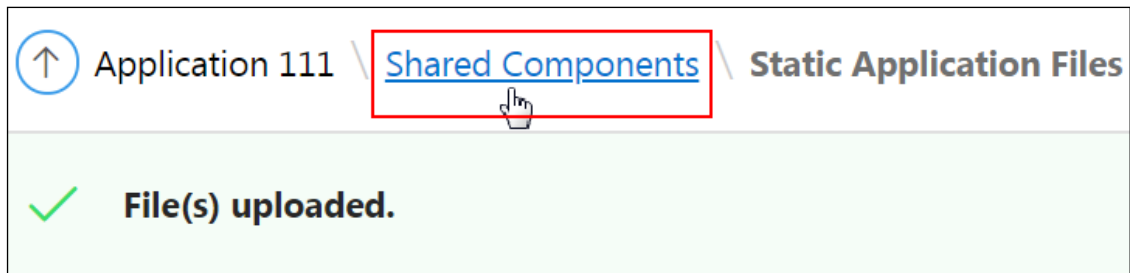
✓ **File(s) uploaded.**

Go Reset

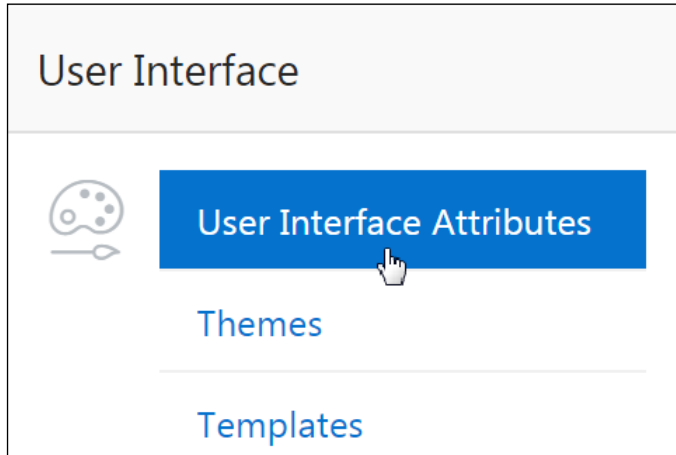
Actions

File Name	Mime Type	File Size	Reference
app-icon.css	text/css	177	<code>#APP_IMAGES#app-icon.css</code>
app-icon.svg	image/svg+xml	2KB	<code>#APP_IMAGES#app-icon.svg</code>
logo.png	image/png	6KB	<code>#APP_IMAGES#logo.png</code>

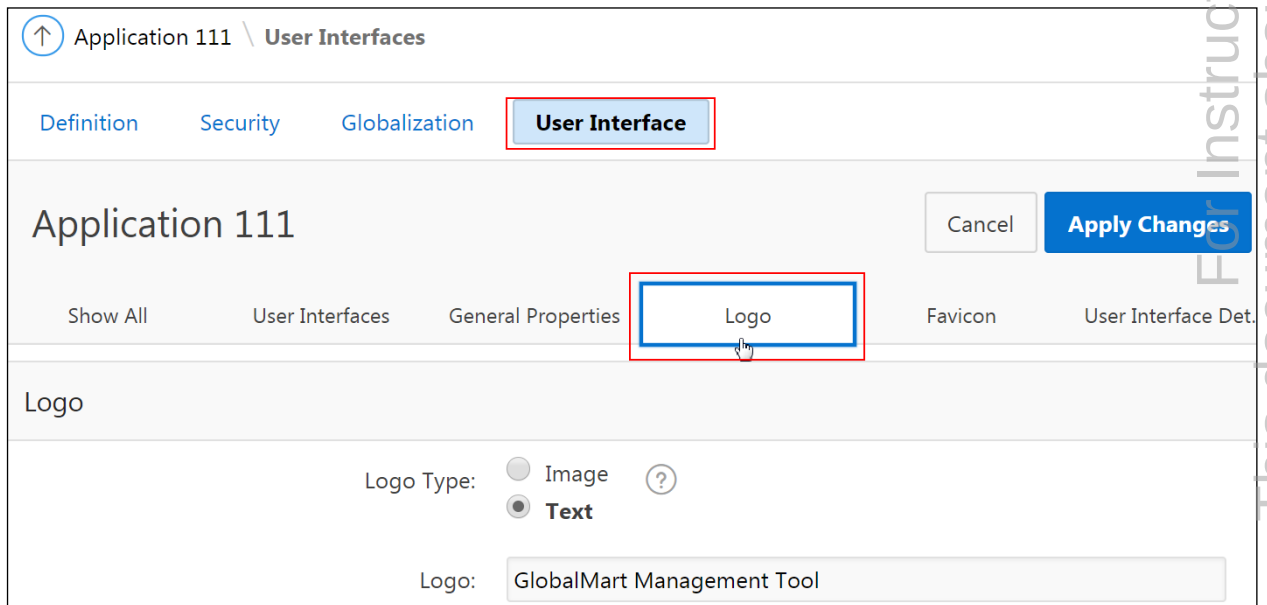
- f. You will now add the logo to your application. Click the **Shared Components** breadcrumb.



- g. Under **User Interface**, click the **User Interface Attributes** link.



- h. Make sure that the **User Interface** subtab is selected and click **Logo**.



- i. Select the **Logo Type** as **Image** and enter the reference that you copied for **Logo**. Click **Apply Changes**.

#APP_IMAGES#logo.png

Application 111

Cancel **Apply Changes**

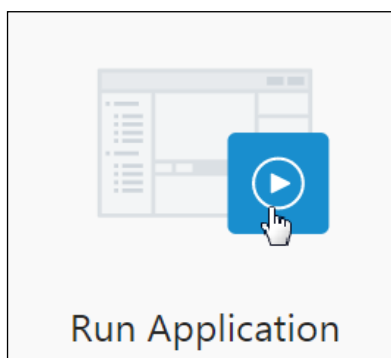
Show All User Interfaces General Properties **Logo** Favicon User Interface Det...

Logo

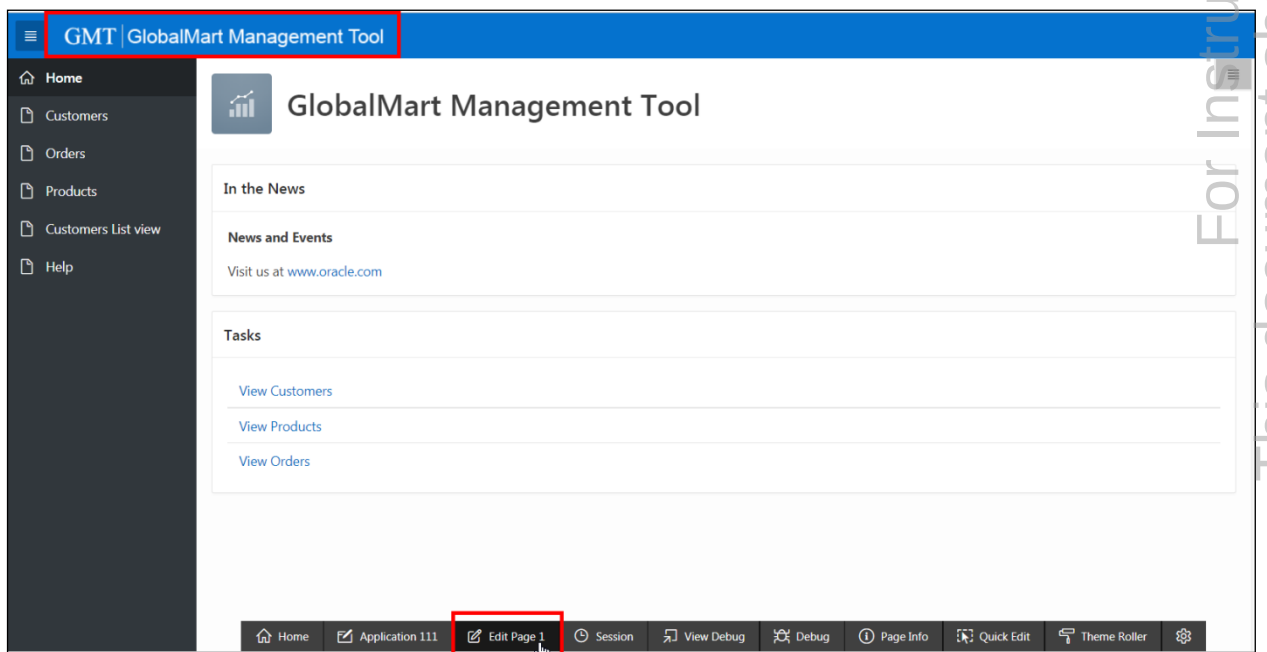
Logo Type: ☒ **Image** ☐ Text ?

Logo: #APP_IMAGES#logo.png

- j. Run the page. Click the **Run Application** icon.

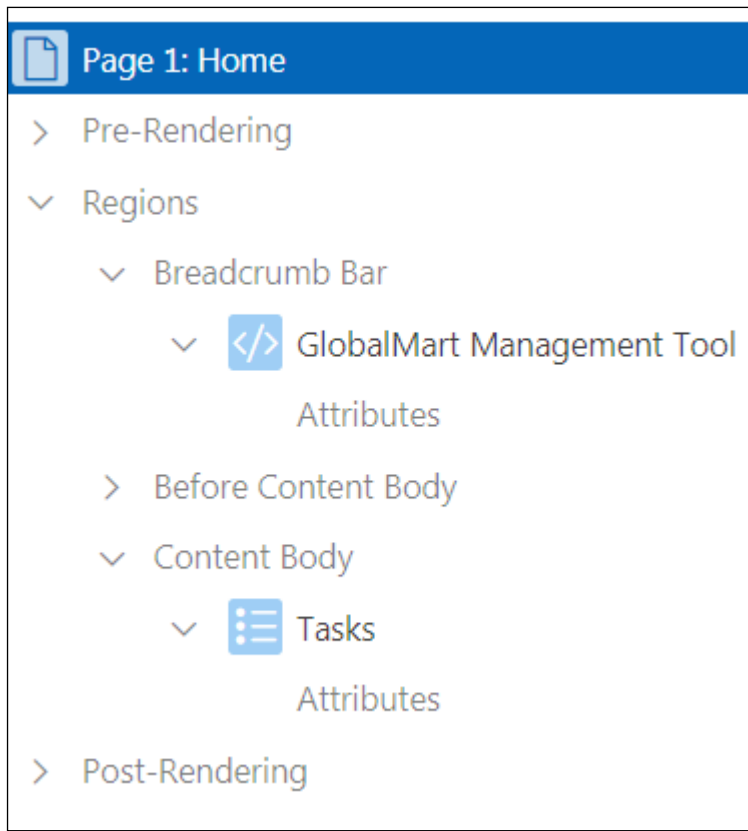


- k. Note that all your pages display the logo. Make sure that you are on the Home page and click the **Edit Page 1** link on the Developer toolbar.

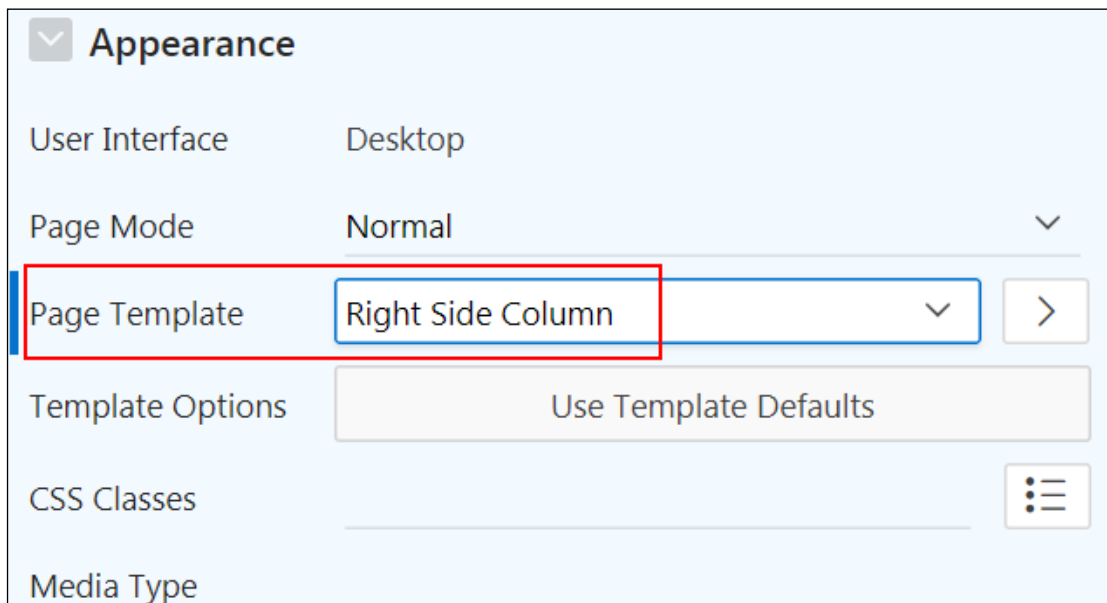


2. On your **Home** page (Page 1), change the Tasks list to appear on the right side of the page, vertically.

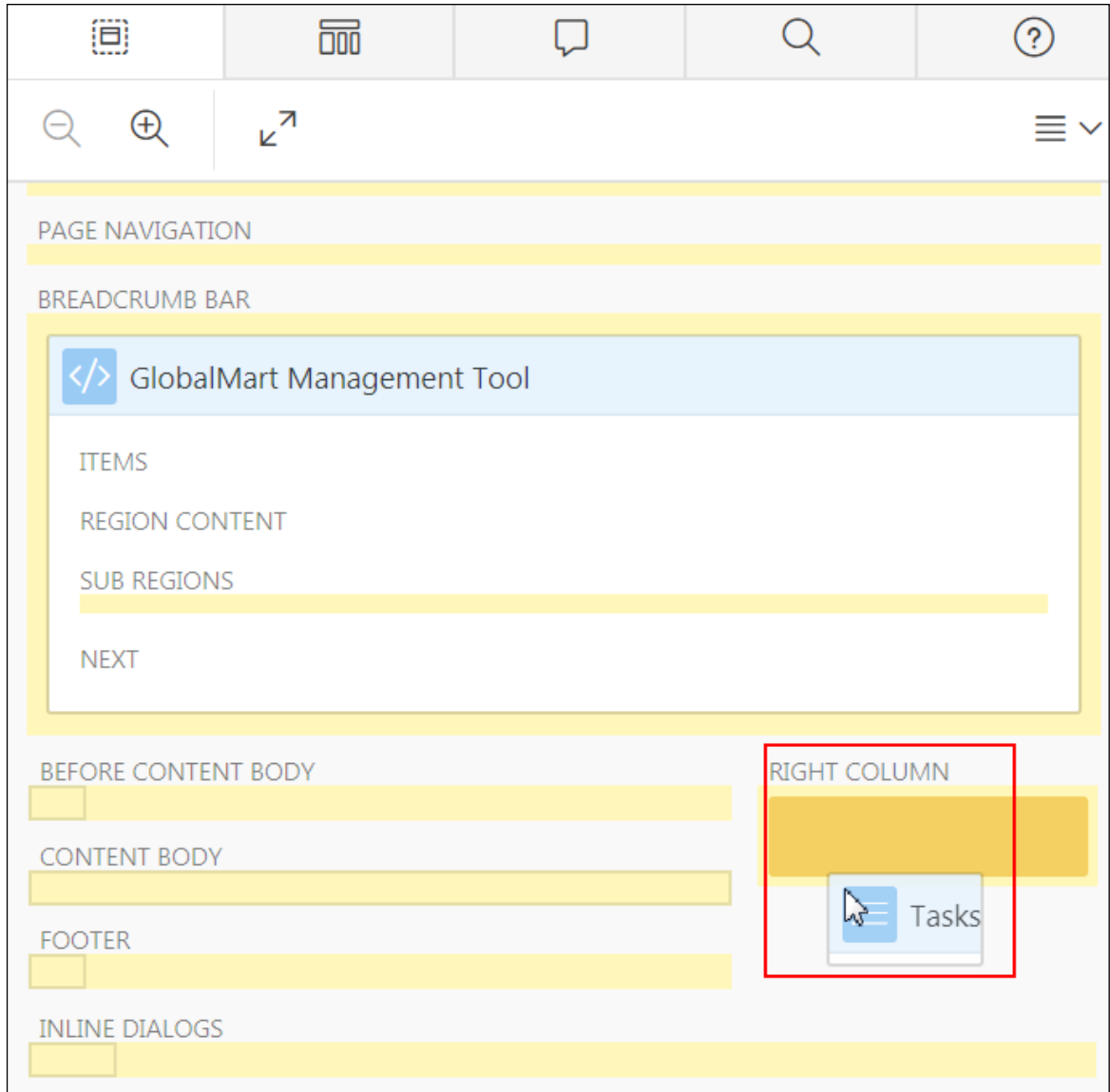
- a. You want to modify the Tasks region. Before modifying that, you need to change the page property. On the Rendering tab, select **Page 1: Home**.



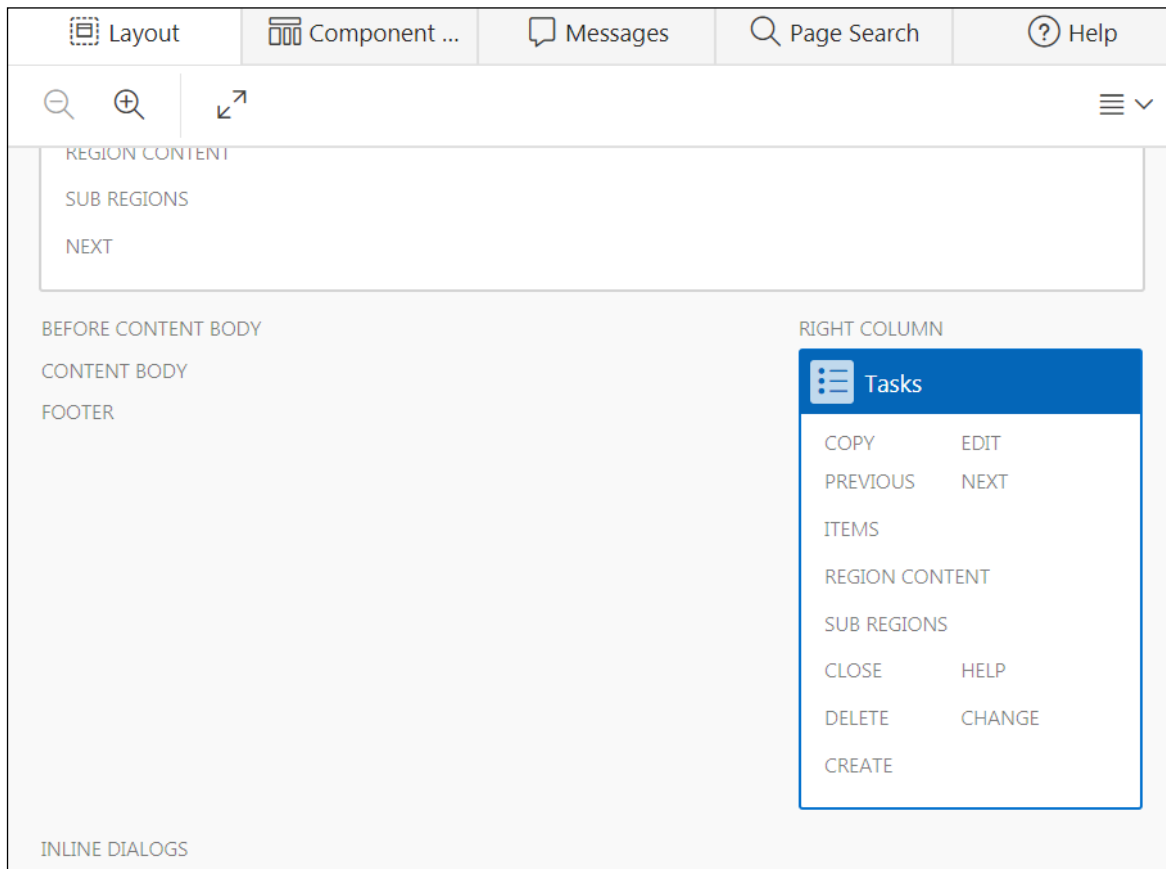
- b. On the Page tab, select **Right Side Column for Appearance > Page Template**.



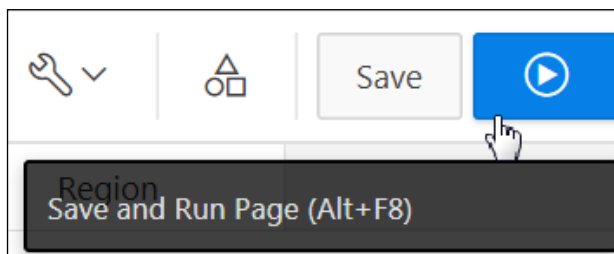
- c. In the Layout, drag the **Tasks** region to inside the right column/right side column area.



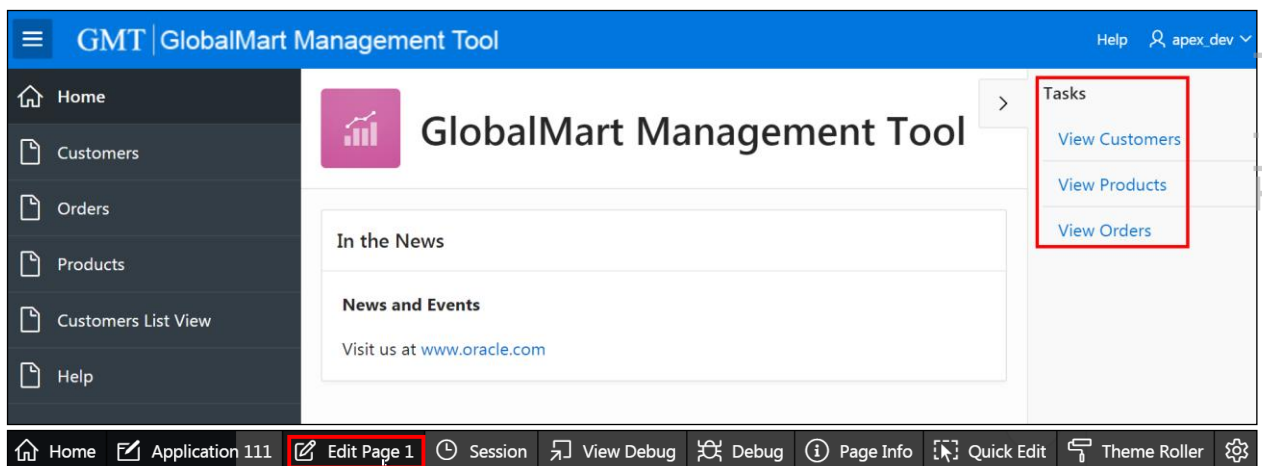
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- d. Click the **Save and Run Page** icon.



- e. The Tasks region is in place. Click the **Edit Page 1** link on the Developer toolbar.



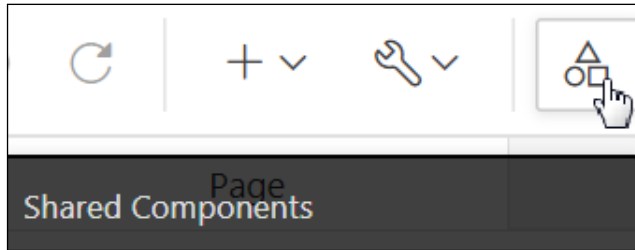
Solution 15-4: Uploading and Applying a CSS

Overview

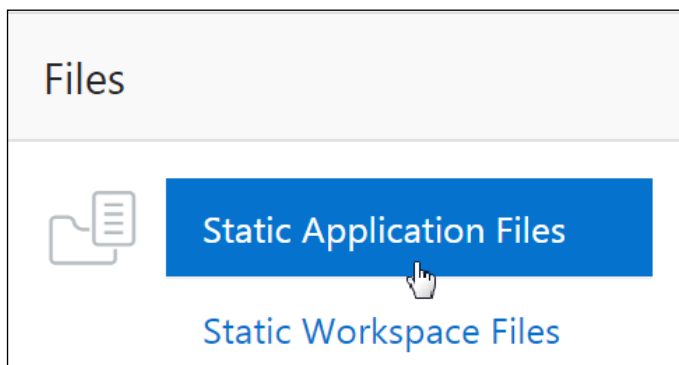
In this practice solution, steps are provided for uploading the CSS file and applying it to the pages of the GMT application to enhance its look and feel.

Steps

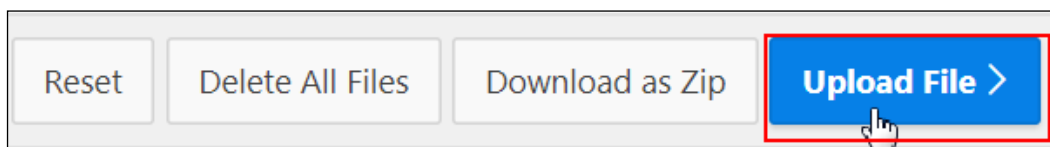
1. Upload the `apexstyle.css` file from the `/home/oracle/labs/labs/` directory.
 - a. Click the **Shared Components** icon.



- b. Under Files, click the **Static Application Files** link.



- c. Click **Upload File >**.



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- d. Select the **apexstyle.css** file in your `/home/oracle/labs/labs/` folder for File and click **Upload**. Note that if you view the CSS file, you will notice that the name of the style is bigblue.

Upload Static Application File(s)

Use this page to associate files like images, CSS or Javascript files with your application. To associate a file with your application, select the file(s), and click Upload.

Directory

* File(s) apexstyle.css

* File Character Set

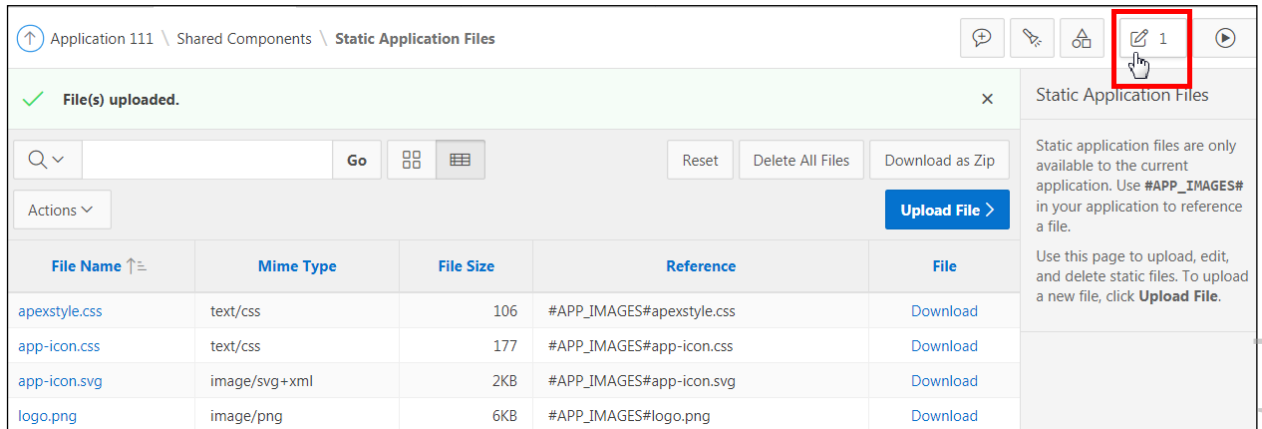
* Unzip File

- e. Notice that the file is uploaded. Copy the reference of the file.

✓ **File(s) uploaded.**

File Name	Mime Type	File Size	Reference
apexstyle.css	text/css	106	#APP_IMAGES#apexstyle.css
app-icon.css	text/css	177	#APP_IMAGES#app-icon.css
app-icon.svg	image/svg+xml	2KB	#APP_IMAGES#app-icon.svg
logo.png	image/png	6KB	#APP_IMAGES#logo.png

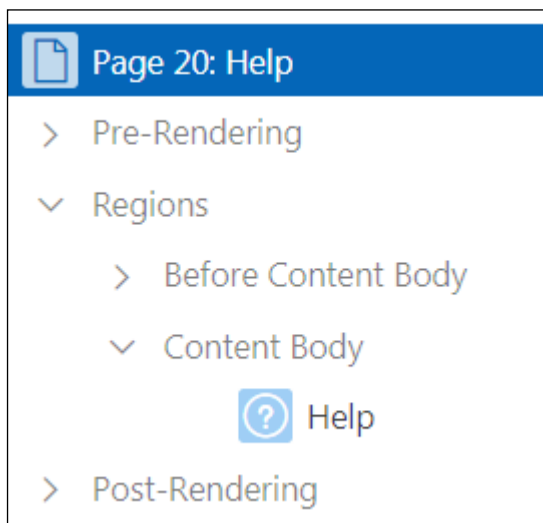
- f. Click the Edit Page icon on the top right corner.



2. Add the cascading style sheet to the Help region on the Help page.
- Navigate to the Help page. Enter the page number of the help page (Page 20) in the **Page Finder** field and click **Go**.



- You first need to add the style sheet to the Help page. On the Rendering tab, ensure that **Page 20: Help** is selected.



- c. In the **CSS > File URLs** field, enter the reference to the uploaded file.

#APP_IMAGES#apexstyle.css

▼ CSS

File URLs

#APP_IMAGES#apexstyle.css

Inline

3. Change the Help text to use the new style, **bigblue**, by using the following script (located in /home/oracle/labs/labs/lab_15_04.txt):

```
<span class="bigblue">No help is available for this page.</span>
```

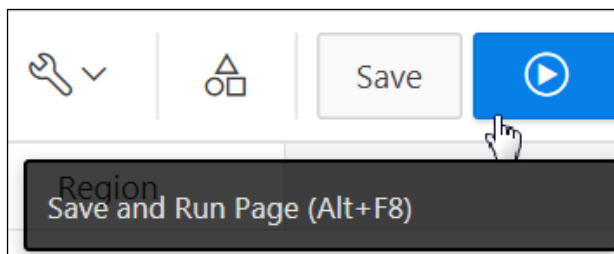
- a. On the Page tab, enter the following for the **Help Text** field (located in /home/oracle/labs/labs/lab_15_04.txt):

```
<span class="bigblue">No help is available for this page.</span>
```

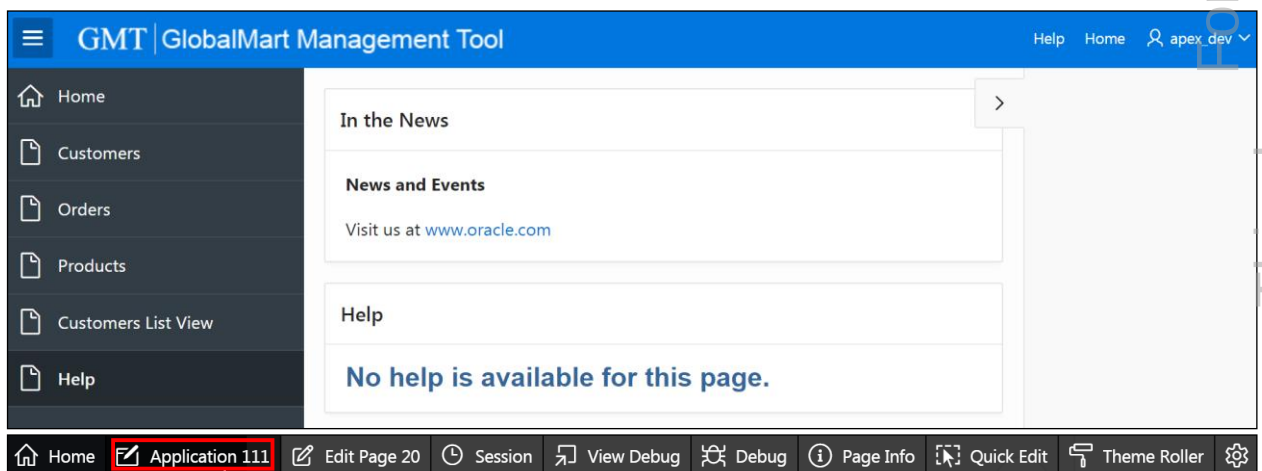


Note: Do not copy the script from this document. Copy the script from the TXT file only.

- b. Click the **Save and Run Page** icon.



- c. The style sheet is successfully applied to the **Help** page. Click the **Application 111** link on the Developer toolbar.



Practices for Lesson 16: Implementing Security

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Practices for Lesson 16: Overview

Overview

In these practices, you examine how to implement security within your application.

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Practice 16-1: Creating an Authentication Scheme

Overview

In this practice, you create an authentication scheme and then make it the current default.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create an authentication scheme called **OpenDoor** based on the Open Door credentials.
2. Switch the current authentication scheme to Application Express.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 16-1**.

If you haven't completed **Practice 15-4**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_15_04.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Practice 16-2: Restricting Users by Using Access Control

Overview

In this practice, you create an access control page, set the application mode to restrict access, create the access control list, and assign application components to an authorization scheme.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create the following users to add to the access control list.

Name	User Type
brad.knight	Developer
susie.parker	Workspace Administrator
john.bell	End User

2. Create an access control page for the GMT application.
3. Set the access control mode to No to give restricted access to only users defined in the access control list to use the application.
4. Add users to the access control list:
 - **john.bell** can only view the information in the application; he cannot make any changes to application data.
 - **brad.knight** should be allowed to edit application data, but he cannot change application administration settings (application mode and the access control list).
 - **susie.parker** is the administrator of the application, so she can change anything in addition to changing user privileges.
 - **apex** is also an administrator of the application.
5. Define and apply the authorization schemes to each application component. This will restrict access to application pages and components.
 - Users with the **Reader** privilege can review customer information, but cannot change or create information.
 - Users with the **Contributor** privilege can make changes to customer information but cannot make changes to the application mode and the access control list.
 - Users with the **Administrator** privilege can make any changes, including administering the GMT application.
6. Run the application and sign in as a different user. What do you observe?

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Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 16-2**.

If you haven't completed **Practice 16-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_16_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Practice 16-3: Enabling Session State Protection

Overview

In this practice, you enable Session State Protection, set the Session State Protection for a particular item, and review the various options for page access protection so that you can examine their differences.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Set the Page Access Protection for the Employee Commission page to **No URL Access**.
2. Add a **Commission** button to the **Top Tier Salary** page that redirects to the **Employee Commission** page. Run the page to see that there is an error because of the branch.
3. Edit the **Commission** button to submit the page and create a **Branch to Page** branch that will go to the **Employee Commission** page without passing the URL.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- a. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- b. Click the **Upload >** button.
- c. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- d. Run the script by clicking the **Run** icon.
- e. Click **Run Now**.
- f. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 16-3**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.

- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. On the Application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Log out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

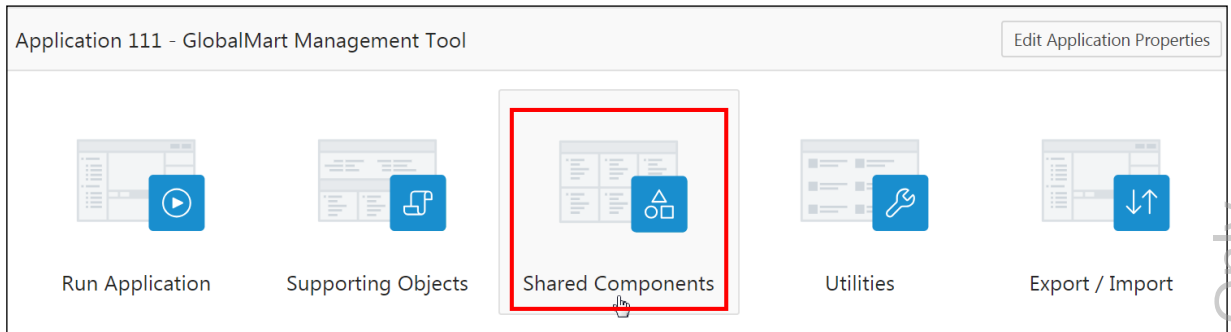
Solution 16-1: Creating an Authentication Scheme

Overview

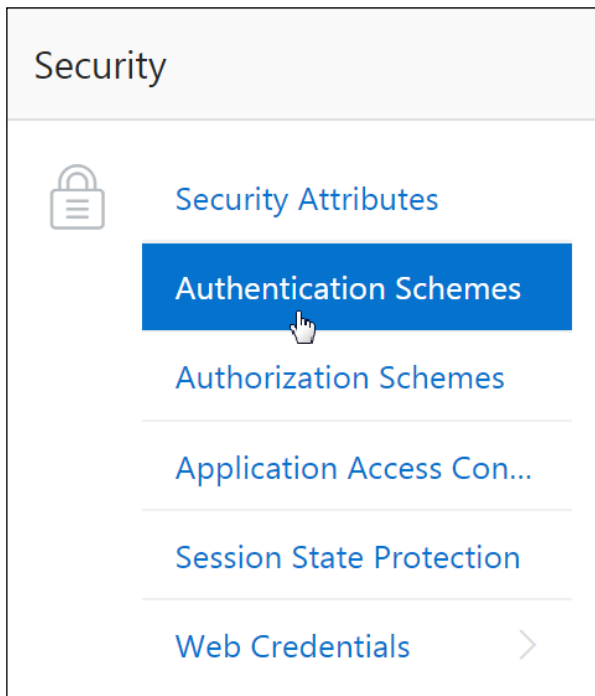
In this practice solution, steps are provided for creating an authentication scheme for GlobalMart Management Tool.

Steps

1. Create an authentication scheme called **OpenDoor** based on the Open Door credentials.
 - a. On the application home page, click **Shared Components**.



- b. Under **Security**, click the **Authentication Schemes** link.



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- c. You will see that Application Express Authentication is set as the Current Authentication scheme. Click **Create >**.

The screenshot shows the 'Authentication Schemes' page. At the top, there are tabs for 'Authentication Schemes', 'Subscription', and 'History'. Below the tabs is a search bar with a magnifying glass icon, a 'Go' button, and a table view icon. To the right of the search bar are buttons for 'Actions', 'Reset', and a highlighted 'Create >' button. Below these buttons is a table with the following columns: 'Name', 'Scheme Type', 'Subscribed From', and 'Subscribers'. The first row of the table shows 'Application Express Authentication - Current' as the Name, 'Application Express Accounts' as the Scheme Type, and empty cells for the other two columns.

- d. Leave the default value of **Based on a pre-configured scheme from the gallery** and click **Next >**.

The screenshot shows the 'Create Authentication Scheme' dialog. At the top, there is a progress bar with a blue dot and the label 'Method'. Below the progress bar is a text area that says: 'When you create a new authentication scheme, you have several options. Most let you reuse implementations that already exist in your application or in other applications within your workspace. There are even some pretested schemes you can copy to get you up and running immediately.' Below the text area is a 'Create Scheme:' section with two radio buttons. The first radio button is selected and is labeled 'Based on a pre-configured scheme from the gallery'. The second radio button is labeled 'As a copy of an existing authentication scheme'. Below the radio buttons is an 'Information' section with a right-pointing arrow. At the bottom of the dialog are two buttons: 'Cancel' and a highlighted 'Next >' button.

- e. For **Name**, enter `OpenDoor` and select the **Open Door Credentials** option for the **Scheme Type**. Click **Create Authentication Scheme**.

The screenshot shows the 'Authentication Scheme' form. At the top, there is a back arrow, the title 'Authentication Scheme', a 'Cancel' button, and a highlighted 'Create Authentication Scheme' button. Below the title bar is a 'Name' field with the text 'OpenDoor' and a question mark icon. Below the 'Name' field is a 'Scheme Type' dropdown menu with the text 'Open Door Credentials' and a question mark icon. The 'Name' and 'Scheme Type' labels are marked with red asterisks. The 'Create Authentication Scheme' button is highlighted with a red box.

The OpenDoor Authentication scheme is created, and it becomes the Current Authentication scheme.

Application 111 \ Shared Components \ **Authentication Schemes**

✓ **Action processed. Authentication scheme activated as current authentication scheme.**

Authentication Schemes Subscription History

Q ▾ Go   Actions ▾ Reset

Name ↑ ▾	Scheme Type	Subscribed From
Application Express Authentication	Application Express Accounts	
OpenDoor - Current	Open Door Credentials	

- f. Click **Application 111** breadcrumb and click **Run Application**.

Application 111 \ Shared Components \ **Authentication Schemes**



Run Application

- g. To sign into the system, enter your name in the **Username** field and then click **Login**. Notice that the Sign in page uses the Open Door Credential Authentication scheme and does not prompt you to enter the password. If you are already logged in, click **Sign Off** and perform this step.

Help  apex_dev ▾



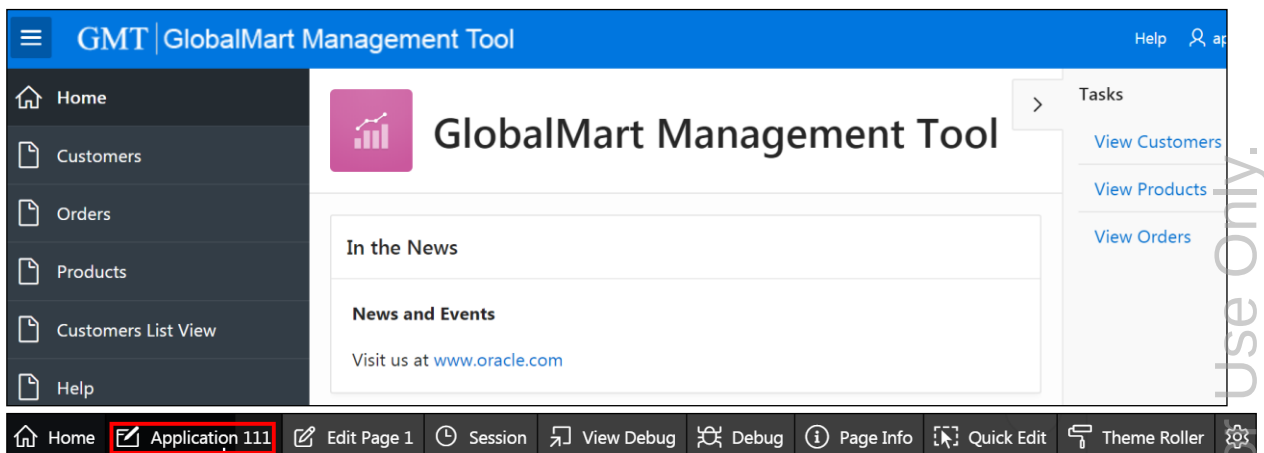
Log In to Application 111

Enter your credentials in this form to start a new session in this Application Express application.

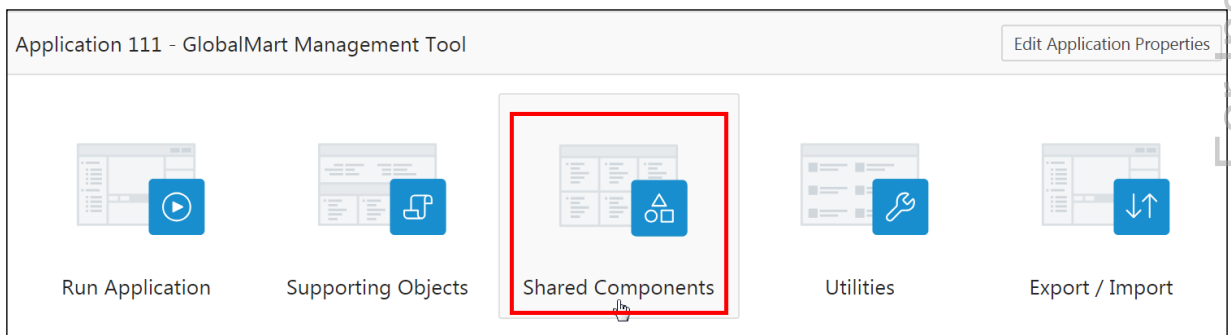
* Username

Login

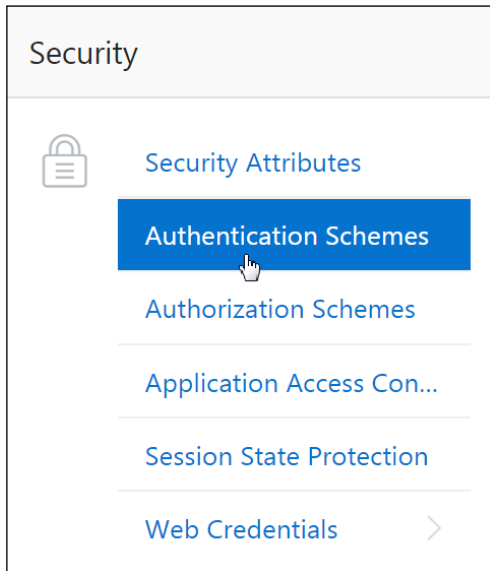
- h. The Home page is displayed. In the next practice, you will use access control to set up authorization to certain pages and buttons. Click the **Application 111** link on the Developer toolbar.



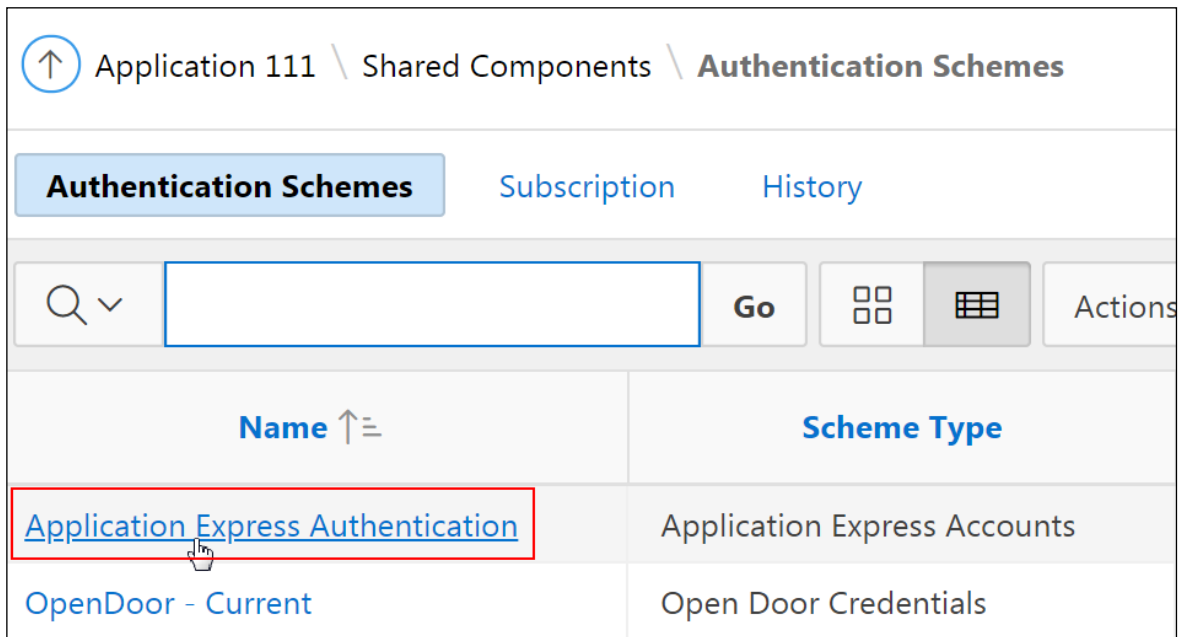
2. Switch the current authentication scheme to Application Express.
 - a. Navigate to the Shared Components page. Click the **Shared Components** icon.



- b. Under **Security**, select **Authentication Schemes**.



- c. Click the **Application Express Authentication** link.



- d. Click **Make Current Scheme**.

The 'Authentication Scheme' dialog box has a title bar with 'Authentication Scheme' and three buttons: 'Cancel', 'Delete', and 'Make Current Scheme' (highlighted with a red rectangle). Below the title bar is a tabbed interface with 'Show All', 'Name', 'Subscript...', 'Source', 'Session N...', 'Login Pro...', 'Post-Log...', 'Session S...', and 'Comments'. The 'Name' tab is active, showing a form with two fields: 'Name' (with a red asterisk) containing 'Application Express Authentication' and 'Scheme Type' (with a red asterisk) set to 'Application Express Accounts'. Both fields have help icons.

- e. You get the following message. Click **OK**.

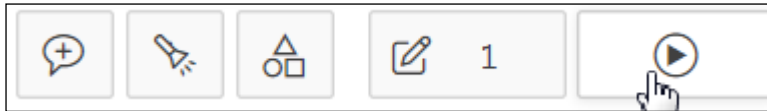
A confirmation dialog box with the text: '10.237.16.240:8080 says' and 'Would you like to make 'Application Express Authentication' the current authentication scheme of the application?'. At the bottom right are two buttons: 'OK' (highlighted with a red rectangle) and 'Cancel'.

Application Express Authentication is now the current scheme.

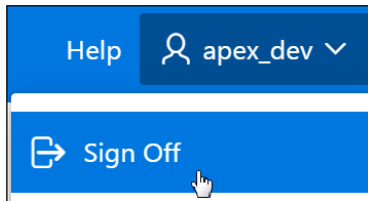
The 'Authentication Schemes' page shows a success message: 'Authentication scheme activated as current authentication scheme.' Below this is a table with two tabs: 'Authentication Schemes' (active) and 'History'. The table has a search bar and a 'Go' button. The table columns are 'Name' and 'Scheme Type'.

Name ↑	Scheme Type
Application Express Authentication - Current	Application Express Accounts
OpenDoor	Open Door Credentials

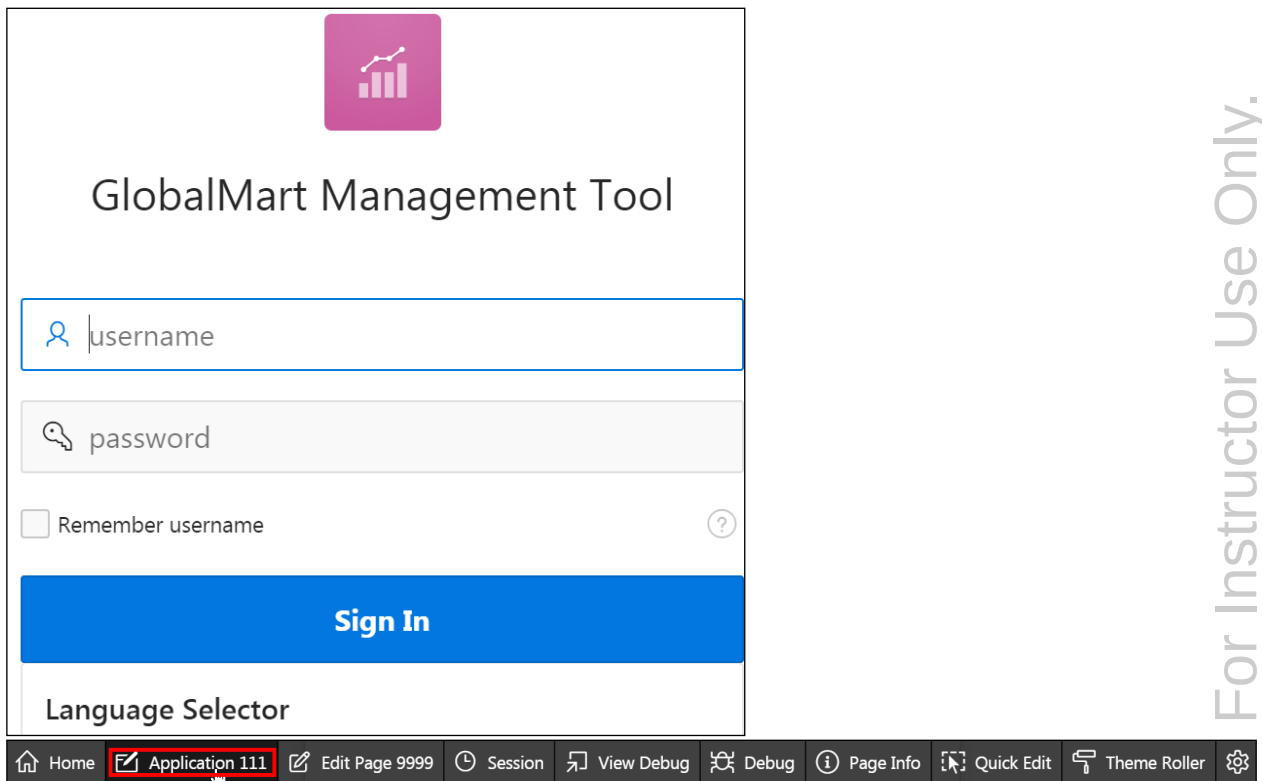
- f. Click the **Run Page 1** icon.



- g. Click **Sign Off**.



- h. Notice that the Sign in page uses the Application Express authentication login now instead of the OpenDoor login. Click the **Application 111** link on the Developer toolbar.



Solution 16-2: Restricting Users by Using Access Control

Overview

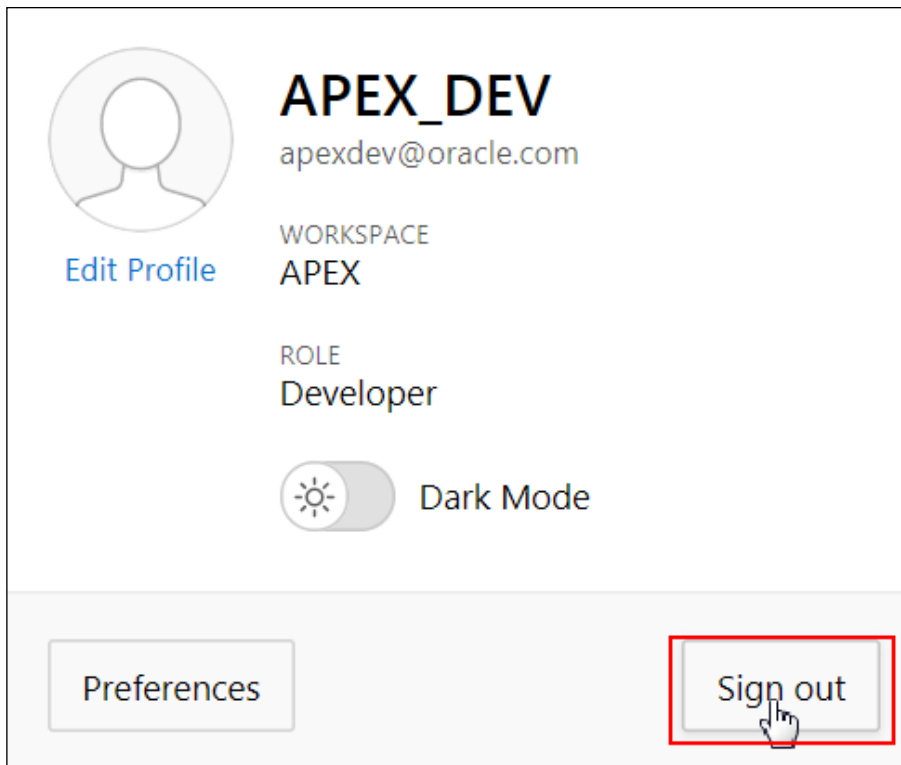
In this practice solution, steps are provided for restricting users to access GlobalMart Management Tool by using Access Control.

Steps

1. Create the following users to add to the access control list:

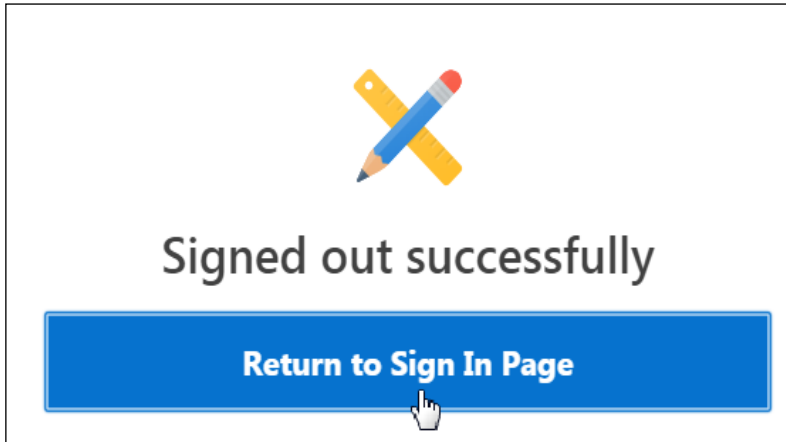
Name	User Type
brad.knight	Developer
susie.parker	Workspace Administrator
john.bell	End User

- a. Sign out of Application Express. Click the **Account Menu** icon at the top-right section of the page and click **Sign out**.



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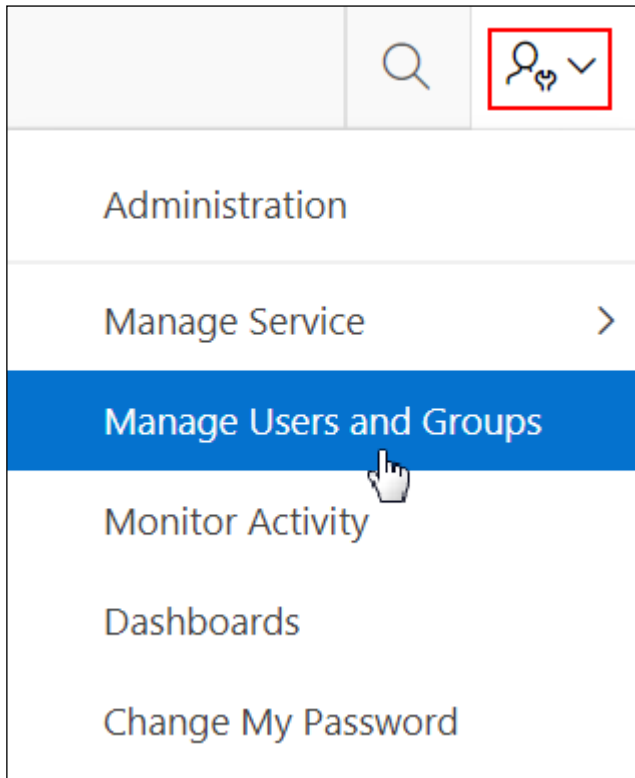
- b. Click the **Return to Sign In Page** button.



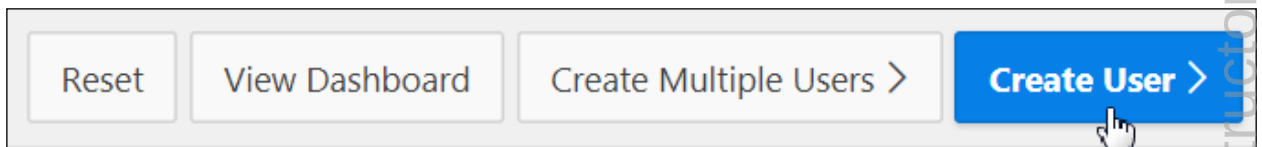
- c. Sign in as the **apex_admin** user (workspace administrator).

A screenshot of the Oracle Application Express sign-in page. At the top is the same crossed-pencils 'X' icon. Below it, the title 'Oracle Application Express' is centered in a bold black font. The sign-in form consists of three input fields, each with a green checkmark on the right: the first field has a database icon and the text 'apex'; the second field has a person icon and the text 'apex_admin'; the third field has a key icon and a masked password '.....'. Below these fields is a checkbox labeled 'Remember workspace and username' with a help icon. At the bottom is a large blue 'Sign In' button with a mouse cursor clicking on it.

- d. Click the **Administration** icon and select **Manage Users and Groups**.



- e. Click **Create User >**.



- f. Enter the following information and click **Create and Create Another**:

- Username: brad.knight
- Email Address: brad.knight@oracle.com
- Default Schema: OEHR
- Select **No** for the **User is a workspace administrator** option.
- Select **Yes** for the **User is a developer** option.
- Password: <refer the password document>
- Confirm Password: <refer the password document>
- Select **No** for **Require Change of Password on First Use**.

User Identification

* Username ?

* Email Address ?

First Name ?

Last Name ?

Account Privileges

Default Schema ?

Accessible Schemas (null for all) ?

User is a workspace administrator: ☐ Yes ☒ No ?

User is a developer: ☒ Yes ☐ No ?

App Builder Access ?

SQL Workshop Access ?

Password

* Password Passwords are case sensitive ?

* Confirm Password ?

Require Change of Password on First Use ?

In the Group Assignments section, click >> to assign the OAuth2 Client Developer, RESTful Services, and SQL Developer roles to the user.

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Group Assignments

Group Assignments:

OAuth2 Client Deve
RESTful Services
SQL Developer

Cancel Create and Create Another Create User

g. Enter the following information and click **Create and Create Another**:

- Username: `susie.parker`
- Email Address: `susie.parker@oracle.com`
- Default Schema: OEHR
- Select **Yes** for the **User is a workspace administrator** option.
- Password: <refer the password document>
- Confirm Password: <refer the password document>
- Select **No** for **Require Change of Password on First Use**.

User Identification

* Username `susie.parker` ?

* Email Address `susie.parker@oracle.com` ?

First Name ?

Last Name ?

Description

Account Privileges

Default Schema OEHR ?

Accessible Schemas (null for all) ?

User is a workspace administrator: ☒ **Yes** ☐ No ?

User is a developer: ☒ **Yes** ☐ No ?

App Builder Access Yes ?

SQL Workshop Access Yes ?

Password

* Password Passwords are case sensitive ?

* Confirm Password ?

Require Change of Password on First Use No ?

In the Group Assignments section, click >> to assign the OAuth2 Client Developer, RESTful Services, and SQL Developer roles to the user.

Group Assignments

Group Assignments:

< >> > < <<

OAuth2 Client Deve
RESTful Services
SQL Developer

Cancel Create and Create Another Create User

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h. Enter the following information and click **Create User**:

- Username: john.bell
- Email Address: john.bell@oracle.com
- Default Schema: OEHR
- Select **No** for the **User is a workspace administrator** option.
- Select **No** for the **User is a developer** option.
- Password: <refer the passwords document>
- Confirm Password: <refer the passwords document>
- Select **No** for **Require Change of Password on First Use**.

User Identification

* Username

john.bell

?

* Email Address

john.bell@oracle.com

?

First Name

?

Last Name

?

Description

Account Privileges

Default Schema

OEHR

▼

?

Accessible Schemas (null for all)

?

User is a workspace administrator:

☐ Yes

☒ No

?

User is a developer:

☐ Yes

☒ No

?

App Builder Access

No

▼

?

SQL Workshop Access

No

▼

?

Password

* Password Passwords are case sensitive (?)

* Confirm Password (?)

Require Change of Password on First Use (?)

In the Group Assignments section, click >> to assign the OAuth2 Client Developer, RESTful Services, and SQL Developer roles to the user.

Group Assignments

Group Assignments:

< >> > < <<

OAuth2 Client Deve
RESTful Services
SQL Developer

Cancel Create and Create Another **Create User**

The new users are now in the list.

↑ Manage Users and Groups

✓ User created.

Users Groups Group Assignments

Q ▾ Go

Actions ▾

User ↑	Email	Account Type
APEX	apex@oracle.com	Developer
APEX_ADMIN	apex@apex.com	Workspace Administrator
APEX_DEV	apexdev@oracle.com	Developer
BRAD.KNIGHT	brad.knight@oracle.com	Developer
JOHN.BELL	john.bell@oracle.com	End User
SUSIE.PARKER	susie.parker@oracle.com	Workspace Administrator

- i. Click the **View Icons** button.

Users
Groups
Group Assignments

APEX

APEX ADMIN

APEX DEV

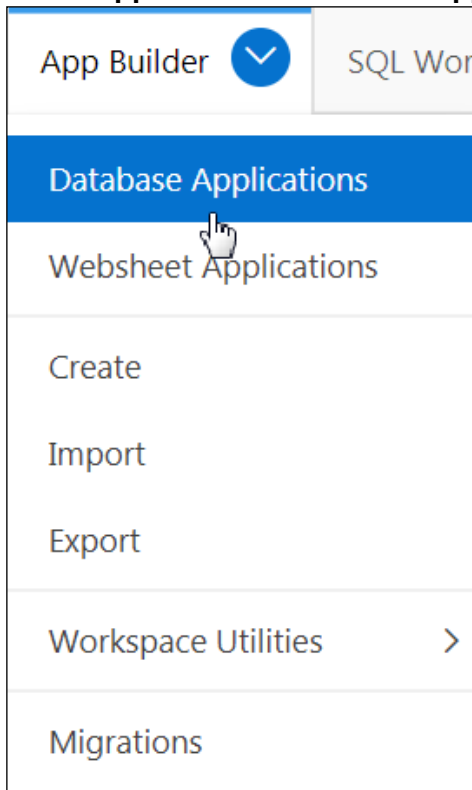
BRAD. KNIGHT

JOHN. BELL

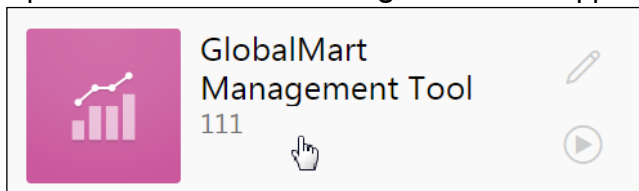
SUSIE. PARKER

2. Create an access control page for the GMT application.

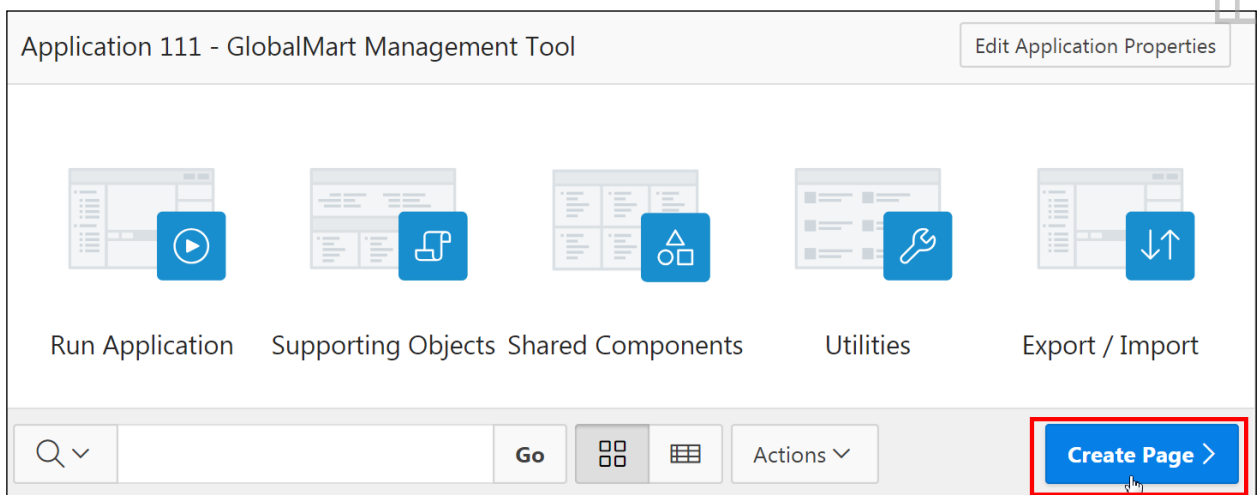
a. Select **App Builder > Database Applications**.



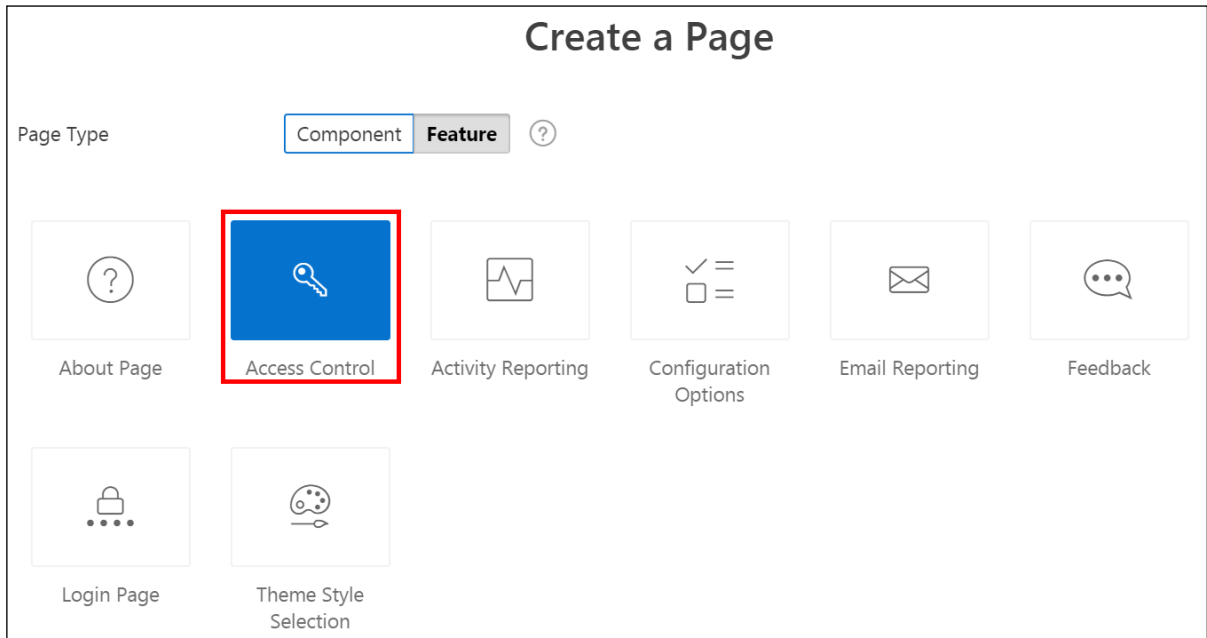
b. Open **111 GlobalMart Management Tool** application.



c. Ensure that you are signed in as an Administrator (username: apex_admin) and click **Create Page >**.



- d. Click the **Feature** tab and select **Access Control**.



- e. On the Create Access Control page, enter the following values and click **Next**:
- Starting Page Number: Enter 10010.
 - Administration Page Reference: Select **Create a new page**.
 - Administration Page Number: Enter 10000.
 - Administration page Name: Enter Access Control Administration Page.
 - Administration Page Navigation Preference: Select **Create a new navigation menu entry**.
 - New Navigation Menu Entry Name: Enter Admin.
 - Parent Navigation Menu Entry Select: -No parent selected -

Create Access Control Pages

* Starting Page Number ?

Page Group ?

Build Option **Feature: Access Control** ?

* Administration Page Preference ☒ **Create a new page** ?
☐ Identify an existing page

* Administration Page Number ?

* Administration Page Name ?

Administration Page Navigation Preference ☐ Do not associate this page with a navigation menu entry ?
☒ **Create a new navigation menu entry**
☐ Identify an existing navigation menu entry for this page

* New Navigation Menu Entry ?

Parent Navigation Menu Entry ?
Home
Customers
Orders
Products
Customers List View

< Cancel

Next

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- f. The **Create Access Control Pages - Confirmation** page appears. Review the details and click **Create**.

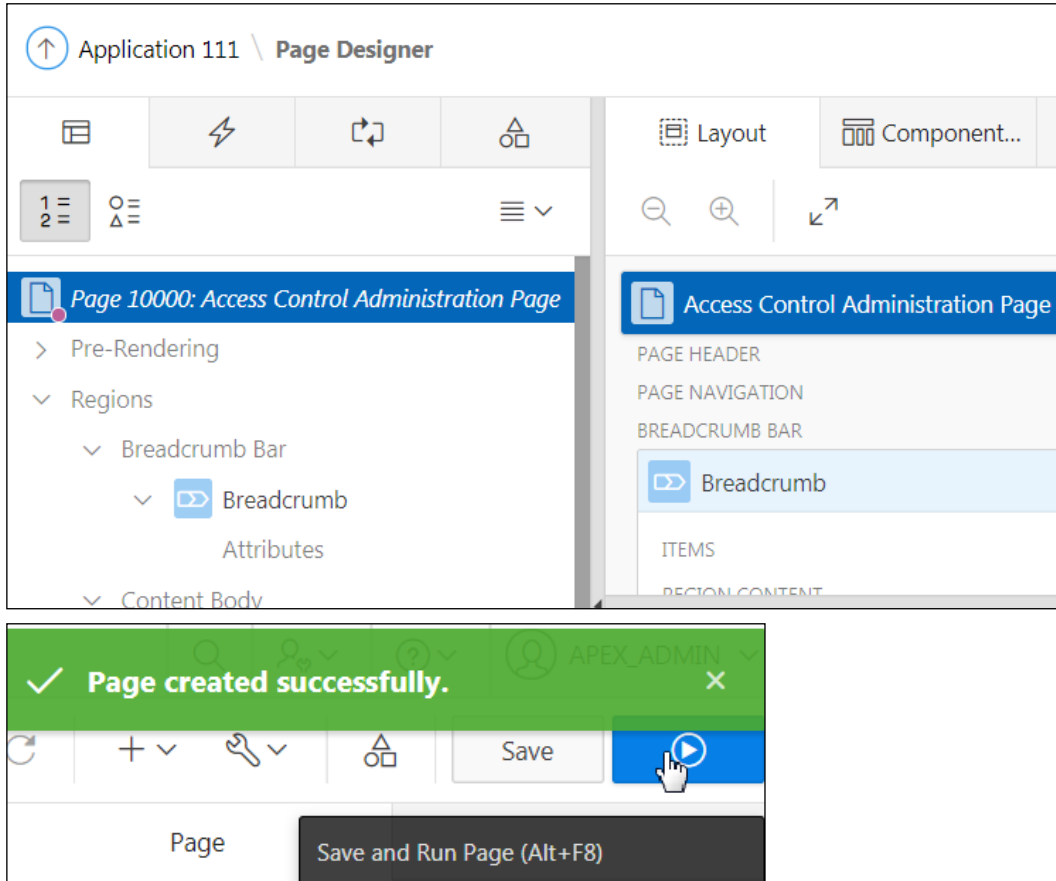
Create Access Control Pages - Confirmation

You have requested to create a page with the following attributes. Please confirm your selections.

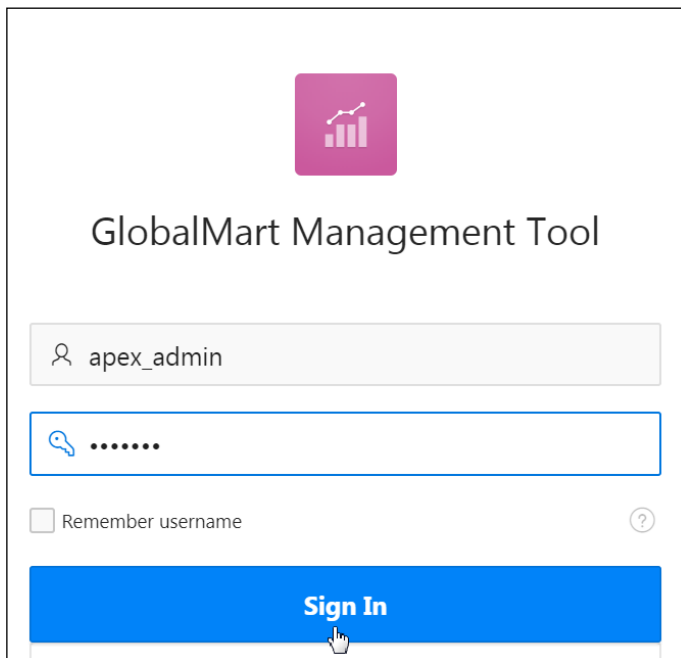
- Multiple pages will be created, starting with Page: 10010
- A new Administration page will be added, Page: 10000 - Access Control Administration Page
- The following Application Access Role(s) will be created:
 - Administrator
 - Contributor
 - Reader
- The following Authorization Scheme(s) will be created:
 - Administration Rights
 - Contribution Rights
 - Reader Rights
- The following Build Option(s) will be created:
 - Feature: Access Control
- The following Application Setting(s) will be created:
 - ACCESS_CONTROL_SCOPE

< Cancel Create

- g. The Page **10000: Access Control Administration Page** opens in Page Designer view. Click the **Save and Run Page** icon.



- h. Sign in as the **apex_admin** user (workspace administrator).



The **Access Control Administration Page** with links to **Users** and **Access Control** loads with the message “Only users defined in the application access control list may access this application.”

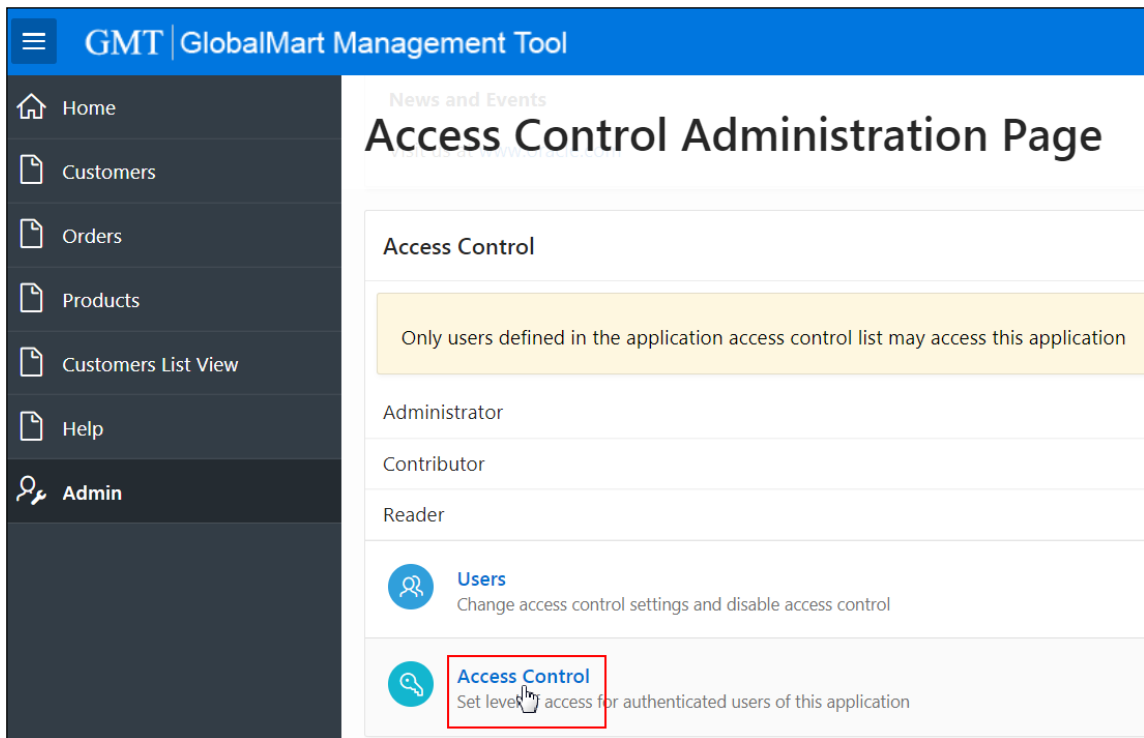
The screenshot shows the GMT GlobalMart Management Tool interface. The top navigation bar is blue with the GMT logo and the text "GlobalMart Management Tool". The left sidebar is dark grey with a list of menu items: Home, Customers, Orders, Products, Customers List view, Help, and Admin (highlighted with a magnifying glass icon). The main content area is white and titled "Access Control Administration Page". Below the title is a "News and Events" section with a link to "www.oracle.com". The "Access Control" section contains a yellow warning box with the message "Only users defined in the application access control list may access this application". Below this are three rows of user roles: Administrator, Contributor, and Reader. At the bottom are two links: "Users" (with a person icon) and "Access Control" (with a key icon), both with descriptive text.

Access Control	
Only users defined in the application access control list may access this application	
Administrator	
Contributor	
Reader	
 Users	Change access control settings and disable access control
 Access Control	Set level of access for authenticated users of this application

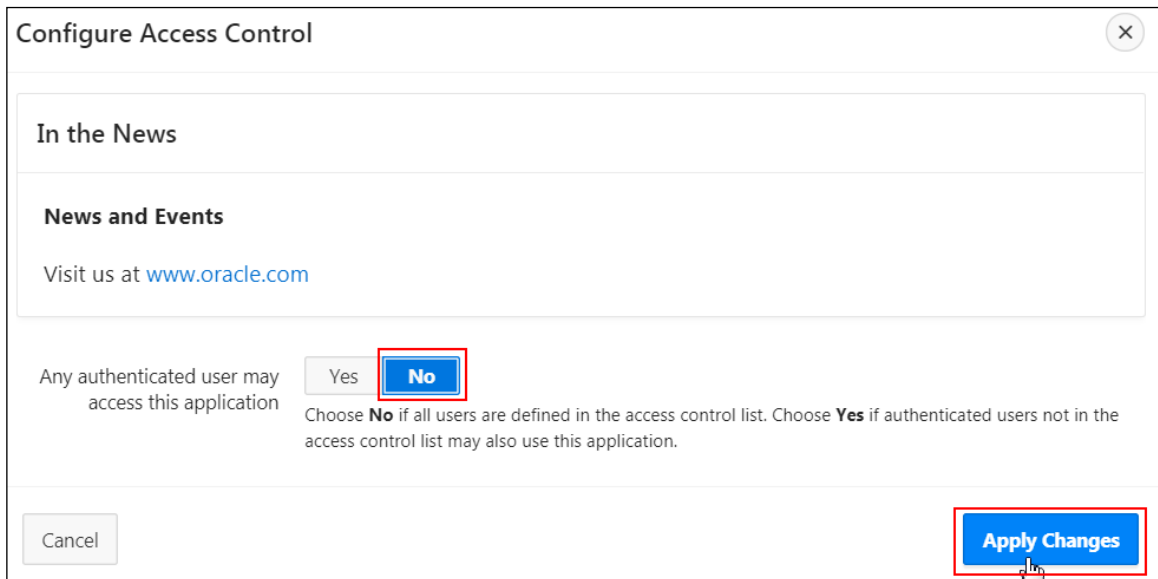
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3. Set Access Control for access to the authenticated users.

a. Click the **Access Control** link.



b. The Configure Access Control page appears. You want to give restricted privileges to only the users defined in the access control list. Select **No**. Click **Apply Changes**.



The message "Access Control settings saved" appears.



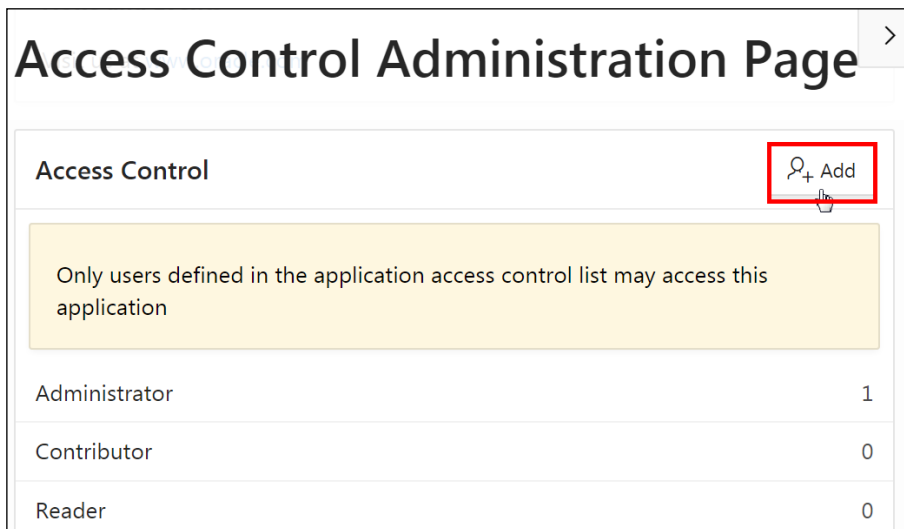
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4. Add users to the access control list:

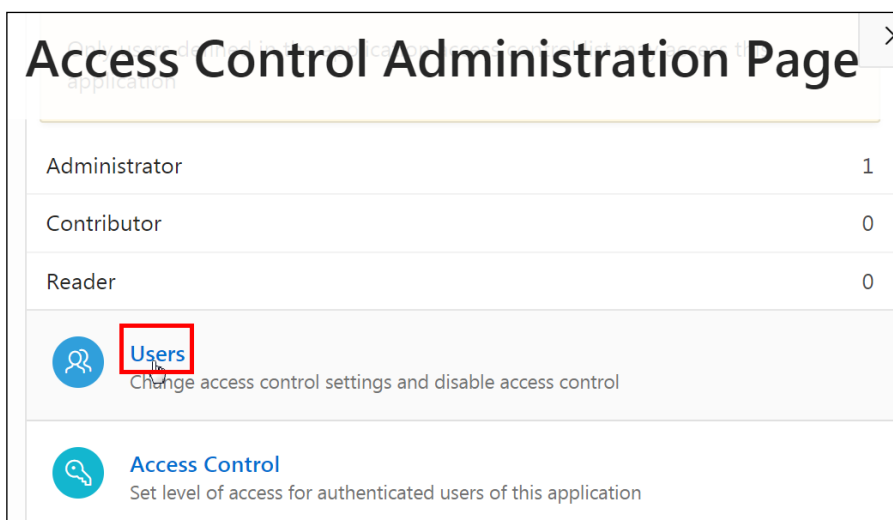
- **john.bell** can only view the information in the application; he cannot make any changes to application data.
- **brad.knight** should be allowed to edit application data, but he cannot change application administration settings (application mode and the access control list).
- **susie.parker** is the administrator of the application, so she can change anything in addition to changing the user privileges.
- **apex_dev** user is already created, but you must grant **apex_dev** user to the access control list.

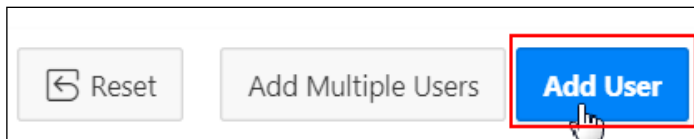
[**apex_admin** is already existing as an administrator of the application.]

- a. The access control mode has been set. Now, users whose credentials are defined in the access control list can access your GlobalMart Management Tool application. To add your privileged users, click **Add User**.



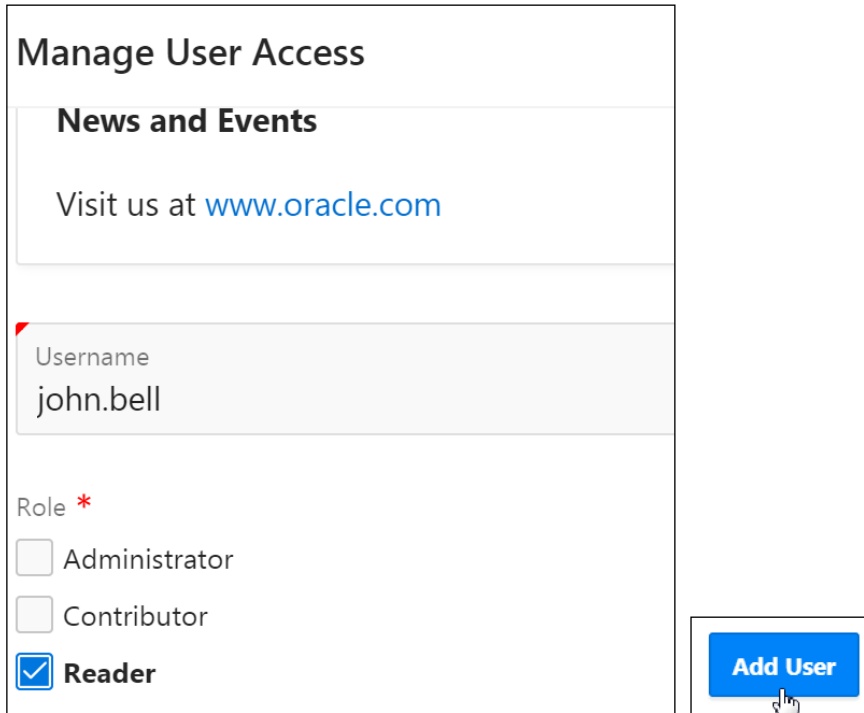
Note that alternatively, you can add a user by clicking the **User** link and then clicking **Add User**.





Reset Add Multiple Users **Add User**

- b. You want John Bell to only be able to view the information in your application. He cannot make any changes to application data. Enter `john.bell` as **Username** and select **Reader** for **Role**. Then click **Add User**.



Manage User Access

News and Events

Visit us at www.oracle.com

Username
john.bell

Role *

☐ Administrator

☐ Contributor

☒ **Reader**

Add User

- c. You want Brad Knight to be able to edit the information in your application, but not be able to change any of the administration options. Enter `brad.knight` as **Username** and select **Contributor** for **Role**. Then click **Add User**.

The screenshot shows a web form titled "Manage User Access". At the top, there is a section "News and Events" with the text "Visit us at www.oracle.com". Below this is a form for adding a user. The "Username" field contains "brad.knight". The "Role" section has three radio buttons: "Administrator", "Contributor" (which is selected), and "Reader". A red asterisk is next to the "Role" label. To the right of the form is a blue "Add User" button with a mouse cursor pointing at it.

- d. You want Susie Parker to be able to edit the information in your application as well as update the administration access control list. Enter `susie.parker` as **Username** and select **Administrator** for **Role**. Click **Add User**.

The screenshot shows the same "Manage User Access" form. The "Username" field now contains "susie.parker". In the "Role" section, the "Administrator" radio button is selected, while "Contributor" and "Reader" are unselected. The "Add User" button is still present with a mouse cursor pointing at it.

- e. You want apex_dev user (that you had created in the initial practices) to be able to edit the information in your application, but not be able to change any of the administration options. Enter apex_dev as **Username** and select **Contributor** for **Role**. Then click **Add User**.

Manage User Access

News and Events

Visit us at www.oracle.com

Username

apex_dev

Role *

☐ Administrator

☒ **Contributor**

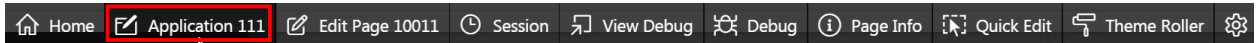
☐ Reader

Add User

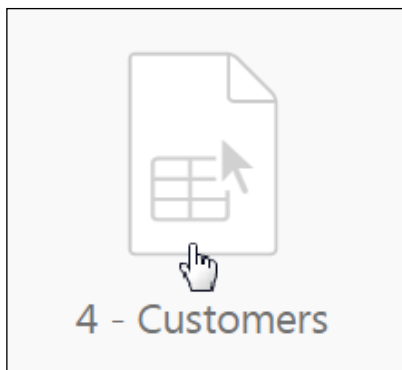
The following users with their respective roles/privileges are created:

Manage User Access		
Visit us at www.oracle.com		
Q ▾ Go Actions ▾		
	Username	Roles
	apex_admin	Administrator
	apex_dev	Contributor
	brad.knight	Contributor
	john.bell	Reader
	susie.parker	Administrator

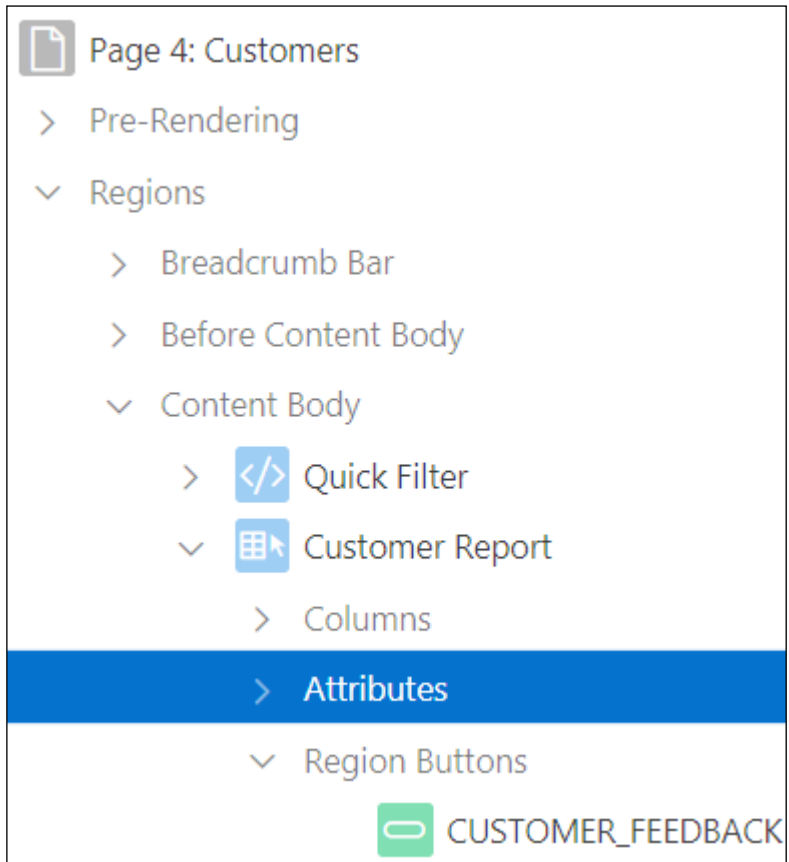
5. Define and apply the authorization schemes to each application component. This will restrict access to application pages and components.
- Users with the **Reader** privilege can review customer information, but cannot change or create information.
 - Users with the **Contributor** privilege can make changes to customer information but cannot make changes to the access control mode and the access control list.
 - Users with the **Administrator** privilege can make any changes, including administering the GMT application.
- a. You can define which areas of the application are restricted. Click the **Application 111** link on the Developer toolbar.



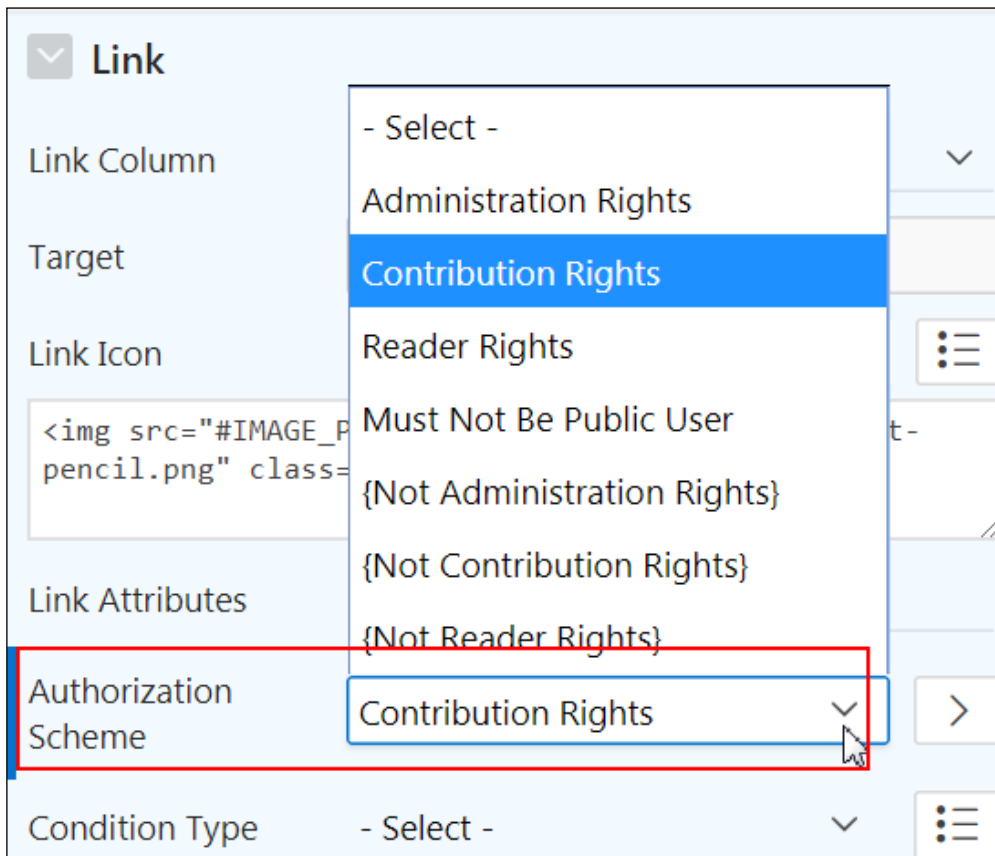
- b. You can restrict access for users with the view privilege to customer information. On the **GlobalMart Management Tool** application home page, click **4- Customers** page.



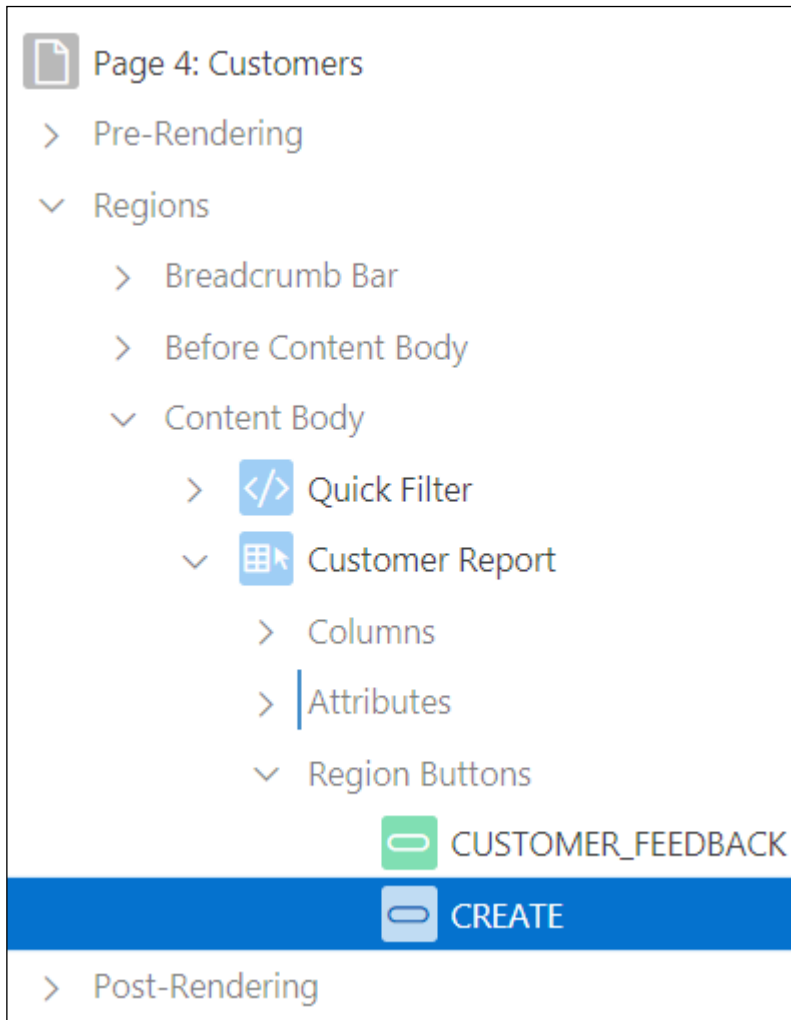
- c. On the Rendering tab, select **Regions > Content Body > Customer Report > Attributes**.



- d. In the Property Editor, select **Contribution Rights** for **Authorization Scheme** under **Link**.

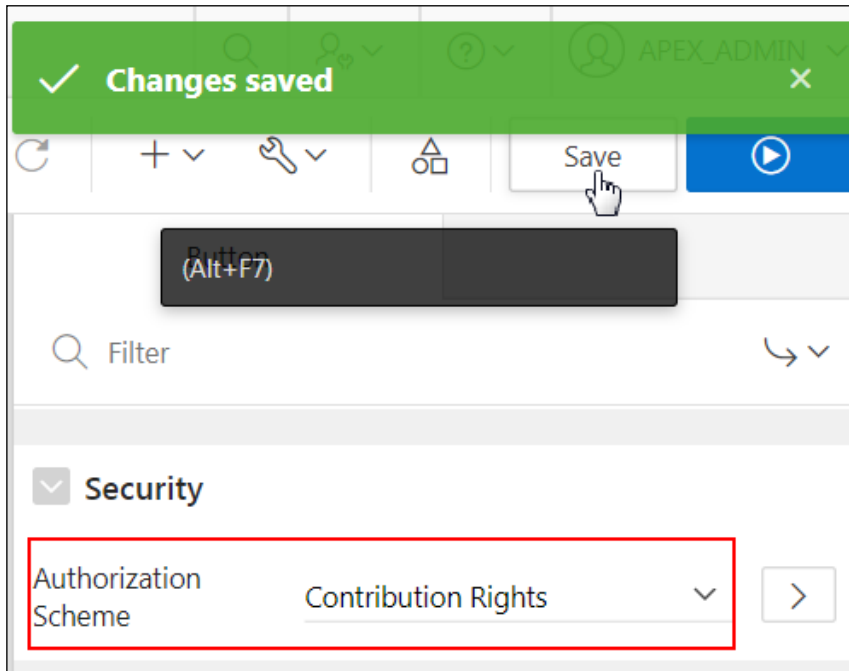


- e. You want the Create Button to appear only if the user has the **Contributor** or **Administrator** privilege. To achieve this, you must set the authorization scheme accordingly. Under **Buttons** for the Customer Report region, select **CREATE**.



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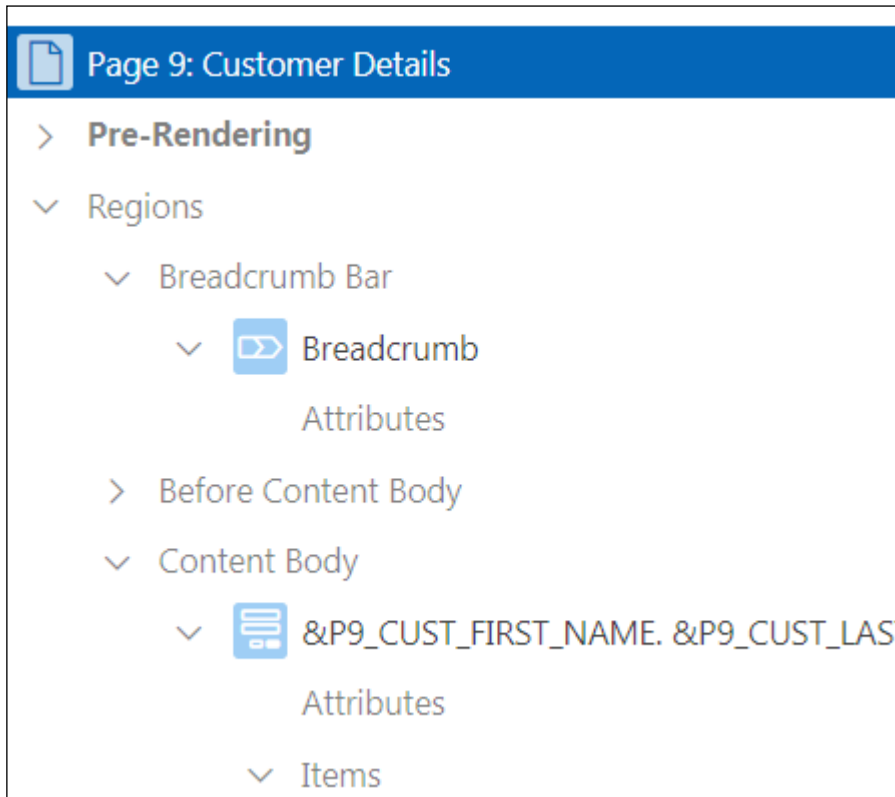
- f. On the Button tab, scroll down to the **Security** tab. Select **Contribution Rights** for **Authorization Scheme** and click **Save**.



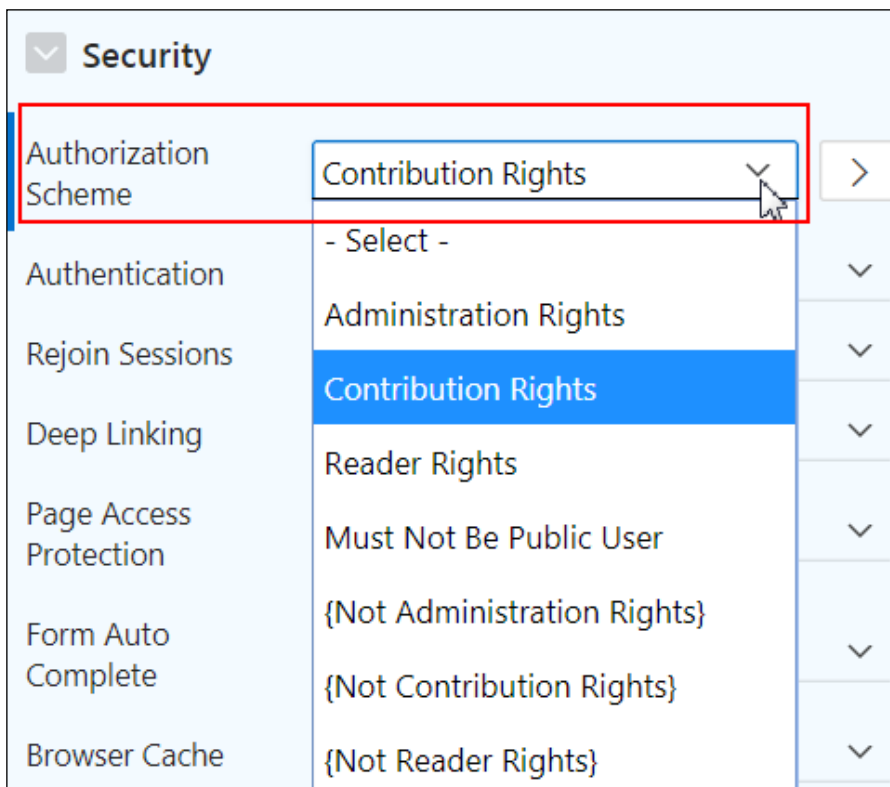
- g. You also want to protect against direct access to a page. Even though you restricted a user who did not have the Contributor privilege from editing or creating users on the Customers page, he or she can still access Customer Details if the correct URL is entered. To prevent this from happening, you must restrict Customer Details to only users with the Contributor (Edit) privilege. In Page Finder, enter the page number of the **Customer Details** page (in this case 9) and click **Go**.



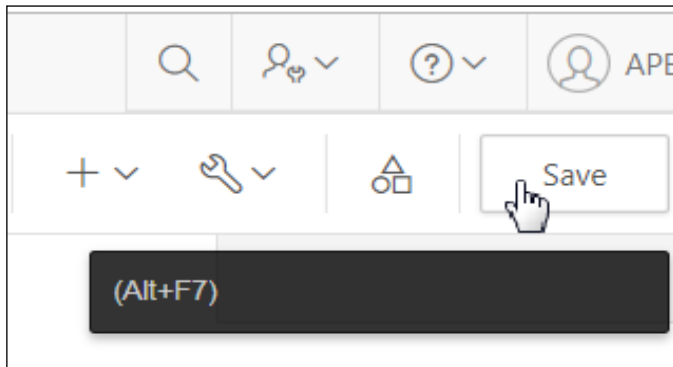
- h. On the Rendering tab, select **Page 9: Customer Details**.



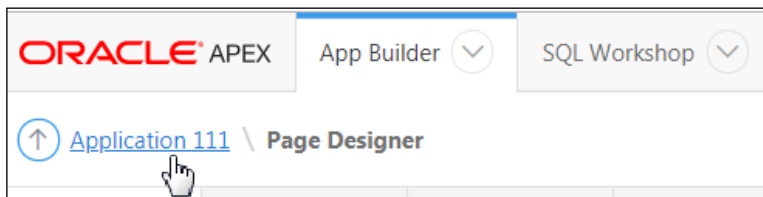
- i. On the Page tab, select **Contribution Rights** for **Authorization Scheme** under **Security**.



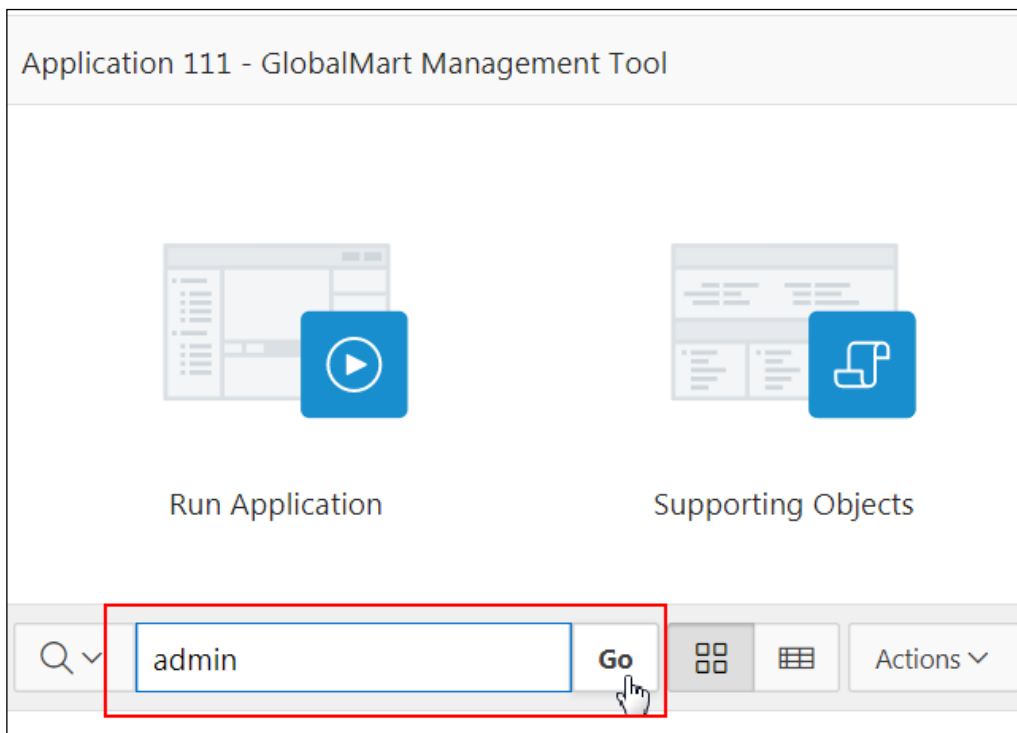
- j. Click the **Save** button.



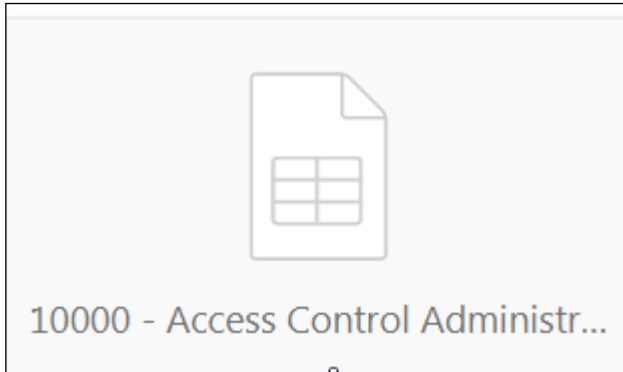
- k. Click the **Application 111** breadcrumb to return to the GlobalMart Management Tool application home page.



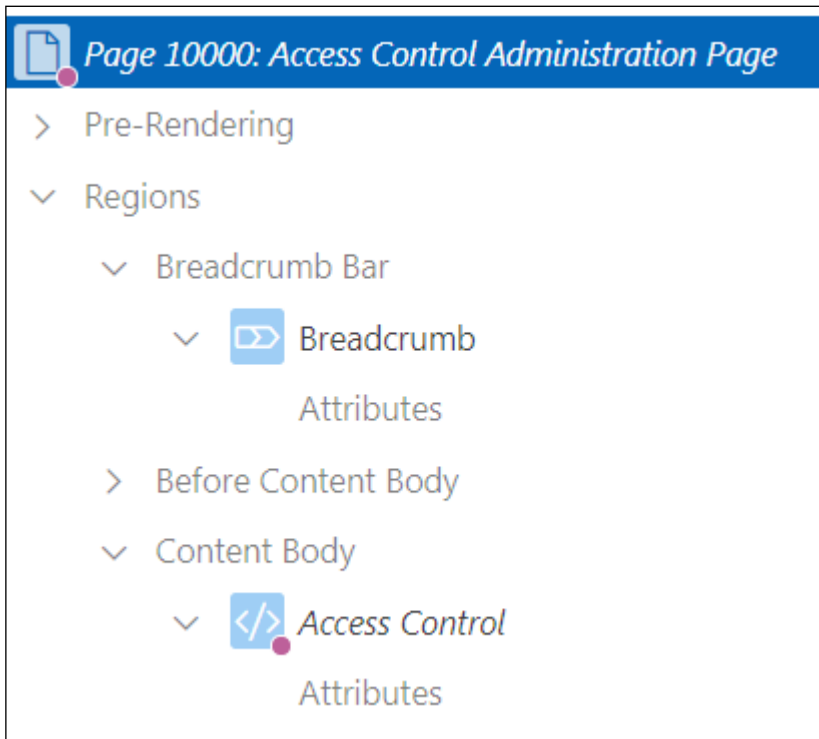
- l. Enter `admin` in the search area and click **Go**.



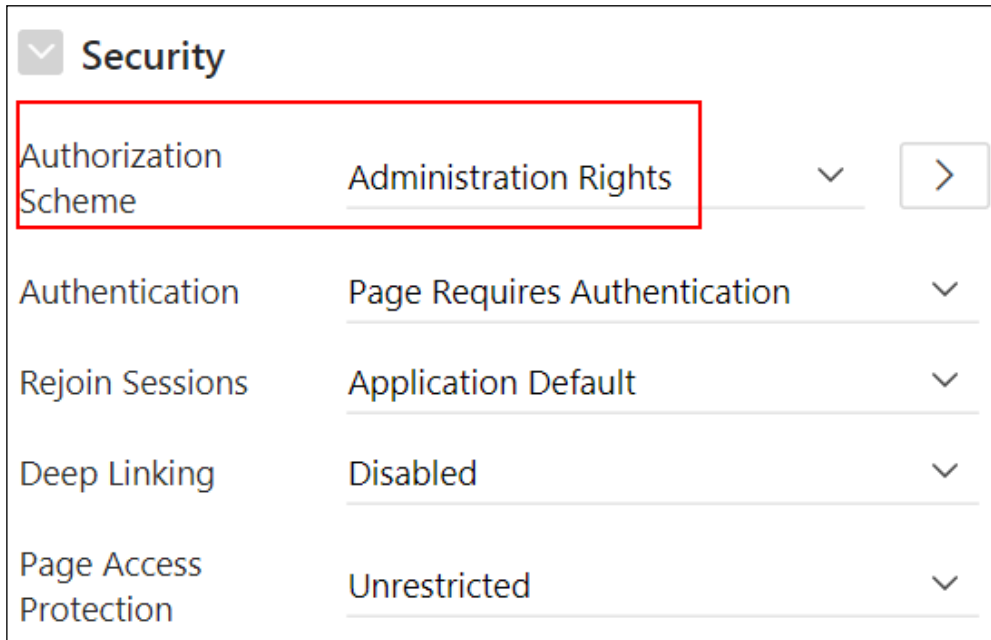
- m. Because users with the Administrator privilege are only allowed to make changes to the access control list, you must review the authorization scheme for the Access Control Administration page. Click **Access Control Administration Page**.



- n. On the Rendering tab, select **Page 10000: Access Control Administration Page**.

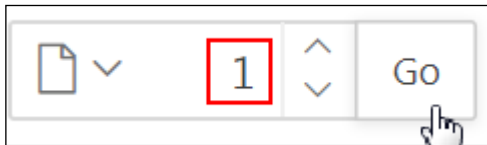


- o. On the Page tab, you see that Administration Rights is already selected by default for the **Authorization Scheme** under **Security** and click **Save**.



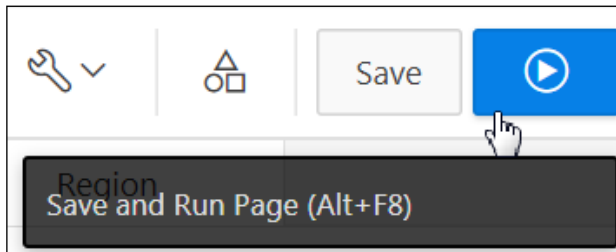
Security	
Authorization Scheme	Administration Rights
Authentication	Page Requires Authentication
Rejoin Sessions	Application Default
Deep Linking	Disabled
Page Access Protection	Unrestricted

6. Run the application and sign in as a different user. What do you observe?
- a. Now you are ready to run the application. In the Page Finder field, enter 1 for Page and click **Go**.



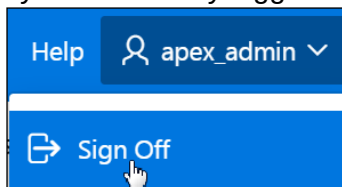
Page Finder: 1 Go

- b. Click the **Save and Run Page** icon.



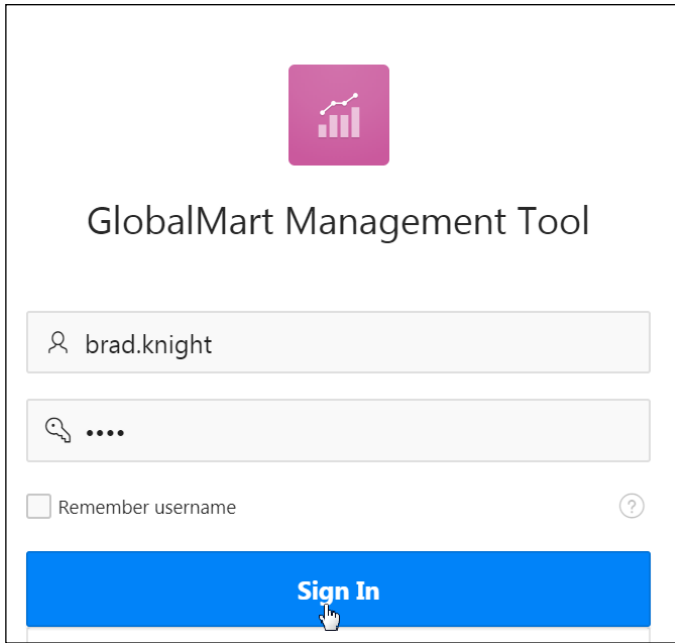
Save and Run Page (Alt+F8)

- c. If you are already logged in, click **Sign Off**.



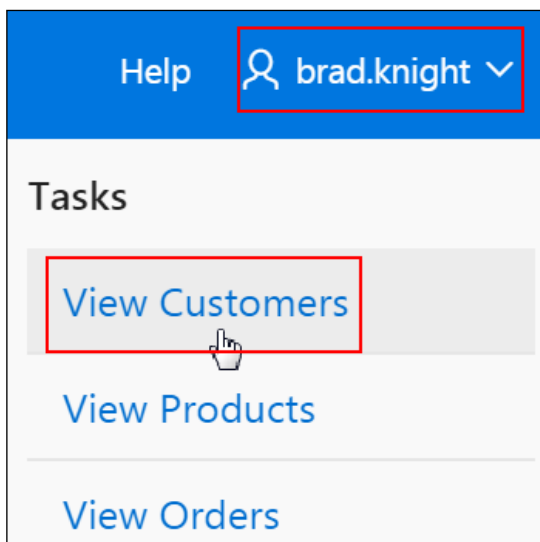
Sign Off

- d. Enter login credentials of `brad.knight` and click **Sign in**.

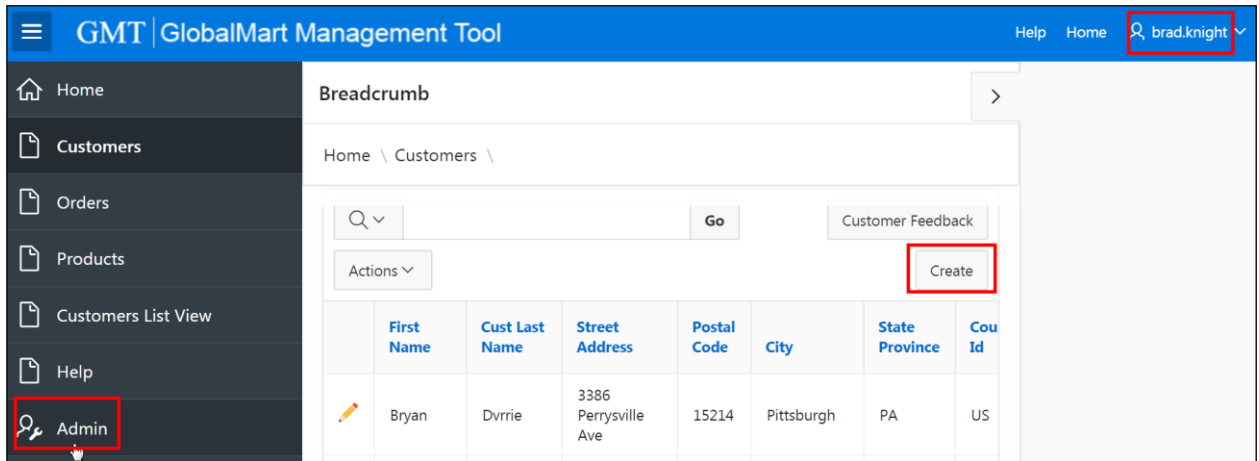


The image shows the login interface for the GlobalMart Management Tool. At the top is a purple square icon with a white bar chart. Below it is the title "GlobalMart Management Tool". There are two input fields: the first contains the username "brad.knight" and the second contains masked characters "....". Below the input fields is a checkbox labeled "Remember username" and a help icon. At the bottom is a large blue button labeled "Sign In" with a mouse cursor pointing at it.

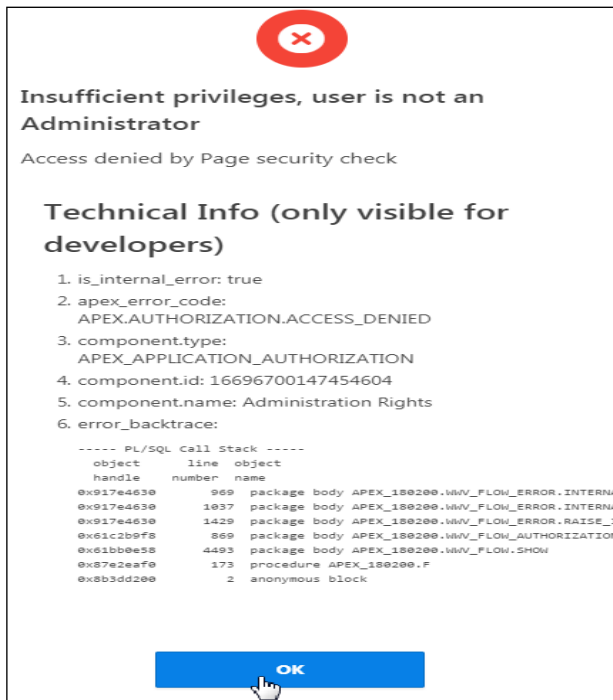
- e. On the Tasks tab, click **View Customers**.



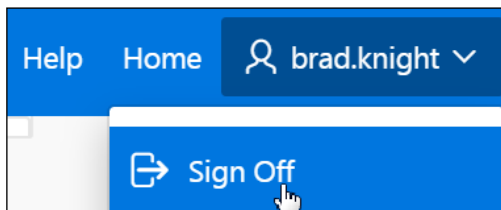
- f. Notice that Brad can edit customer information and can see the **Create** button. Click the **Admin** tab.



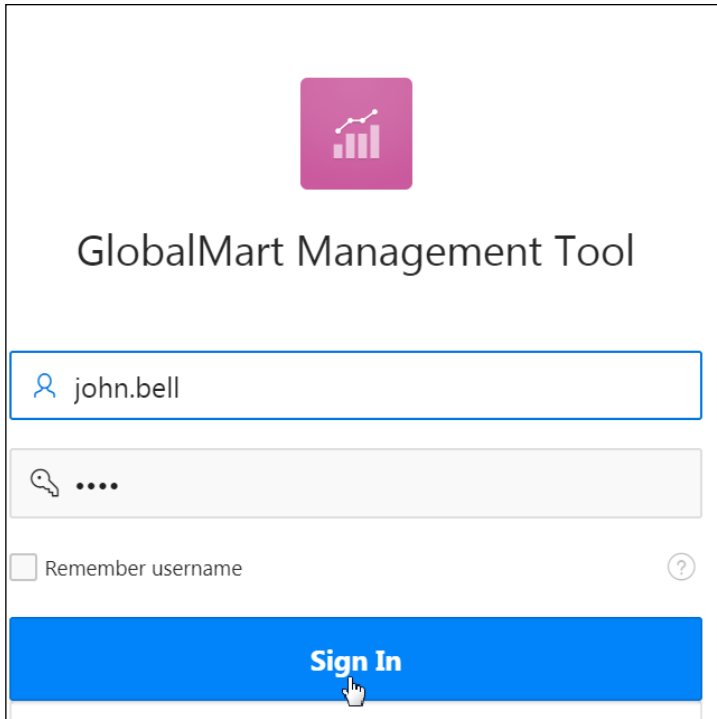
- g. You receive an error because you need the Administrator privilege to access this page. Click **OK**.



- h. Click **Sign Off**.

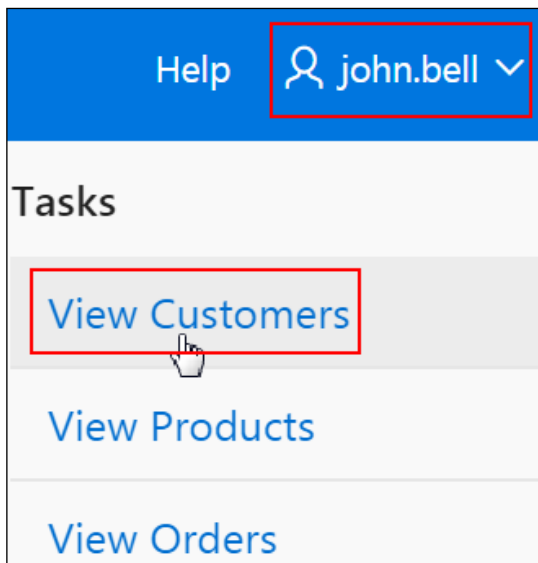


- i. Enter the login credentials of `john.bell` and click **Sign in**.



The login screen for the GlobalMart Management Tool. It features a purple square icon with a white bar chart and an upward arrow. Below the icon is the title "GlobalMart Management Tool". There are two input fields: the first contains the username "john.bell" and the second contains a masked password "....". Below the password field is a checkbox labeled "Remember username" and a help icon. A blue "Sign In" button is at the bottom, with a mouse cursor clicking on it.

- j. Click **View Customers**.



- k. John has the Reader privilege only and, therefore, cannot edit customer information. Also, the **Create** button is not displayed for him.

Go Actions Customer Feedback											
Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Country Id	Phone Number	Nls Language	Nls Territory	Credit Limit	Cust Email
Eddie	Boyer	411 E Wisconsin Ave # 2550	53202	Seattle	WI	US	+1 414 123 4369	us	AMERICA	700	-
Eddie	Stern	808 3Rd St # 100	54403	Wausau	WI	US	+1 715 123 4372	us	AMERICA	700	Eddie.Stern@GODW
Ernest	Weaver	300 Crooks St	54301	Green Bay	WI	US	+1 414 123 4373	us	AMERICA	900	Ernest.Weaver@GRC
Ernest	George	122 E	53703	Madison	WI	US	+1 608 123	us	AMERICA	900	Ernest.George@LAPT

- l. Change the page number in your URL to try to access the Customer Details page. Press the **Enter** key:

Example url .../f?p=111:4:8958804941859::::

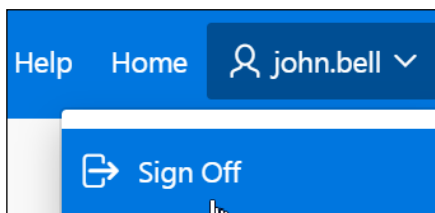
Change to .../f?p=111:9:8958804941859::::

10.237.16.240:8080/apex/f?p=111:9:7451934953081:::

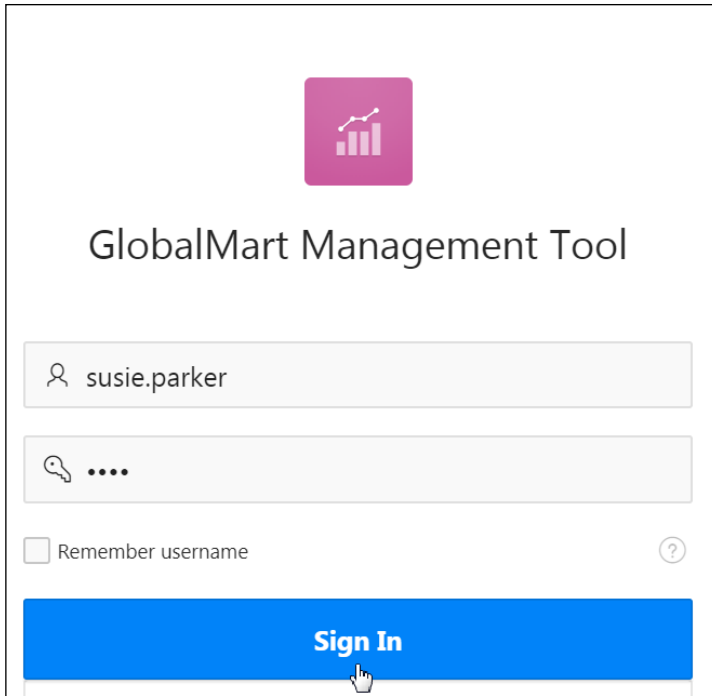
- m. Notice the change of page number in the URL. You receive a message denying access to the page, because access to the Customer Details page is restricted to users with the Contributor privilege and John Bell has only the Reader privilege assigned to him. Click **OK**.



- n. Lastly, you want to sign in as Susie Parker, who is an administrator. Click **Sign Off**.

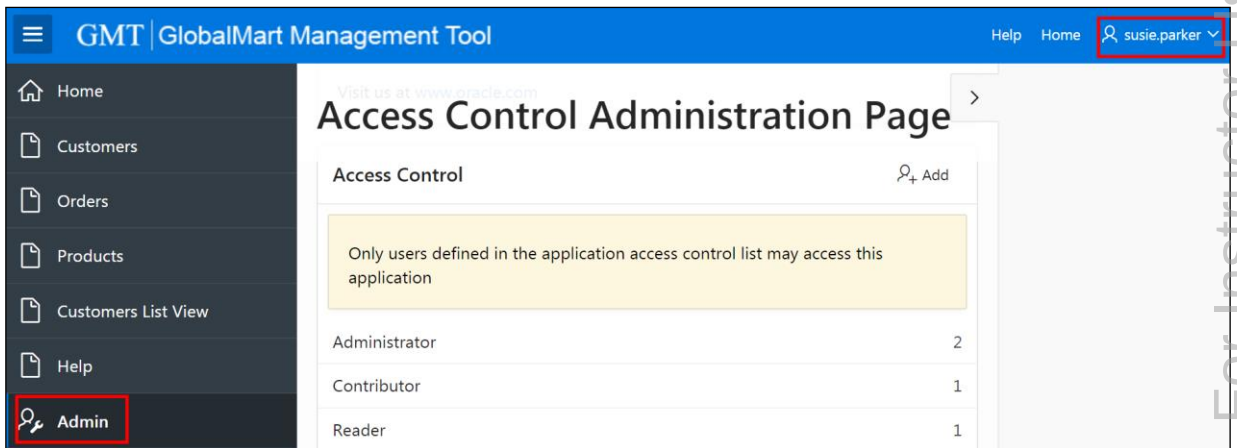


- o. Enter the login credentials of `susie.parker` and click **Sign in**.



The image shows the login page for the GlobalMart Management Tool. At the top center is a purple square icon with a white bar chart. Below the icon is the title "GlobalMart Management Tool". Underneath the title are two input fields: the first contains the username "susie.parker" and the second contains masked characters "....". Below these fields is a checkbox labeled "Remember username" and a question mark icon. At the bottom is a large blue button labeled "Sign In". A mouse cursor is pointing at the "Sign In" button.

- p. Click the **Admin** tab.



The image shows the "Access Control Administration Page" in the GlobalMart Management Tool. The top navigation bar is blue and contains the GMT logo, the title "GlobalMart Management Tool", and links for "Help" and "Home". The user "susie.parker" is logged in, as indicated by a dropdown menu in the top right. The left sidebar is dark gray and contains a list of navigation items: "Home", "Customers", "Orders", "Products", "Customers List View", "Help", and "Admin". The "Admin" item is highlighted with a red box. The main content area is white and has the title "Access Control Administration Page". Below the title is a section labeled "Access Control" with an "Add" button. A yellow warning box states: "Only users defined in the application access control list may access this application". Below the warning box is a table with the following data:

Role	Count
Administrator	2
Contributor	1
Reader	1

- q. Notice that because Susie is an Administrator, you can access this page. Click the **Application 111** link on the Developer toolbar.



The image shows the Developer toolbar at the bottom of the application. It contains several icons and labels: "Home", "Application 111" (highlighted with a red box), "Edit Page 10000", "Session", "View Debug", "Debug", "Page Info", "Quick Edit", and "Theme Roller".

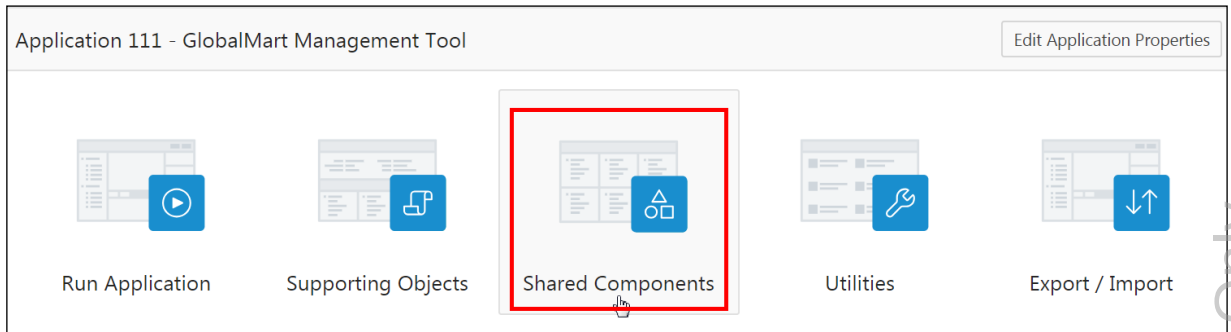
Solution 16-3: Enabling Session State Protection

Overview

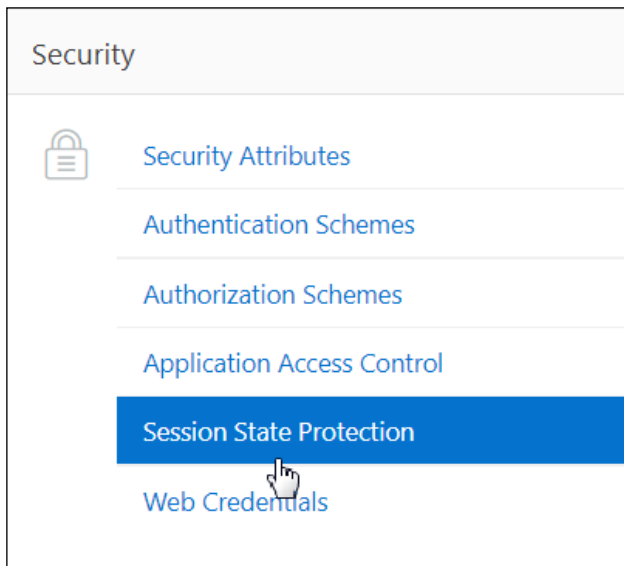
In this practice solution, steps are provided for enabling Session State Protection in GlobalMart Management Tool.

Steps

1. Set the Page Access Protection for the Employee Commission page to **No URL Access**.
 - a. Sign in as Click **Shared Components**.



- b. Under Security, select **Session State Protection**.



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- c. Click the right arrow (>) next to Pages.

Application Session State Protection Controls

Enabling Session State Protection can prevent hackers from tampering with the URLs within your application. URL tampering can adversely affect program logic, session state contents, and information privacy.

To enable, disable, or configure Session State Protection using a wizard, click **Set Protection**.

Application: **111 - GlobalMart Management Tool** ?

Session State Protection: **Enabled** ?

Existing Session State Protection Settings

Pages		Page Items		Application Items
Page Access	Pages	Item Access Level	Items	
Arguments Must Have Checksum	10	Unrestricted	84	
No URL Access	1	Checksum Required - Session Level	8	
Unrestricted	19			

- d. Click the Page link for the **Employee Commission** page.

Application: **111 - GlobalMart Management Tool** ?

Session State Protection: **Enabled** ?

Q ▾ Go [Table Icon] [Bar Chart Icon] Actions ▾

Page ↑≡	Name	Page Access Protection
0	Global Page - Desktop	No URL Access
1	Home	Unrestricted
2	Top Tier Salary	Unrestricted
3	Customer Address List	Unrestricted
4	Customers	Unrestricted
5	Employee Commission	Unrestricted
6	Product Information Details	Unrestricted

- e. Select **No URL Access** for **Page Access Protection** and click **Apply Changes**.

Set Page and Item Protection

Application: **111 - GlobalMart Management Tool** ?

Session State Protection: **Enabled** ?

Page: **5** ?

Name: **Employee Commission** ?

Page Access Protection: **No URL Access** ?

Display Item Type: **No URL Access** ?

Page Items (Form Items)

Cancel **Apply Changes**

You get the message **Action Processed**.

✓ **Action Processed.**

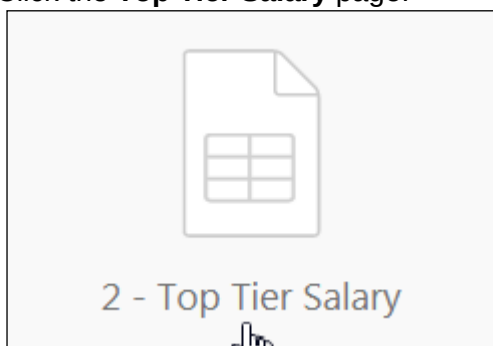
Set Page and Item Protection

Set Page and Item Protection

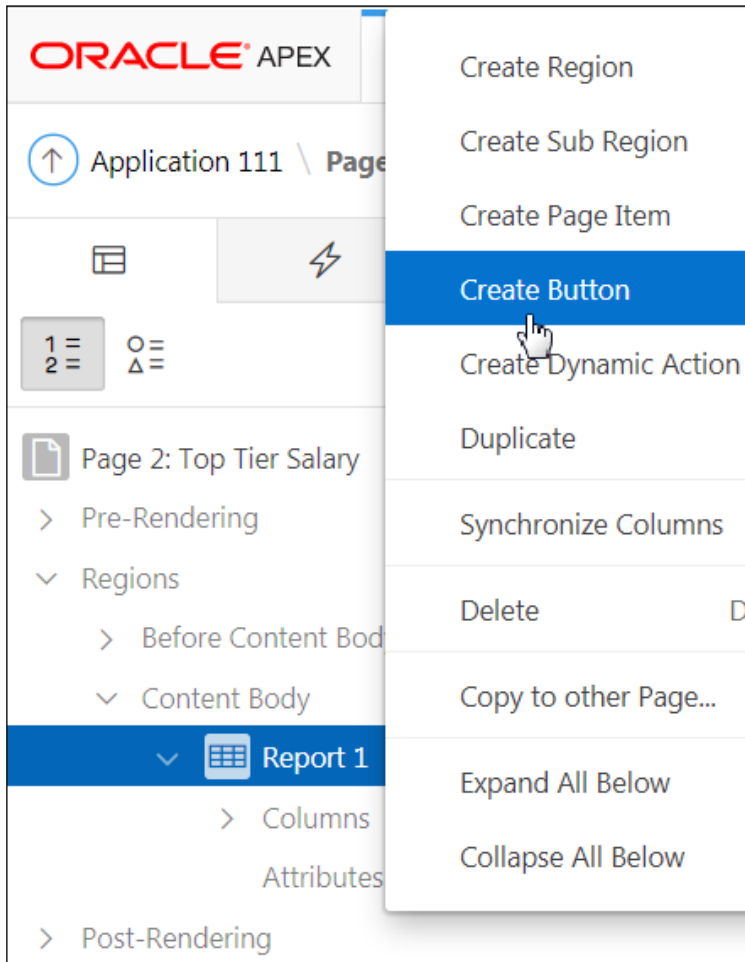
2. Add a Commission button to the Top Tier Salary page that redirects to the Employee Commission page. Run the page to see that there is an error because of the branch.
- a. Click the **Application 111** breadcrumb.

Application 111 \ Shared Components \ Session State Protection \ Set Page and Item Protection

- b. Click the **Top Tier Salary** page.



- c. On the Rendering tab, right-click **Report 1** and select **Create Button**.



- d. On the Rendering tab, make sure that the newly created button is selected. On the Button tab, enter the following values for the respective fields:
- **Identification > Button Name:** Enter `Commission`
 - **Identification > Label:** `Commission` (This value will be automatically populated.)
 - **Layout > Button Position:** Select **Top of Region**
 - **Appearance > Hot:** Select **Yes**
 - **Behavior > Action:** Select **Redirect to Page in this Application**
 - **Behavior > Target > Page:** Select `5` (Page number of the Employee Commission page) and click **OK**.

Behavior

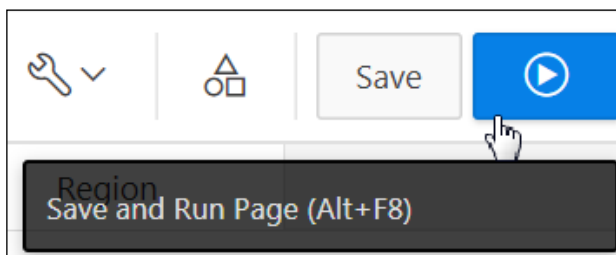
Action Redirect to Page in this Applic

Target Page 5

Warn on Unsaved Changes Do Not Check

Database Action - Select -

- e. Click the **Save and Run Page** icon.



- f. Click **Commission**.

Report 1

Commission

Last Name ↑	Email	Salary
Abel	EABEL	11000
Ande	SANDE	6400
Baer	HBAER	10000
Banda	ABANDA	6200
Bates	EBATES	7300
Bernstein	DBERNSTE	9500

- g. Notice you receive an error indicating that you must create a Branch to the Page branch that generates a page without a URL. Click **OK**.



3. Edit the **Commission** button to submit the page and create a Branch to Page branch that will go to the **Employee Commission** page without passing the URL.
 - a. Click the **Edit Page 2** link on the Developer toolbar.

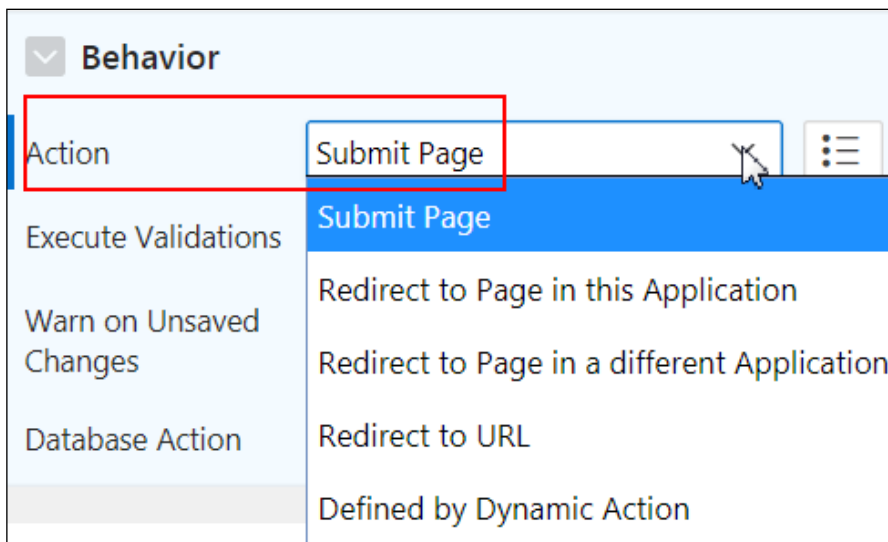
The screenshot shows the APEX application interface. On the left is a navigation menu with items: Orders, Products, Customers List View, Help, and Admin. The main content area displays a table with employee data. A blue button labeled 'Commission' is in the top right. At the bottom, the Developer toolbar is visible, with the 'Edit Page 2' link highlighted by a red box.

Last Name ↑	Email	Salary
Abel	EABEL	11000
Ande	SANDE	6400
Baer	HBAER	10000
Banda	ABANDA	6200
Bates	EBATES	7300
Bernstein	DBERNSTE	9500
Bloom	HBLOOM	10000

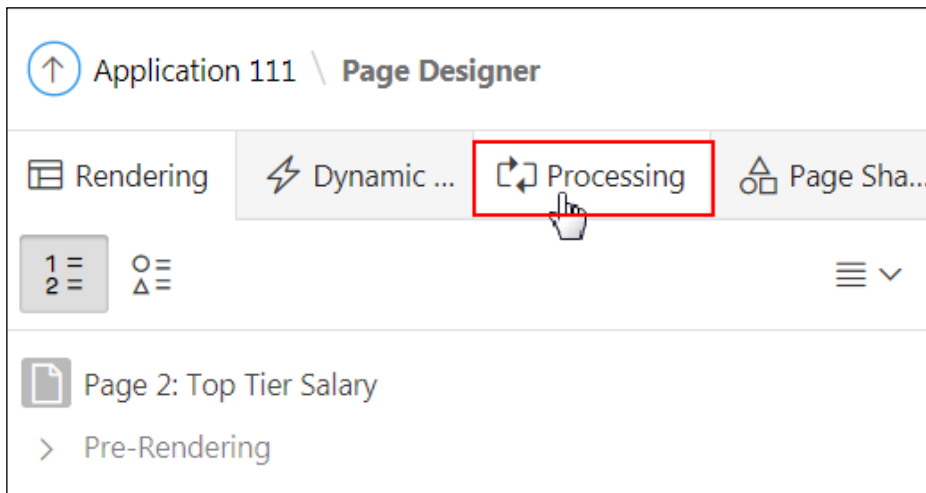
- b. On the Rendering tab, make sure that the **Commission** button is selected. On the **Button** tab, select **Submit Page** for **Behavior > Action**.

The screenshot shows the APEX page editor for 'Page 2: Top Tier Salary'. The left sidebar shows a tree view with 'Regions' expanded, containing 'Before Content Body' and 'Content Body'. Under 'Content Body', 'Report 1' is expanded, showing 'Columns', 'Attributes', and 'Region Buttons'. The 'Commission' button is selected at the bottom of the editor.

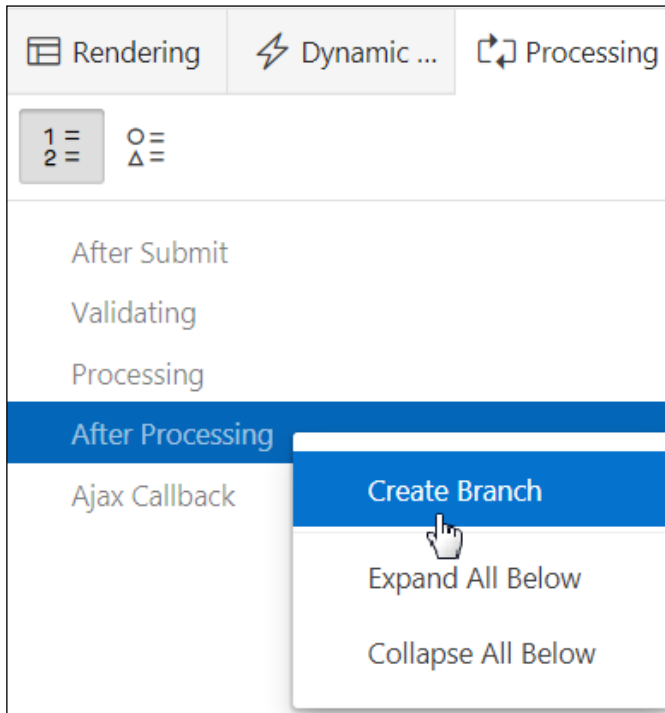
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- c. Now you need to create a **Branch to Page** branch. In Page Designer, click the **Processing** tab.



- d. Right-click **After Processing** and select **Create Branch**.



- e. On the Branch tab, enter the following values for the respective fields:
- **Identification > Name:** Enter `Go to Page 5`
 - **Execution Options > Point:** Select **After Processing** (by default)
 - **Behavior > Type:** Select **Page (Show only)**
 - **Behavior > Page Number:** Select 5 (Page number of the Employee Commission page)
 - **Condition > When Button Pressed:** Select **Commission**

Identification

Name
Go to Page 5

Execution Options

Sequence
10

Point
After Processing

Behavior

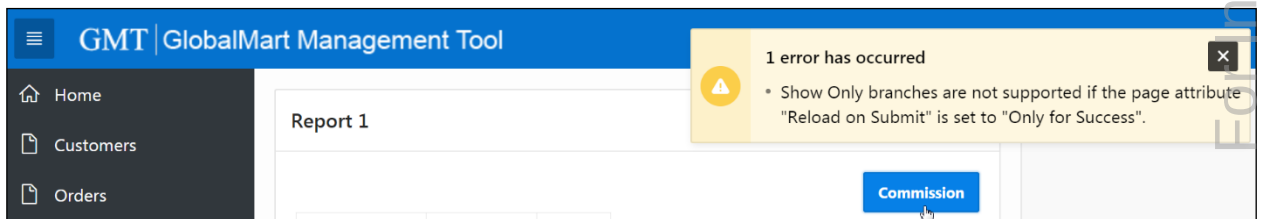
Type
Page (Show only)

Page Number
5

Server-side Condition

When Button Pressed
Commission

- f. If you save and run the page and click the **Commission** button now, you will get the following error as the **Reload on Submit** under **Advanced** is currently set as **Only for Success** for Page 2: Top Tier Page:



To change it:

- i. Click **Edit Page 2** on the Developer toolbar.

GMT | GlobalMart Management Tool

Home Customers Orders Products Customers List View Help Admin

Report 1

Last Name ↑	Email	Salary
Abel	EABEL	11000
Ande	SANDE	6400
Baer	HBAER	10000
Banda	ABANDA	6200
Bates	EBATES	7300
Bernstein	DBERNSTE	9500
Bloom	HBLOOM	10000

1 error has occurred

- Show Only branches are not supported if the page attribute "Reload on Submit" is set to "Only for Success".

Home Application 111 Edit Page 2 Session View Debug Debug Page Info Quick Edit Theme Roller

- ii. **Page 2: Top Tier Salary** opens in Page Designer view. Select **Page 2: Top Tier Salary** in the Rendering pane and select **Always** for **Reload on Submit** under **Advanced**.

Page 2: Top Tier Salary

- > Pre-Rendering
- > Regions
 - > Before Content Body
 - > Content Body
 - > Report 1
 - > Columns
 - Attributes
 - > Region Buttons
 - Commission
 - > Post-Rendering

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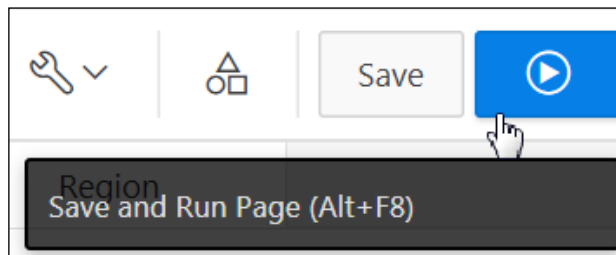
☒ **Advanced**

Enable duplicate page submissions Yes - Enable page to be re-posted ☐

Reload on Submit Always

☒ **Server Cache** Only for Success

iii. Click the **Save and Run Page** icon.



g. Click **Commission**.

GMT | GlobalMart Management Tool

Home Customers Orders Products Customers List View Help Admin

Report 1

Commission

Last Name ↑	Email	Salary
Abel	EABEL	11000
Ande	SANDE	6400
Baer	HBAER	10000
Banda	ABANDA	6200
Bates	EBATES	7300

- h. The **Employee Commission** page is displayed. Review the URL. Notice that it no longer passes the `f?p=` in the URL, and it is rendered successfully. Click the **Application 111** link on the Developer toolbar.

Not secure | 10.237.16.240:8080/apex/www_flow.accept

GMT | GlobalMart Management Tool

Home

Customers

Orders

Products

Customers List View

Help

Admin

Home \

Go

1. Primary Report

>

Employee Commission

First Name	Last Name	Commission Pct	Manager
John	Russell	.4	Steven King
Karen	Partners	.3	Steven King
Alberto	Errazuriz	.3	Steven King
Gerald	Cambrault	.3	Steven King
Eleni	Zlotkey	.2	Steven King
Peter	Tucker	.3	John Russell

Home

Application 111

Edit Page 5

Session

View Debug

Debug

Page Info

Quick Edit

Theme Roller

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**Practices for Lesson 17:
Managing Application
Navigation**

Practices for Lesson 17: Overview

Practices Overview

In these practices, you build a hierarchical list with images, a database-driven report, and a site map and enforce authorization on the site map.

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Practice 17-1: Building a Hierarchical List with Images

Overview

In this practice, you upload the images to the Home Page List that you have created in the previous practices, assign the pull-down menu with the images template to the list, change the attributes of the Tasks region, and associate the region with each Parent List Entry.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Change the Home Page List to the following list structure and add icons for each parent entry:

Sequence	List Entry Label	Parent List Entry	Page Number
10	Manage Customers		
12	View Customers	Manage Customers	4
20	Manage Products		
22	View Products	Manage Products	7
30	Manage Orders		
32	View Orders	Manage Orders	12

2. In the Home page, change the region template of **Tasks** to no template, the **List Template** to **Navigation Bar**, and the **Page Template** to **Theme Default**.
3. Run the page to see the Home Page List. Click the parent entry to show the sublist entries.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 17-1**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. From the application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 16-3**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_16_03.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).

- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

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Practice 17-2: Building a Database-Driven Report

Overview

In this practice, you build a report based on the data in a table and then navigate to the detail. You also change the template to the Navigation Region template.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a new page with a Classic Report called **List of Customers**.
2. Hide the `CUSTOMER_ID` column in the report and turn off pagination. Display the Customer Name column heading in blue by editing the column definition and entering the following for Column Heading.

`<font color="blue">CUSTOMERS</font>`

3. Create a link to the Customer Details page on the `NAME` column. Pass the `P9_CUSTOMER_ID` item from the Customer Details page with the value `#CUSTOMER_ID#` from this page.
4. Run the page to see the database-driven report.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 17-2**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.

- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. From the application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 17-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_17_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Practice 17-3: Building a Site Map

Overview

In this practice, you build a site map page that is added as a navigation bar entry.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Assign each page that you want in the site map to the same page group. Create a page group called Site Map and assign the following pages to the Site Map page group.

Pages
Customers
List of Orders
Products
Customer Feedback
Access Control Administration Page

2. Create a new page called **Site Map** with a Classic Report that shows only the pages in the Site Map page group.
3. Change the **Pagination Scheme** to **None** and do not show the `PAGE_ID` column. Link the `PAGE_NAME` column to the `#PAGE_ID#` so that it opens the corresponding page when clicked.
4. Add this page to your navigation bar and call it **Site Map**.
5. Run the page and click one of the page names to view the corresponding page.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 17-3**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. From the application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles were imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 17-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_17_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.

- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

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Practice 17-4: Enforcing Authorization in Site Map

Overview

In this practice, you add a function that determines whether you are authorized to see a particular page in your Site Map. You also change the SQL Report query for the Site Map to make sure that the page is selected only if you are authorized to see it.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Navigate to **SQL Workshop > SQL Commands**. Copy and paste the following code (located in the `/home/oracle/labs/labs/lab_17_04_01.txt` file) to create the `authorization_check` function.

```
create or replace function authorization_check(
p_scheme in varchar2)
return varchar2
is
begin
if apex_util.public_check_authorization(p_scheme) then
return 'true';
else
return 'false';
end if;
end;
```

2. For the Site Map, change the Source for your Classic Report to the following code (also located in the `/home/oracle/labs/labs/lab_17_04_02.txt` file) so that only the pages that are authorized will be shown based on the user logged in.

```
select distinct PAGE_ID, PAGE_NAME
from APEX_APPLICATION_PAGES p, apex_application_authorization a
where p.APPLICATION_ID = :APP_ID
and a.application_id = p.application_id
and (p.authorization_scheme is null or
(a.authorization_scheme_id = p.authorization_scheme_id and
authorization_check(a.authorization_scheme_name) = 'true'))
and PAGE_GROUP = 'Site Map'
```

3. You do not want end users or developers to see the Access Control Administration page. Make sure that the **Authorization Schemes** for this page is set to **access control - Administrator**. Run the page to see if only an authorized user can view it.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 17-4**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. From the application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.

- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 17-3**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_17_03.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Solution 17-1: Building a Hierarchical List with Images

Overview

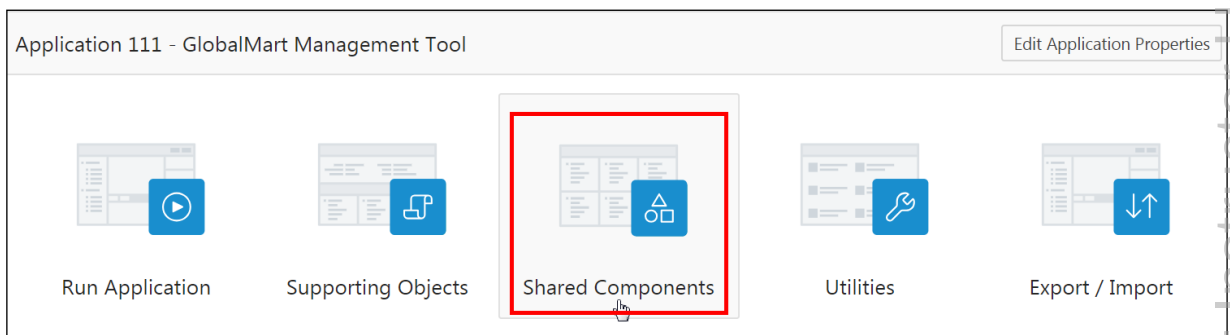
In this practice solution, steps are provided for building a hierarchical list with images.

Steps

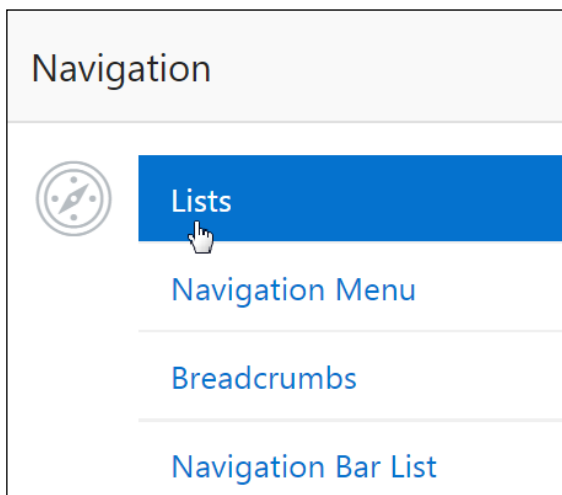
1. Change the Home Page List to the following list structure and add icons for each parent entry:

Sequence	List Entry Label	Parent List Entry	Page Number
10	Manage Customers		
12	View Customers	Manage Customers	4
20	Manage Products		
22	View Products	Manage Products	7
30	Manage Orders		
32	View Orders	Manage Orders	12

- a. Sign out and sign in as the **apex_dev** user (workspace developer).
- b. Click the **Shared Components** icon.



- c. Under **Navigation**, select **Lists**.



d. Select **Home Page List**.

Lists List Details Unused Conditional Entries			
Q ▼ Go □ □			
Name 	Type	Entries	References
Access Control	Static	2	1
Desktop Navigation Bar	Static	5	0
Desktop Navigation Menu	Static	7	0
Home Page List	Static	3	1
List of Products	Dynamic	0	1

- e. Click **View Customers** to edit this List Entry detail.

Application 111 \ Shared Components \ **List Details**

Lists **List Details** Unused Conditional Entries Utilization

List Home Page List ?

Search Go Rows 50 Acti

Sequence ↑	Name	Parent Entry	Target
10	View Customers	-	f? p=&APP_ID.:4:&SESSION.:
20	View Products	-	f? p=&APP_ID.:7:&SESSION.:
30	View Orders	-	f? p=&APP_ID.:12:&SESSION.:

- f. In the **Entry** section, retain 10 for **Sequence**; change **View Customers** to Manage Customers for **List Entry Label**. In the **Target** section, change the **Target type** to - No Target -. In the **Entry** section, click the up arrow for the **Image/Class** field.

< > List Entry Cancel Delete Apply Changes

Show All **Entry** Target Current List... Conditions Authorizati... Configurati... Click Count... User Define... Developer...

Image/Class ?

Attributes ?

Alt Attribute ?

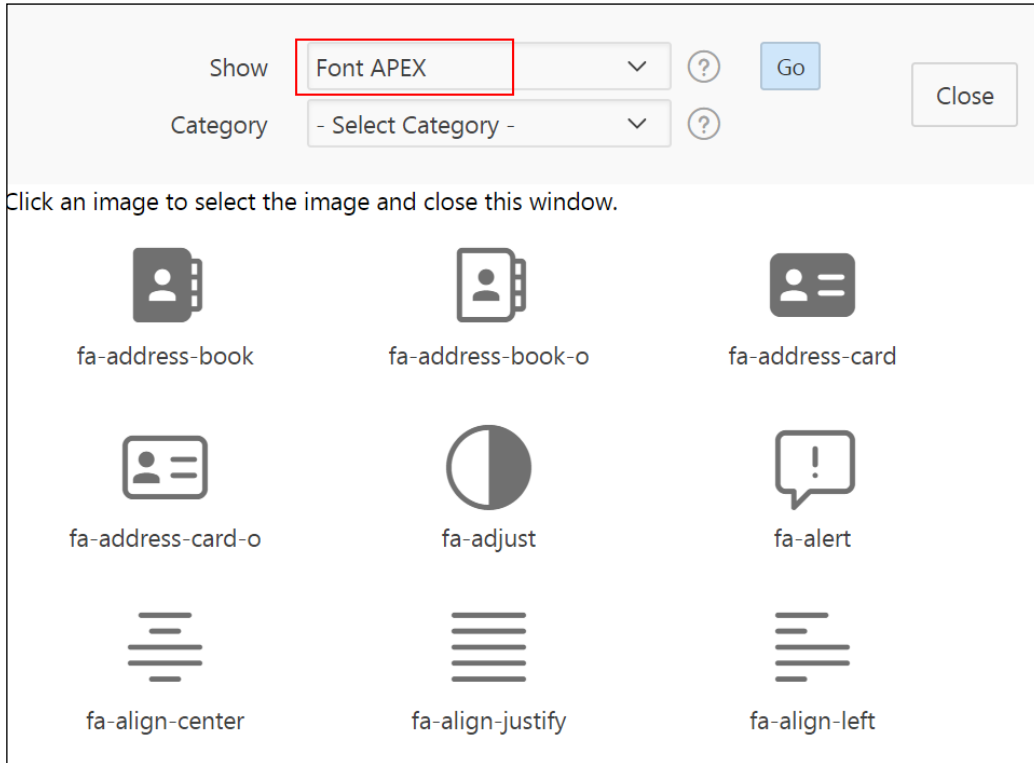
* List Entry Label Manage Customers ?

Target

Target type - No Target - ?

* Page 4 ?

- g. Make sure that **Font APEX Icons** is selected in the **Show** drop-down list.



- h. Select **fa-user** from the list.



- i. Click **Apply Changes**.

The screenshot shows a 'List Entry' dialog box. At the top, there are navigation buttons '<' and '>', the title 'List Entry', and buttons 'Cancel', 'Delete', and 'Apply Changes'. The 'Apply Changes' button is highlighted with a red box. Below the title bar, there are tabs: 'Show All', 'Entry' (selected), 'Target', 'Current ...', 'Conditio...', 'Authoriz...', 'Configur...', 'Click Co...', 'User Def...', and 'Develop...'. In the 'Entry' tab, there is a text field labeled 'Image/Class' containing the text 'fa-user'. Below this field are two icons: a list icon and a help icon.

- j. Click **Create Entry >**.

The screenshot shows a 'List Details' dialog box. At the top, there is a green bar with a checkmark and the text 'Action processed.'. Below this, there are tabs: 'Lists', 'List Details' (selected), 'Unused', 'Conditional Entries', 'Utilization', and 'History'. In the 'List Details' tab, there is a 'List' dropdown menu showing 'Home Page List'. To the right of the dropdown are 'Grid Edit' and 'Edit List' buttons. The 'Create Entry >' button is highlighted with a red box.

- k. Select **Manage Customers** for **Parent List Entry**, enter 12 for **Sequence**, and enter **View Customers** for **List Entry Label**. In the **Target** section, make sure that **Target type** is set to **Page in this Application** and the **Page** is 4 (the page number corresponding to the Customer page). Then click **Create List Entry**.

The screenshot displays the 'List Entry' configuration window. The 'Entry' tab is selected, showing the following fields:

- Parent List Entry:** Manage Customers (dropdown menu)
- Sequence:** 12 (text input)
- Image/Class:** (empty text input)
- Attributes:** (empty text input)
- Alt Attribute:** (empty text input)
- * List Entry Label:** View Customers (text input)

The 'Target' tab is also visible, showing the following fields:

- Target type:** Page in this Application (dropdown menu)
- * Page:** 4 (text input)
- ☐ reset pagination for this page
- ☐ Printer Friendly
- Request:** (empty text input)
- Clear Cache:** (empty text input)

The 'Create List Entry' button is highlighted in red in the top right corner.

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- I. Select **View Products**.

Lists **List Details** Unused Conditional Entries Utilization History

List Home Page List ?

Q Go Rows 50 Actions

Sequence ↑	Name	Parent Entry	Target
10	Manage Customers	-	-
12	View Customers	Manage Customers	f?p=&APP_ID.:4:&SESSION...&DEBUG.....
20	View Products	-	f?p=&APP_ID.:7:&SESSION.:
30	View Orders	-	f?p=&APP_ID.:12:&SESSION.:

- m. In the **Entry** section, retain 20 for **Sequence** and enter Manage Products for **List Entry Label**. In the **Target** section, set the **Target type** to - No Target -. In the **Entry** section, click the up arrow for the **Image/Class** field.

Entry Target Current List... Conditions Authorizati... Configurati... Click Count... U

Image/Class ⋮

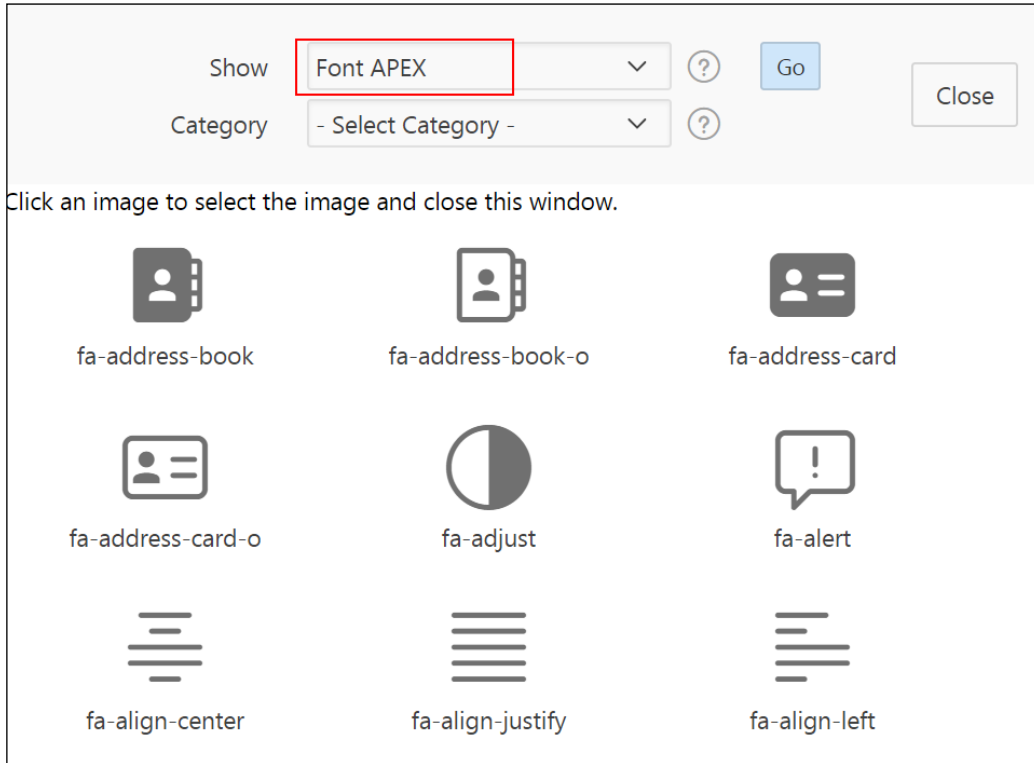
Attributes ⋮ ?

Alt Attribute ?

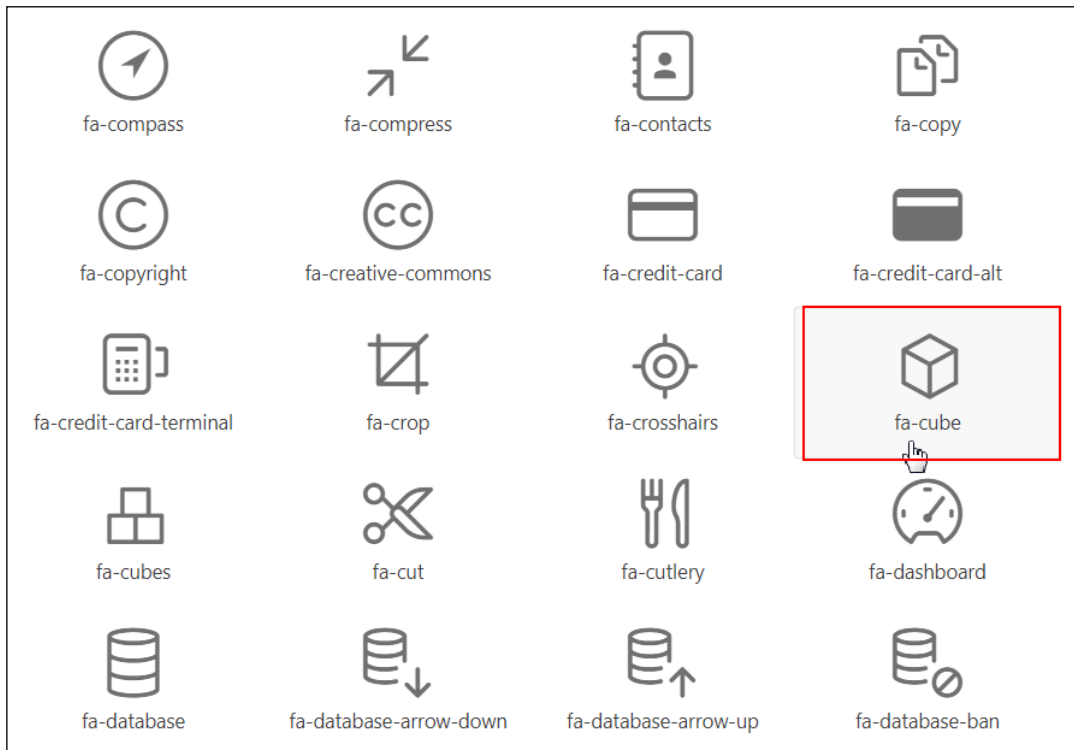
* List Entry Label Manage Products ?

Target type - No Target - ?

- n. Make sure that **Font APEX Icons** is selected in the **Show** drop-down list.



- o. Select **fa-cube** from the list.



- p. Click **Apply Changes**.

The screenshot shows the 'List Entry' form with the 'Apply Changes' button highlighted in red. The form includes a 'List Entry' header, a 'Cancel' button, a 'Delete' button, and an 'Apply Changes' button. Below the header, there are tabs for 'Show All', 'Entry', 'Target', 'Current List...', 'Conditions', 'Authorizati...', 'Configurati...', 'Click Count...', 'User Define...', and 'Developer...'. The 'Image/Class' field is set to 'fa-cube'.

- q. Click **Create Entry >**.

The screenshot shows the 'List Details' form with the 'Create Entry >' button highlighted in red. The form includes a 'List Details' header, a 'Grid Edit' button, an 'Edit List' button, and a 'Create Entry >' button. Below the header, there are tabs for 'Lists', 'List Details', 'Unused', 'Conditional Entries', 'Utilization', and 'History'. The 'List' dropdown is set to 'Home Page List'.

- r. Select **Manage Products** for **Parent List Entry**, enter 22 for **Sequence**, and enter View Products for **List Entry Label**. In the **Target** section, make sure that **Target type** is set to **Page in this Application** and the **Page** is 7 (the page number corresponding to the Products page). Then click **Create List Entry**.

The screenshot shows the 'List Entry' form with the 'Create List Entry' button highlighted in red. The form includes a 'List Entry' header, a 'Cancel' button, a 'Create and Create Another' button, and a 'Create List Entry' button. Below the header, there are tabs for 'Show All', 'Entry', 'Target', 'Current List...', 'Conditions', 'Authorizati...', 'Configurati...', 'Click Count...', 'User Define...', and 'Developer...'. The 'List' dropdown is set to 'Home Page List'. The 'Parent List Entry' dropdown is set to 'Manage Products'. The 'Sequence' field is set to 22. The 'Image/Class' field is empty. The 'Attributes' field is empty. The 'Alt Attribute' field is empty. The '* List Entry Label' field is set to 'View Products'.

The screenshot shows the 'Target' section of the 'List Entry' form. The 'Target type' dropdown is set to 'Page'. The 'Page' field is set to 7. The 'reset pagination for this page' checkbox is unchecked. The 'Printer Friendly' checkbox is unchecked.

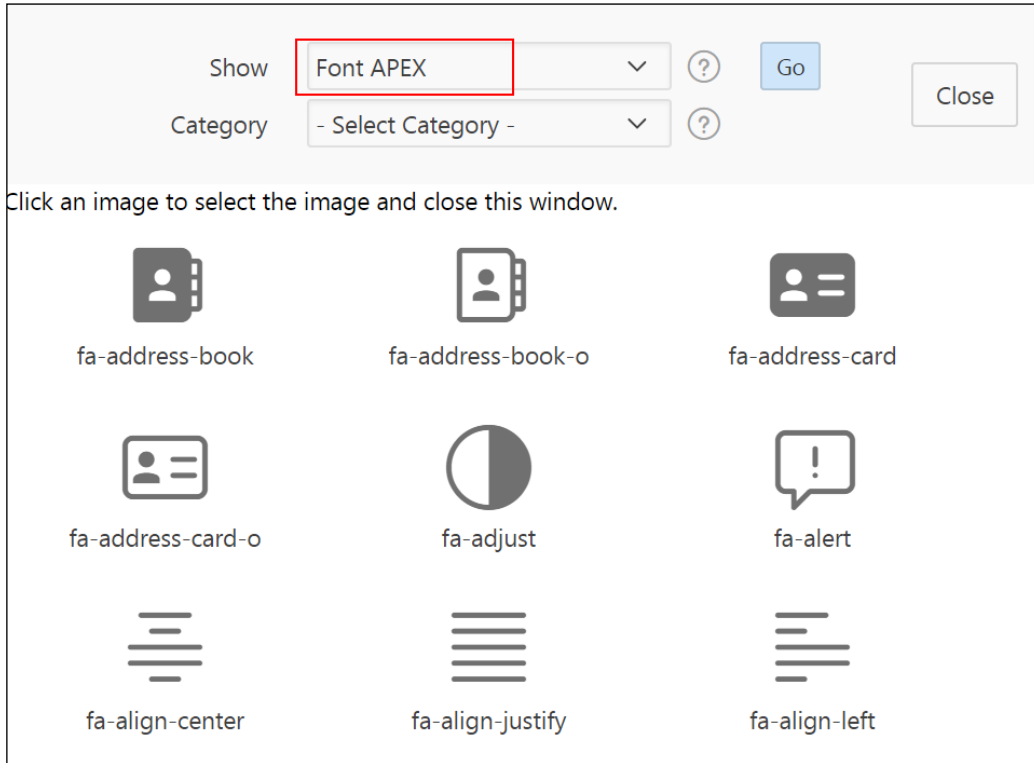
- s. Select **View Orders**.

List Home Page List ?			
Q Go Rows 50 Actions			
Sequence ↑≡	Name	Parent Entry	Target
10	Manage Customers	-	-
12	View Customers	Manage Customers	f?p=&APP_ID.:4:&SESSION.::&DEBUG.::
20	Manage Products	-	-
22	View Products	Manage Products	f?p=&APP_ID.:7:&SESSION.::&DEBUG.::
30	View Orders	-	f?p=&APP_ID.:12:&SESSION.::

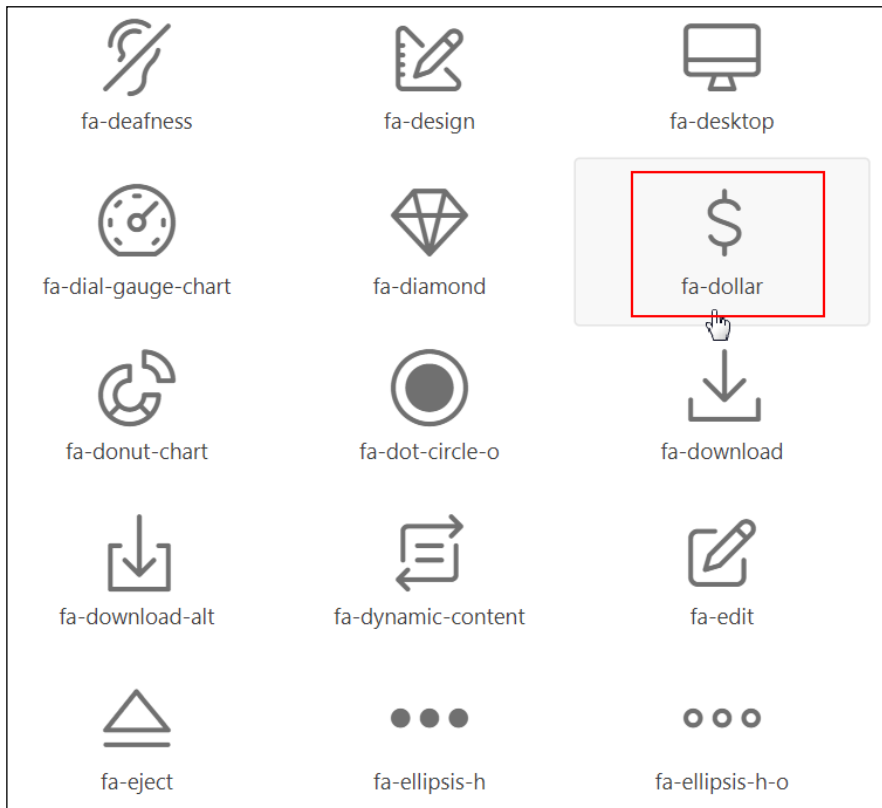
- t. In the **Entry** section, retain 30 for **Sequence** and enter **Manage Orders** for **List Entry Label**. In the **Target** section, set the **Target type** to **- No Target -**. In the **Entry** section, click the up arrow for the **Image/Class** field.

Entry	Target	Current List...	Conditions	Authorizati...	Configurati...	Click Count...	U
Image/Class							
Attributes							
Alt Attribute							
* List Entry Label	Manage Orders						
Target type	- No Target -						

- u. Make sure that **Font APEX Icons** is selected in the **Show** drop-down list.



- v. Select **fa-dollar** from the list.



- w. Click **Apply Changes**.

The screenshot shows the 'List Entry' form. At the top, there are navigation buttons '<' and '>', the title 'List Entry', and buttons 'Cancel', 'Delete', and 'Apply Changes'. The 'Apply Changes' button is highlighted with a red box. Below the title bar, there are tabs: 'Show All', 'Entry' (selected), 'Target', 'Current ...', 'Conditio...', 'Authoriz...', 'Configur...', 'Click Co...', 'User Def...', and 'Develop...'. The 'Image/Class' field contains 'fa-dollar'. Below it is an 'Attributes' field.

- x. Click **Create Entry >**.

The screenshot shows the 'List Details' form. At the top, there is a green message bar that says 'Action processed.' with a checkmark icon and a close button 'x'. Below the message bar, there are tabs: 'Lists', 'List Details' (selected), 'Unused', 'Conditional Entries', 'Utilization', and 'History'. Below the tabs, there is a 'List' dropdown menu showing 'Home Page List'. To the right of the dropdown are buttons 'Grid Edit', 'Edit List', and 'Create Entry >'. The 'Create Entry >' button is highlighted with a red box.

- y. Select **Manage Orders** for **Parent List Entry**, enter 32 for **Sequence**, and enter View Orders for **List Entry Label**. In the **Target** section, make sure that **Target type** is set to **Page in this Application** and the **Page** is 12 (the page number corresponding to the List of Orders page). Then click **Create List Entry**.

The screenshot shows the 'List Entry' form. At the top, there are navigation buttons '<' and '>', the title 'List Entry', and buttons 'Cancel', 'Create and Create Another', and 'Create List Entry'. The 'Create List Entry' button is highlighted with a red box. Below the title bar, there are tabs: 'Show All', 'Entry' (selected), 'Target', 'Current List...', 'Conditions', 'Authorizati...', 'Configurati...', 'Click Count...', 'User Define...', and 'Developer...'. The 'Entry' tab is active. Below the tabs, there is a 'List' dropdown menu showing 'Home Page List'. Below the 'List' dropdown, there are fields for 'Parent List Entry' (set to 'Manage Orders'), 'Sequence' (set to '32'), 'Image/Class', 'Attributes', and 'Alt Attribute'. Below these fields, there is a field for '* List Entry Label' (set to 'View Orders').

Target

Target type

*
Page

Page in this Application

12

?

☐ reset pagination for this page

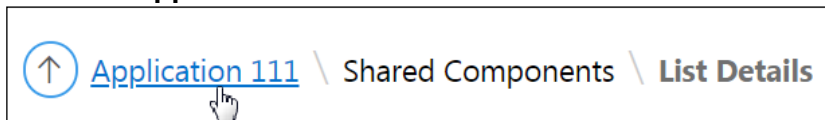
?

☐ Printer Friendly

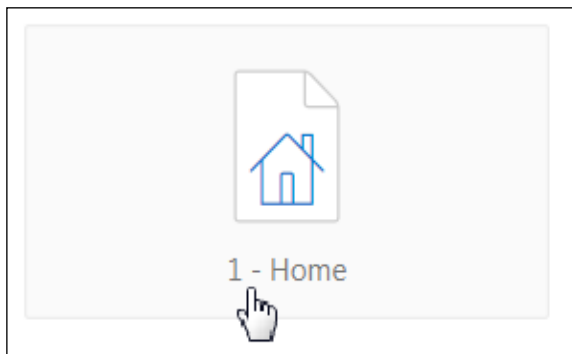
?

2. In the Home page, change the region template of **Tasks** to no template, the **List Template** to **Navigation Bar**, and the **Page Template** to **Theme Default**.

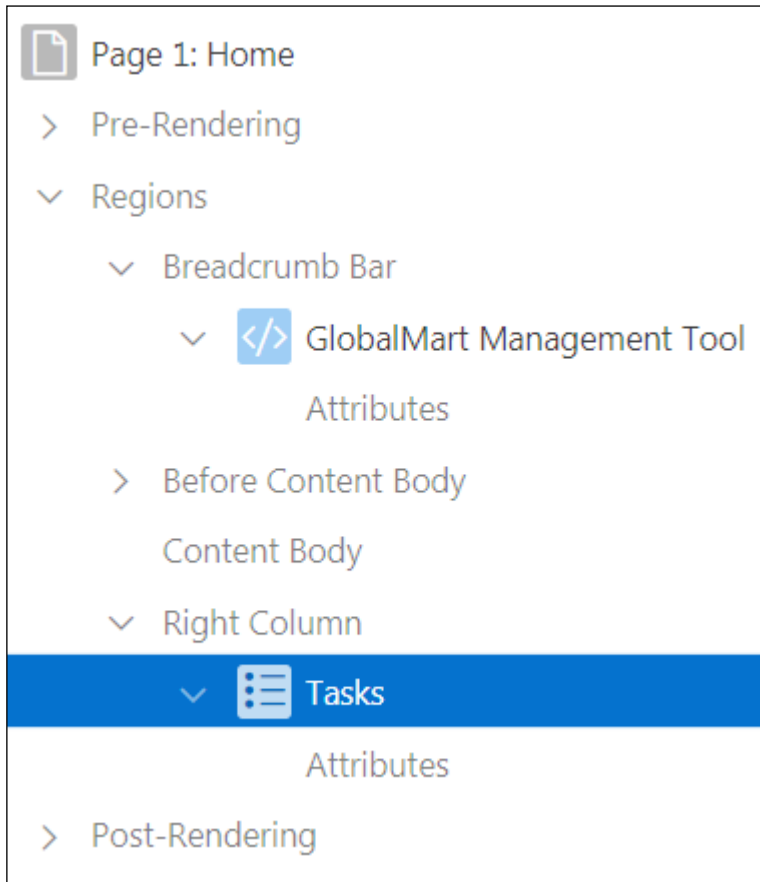
- a. Click **111 Application** Breadcrumb.



- b. Open the **Home Page**.



- c. In the Rendering tab, select the **Tasks** region.



- d. In the Region tab, make the following changes:
- **Layout > Sequence:** Enter 5
 - **Layout > Position:** Select **Content Body**
 - **Appearance > Template:** Select - Select -

Layout

Sequence

5

Parent Region

- Select -

>

Position

Content Body

>

Start New Row

Yes

No

Column

Automatic

>

Column Span

Automatic

>

Column CSS Classes

>

Column Attributes

>

Appearance

Template

- Select -

>

Item Display Position

Above Content

>

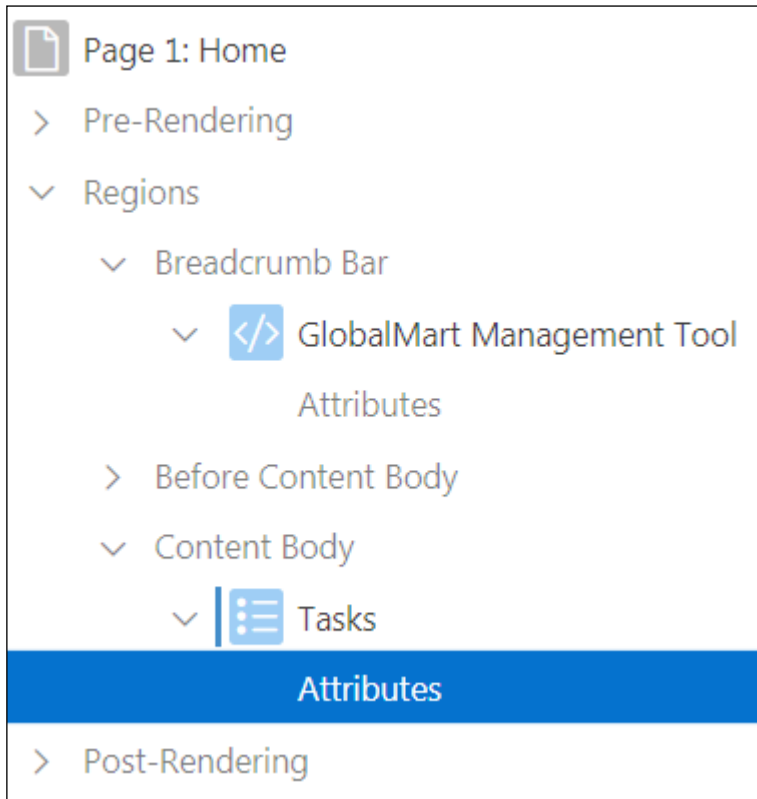
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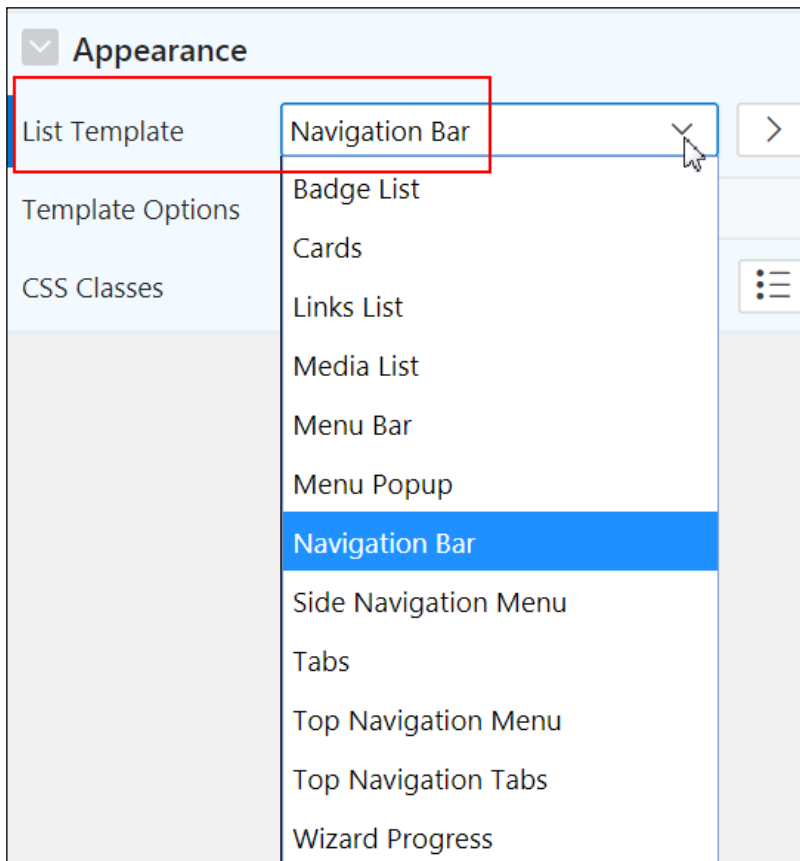
664

Practices for Lesson 17: Managing Application Navigation

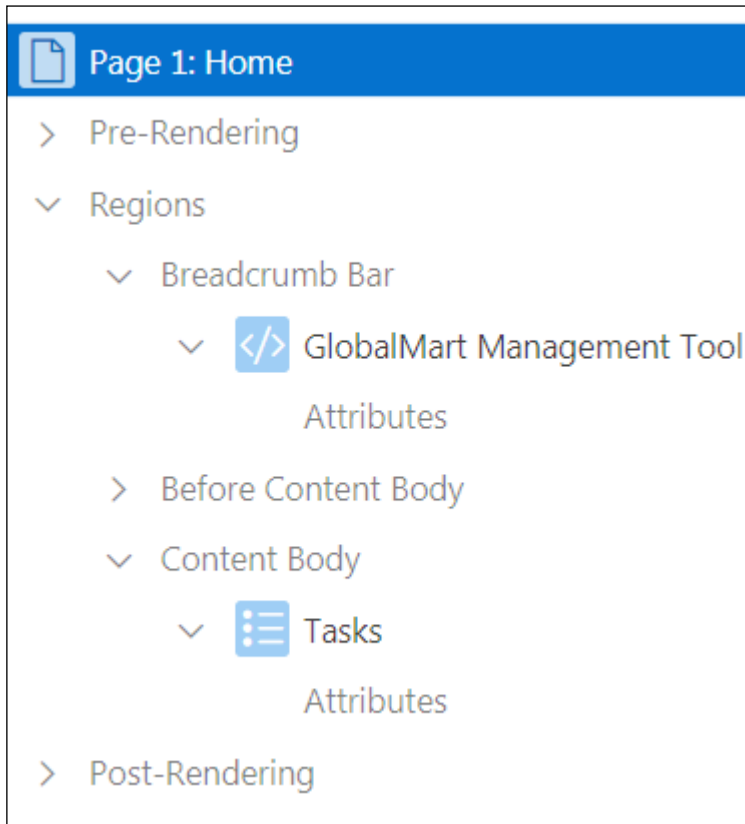
- e. In the Rendering tab, select **Tasks > Attributes**.



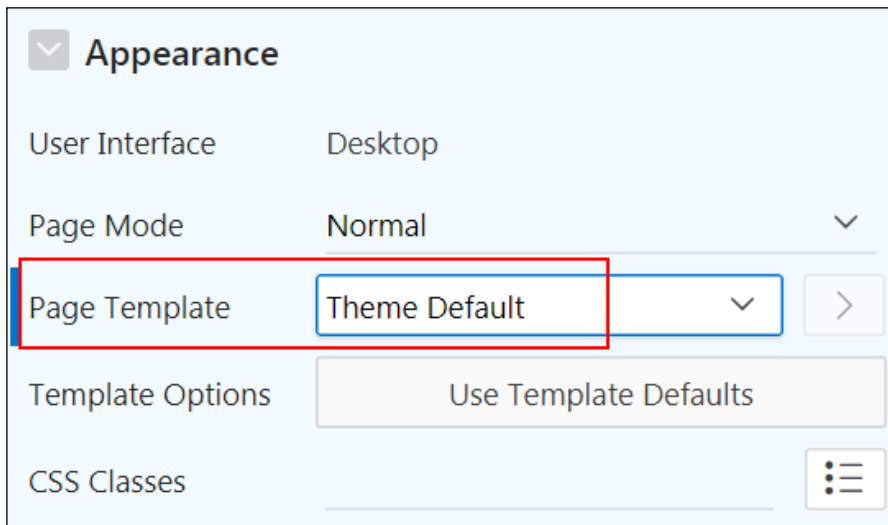
- f. In the Attributes tab, change the **List Template** value to **Navigation Bar**.



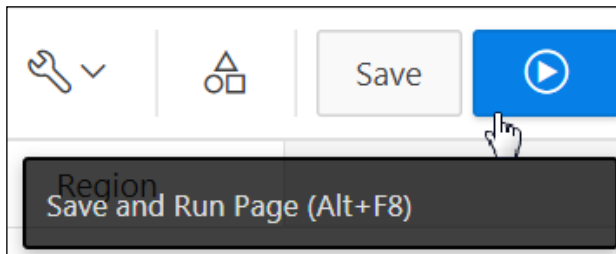
- g. In the Rendering tab, select **Page 1: Home**.



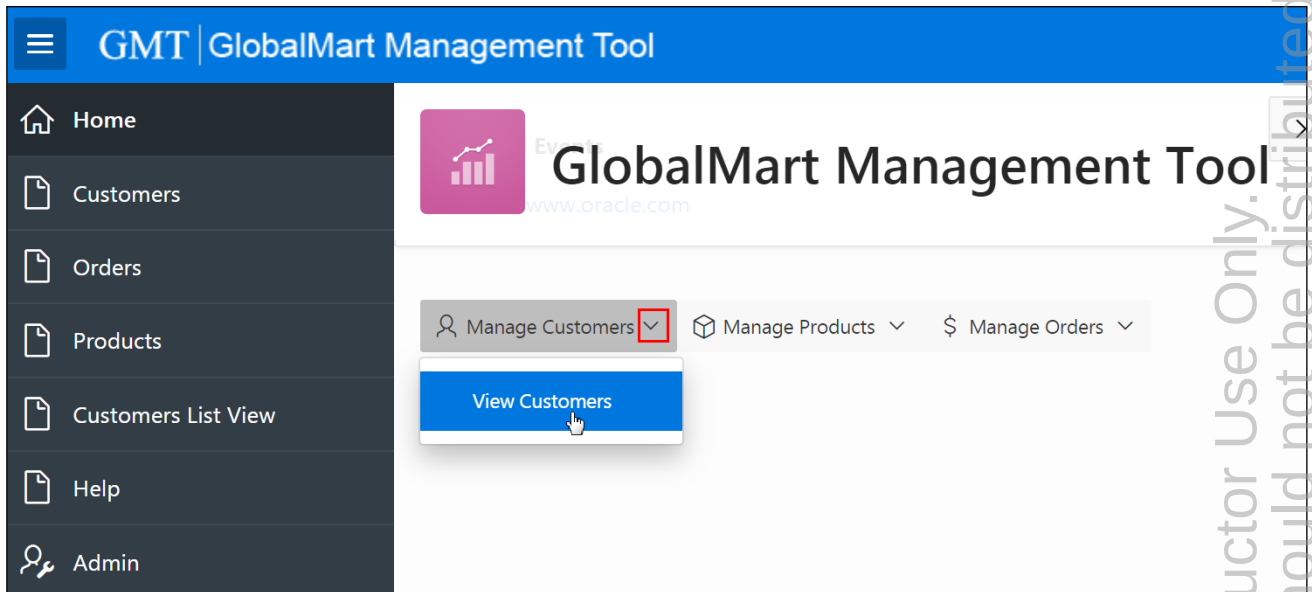
- h. In the Page tab, change the **Appearance > Page Template** value to **Theme Default**.



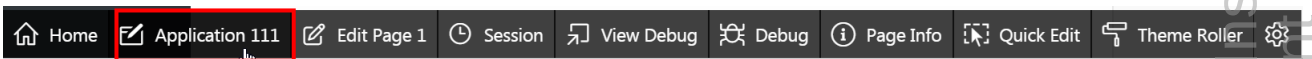
3. Run the page to see the Home Page List. Click the parent entry to show the sublist entries.
 - a. Click the **Save and Run Page** icon.



- b. The page is displayed. Click any parent entry to see the sublist.



- c. Click the **Application 111** link on the Developer toolbar.



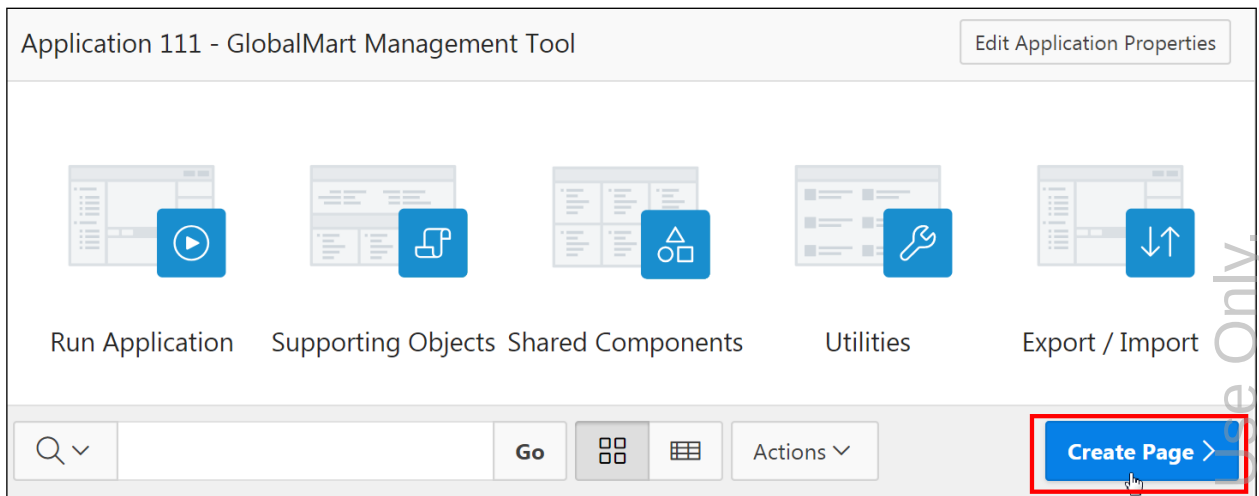
Solution 17-2: Building a Database-Driven Report

Overview

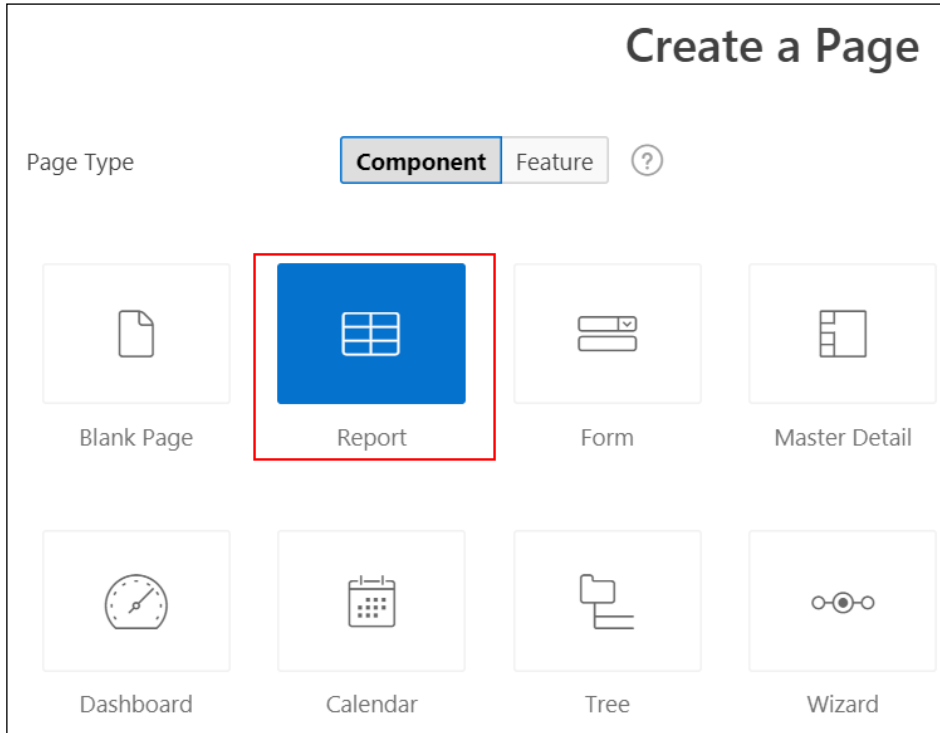
In this practice solution, steps are provided for building a database-driven navigation report.

Steps

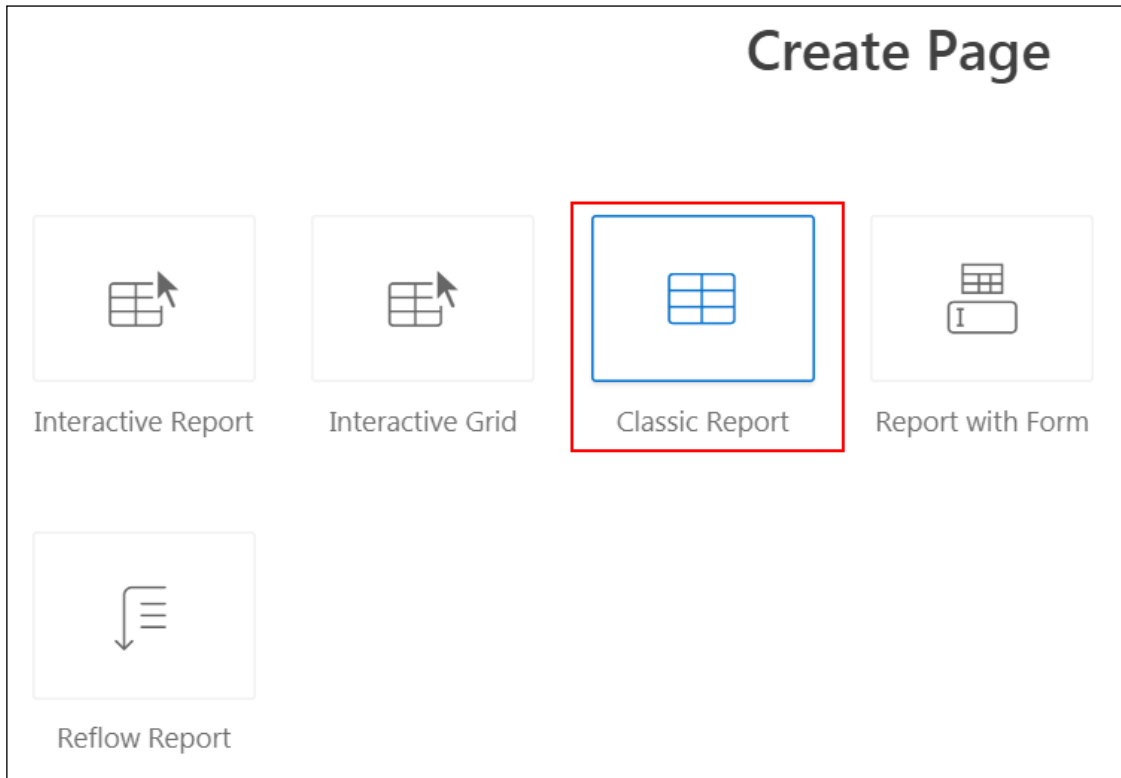
1. Create a new page with a Classic Report called **List of Customers**.
 - a. In the GlobalMart Management Tool application home page, click **Create Page >**.



- b. Select **Report**.



- c. Select **Classic Report**.



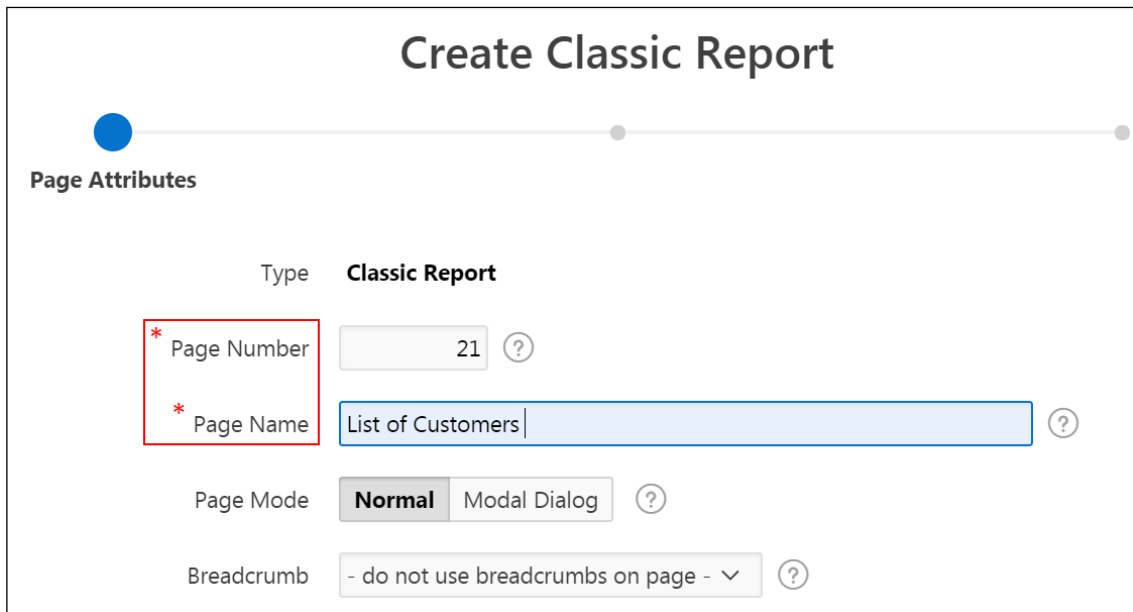
The 'Create Page' dialog box displays five options: Interactive Report, Interactive Grid, Classic Report, Report with Form, and Reflow Report. The 'Classic Report' option is highlighted with a red border.

Create Page

Interactive Report Interactive Grid **Classic Report** Report with Form

Reflow Report

- d. Enter 21 for **Page Number** and List of Customers for **Page Name**. Click **Next >**.



The 'Create Classic Report' dialog box shows the 'Page Attributes' section. The 'Page Number' is set to 21 and the 'Page Name' is 'List of Customers'. The 'Page Mode' is set to 'Normal' and the 'Breadcrumb' is set to '- do not use breadcrumbs on page -'. The 'Page Number' and 'Page Name' fields are highlighted with a red border.

Create Classic Report

Page Attributes

Type **Classic Report**

* Page Number 21 ?

* Page Name List of Customers ?

Page Mode **Normal** Modal Dialog ?

Breadcrumb - do not use breadcrumbs on page - ?

- e. Accept the default and click **Next >**.

Create Classic Report

✓

●

●

Navigation Menu

Navigation Preference  **Do not associate this page with a navigation menu entry** 

- ☐ Create a new navigation menu entry
- ☐ Identify an existing navigation menu entry for this page

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- f. For Report Source, select **SQL Query** and enter the following query (located in the /home/oracle/labs/labs/lab_17_02.txt file). Click **Create**.

```
SELECT cust_first_name||' '|| cust_last_name as Name, customer_id  
FROM oehr_customers
```

Create Classic Report

Report Source

Data Source: **Local Database** REST Enabled SQL Service Web Source

* Source Type: ☒ **SQL Query**

* Enter a SQL SELECT statement

Validation successful

1 SELECT cust_first_name||' '|| cust_last_name as Name, customer_id FROM oehr_customers

Cancel Create

The List of Customers Classic Report opens in Page Designer view.

ORACLE App Builder SQL Workshop Team Development App Gallery

Application 111 \ Page Designer

Page 21: List of Customers

Pre-Rendering

Regions

Content Body

In the News [Global Page]

Report 1

Columns

Attributes

List of Customers

PAGE HEADER

PAGE NAVIGATION

BREADCRUMB BAR

MASTER DETAIL

CONTENT BODY

RIGHT SIDE COLUMN

Report 1

COPY EDIT

Page

Filter

Identification

Name List of Customers

Page Alias

Title List of Customers

Page Group - Select -

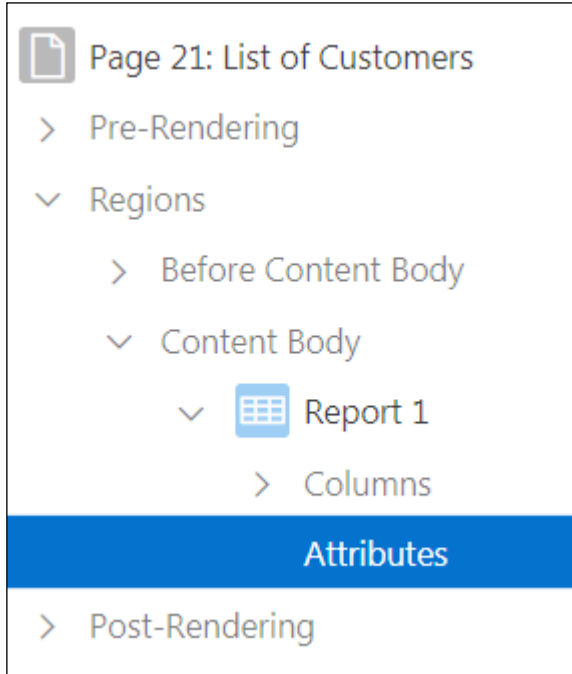
Appearance

Report page created successfully.

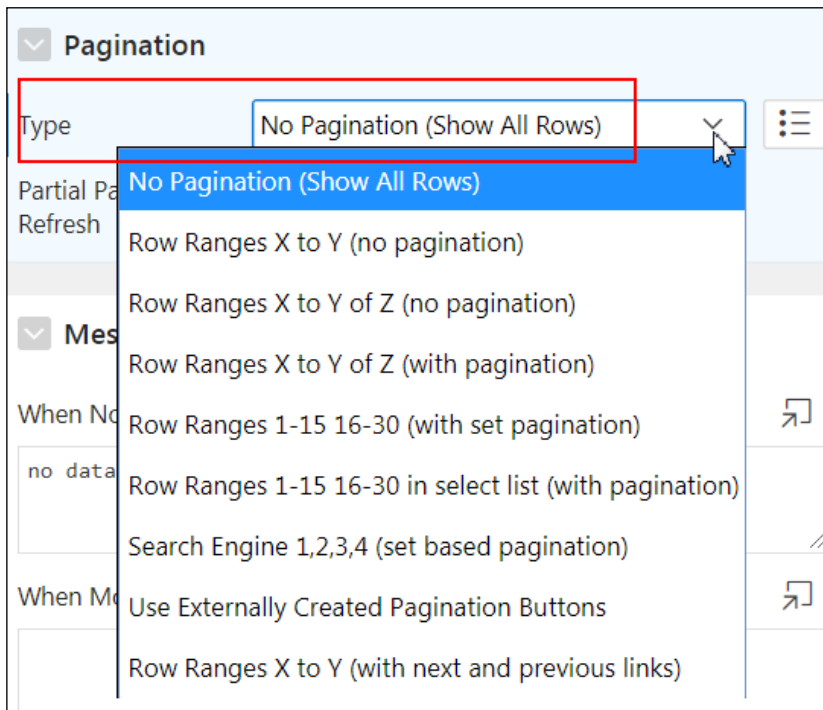
2. Hide the `CUSTOMER_ID` column in the report and turn off pagination. Display the Customer Name column heading in blue by editing the column definition and entering the following for Column Heading.

```
<font color="blue">CUSTOMERS</font>
```

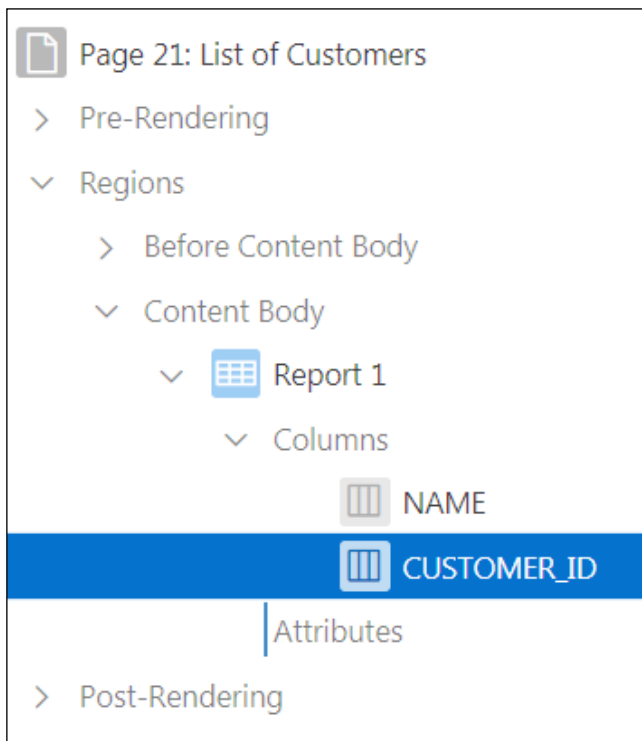
- a. In the Rendering tab, select **Regions > Content Body > Report 1 > Attributes**.



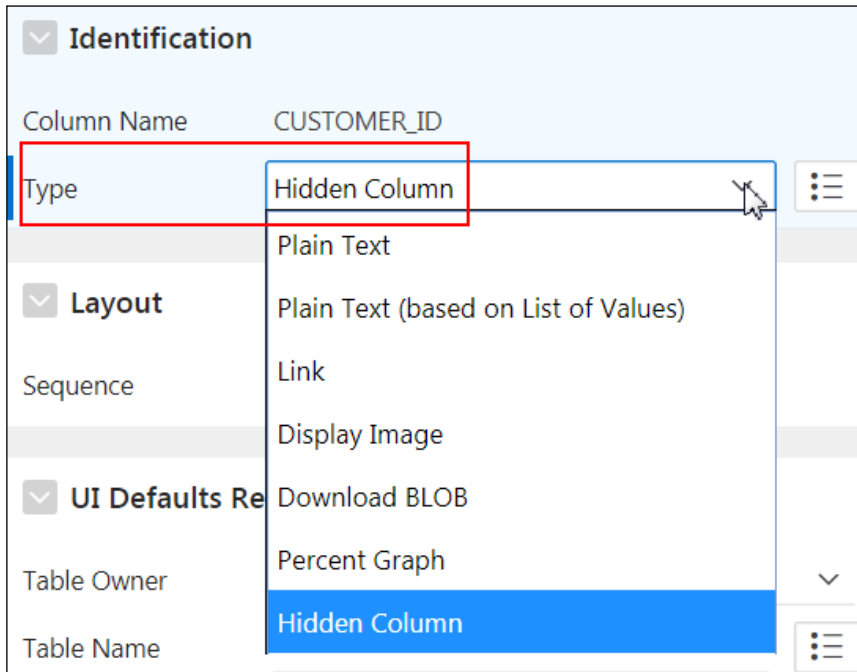
- b. In the Attributes tab, select **No Pagination (Show All Rows)** from the **Pagination > Type** drop-down list.



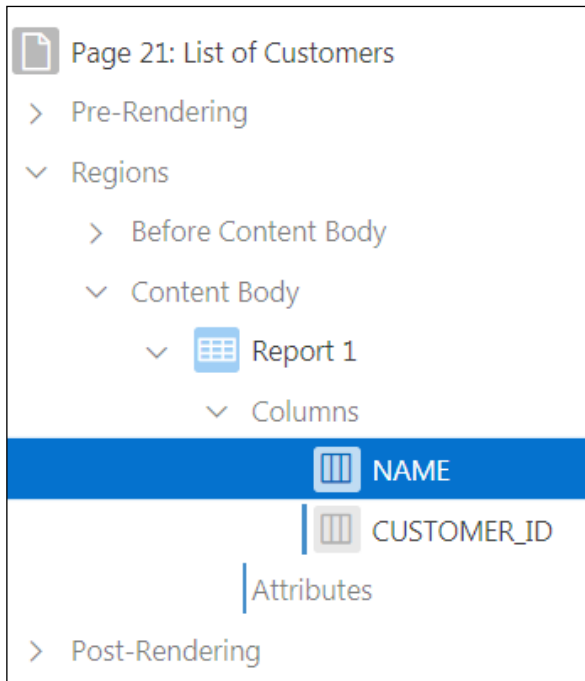
- c. In the Rendering tab, select **Regions > Content Body > Report 1 > Columns > CUSTOMER_ID**.



- d. In the Column tab, select **Identification > Type** as **Hidden Column**.



- e. In the Rendering tab, select the `NAME` column.

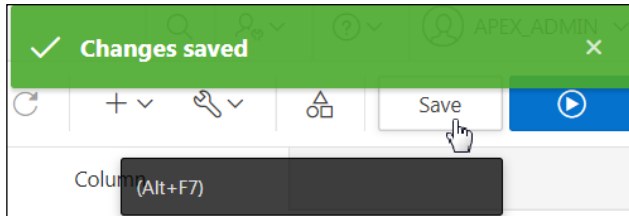


- f. In the Column tab, change the **Identification > Heading** to:

```
<font color="blue">CUSTOMERS</font>
```

The screenshot shows the 'Heading' property in the Column Editor. The text input field contains the HTML code: `<font color="blue">CUSTOMERS</font>`. The 'Heading' tab is selected, and the 'Alignment' options are visible below.

- g. Click **Save**.



3. Create a link to the Customer Details page on the `NAME` column. Pass the `P9_CUSTOMER_ID` item from the Customer Details page with the value `#CUSTOMER_ID#` from this page.
- a. Ensure that the `NAME` column remains selected. In the Column tab, change **Identification > Type** to **Link**.

The screenshot shows the 'Identification' tab in the Column Editor. The 'Type' property is highlighted with a red box, and the value 'Link' is selected from the dropdown menu. The 'NAME' column is selected in the list above.

- b. Click the **No Link Defined** button under the **Link** subtab.

The screenshot shows the 'Link' subtab in the Column Editor. The 'No Link Defined' button is highlighted with a mouse cursor.

- c. Select the **Customer Details** page for **Page** (in this case 9) by clicking the arrow to the right of the Page field. Under **Set Items**, select P9_CUSTOMER_ID for **Name** and CUSTOMER_ID for **Value**. Click **OK**.

Link Builder - Target

Target

Type Page in this application

Page 9

Set Items

Name P9_CUSTOMER_ID

Value #CUSTOMER_ID#

Clear Session State

Clear Cache

Reset Pagination Yes No

Cancel Clear OK

- d. Under the **Link** section, ensure #NAME# appears for **Link Text**.

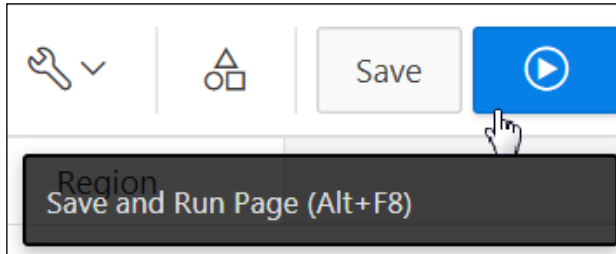
Link

Target Page 9

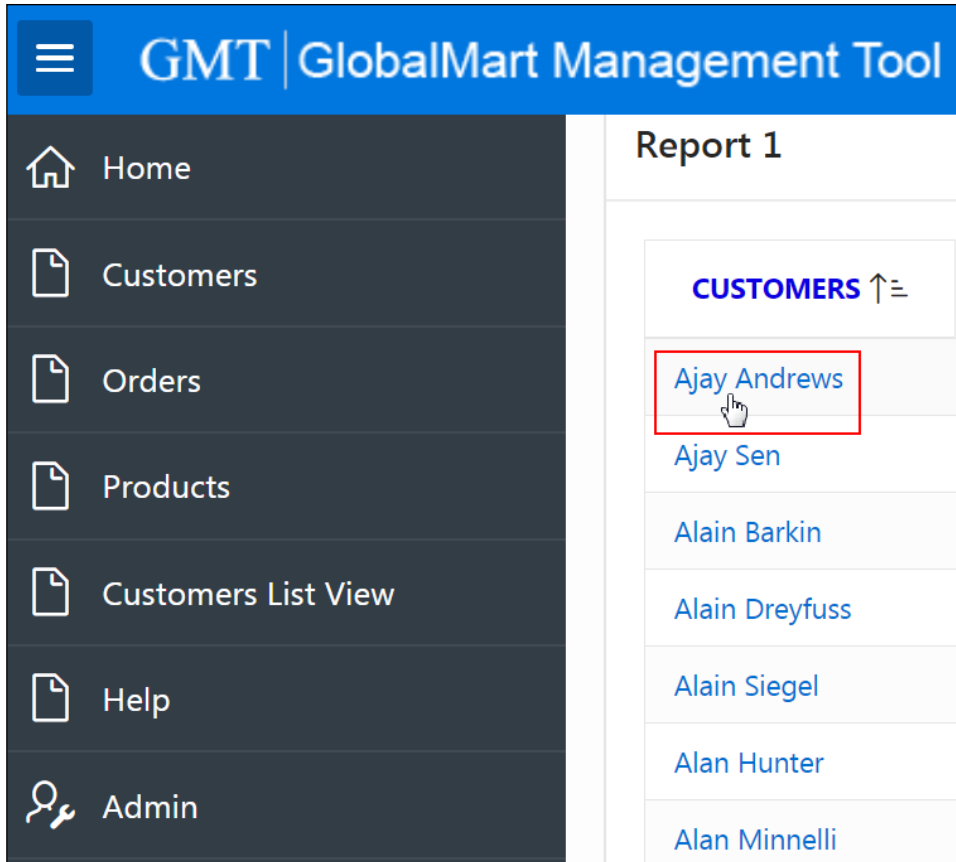
Link Text #NAME#

Link Attributes

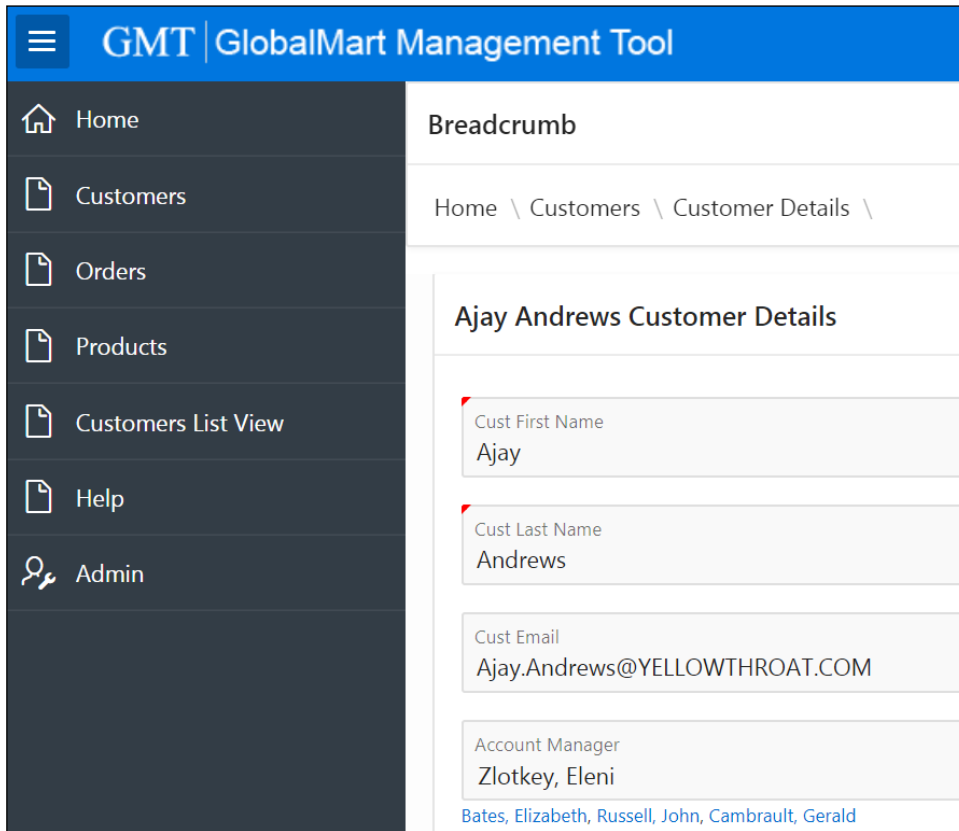
4. Run the page to see the database-driven report.
 - a. Click the **Save and Run Page** icon.



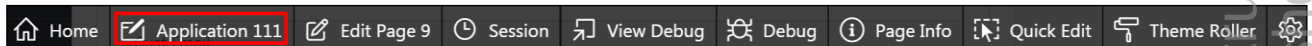
- b. Click any one customer to view the details.



- c. The Customer Details page is displayed for the customer.



- d. Click the **Application 111** link on the Developer toolbar.



Solution 17-3: Building a Site Map

Overview

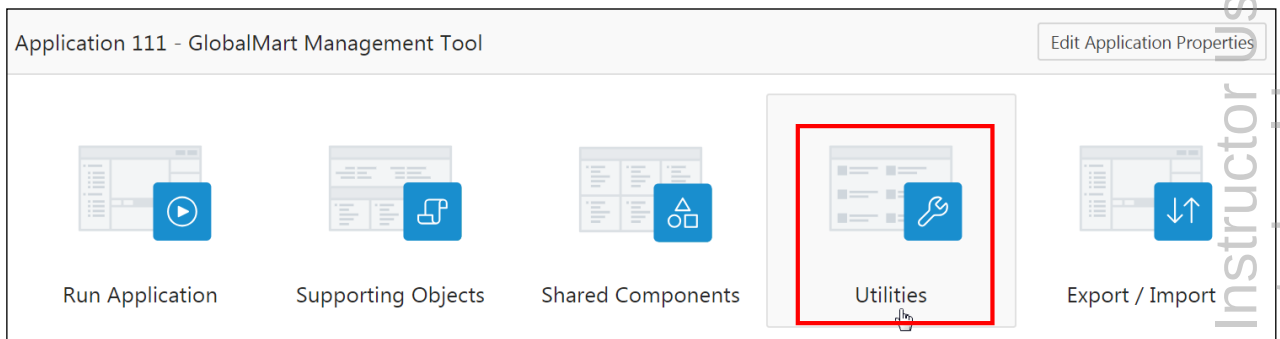
In this practice solution, steps are provided for building a site map.

Steps

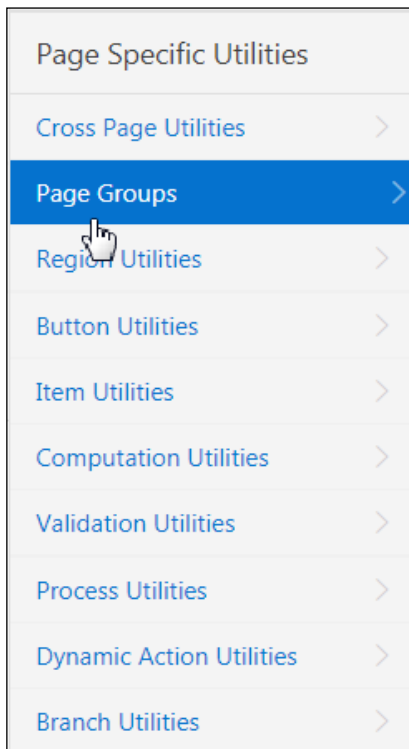
1. Assign each page that you want in the site map to the same page group. Create a page group called Site Map and assign the following pages to the Site Map page group.

Pages
Customers
Products
List of Orders
Customer Feedback
Access Control Administration Page

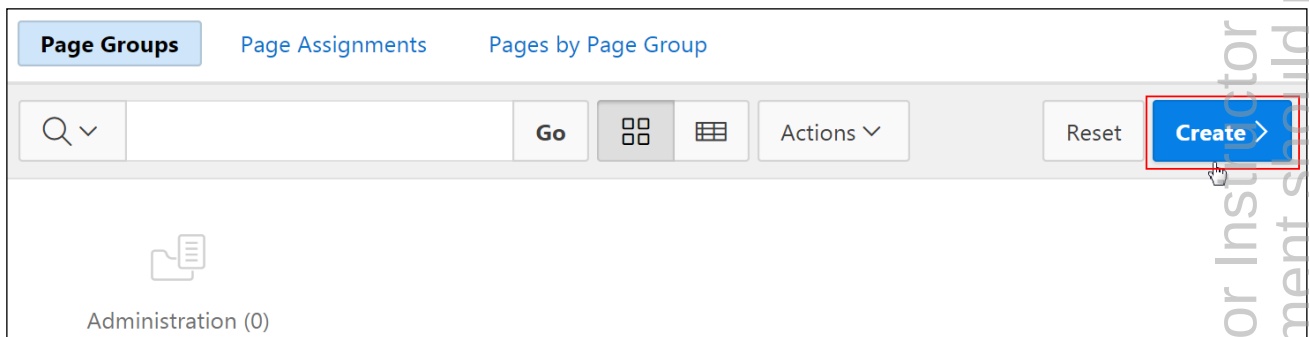
- a. In the GlobalMart Management Tool home page, click **Utilities**.



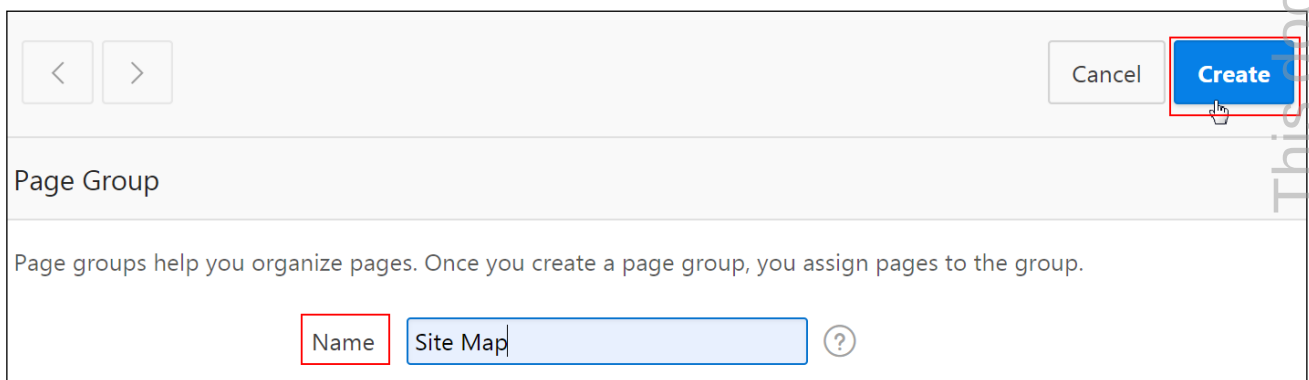
- b. Select **Page Groups** under **Page Specific Utilities**.



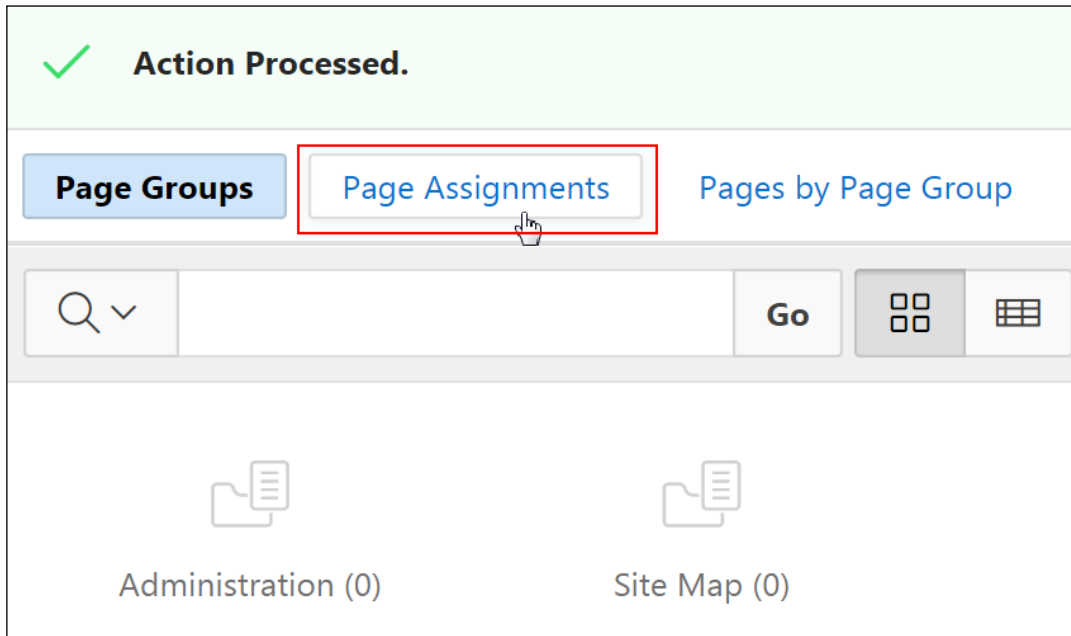
- c. Click **Create >**.



- d. Enter **Site Map** for **Name** and click **Create**.



- e. Click the **Page Assignments** tab.

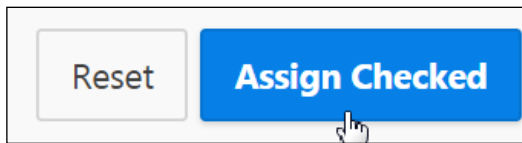


- f. Select **Site Map** from the **New Group** drop-down list. Then select the check box next to each of the following pages and click **Assign Checked**.

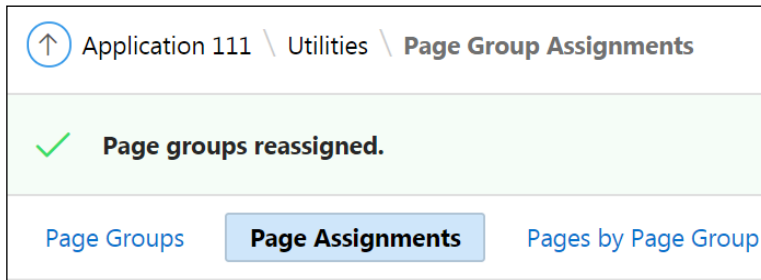
Pages
Customers
Products
List of Orders
Customer Feedback
Access Control Administration Page

New Group			
Site Map			
Go Actions			
☐	Page	Name	Group
☐	0	Global Page - Desktop	Unassigned
☐	1	Home	Unassigned
☐	2	Top Tier Salary	Unassigned
☐	3	Customer Address List	Unassigned
☒	4	Customers	Unassigned
☐	5	Employee Commission	Unassigned
☐	6	Product Information Details	Unassigned
☒	7	Products	Unassigned
☒	12	List of Orders	Unassigned
☐	13	Oehr Orders	Unassigned
☐	14	Oehr Order Items	Unassigned
☐	15	Orders List	Unassigned
☐	16	Customer Master Information	Unassigned
☐	17	Customer Order History	Unassigned
☒	18	Customer Feedback	Unassigned
☐	19	Employee Details	Unassigned
☐	20	Help	Unassigned
☐	21	List of Customers	Unassigned
☐	201	Employee List	Unassigned
☐	202	Product Information	Unassigned
☐	9999	Login Page	Unassigned
☒	10000	Access Control Administration Page	Unassigned

Note that the image is truncated and not the complete list.

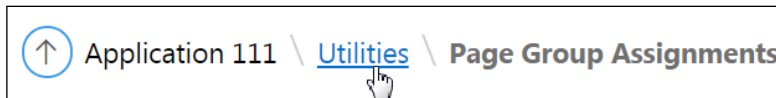


You get the **Page Groups reassigned** message.

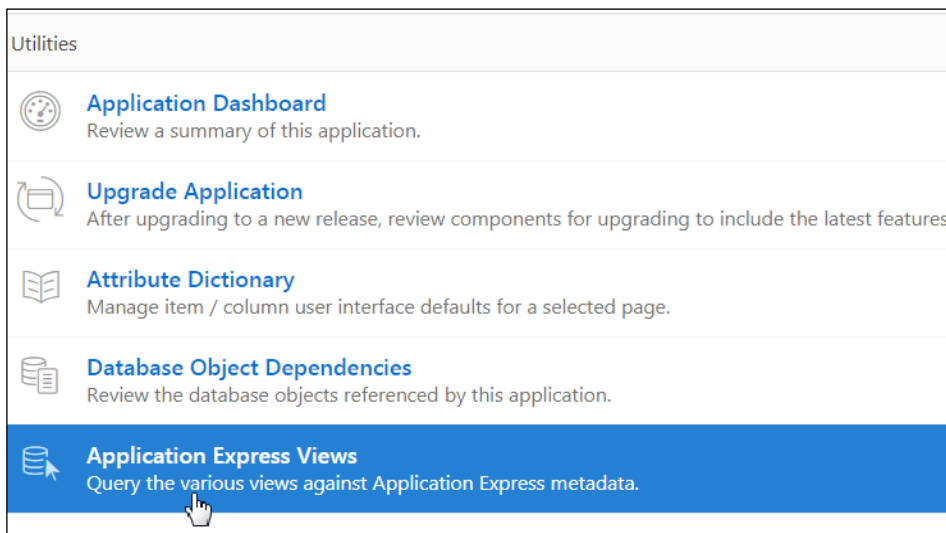


2. Create a new page called **Site Map** with a Classic Report that shows only the pages in the Site Map page group.

- a. Click the **Utilities** breadcrumb.



- b. Select **Application Express Views**.



- d. Select the **APEX_APPLICATION_PAGES** view.

View 	Comment	Parent View
APEX_APPLICATION_NAV_BAR	Identifies navigation bar entries displayed on pages that use a Page Template that include a #NAVIGATION_BAR# substitution string	APEX_APPLICATIONS
APEX_APPLICATION_PAGES	A Page definition is the basic building block of page. Page components including regions, items, buttons, computations, branches, validations, and processes further define the definition of a page.	APEX_APPLICATIONS
APEX_APPLICATION_PAGE_BRANCHES	Identifies branch processing associated with a page. A branch is a directive to navigate to a page or URL which is run at the conclusion of page accept processing.	APEX_APPLICATION_PAGES

- e. On the Select Columns tab, if **PAGE_NAME** is not included in the list on the right, then select **PAGE_NAME** column and click the right arrow (**>**) to move it to the columns selected list. Similarly, you do not want to display the **WORKSPACE**, **WORKSPACE_DISPLAY_NAME**, and **APPLICATION_NAME**. Select **WORKSPACE**, **WORKSPACE_DISPLAY_NAME**, and **APPLICATION_NAME** and click the left arrow (**<**) to move it to the column list on the left. Click **Filter >**.

Selected View: **APEX_APPLICATION_PAGES** 

< Views **Filter >** **Results >**

Select Columns

Select Columns

PAGE_TITLE
PAGE_MODE
USER_INTERFACE_ID
MEDIA_TYPE
TAB_SET
PAGE_ALIAS
PAGE_FUNCTION
ALLOW_DUPLICATE_SUBMISSIONS
INCLUDE_APEX_CSS_JS_YN


>>
>
<
<<

APPLICATION_ID
PAGE_ID
PAGE_NAME

- f. Select **APPLICATION_ID** for **Column** and enter **111** for **Value** (note that you need to replace **111** with your own application ID in case your application ID varies). In the next line, enter **PAGE_GROUP** for **Column** and enter 'Site Map' for **Value**. Click **Results >**.

Report View Tree View Select Columns **Filter** Results

Selected View: **APEX_APPLICATION_PAGES** 

< Columns **Results >**

Filter

Column	APPLICATION_ID 	Condition	= 	Value	111 
	PAGE_GROUP		=		'Site Map'

- g. Notice that the pages you selected earlier are on the list. Expand **Query**.

Report ViewTree ViewSelect ColumnsFilterResults

Selected View: APEX_APPLICATION_PAGES ?

APPLICATION_ID ↑	PAGE_ID	PAGE_NAME
111	4	Customers
111	7	Products
111	12	List of Orders
111	18	Customer Feedback
111	10000	Access Control Administration Page

Download

 Query

- h. Expand **Query** and select the query and copy it to your clipboard.

▼ Query

```
select APPLICATION_ID,PAGE_ID,PAGE_NAME
from APEX_APPLICATION_PAGES
where APPLICATION ID = 111
and PAGE_GROUP = 'Site Map'
```

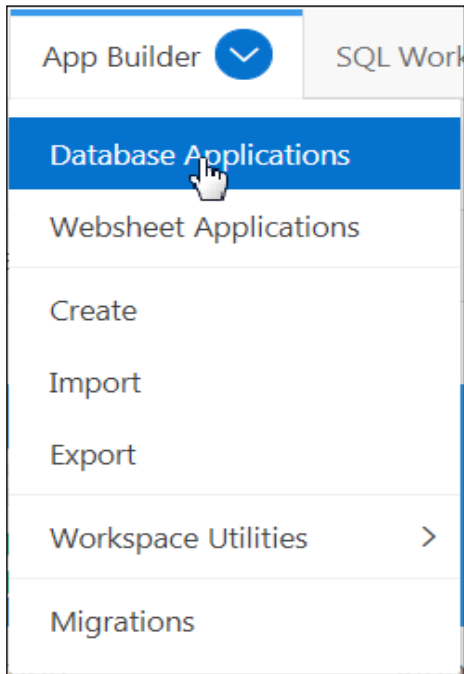
Copy

Search Search for Chrome for "select APPLICATION_ID,PAGE_ID,PAGE_NAME from..."

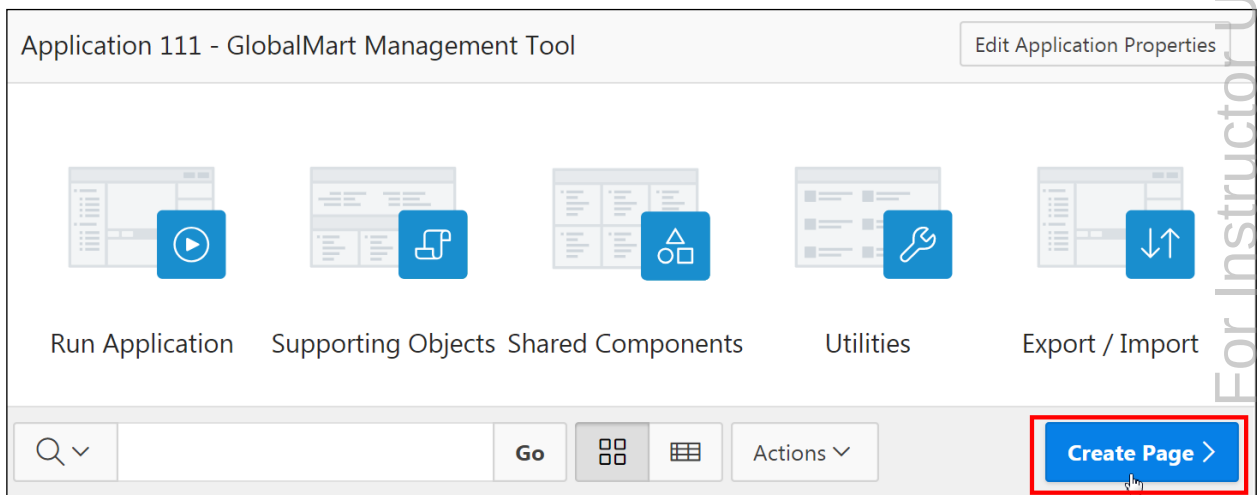
Print...

Inspect

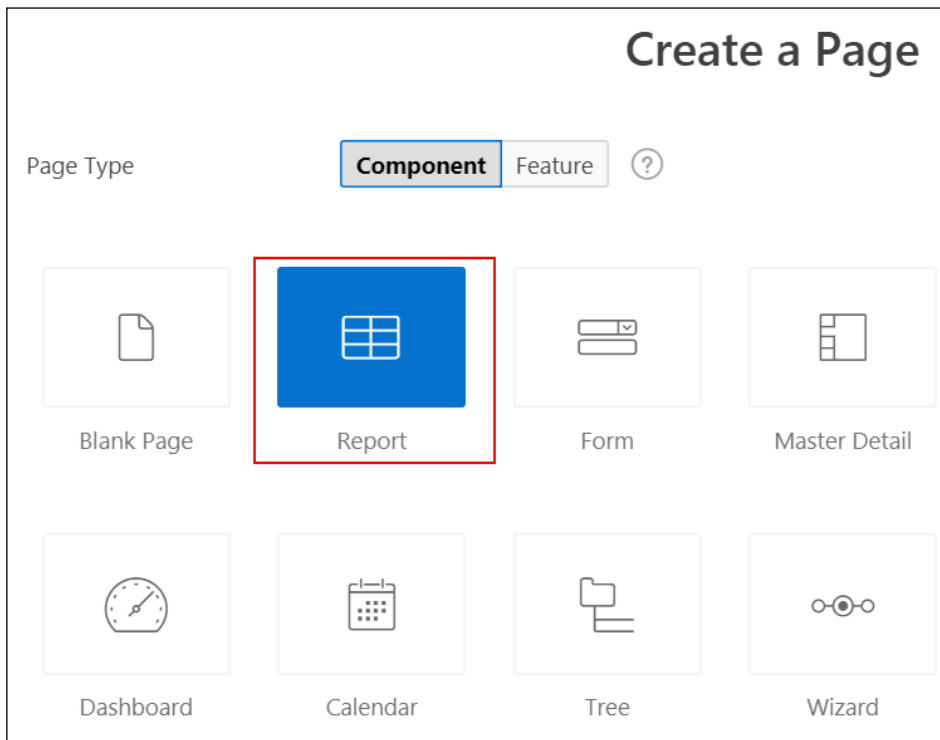
- i. Click the **Application Builder** pull-down menu and select **Database Applications**.



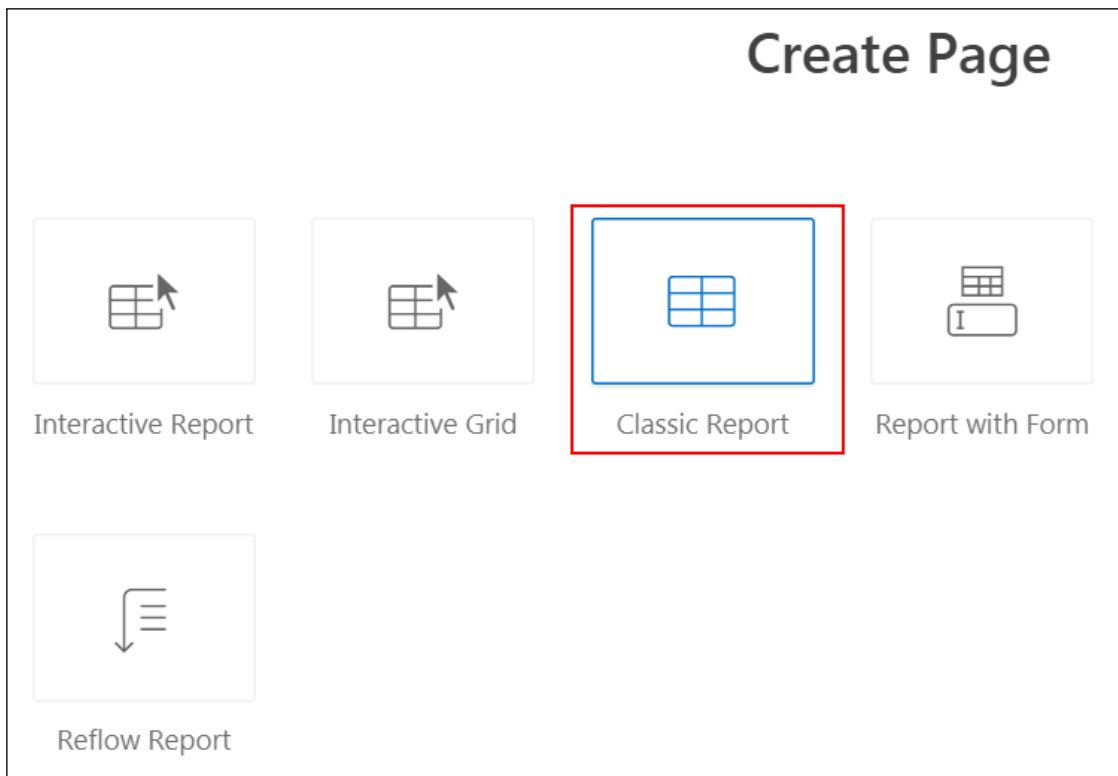
- j. Select **GlobalMart Management Tool** application.
k. Click **Create Page >**.



I. Select **Report**.



m. Select **Classic Report**.



- n. Enter 22 for **Page Number** and Site Map for **Page Name** and click **Next >**.

Create Classic Report

Page Attributes

Type **Classic Report**

* Page Number ?

* Page Name ?

Page Mode **Normal** Modal Dialog ?

Breadcrumb - do not use breadcrumbs on page - v ?

- o. Accept the default and click **Next >**.

Create Classic Report

Navigation Menu

Navigation Preference ☒ **Do not associate this page with a navigation menu entry** ?

☐ Create a new navigation menu entry

☐ Identify an existing navigation menu entry for this page

- p. Select **SQL Query** for **Source Type**. Paste your query in the SQL area and click **Create**. You can also copy and paste the query from the /home/oracle/labs/labs/lab_17_03.txt file. Note that you need to replace 111 with your own application ID in the script.

```
select APPLICATION_ID,PAGE_ID,PAGE_NAME
  from APEX_APPLICATION_PAGES
 where APPLICATION_ID = 111
    and PAGE_GROUP = 'Site Map'
```

Create Classic Report

Report Source

Data Source: **Local Database** | REST Enabled SQL Service | Web Source

* Source Type: ☒ **SQL Query** | ☐ Table

* Enter a SQL SELECT statement

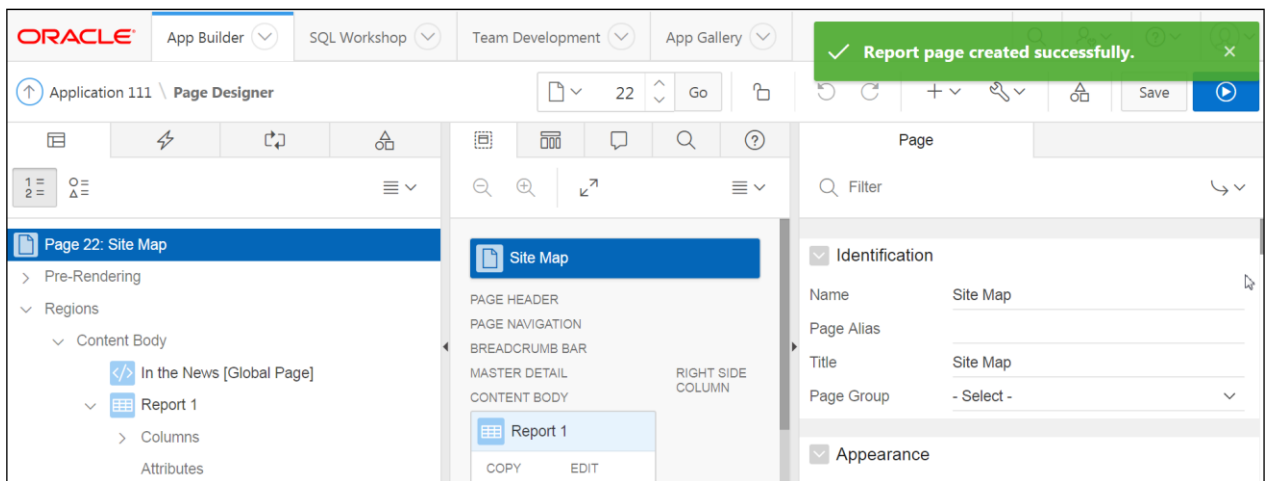
Validation successful

```
1 select APPLICATION_ID,PAGE_ID,PAGE_NAME
2   from APEX_APPLICATION_PAGES
3  where APPLICATION_ID = 111
4    and PAGE_GROUP = 'Site Map'
```

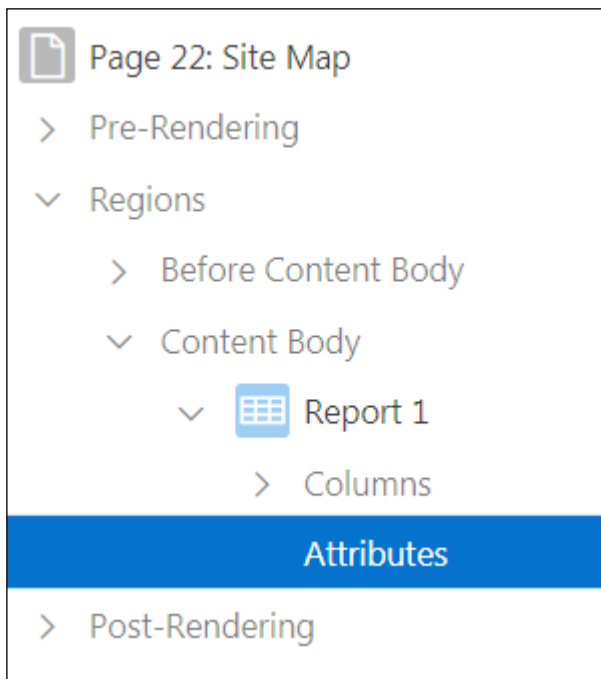
SQL Query Example

< Cancel **Create**

The Site Map page opens in Page Designer view.

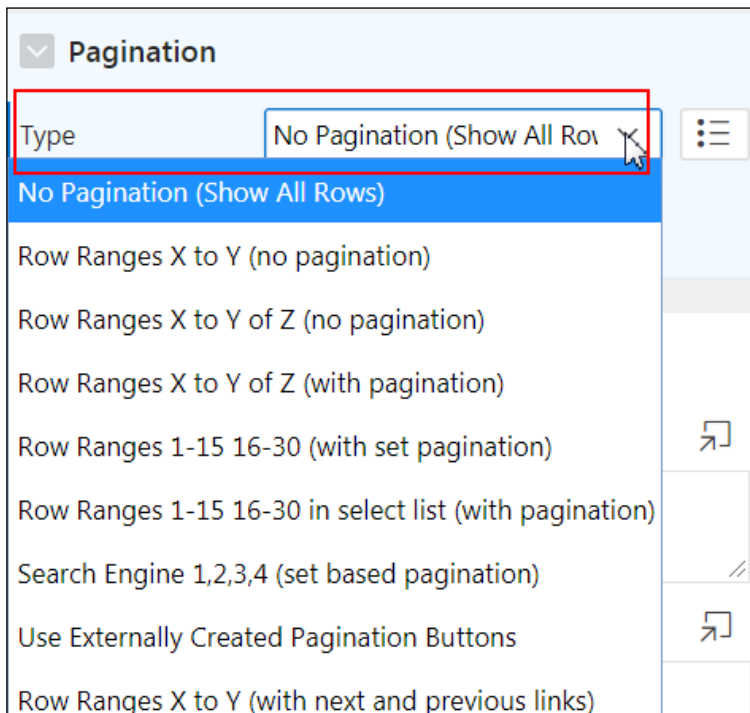


3. Change the **Pagination Scheme** to **None** and do not show the `PAGE_ID` column. Link the `PAGE_NAME` column to the `#PAGE_ID#` so that it opens the corresponding page when clicked.
 - a. In the Rendering tab, under **Regions**, select **Report 1 > Attributes**.

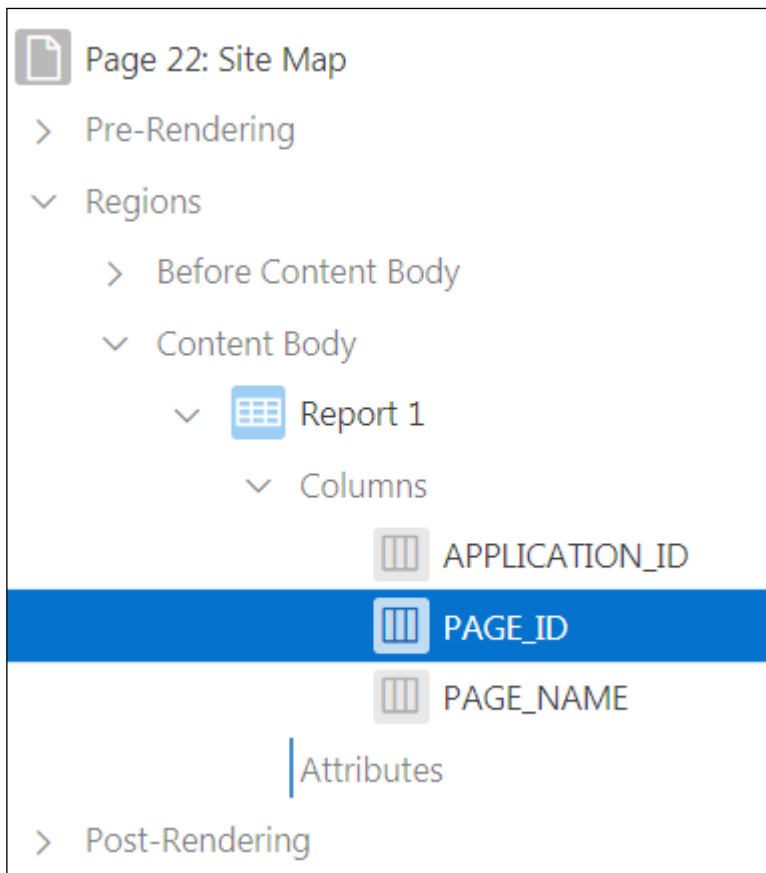


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- b. Select **No Pagination (Show All Rows)** from the **Pagination > Type** drop-down list.



- c. In the Rendering tab, under **Regions**, select **Columns > PAGE_ID**.



- d. In the Column tab, select **Identification > Type** as **Hidden Column**.

The screenshot shows a configuration window with a left sidebar and a main content area. The sidebar has three expandable sections: 'Identification' (expanded), 'Layout', and 'UI Defaults Re'. The 'Identification' section contains the following items: 'Column Name' (PAGE_ID), 'Type' (selected), 'Sequence', 'Table Owner', and 'Table Name'. The 'Type' dropdown is open, displaying a list of options: 'Plain Text', 'Plain Text (based on List of Values)', 'Link', 'Display Image', 'Download BLOB', 'Percent Graph', and 'Hidden Column' (highlighted in blue). A red rectangle highlights the 'Type' dropdown and its list of options.

Section	Item	Value
Identification	Column Name	PAGE_ID
	Type	Hidden Column
	Sequence	
	Table Owner	
	Table Name	
Layout		
UI Defaults Re		

- Plain Text
- Plain Text (based on List of Values)
- Link
- Display Image
- Download BLOB
- Percent Graph
- Hidden Column

- e. In the Rendering tab, under **Regions**, select **Columns** > APPLICATION_ID. In the Column tab, select **Identification** > Type as **Hidden Column**.

The screenshot shows the Oracle APEX page editor interface. The top pane displays the page structure: Page 22: Site Map > Pre-Rendering > Regions > Before Content Body > Content Body > Report 1 > Columns. The 'Columns' section is expanded, showing a list of columns: APPLICATION_ID, PAGE_ID, and PAGE_NAME. The 'APPLICATION_ID' column is selected and highlighted in blue.

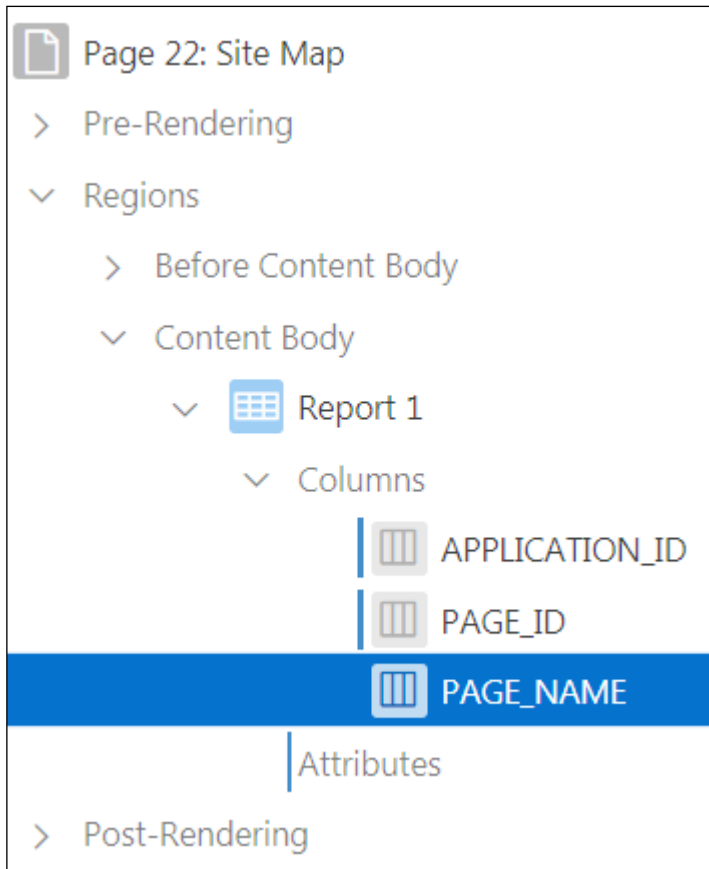
The bottom pane shows the 'Identification' tab for the selected column. It contains a table with the following data:

Column Name	APPLICATION_ID
Type	Hidden Column

The 'Type' row is highlighted with a red border. A dropdown menu is open for the 'Type' field, showing the following options: Plain Text, Plain Text (based on List of Values), Link, Display Image, Download BLOB, Percent Graph, and Hidden Column. The 'Hidden Column' option is selected and highlighted in blue.

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- f. In the Rendering tab, under **Regions**, select **Columns** > `PAGE_NAME`.



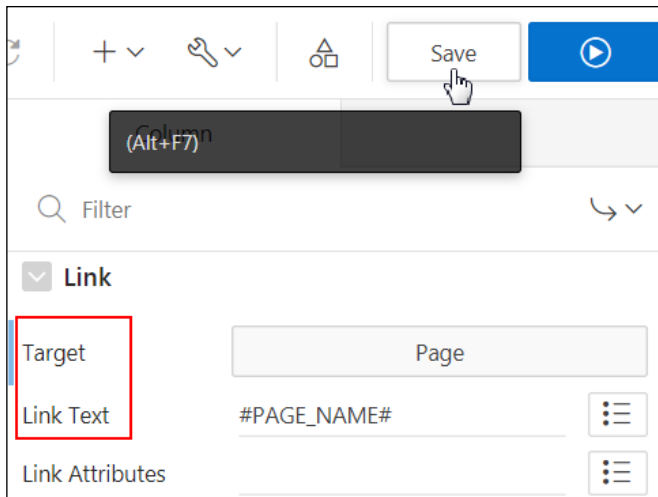
- g. In the Column tab, select **Identification > Type** as **Link**. Click the **No Link Defined** button under **Link > Target**.

The screenshot shows a configuration window with several tabs: Identification, Heading, Layout, Appearance, and Link. The 'Identification' tab is active. It contains a 'Column Name' field with the value 'PAGE_NAME'. Below it, the 'Type' dropdown is set to 'Link'. At the bottom, under the 'Link' tab, the 'Target' field is set to 'No Link Defined'. A red box highlights the 'Type' dropdown and the 'No Link Defined' button.

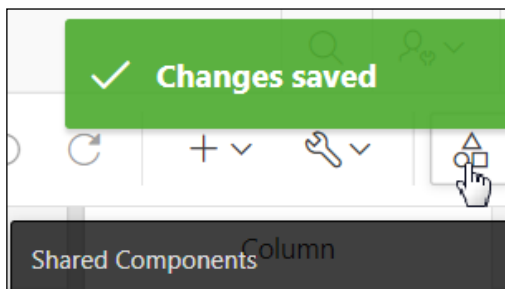
- h. Make sure that **Page in this application** is selected for **Target > Type** and enter **#PAGE_ID#** in the **Page** text box. Click **OK**.

The screenshot shows a 'Link Builder - Target' dialog box. It has a 'Target' section with a 'Type' dropdown set to 'Page in this application' and a 'Page' text box containing '#PAGE_ID#'. Below this is a 'Set Items' section with 'Name' and 'Value' fields. Further down is a 'Clear Session State' section with a 'Clear Cache' checkbox and a 'Reset Pagination' section with 'Yes' and 'No' buttons. At the bottom, there is an 'Advanced' section and three buttons: 'Cancel', 'Clear', and 'OK'. The 'OK' button is highlighted with a red box.

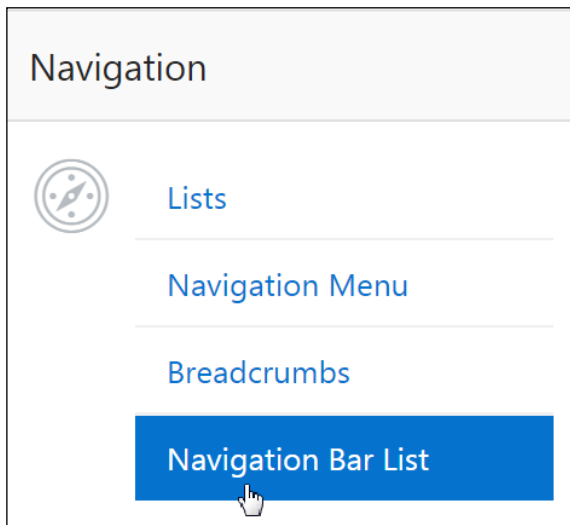
- i. Ensure `#PAGE_NAME#` appears for **Link Text** and click the **Save** button.



4. Add this page to your navigation bar and call it **Site Map**.
- a. Click the **Shared Components** icon.



- b. Select **Navigation Bar List** under **Navigation**.



- c. Click **Desktop Navigation Bar**.

Application 111 \ Shared Components \ Lists

Lists List Details Unused Condition

Navigation Bar List

Name ↑	Type	Entries
Desktop Navigation Bar	Static	5

- d. Click **Create Entry >**.

Grid Edit Edit List Create Entry >

- e. Enter **Site Map** for **List Entry Label**. Select your Site Map page number for **Page** (Page 22) and click **Create List Entry**.

List Entry

Cancel Create and Create Another Create List Entry

Show All Entry Target Current List... Conditions Authorizati... Configurati... Click Count... User Define... Developer ...

* List Entry Label Site Map

Target

Target type Page in this Application

* Page 22

reset pagination for this page

- f. Change the Sequence of the Site Map such that it appears before Sign Off. Click **Grid Edit**.

✓ Action processed.

Lists

List Details

Unused

Conditional Entries

Utilization

History

List Desktop Navigation Bar ?

Q ▾

Go

Rows 50 ▾

Actions ▾

Sequence ↑ ▾	Name	Parent Entry	Target
5	Help	-	f?p=&APP_ID.:20:&SESSION.:&APP_PAGE_ID.:8
7	Home	-	f?p=&APP_ID.:1:&SESSION.:&DEBUG.::::
10	&APP_USER.	-	#
20	---	&APP_USER.	separator
30	Sign Off	&APP_USER.	&LOGOUT_URL.
40	Site Map	-	f?p=&APP_ID.:22:&SESSION.:&DEBUG.::::

Grid Edit

Edit List

Create Entry >

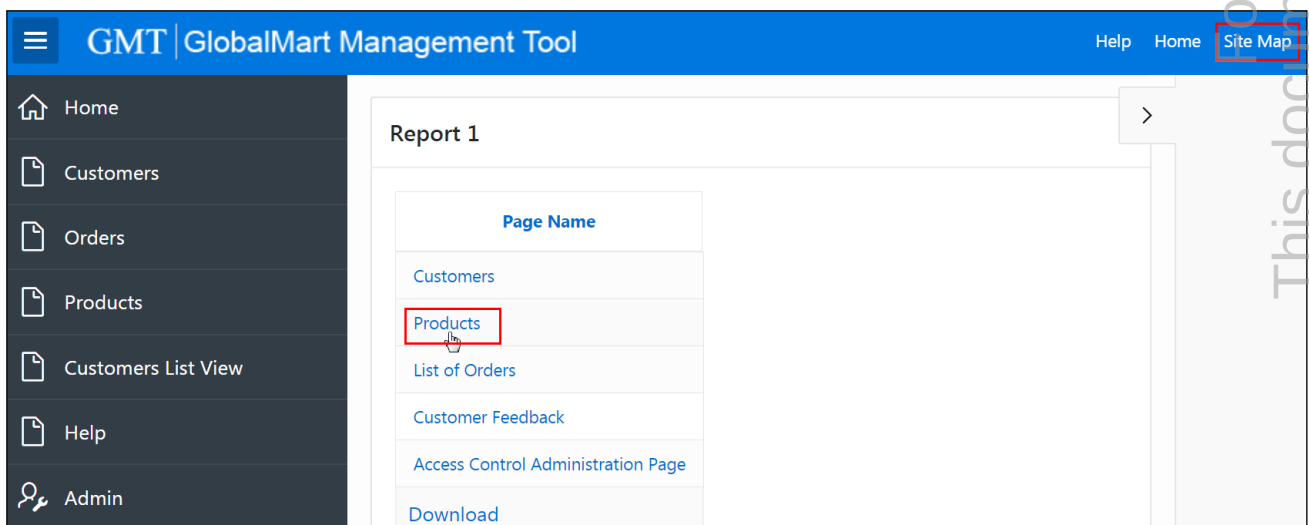
- g. Change the **Sequence** of **Site Map** to 8 and click **Save**.

List: Desktop Navigation Bar				
Search: All Text Columns Go Actions Edit Save				
☐		Sequence	Text	Target
☐		5	Help	f?p=&APP_ID.:20:&SESSION.:8
☐		7	Home	f?p=&APP_ID.:1:&SESSION.:&
☐		10	&APP_USER.	#
☐		20	---	separator
☐		30	Sign Off	&LOGOUT_URL.
☒		8	Site Map	f?p=&APP_ID.:22:&SESSION.:

5. Run the page and click one of the page names to view the corresponding page.
- a. Click the **Run Page 22** icon.



- b. Click one of the page names to view the corresponding page.



GMT GlobalMart Management Tool
Help Home Site Map

Home
Customers
Orders
Products
Customers List View
Help
Admin

Products List

Product Id ↑	Product Name	Category Id	Product Status	List Price
1726	LCD Monitor 11/PM	11	under development	259
1729	Chemicals - RCP	39	orderable	80
1733	PS 220V /UK	19	orderable	89
1734	Cable RS232 10/AM	19	orderable	6
1737	Cable SCSI 10/FW/ADS	19	orderable	8

- c. Click the **Application 111** link on the Developer toolbar.

Home
Application 111
Edit Page 7
Session
View Debug
Debug
Page Info
Quick Edit
Theme Roller

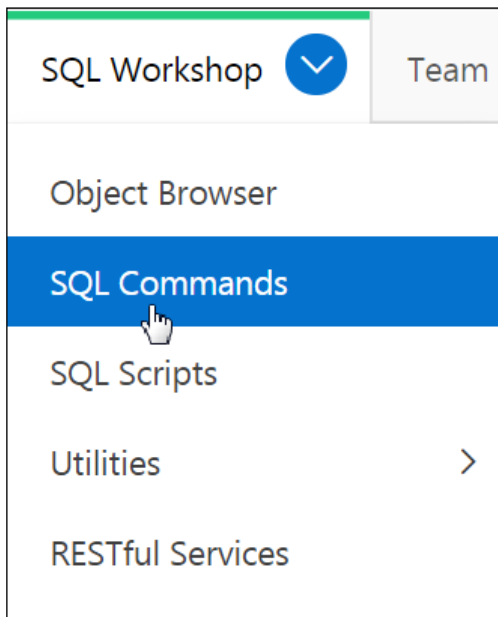
Solution 17-4: Enforcing Authorization in Site Map

Overview

In this practice solution, steps are provided for enforcing authorization in Site Map.

Steps

1. Navigate to **SQL Workshop > SQL Commands**. Copy and paste the code provided in step b (also located in the `/home/oracle/labs/labs/lab_17_04_01.txt` file) to create the `authorization_check` function.
 - a. Click the **SQL Workshop** pull-down menu and select **SQL Commands**.



- b. Paste the following code (located in the `/home/oracle/labs/labs/lab_17_04_01.txt` file) to create the `authorization_check` function. Click **Run**.

```
create or replace function authorization_check(  
p_scheme in varchar2)  
return varchar2  
is  
begin  
if apex_util.public_check_authorization(p_scheme) then  
return 'true';  
else  
return 'false';  
end if;  
end;
```

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↑ SQL Commands

Rows 10 ? Clear Command Find Tables

```
create or replace function authorization_check(
p_scheme in varchar2)
return varchar2
is
begin
if apex_util.public_check_authorization(p_scheme) then
return 'true';
else
return 'false';
end if;
end;
```

Schema OEHR ?

Save Run

- c. The function is created successfully.

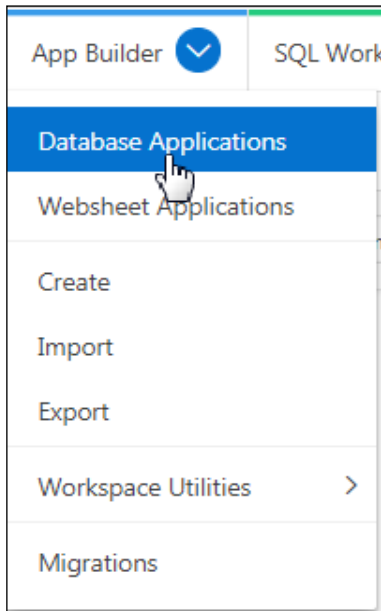
↑ SQL Commands

Rows 10 ? Clear Command

```
create or replace function authorization_check(
p_scheme in varchar2)
return varchar2
is
begin
if apex_util.public_check_authorization(p_scheme) then
return 'true';
else
```

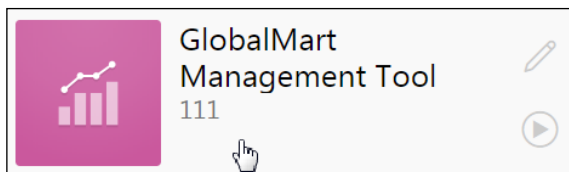
Results	Explain	Describe	Saved SQL	History
Function created.				
0.09 seconds				

- d. Click the **Application Builder** pull-down menu and select **Database Applications**.

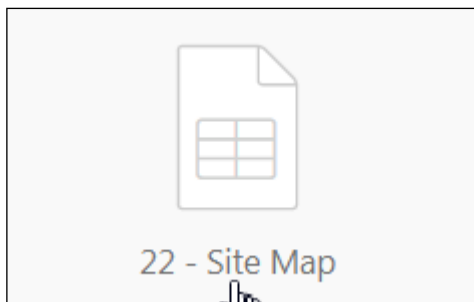


2. For Site Map, change the Source for your Classic Report to the code in step d (also located in the `/home/oracle/labs/labs/lab_17_04_02.txt` file) so that only the pages that are authorized will be shown based on the user logged in.

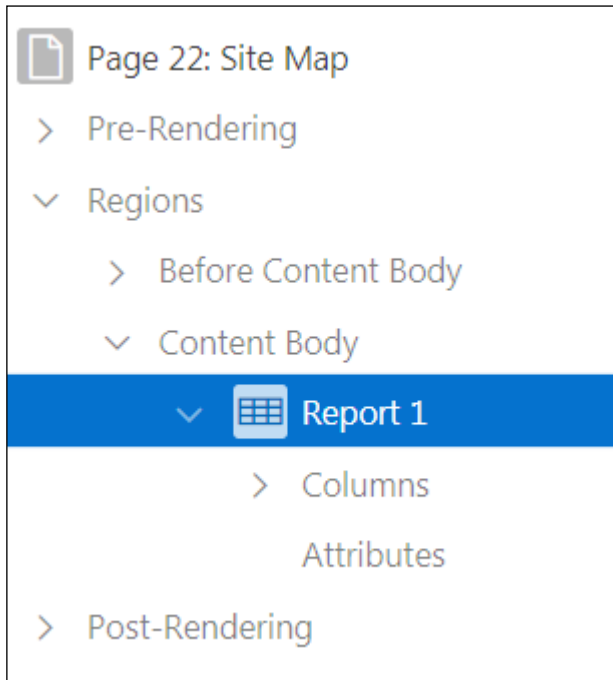
- a. Click the **GlobalMart Management Tool** application icon.



- b. Select the **Site Map** page.



- c. Under **Regions**, select **Report 1**.



- d. Paste the following code (located in the /home/oracle/labs/labs/lab_17_04_02.txt file) in the **Source > SQL Query** text box and click **Save**.

```
select distinct PAGE_ID, PAGE_NAME
from APEX_APPLICATION_PAGES p, apex_application_authorization a
where p.APPLICATION_ID = :APP_ID
and a.application_id = p.application_id
and (p.authorization_scheme is null or
(a.authorization_scheme_id = p.authorization_scheme_id and
authorization_check(a.authorization_scheme_name) = 'true'))
and PAGE_GROUP = 'Site Map'
```

Source

Location

Local Database

Type

SQL Query

SQL Query

```
(a.authorization_scheme_id = p.authorization_scheme_id and  
authorization_check(a.authorization_scheme_name) = 'true'))  
and PAGE_GROUP = 'Site Map'
```

Page Items to Submit

Optimizer Hint

+ v

Save

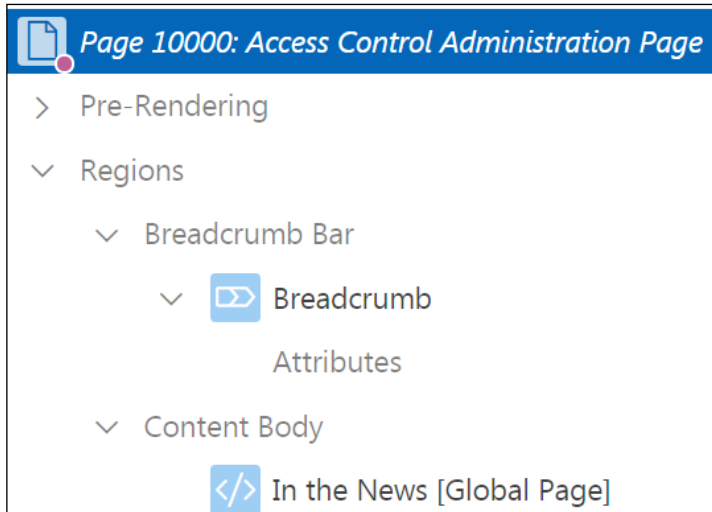
Region

(Alt+F7)

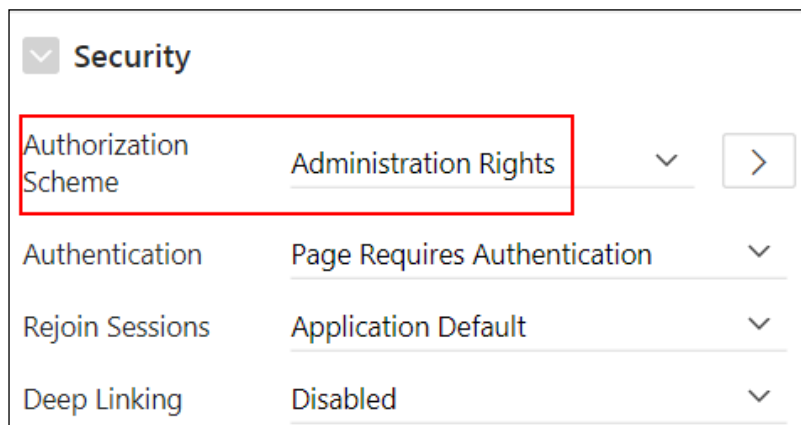
Click **111 Application** Breadcrumb.

[Application 111](#) \ Page Designer

3. You do not want end users or developers to see the Access Control Administration page. Make sure that the **Authorization Schemes** for this page is set to **access control - Administrator**. Run the page to see if only an authorized user can view it.
- a. Select the **Access Control Administration** page and click to open in Page Designer view.



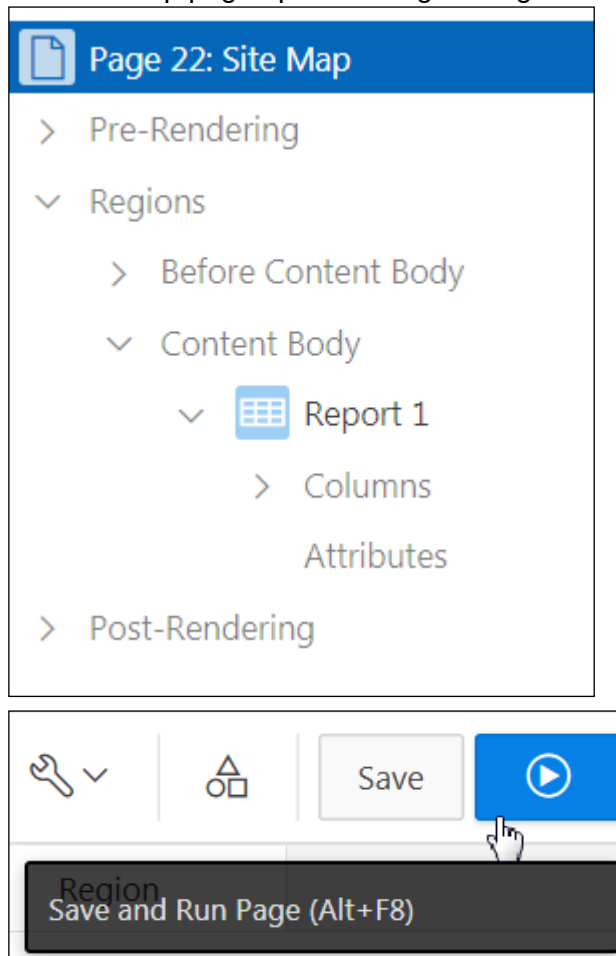
- b. Under **Security > Authorization Scheme**, ensure that **Administration Rights** is selected. Note that Administration Rights is selected by default.



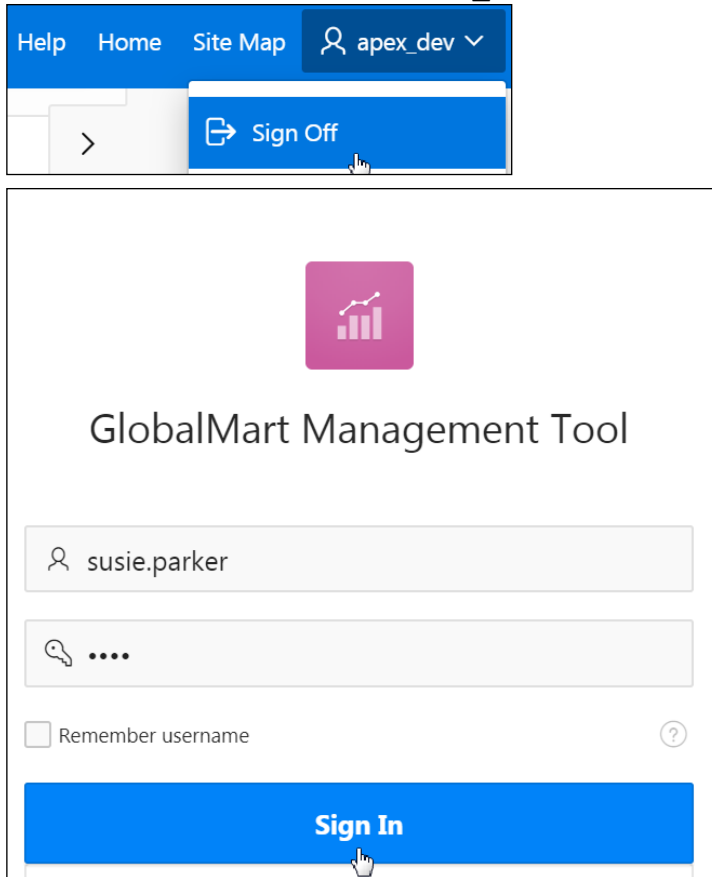
- c. Enter the page number of the Site Map Page and click **Go**.



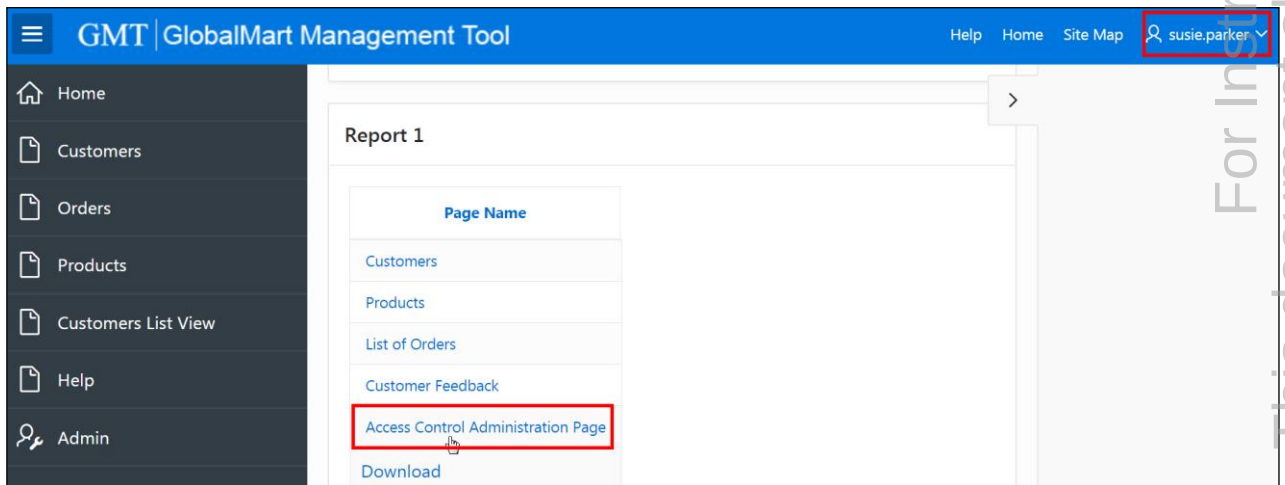
- d. The Site Map page opens in Page Designer view. Click **Save and Run Page** icon.



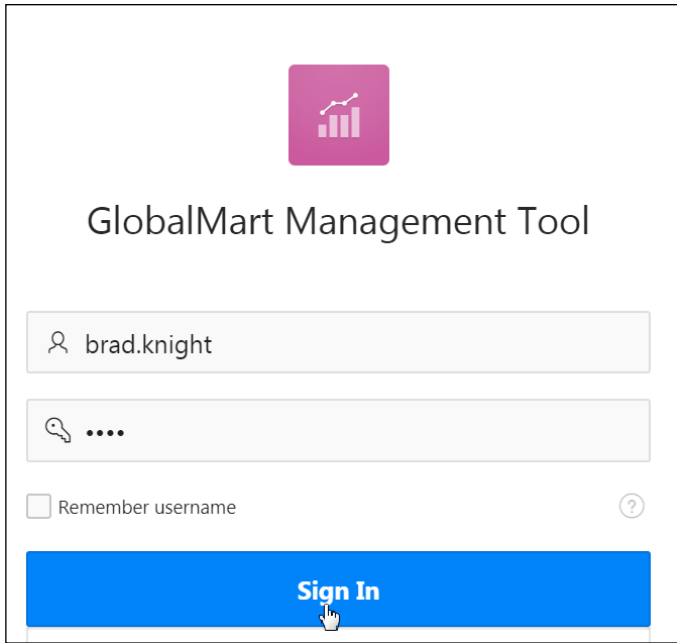
- e. You are currently logged in as apex_dev. Sign off and log in as Susie Parker.



- f. Notice that when you are currently logged in as susie.parker due to which you currently see the **Access Control Administration Page** in the site map list. Click **Sign Off**.

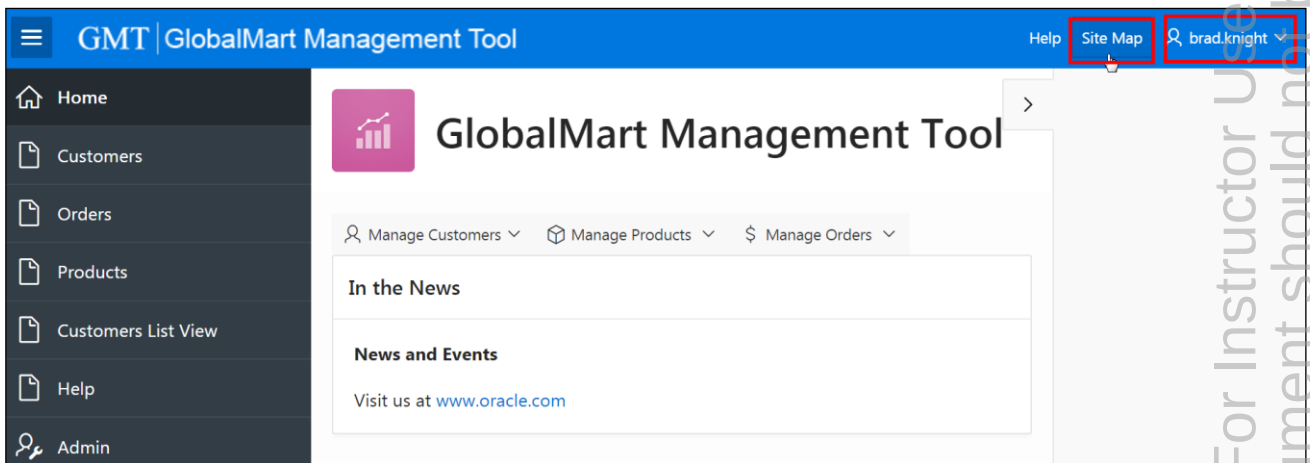


- g. To check the Site Map as a developer, sign in as `brad.knight`.



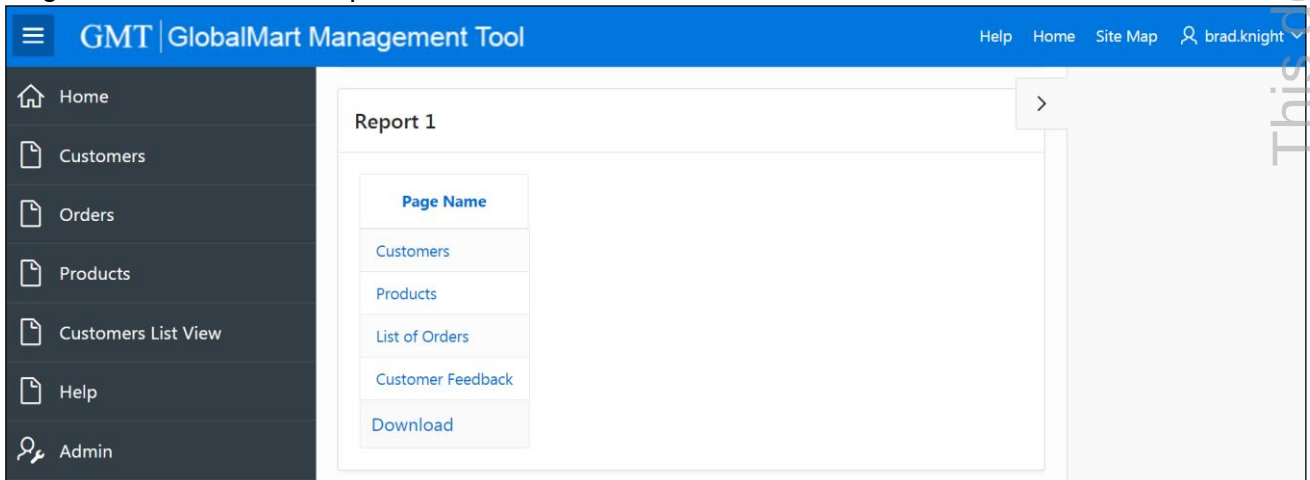
The login screen for the GlobalMart Management Tool. It features a purple square icon with a white bar chart. Below the icon is the title "GlobalMart Management Tool". There are two input fields: the first contains the username "brad.knight" and the second contains a masked password "....". Below these fields is a checkbox labeled "Remember username" and a question mark icon. At the bottom is a large blue button labeled "Sign In". A mouse cursor is pointing at the "Sign In" button.

- h. In the navigation bar, click **Site Map**.



The dashboard of the GlobalMart Management Tool. The top navigation bar is blue and contains the "GMT | GlobalMart Management Tool" logo, a "Help" link, a "Site Map" link (highlighted with a red box), and a user profile dropdown for "brad.knight" (also highlighted with a red box). The left sidebar is dark gray and contains links for Home, Customers, Orders, Products, Customers List View, Help, and Admin. The main content area has a white background and contains the "GlobalMart Management Tool" title, a search bar with filters for "Manage Customers", "Manage Products", and "Manage Orders", and a section titled "In the News" with a link to "Visit us at www.oracle.com".

Brad Knight does not have access to the Access Control Administration page, so you no longer see it in the Site Map list.



The Site Map page in the GlobalMart Management Tool. The top navigation bar is blue and contains the "GMT | GlobalMart Management Tool" logo, a "Help" link, a "Home" link, a "Site Map" link, and a user profile dropdown for "brad.knight". The left sidebar is dark gray and contains links for Home, Customers, Orders, Products, Customers List View, Help, and Admin. The main content area has a white background and contains a section titled "Report 1" with a table of links. The table has a header "Page Name" and the following rows: "Customers", "Products", "List of Orders", "Customer Feedback", and "Download".

- i. Click the **Application 111** link on the Developer toolbar.



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**Practices for Lesson 18:
Extending Your Application**

Practices for Lesson 18: Overview

Practices Overview

In these practices, you add advanced features to your application by creating Data Load Wizard pages and upload and download pages.

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Practice 18-1: Creating Data Load Wizard Pages

Overview

In this practice, you create a series of wizard pages to upload data into the `PRODUCT_INFORMATION` table.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. You create a series of wizard pages to upload data into the `OEHR_PRODUCT_INFORMATION` table.
2. Test the upload by using the following data (located in the `/home/oracle/labs/labs/lab_18_01.txt` file). Use the Copy and Paste Delimited Data area.

PRODUCT_ID	PRODUCT_NAME	PRODUCT_DESCRIPTION	CATEGORY_ID
WEIGHT_CLASS	WARRANTY_PERIOD	SUPPLIER_ID	
PRODUCT_STATUS	LIST_PRICE	MIN_PRICE	
3194	LED Monitor 15/PM	Light Emitting Diode 15 inch passive monitor. The virtually-flat, high-resolution screen delivers outstanding image quality with reduced glare.	11 3 +03-00
102067	under development	259 208	
3195	LCD Monitor 20/PM	Liquid Cristal Display 20 inch passive monitor. Enjoy the productivity that a small monitor can bring via more workspace on your desk. Easy setup with plug-and-play compatibility	11 3 +03-00 102061 orderable 249
206			
3196	Plastic Stock - B/HD	Plastic Stock - Black, high density.	39
1	+03-00	103095 orderable 3 2	

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Log in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 18-1**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Log in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. On the Application's home page click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 17- 4**, perform the following steps:

- a. Log in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_17_04.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Practice 18-2: Adding an Upload and Download Page

Overview

In this practice, you create a form in an HTML region with a file upload item and a button. The button will submit the page and return the user to the same page. Also, you create a report on the document table that has links to download documents. Provide links to download the documents in the report.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a blank page called **Upload Download Files** with an HTML region called **Submit File** that contains a File Browse item type on the source type `APEX_APPLICATION_TEMP_FILES`. Create a **Submit** button and upload the `OracleLogo.png` file from the `/home/oracle/labs/labs` directory.

Note that every time you upload a script in SQL Workshop or upload a file in Shared Components for an application, the reference to the file is placed in the `APEX_APPLICATION_TEMP_FILES` table.

2. Create an Interactive Report subregion called **Uploaded Files** with the following SQL query (located in `/home/oracle/labs/labs/lab_18_02.txt`):

```
SELECT id, filename FROM APEX_APPLICATION_TEMP_FILES order by
created_on desc
```

3. Modify the report to create a link on the ID column to download the file.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Log in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 18-2**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Log in to the Application Express workspace as the `apex_dev` user.

- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. On the Application's home page click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the `apex_admin` user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 18-1**, perform the following steps:

- a. Log in to the Application Express workspace as the `apex` user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_18_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).

- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

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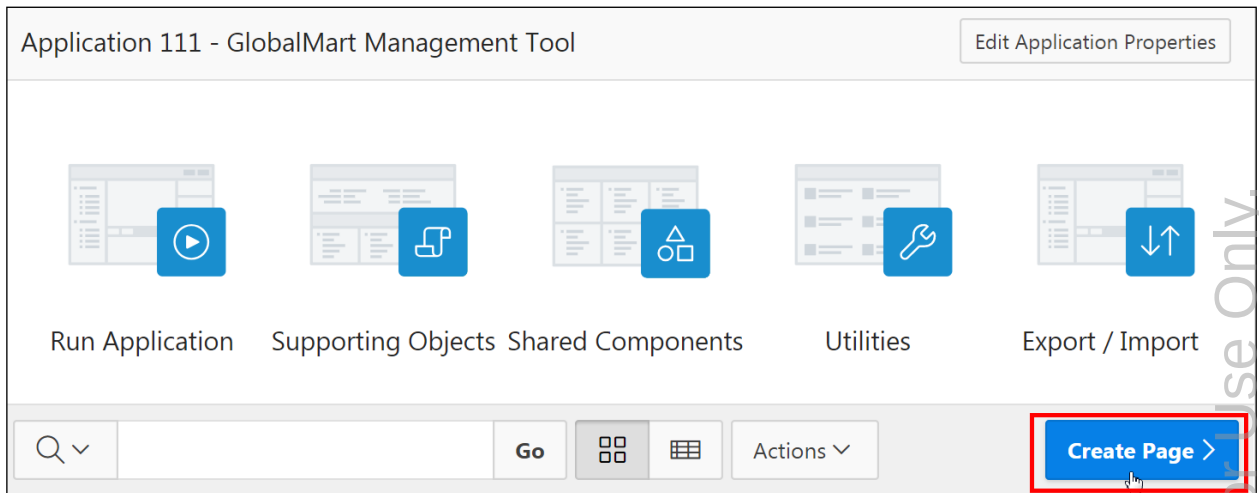
Solution 18-1: Creating Data Load Wizard Pages

Overview

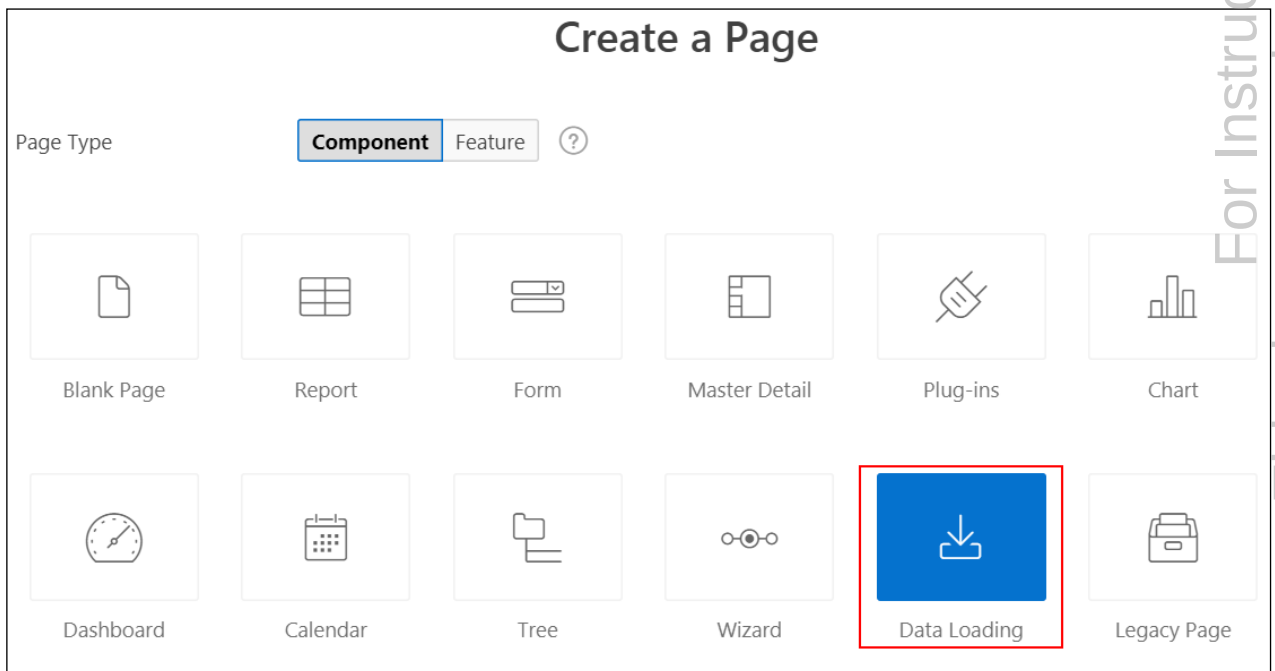
In this practice, you create a series of wizard pages to upload data into the `OEHR_PRODUCT_INFORMATION` table.

Steps

1. You create a series of wizard pages to upload data into the `PRODUCT_INFORMATION` table.
 - a. In the GlobalMart Management Tool application home page, click **Create Page >**.



- b. Select the **Data Loading** page type.



c. Enter the following and click **Next >**.

- **Data Load Definition:** Select **Create New**
- **Definition Name:** Enter `Load Products`
- **Owner:** `OEHR`
- **Table Name:** Select `OEHR_PRODUCT_INFORMATION (table)`
- **Unique Column > Column 1:** Select `PRODUCT_ID (Number)`

Create Data Load Wizard

Data Load Table

Data Loading pages reference a data load definition. To create a new data load definition, specify a name and select a schema owner, target table, and up to three columns that uniquely identify a row.

Data Load Definition: Create New ?

* Definition Name: Load Products ?

* Owner: OEHR ?

* Table Name: OEHR_PRODUCT_INFORMATION (table) ?

Unique Column ? Case Sensitive ?

* Column 1: PRODUCT_ID (Number) No

Column 2: - Select Column - No

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- d. You want to create a rule that will change the Product Name to uppercase. Select **PRODUCT_NAME (Varchar2)** and click the right arrow (>) to select it to bring to the right column. Enter **Product Name in Uppercase** for **Rule Name**. Select **To Upper Case** for **Type**. Click **Add Transformation**.

Create Data Load Wizard

Transformation Rules

Transformation Rules allow you to change the data being uploaded before it is inserted into the base table. You select the column to transform and then the desired rule to apply to it.

Select Column(s) to create a transformation rule

Column	Type
PRODUCT_ID	Number
PRODUCT_DESCRIPTION	Varchar2
CATEGORY_ID	Number
WEIGHT_CLASS	Number
WARRANTY_PERIOD	Interval Year(2)
SUPPLIER_ID	Number
PRODUCT_STATUS	Varchar2
LIST_PRICE	Number
MIN_PRICE	Number
CATALOG_URL	Varchar2

PRODUCT_NAME (Varchar2)

* Rule Name: Product Name in Uppercase

* Sequence: 10

* Type: To Upper Case

< Cancel Add Transformation Next >

Create Data Load Wizard

✓

✓

●

●

●

●

Table Lookups

Table Lookups allow you to match an uploaded value against another table and use the associated key value, instead of the uploaded value.

Add new table lookup for Column

- Select Column -

?

<

Cancel

Next >

- g. Set the page numbers to 23, 24, 25, and 26 (as shown in the screenshot) and click **Next** >.

Create Data Load Wizard

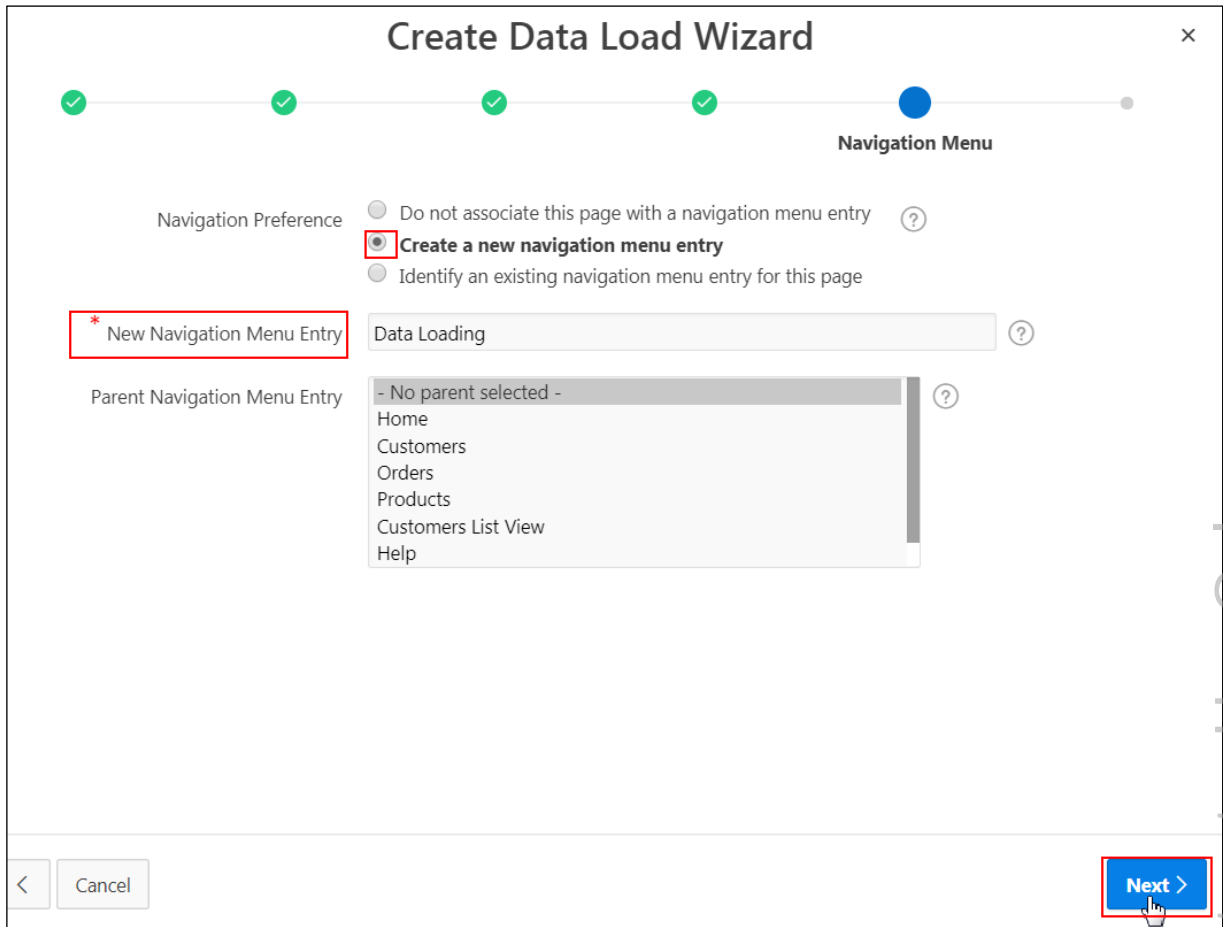
Pages Attributes

This page provides a short descriptive name for each page of the data load wizard to be created. Please provide the page name, number, and region names to use in the data load wizard.

	Page Name	Page Number
* Step 1	Data Load Source ?	23
* Step 2	Data / Table Mapping ?	24
* Step 3	Data Validation ?	25
* Step 4	Data Load Results ?	26
* Page Mode	Normal Modal Dialog ?	
Page Group	- Select Page Group - ?	
Breadcrumb	- do not add breadcrumb region to page - ?	

< Cancel Next >

- h. You want to create a new navigation menu entry. Select **Create a new navigation menu entry** and ensure that **Data Loading** is selected for the New Navigation Menu Entry. Click **Next >**.



The image shows a 'Create Data Load Wizard' dialog box with a progress bar at the top. The progress bar has five steps: the first four are marked with green checkmarks, and the fifth is marked with a blue circle and labeled 'Navigation Menu'. The main content area is divided into two sections. The 'Navigation Preference' section has three radio buttons: 'Do not associate this page with a navigation menu entry' (unselected), 'Create a new navigation menu entry' (selected and highlighted with a red box), and 'Identify an existing navigation menu entry for this page' (unselected). Below this, the 'New Navigation Menu Entry' section (highlighted with a red box) contains a text field with 'Data Loading' and a dropdown menu. The dropdown menu is open, showing options: '- No parent selected -', 'Home', 'Customers', 'Orders', 'Products', 'Customers List View', and 'Help'. The 'Parent Navigation Menu Entry' section is empty. At the bottom, there are three buttons: a back arrow, a 'Cancel' button, and a 'Next >' button (highlighted with a red box and a mouse cursor).

- i. You need to specify which page to go to when you click Cancel and Finish. Enter 1 (Home Page) for both the **Cancel Button Branch to Page** and **Finish button Branch to Page**. You are now ready to create the Data Load Wizard pages. Click **Create**.

Create Data Load Wizard

Buttons and Branching

* Next Button Label: Next

* Previous Button Label: Previous

* Cancel Button Label: Cancel

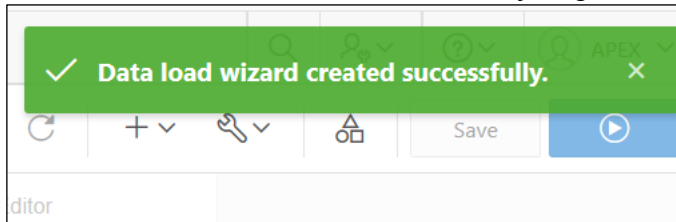
* Cancel Button Branch to Page: 1

* Finish Button Label: Finish

* Finish Button Branch to Page: 1

Cancel Create

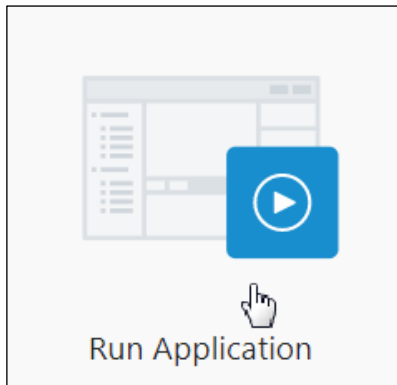
- j. Once the Data Load Wizard is created, you get the following message:



- k. Because Data Load Wizard Pages cannot be run directly, navigate to the GlobalMart Management Tool application home page and then run the application. Click **Application 111** Breadcrumb.



- I. Click **Run Application**.



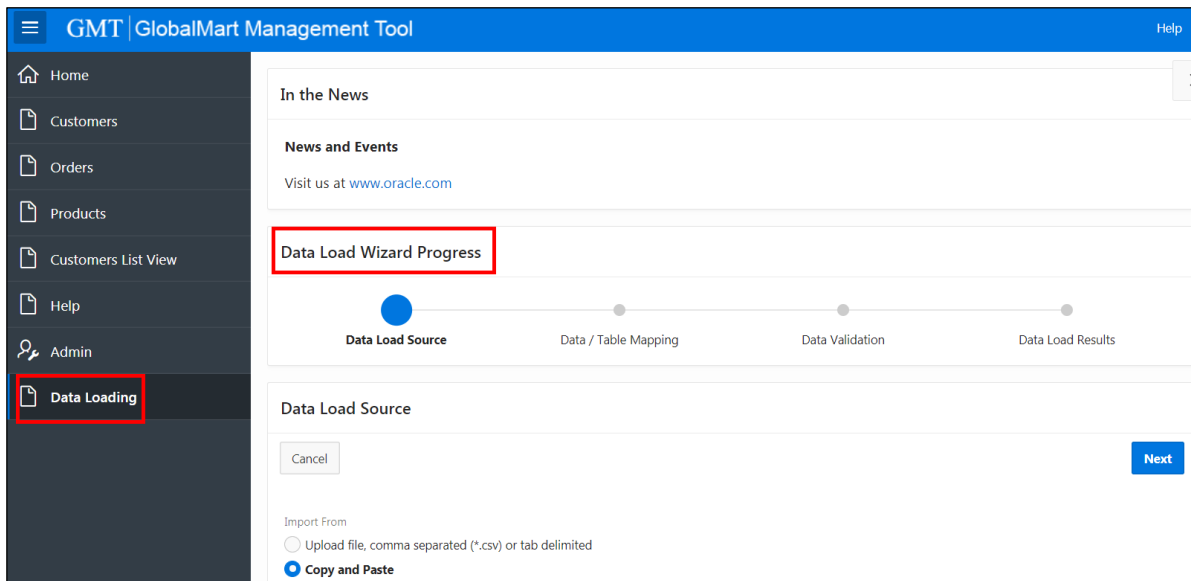
2. Test the upload by using the following data (located in the /home/oracle/labs/labs/lab_18_01.txt file). Use the Copy and Paste Delimited Data area.

PRODUCT_ID	PRODUCT_NAME	PRODUCT_DESCRIPTION	CATEGORY_ID
WEIGHT_CLASS	WARRANTY_PERIOD	SUPPLIER_ID	
PRODUCT_STATUS	LIST_PRICE	MIN_PRICE	
3194	LED Monitor 15/PM	Light Emitting Diode 15 inch passive monitor. The virtually-flat, high-resolution screen delivers outstanding image quality with reduced glare.	11 3 +03-00
102067	under development	259 208	
3195	LCD Monitor 20/PM	Liquid Cristal Display 20 inch passive monitor. Enjoy the productivity that a small monitor can bring via more workspace on your desk. Easy setup with plug-and-play compatibility	11 3 +03-00 102061 orderable 249
206			
3196	Plastic Stock - B/HD	Plastic Stock - Black, high density.	39
1	+03-00 103095	orderable 3 2	

- a. On the Sign in page, enter the apex_dev user credentials and click **Sign In**.

A sign-in page for the GlobalMart Management Tool. It features a purple icon with a bar chart and a line graph. Below the icon is the title "GlobalMart Management Tool". There are two input fields: the first contains the username "apex_dev" and the second contains masked characters "....". Below the input fields is a checkbox labeled "Remember username" and a help icon. At the bottom is a large blue button labeled "Sign In" with a hand cursor pointing at it.

- b. Click **Data Load** in the navigation menu of the GlobalMart Management Tool application home page.



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- c. The first wizard page is where you specify the Data Load Source. You want to copy and paste the following text located in the `/home/oracle/labs/labs/lab_18_01.txt` file into the Copy and Paste Delimited Data area. Ensure that the Separator includes `'\t'` (tab space) and ensure that the check box for **First Row has Column Names** is selected. Then click **Next >**.

Data Load Source

Cancel **Next**

Import From

☐ Upload file, comma separated (*.csv) or tab delimited

☒ **Copy and Paste**

Copy and Paste Delimited Data

PRODUCT_ID	PRODUCT_NAME	PRODUCT_DESCRIPTION	CATEGORY_ID	WEIGHT_CLASS
WARRANTY_PERIOD	SUPPLIER_ID	PRODUCT_STATUS	LIST_PRICE	MIN_PRICE
3194	LED Monitor 15/PM	Light Emitting Diode 15 inch passive monitor. The virtually-flat, high-resolution screen delivers outstanding image quality with reduced glare.	11	3
102067	under development	259	208	+03-00
3195	LCD Monitor 20/PM	Liquid Cristal Display 20 inch passive monitor. Enjoy the productivity that a small monitor can bring via more workspace on your desk. Easy setup with plug-and-play compatibility	11	3
3196	Plastic Stock - B/HD	Plastic Stock - Black, high density.	39	1
orderable	3	2	+03-00	103095

Separator
\t

Optionally Enclosed By
"

☒ **First Row has Column Names** ?

☐ Use Advanced Settings ?

- d. Data/Table Mapping is displayed. Make sure that the Column Names for all the columns are selected. Otherwise, select the corresponding column names from their respective pull-down menus and click **Next**.

Data Load Wizard Progress

Progress bar: Data Load Source (✓) → **Data / Table Mapping** (●) → Data Validation (●) → Data Load Results (●)

Data / Table Mapping

Previous Cancel **Next**

Target Column	PRODUCT_ID - number *	PRODUCT_NAME - varchar2(50)	PRODUCT_DESCRIPTION - varchar2(255)	CATEGORY - varchar2(25)
Source Column	PRODUCT_ID	PRODUCT_NAME	PRODUCT_DESCRIPTION	CATEGORY
Row 1	3194	LED Monitor 15/PM	Light Emitting Diode 15 inch passive matrix, flat, high-resolution screen delivers outstanding picture quality	Computer Monitors
Row 2	3195	LCD Monitor 20/PM	Liquid Cristal Display 20 inch passive matrix, and-play compatibility	Computer Monitors

- e. The Data Validation page is displayed. This page displays the data that will be inserted and/or updated into the database. Notice that the `PRODUCT_NAME` was transformed to all uppercase. Click **Load Data**.

Data Load Wizard Progress

✓ Data Load Source
✓ Data / Table Mapping
Data Validation
• Data Load Results

Data Validation

Sequence	Action	PRODUCT_ID	PRODUCT_NAME	PRODUCT_DESCRIPTION	CATEGORY_ID	WEIGHT_CLASS	WARRANTY_PERIOD	SUPPLIER_ID	PRODUCT
1	Update row	3194	LED MONITOR 15/PM	Light Emitting Diode 15 inch passive monitor. The virtually-flat, high-resolution screen delivers outstanding image quality with reduced glare.	11	3	+03-00	102067	under a
2	Update row	3195	LCD MONITOR 20/PM	Liquid Cristal Display 20 inch passive monitor. Enjoy the productivity that a small monitor can bring via more workspace on your desk. Easy setup with plug-and-play compatibility.	11	3	+03-00	102061	orderabl

- f. Notice that three rows are inserted. Click **Finish**.

Data Load Wizard Progress

✓ Data Load Source
✓ Data / Table Mapping
✓ Data Validation
Data Load Results

Data Load Results

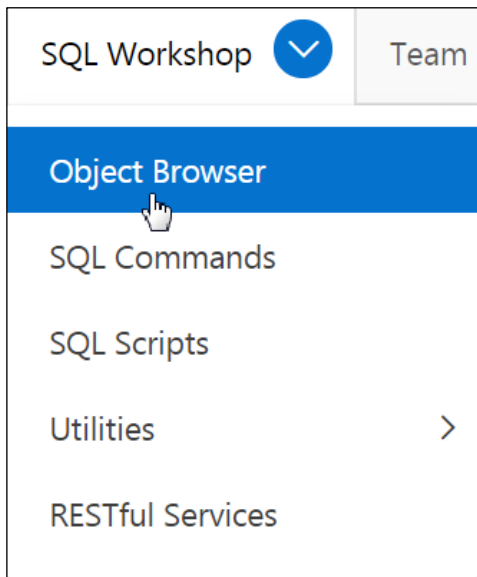
Inserted Row(s):
 3

Updated Row(s):
 0

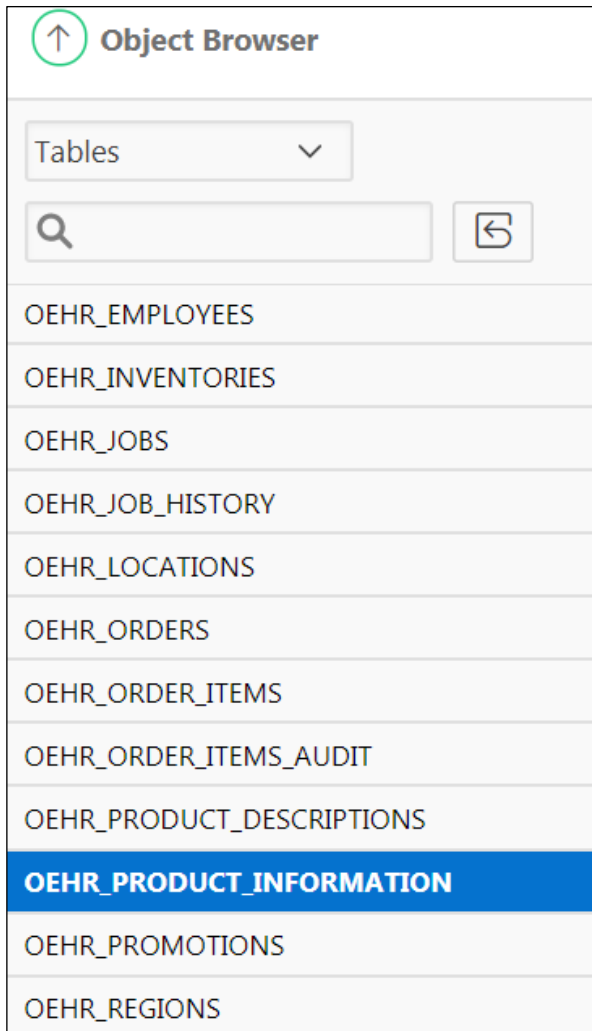
Failed Row(s):
 0

To be Reviewed Row(s):

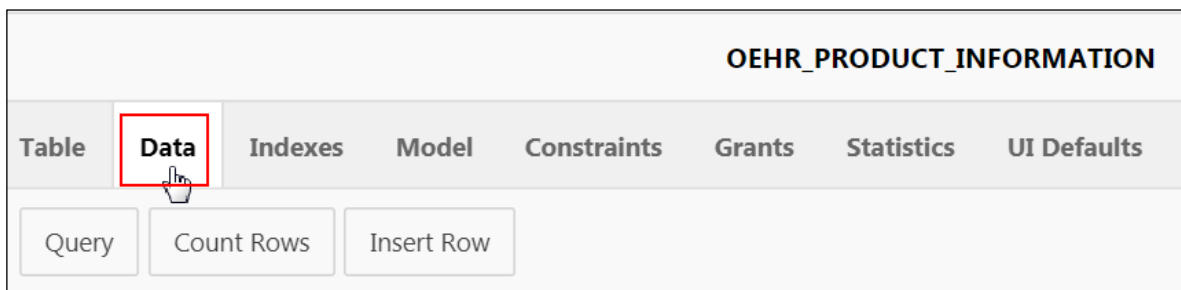
- g. Switch to the Page Designer window in your browser. Click **SQL Workshop** and select **Object Browser**.



- h. In Object Browser, select the `OEHR_PRODUCT_INFORMATION` table.



- i. The `OEHR_PRODUCT_INFORMATION` table definition is displayed. Click the **Data** tab.



- j. Click the **Query** button

OEHR_PRODUCT_INFORMATION

Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults
Query Count Rows Insert Row							

- k. Select all columns and enter 3194 in the **Column Condition** field for `PRODUCT_ID`.

OEHR_PRODUCT_INFORMATION

Schema: **OEHR** ?

Table/View Name: **OEHR_PRODUCT_INFORMATION** ?

Select the columns you wish to view. To restrict specific rows, enter a condition in **Column Condition**. Use "%" for wildcards.

		Column	Type	Column Condition	
☒					
☒	1.	PRODUCT_ID	?89	3194	Primary key column.
☒	2.	PRODUCT_NAME	A		None
☒	3.	PRODUCT_DESCRIPTION	A		Primary language description corresponding to translated_description in oehr_product_descriptions, added to provide non-NLS text columns for OC views to access.
☒	4.	CATEGORY_ID	?89		Low cardinality column, can be used for bitmap index. Schema SH uses it as foreign key

- l. Scroll down and click **Query**.

☒ Ascending ☐ Descending	☒ Ascending ☐ Descending	☒ Ascending ☐ Descending
Query		

- m. The newly inserted row with `PRODUCT_ID` 3194 is displayed. Click **< New Query**.

OEHR_PRODUCT_INFORMATION						
Edit	Product Id	Product Name	Product Description	Category Id	Weight Class	Warranty Period
	3194	LED MONITOR 15/PM	Light Emitting Diode 15 inch passive monitor. The virtually-flat, high-resolution screen delivers outstanding image quality with reduced glare.	11	3	+000000003-00
Download						
1 - 1						
< New Query Cancel						

- n. Make sure that all the columns are selected and enter 3195 in the **Column Condition** field for `PRODUCT_ID`.

OEHR_PRODUCT_INFORMATION					
Schema: OEHR ?					
Table/View Name: OEHR_PRODUCT_INFORMATION ?					
Select the columns you wish to view. To restrict specific rows, enter a condition in Column Condition . Use "%" for wildcards.					
☒		Column	Type	Column Condition	
☒	1.	PRODUCT_ID	789	3195	Primary key column.
☒	2.	PRODUCT_NAME	A		None
☒	3.	PRODUCT_DESCRIPTION	A		Primary language description corresponding to translated_description in oeher_product_descriptions, added to provide non-NLS text columns for OC views to access.
☒	4.	CATEGORY_ID	789		Low cardinality column, can be used for bitmap index. Schema SH uses it as foreign key

- o. Scroll down and click **Query**.

☒ Ascending ☐ Descending	☒ Ascending ☐ Descending	☒ Ascending ☐ Descending
Query		

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- p. The newly inserted row with `PRODUCT_ID` 3195 is displayed. Click < **New Query**.

OEHR_PRODUCT_INFORMATION						
Edit	Product Id	Product Name	Product Description	Category Id	Weight Class	Warranty Period
	3195	LCD MONITOR 20/PM	Liquid Cristal Display 20 inch passive monitor. Enjoy the productivity that a small monitor can bring via more workspace on your desk. Easy setup with plug-and-play compatibility	11	3	+000000003-00

Download

1 - 1

< **New Query** Cancel

- q. Select all columns and enter 3196 in the **Column Condition** field for `PRODUCT_ID`.

OEHR_PRODUCT_INFORMATION					
Schema: OEHR ?					
Table/View Name: OEHR_PRODUCT_INFORMATION ?					
Select the columns you wish to view. To restrict specific rows, enter a condition in Column Condition . Use "%" for wildcards.					
☒		Column	Type	Column Condition	
☒	1.	PRODUCT_ID	789	3196	Primary key column.
☒	2.	PRODUCT_NAME	A		None
☒	3.	PRODUCT_DESCRIPTION	A		Primary language description corresponding to translated_description in oehr_product_descriptions, added to provide non-NLS text columns for OC views to access.
☒	4.	CATEGORY_ID	789		Low cardinality column, can be used for bitmap index. Schema SH uses it as foreign key

- r. Scroll down and click **Query**.

☒ Ascending
☐ Descending

☒ Ascending
☐ Descending

☒ Ascending
☐ Descending

Query

- s. The newly inserted row with `PRODUCT_ID` 3196 is displayed.

OEHR_PRODUCT_INFORMATION

Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers
-------	------	---------	-------	-------------	--------	------------	-------------	----------

Query Result

Edit	Product Id	Product Name	Product Description	Category Id	Weight Class	Warranty Period
	3196	PLASTIC STOCK - B/HD	Plastic Stock - Black, high density.	39	1	+000000003-00

Download

1 - 1

< New Query

Cancel

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Solution 18-2: Adding an Upload and Download Page

Overview

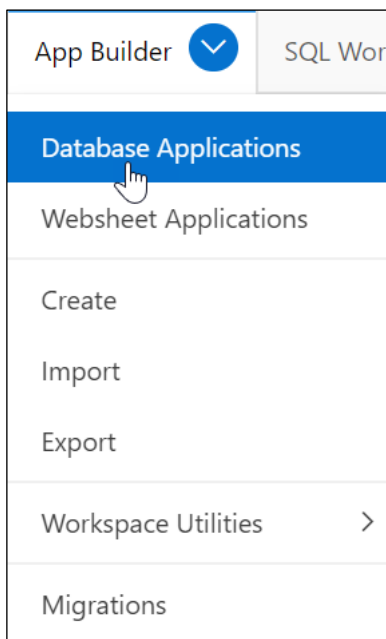
In this solution, you create a form in an HTML region with a File Upload item and a button. The button will submit the page and return the user to the same page. Also, you create a report on the document table that has links to download documents. Provide links to download the documents in the report.

Steps

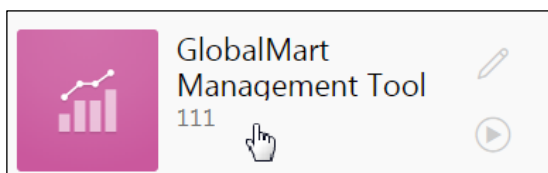
1. Create a blank page called **Upload Download Files** with an HTML region called Submit File that contains a File Browse item type on the source type `APEX_APPLICATION_TEMP_FILES`. Create a Submit button and upload the `OracleLogo.png` file from the `/home/oracle/labs/labs` directory.

Note that every time you upload a script in SQL Workshop or upload a file in Shared Components for an application, the reference to the file is placed in the `APEX_APPLICATION_TEMP_FILES` table.

- a. Click the **Application Builder** pull-down menu in the navigation bar and select **Database Applications**.

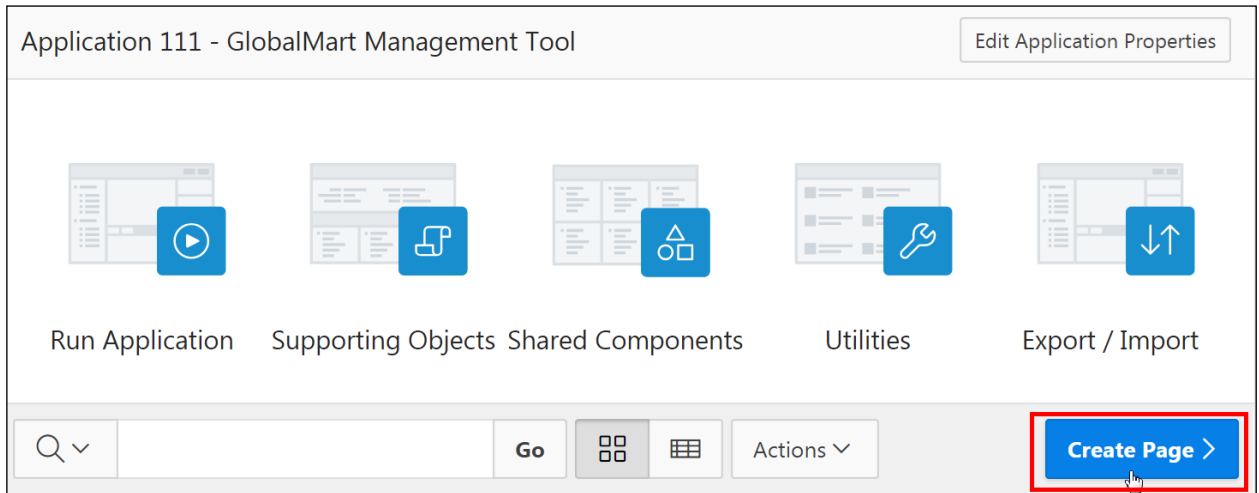


- b. Click the GlobalMart Management Tool application icon.

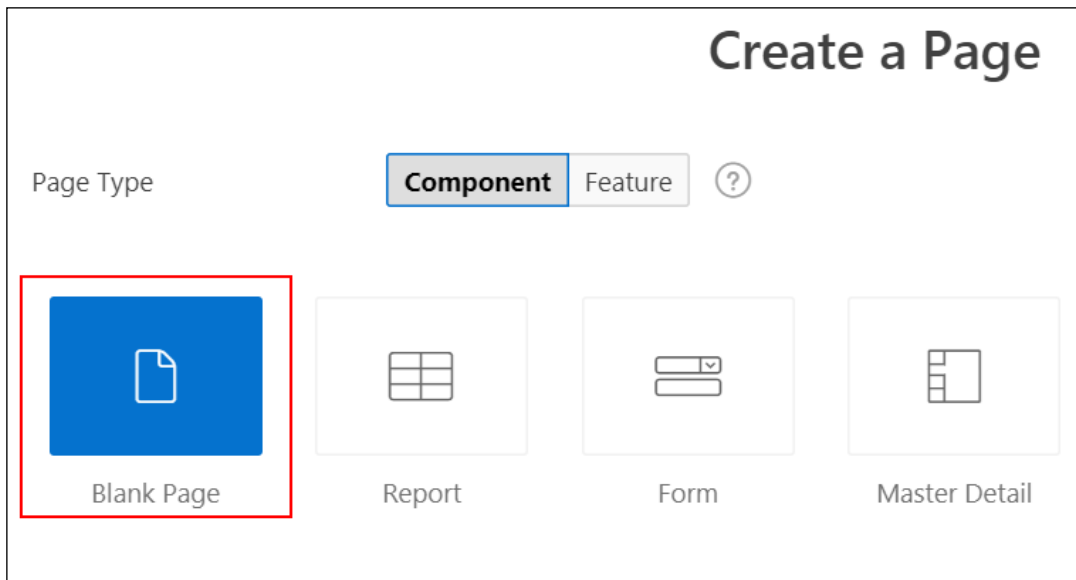


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- c. Click **Create Page >**.



- d. Select the **Blank Page** page type.



- e. Enter **Page Number** as 27. For **Name**, enter Upload Download Files and click **Next** >.

Create a Blank Page

Page Attributes

* Page Number ?

* Name ?

* Page Mode **Normal** Modal Dialog Non-Modal Dialog ?

Page Group - Select Page Group - ?

Breadcrumb - don't use breadcrumbs on page - ?

> Optional Static Content Regions

- f. Leave the defaults for the **Navigation Preference** and click **Next** >.

Create a Blank Page

Navigation Menu

Navigation Preference ☒ **Do not associate this page with a navigation menu entry** ?

☐ Create a new navigation menu entry

☐ Identify an existing navigation menu entry for this page

- g. Click **Finish**.

Create a Blank Page

✓

✓

Confirm

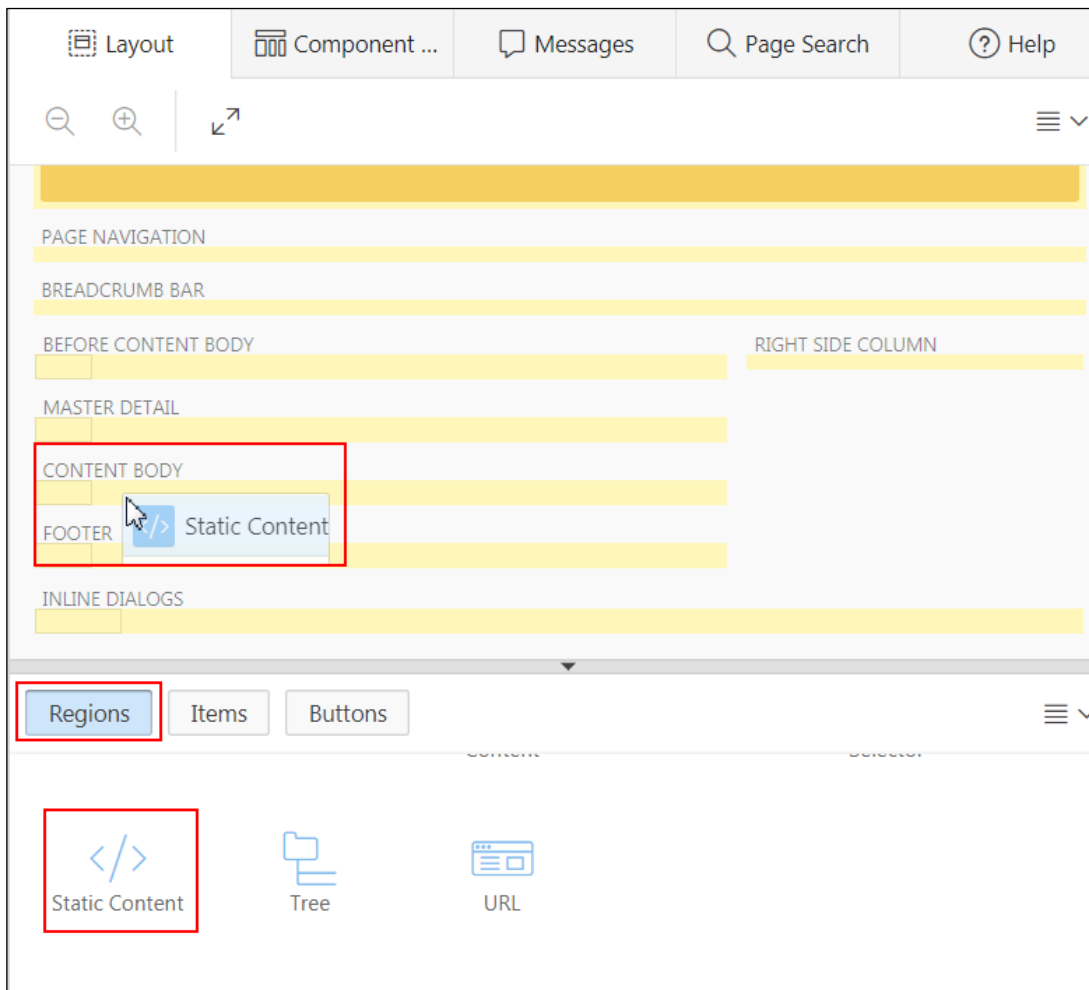
You have requested to create a page with the following attributes. Please confirm your selections.

Application	111
Page	27
Page Name	Upload Download Files
Page Title	Upload Download Files

< Cancel

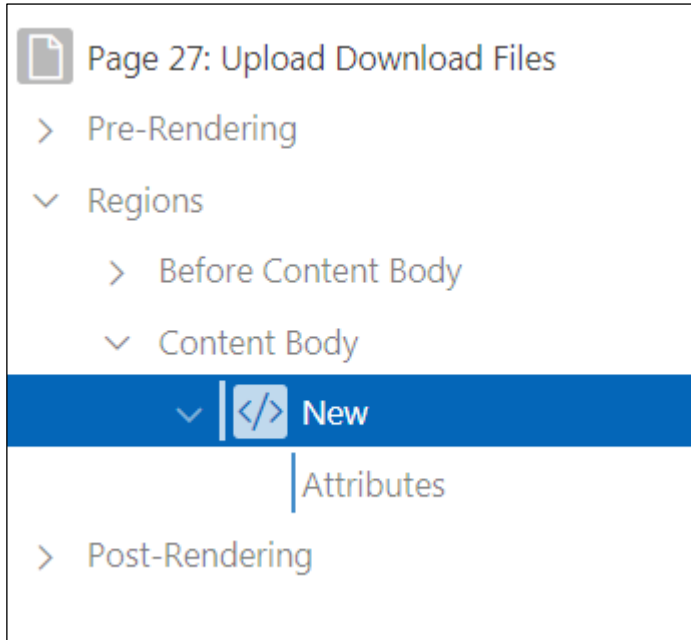
Finish

- h. The **Upload Download Files** page is created, and it opens in the Page Designer view. Drag **Static Content** `</>` region from **Regions gallery** to the **Content Body** of the Layout tab.

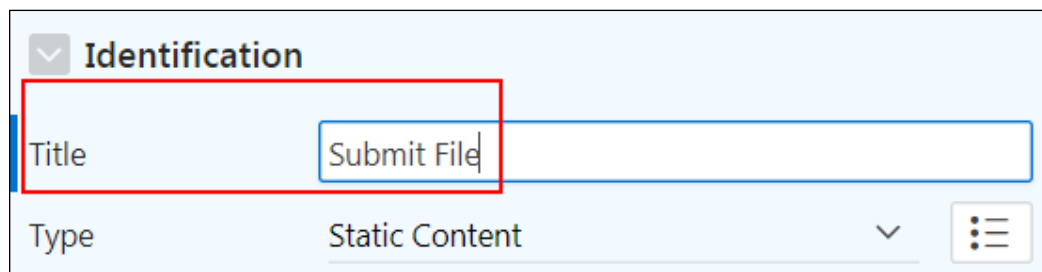


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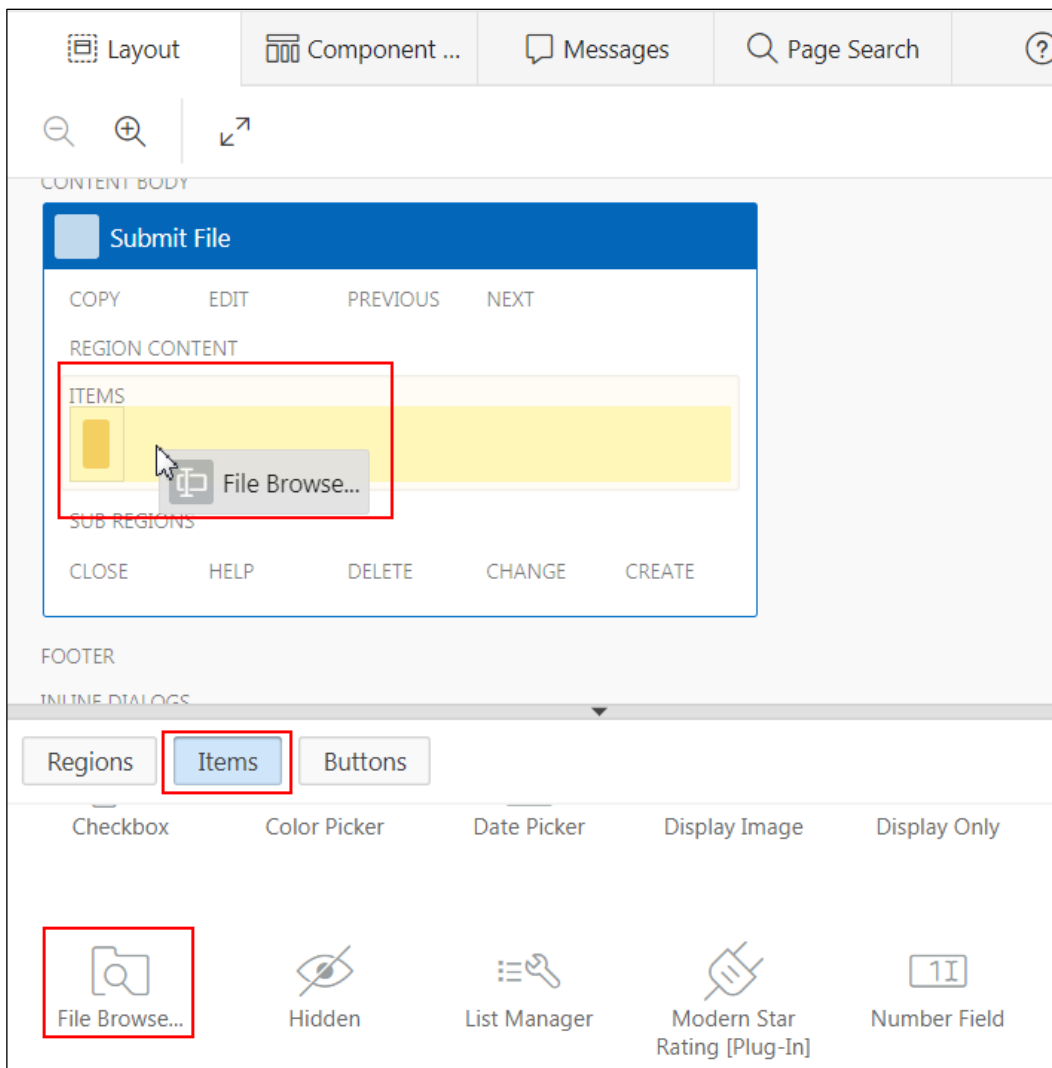
- i. Ensure that the newly created region is selected in the Rendering tab.



- j. In the Region panel, enter `Submit File` for **Identification > Title**.

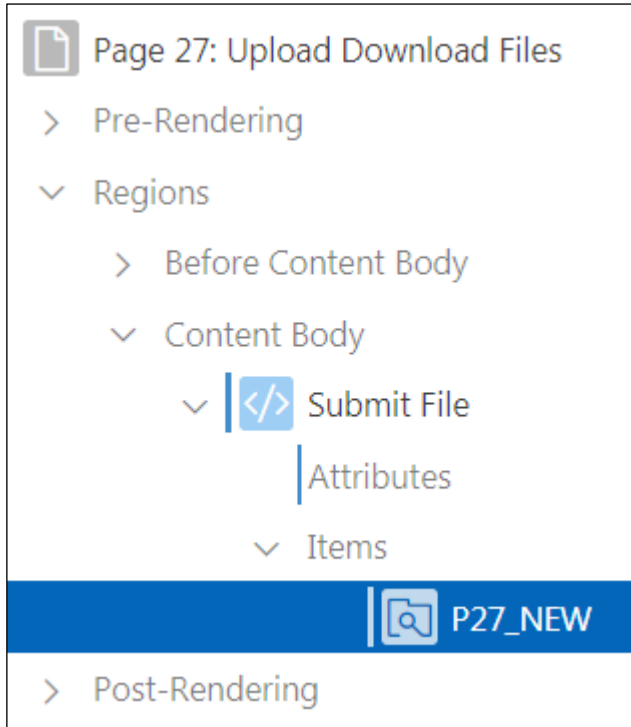


- k. Drag **File Browse** item from Items gallery to **Items** in Content Body on the Layout of the page.



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- I. Ensure that the new item is selected in the Rendering tab.



m. In the Page Item tab, enter the following values for the respective fields:

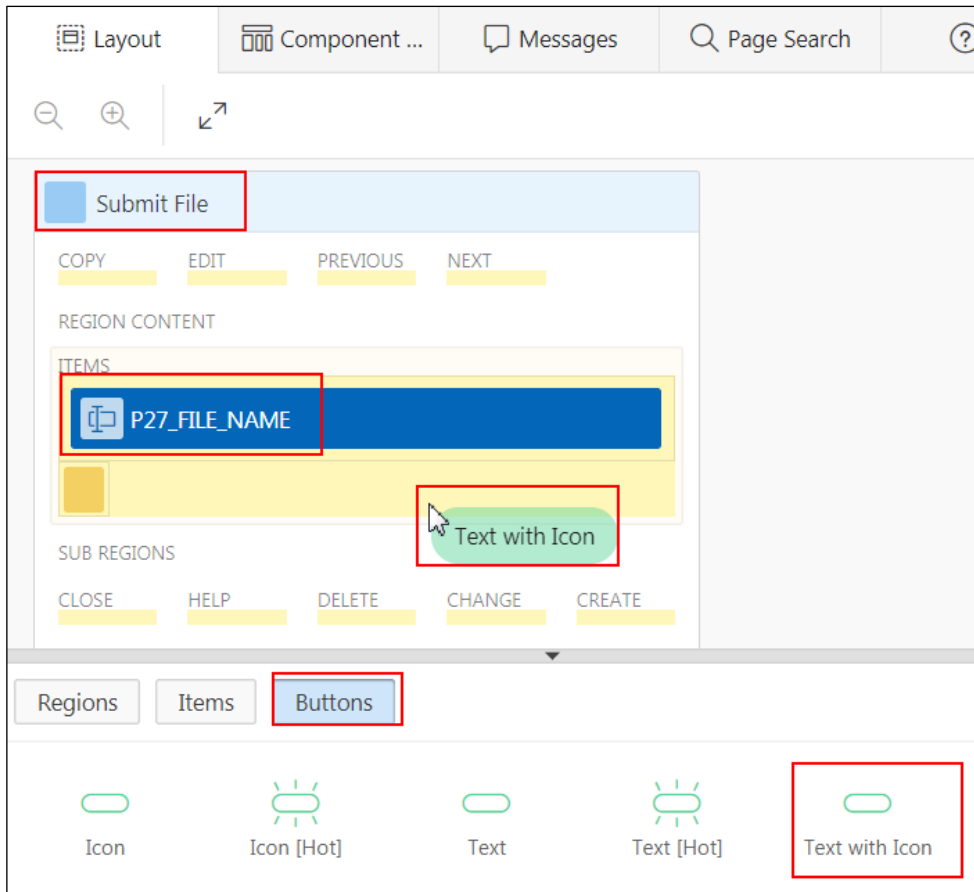
- **Identification > Item Name:** Enter P27_FILE_NAME
- **Settings > Storage Type:** Select Table APEX_APPLICATION_TEMP_FILES (if not selected by default)

The screenshot shows the APEX Page Item configuration dialog. It is divided into three main sections: Identification, Label, and Settings. In the Identification section, the 'Name' field is highlighted with a red box and contains the text 'P27_FILE_NAME'. The 'Type' field is set to 'File Browse...'. In the Label section, the 'Label' field is set to 'File Name'. In the Settings section, the 'Storage Type' field is highlighted with a red box and set to 'Table APEX_APPLICATION_TEMP_FILES'. The 'Purge File at' field is set to 'End of Session'.

After you define the attributes, this is how it looks in the Layout tab:

The screenshot shows the APEX Layout tab. The layout is divided into several regions: PAGE HEADER, PAGE NAVIGATION, BREADCRUMB BAR, BEFORE CONTENT BODY, MASTER DETAIL, and CONTENT BODY. The 'CONTENT BODY' region is expanded, showing a 'Submit File' item. The item is located in the 'ITEMS' section and is labeled 'P27_FILE_NAME'. The item is represented by a blue button with a file icon and the text 'P27_FILE_NAME'. The layout also shows a 'RIGHT SIDE COLUMN' and a 'SUB REGIONS' section with buttons for CLOSE, HELP, DELETE, CHANGE, and CREATE.

- n. You want to create a Submit button that will perform the upload. Drag **Text with Icon** from **Buttons** gallery to the **P27_FILE_NAME** item in the **Submit File** region on the Layout tab.



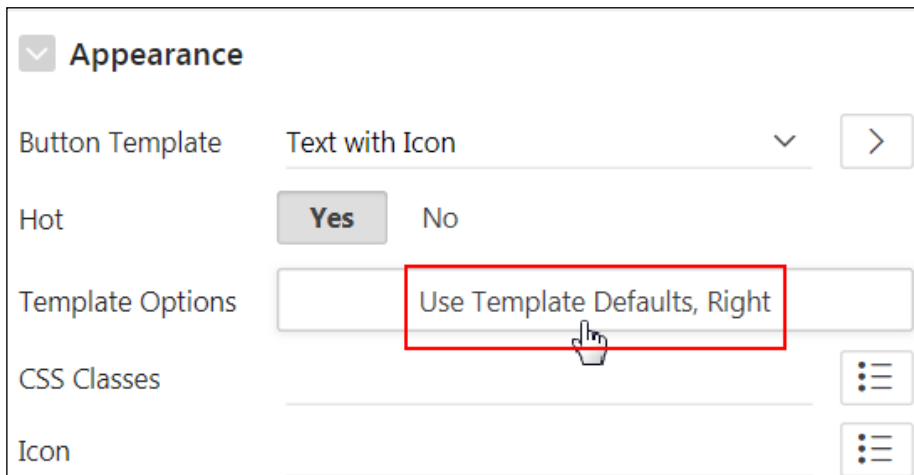
- o. Enter **P27_SUBMIT** for **Identification > Button Name** and **Submit** for **Identification > Label**.

The screenshot shows the 'Identification' configuration panel for a button. It has a dropdown menu labeled 'Identification' with a downward arrow. Below this, there are two input fields: 'Button Name' with the value 'P27_SUBMIT' and 'Label' with the value 'Submit'.

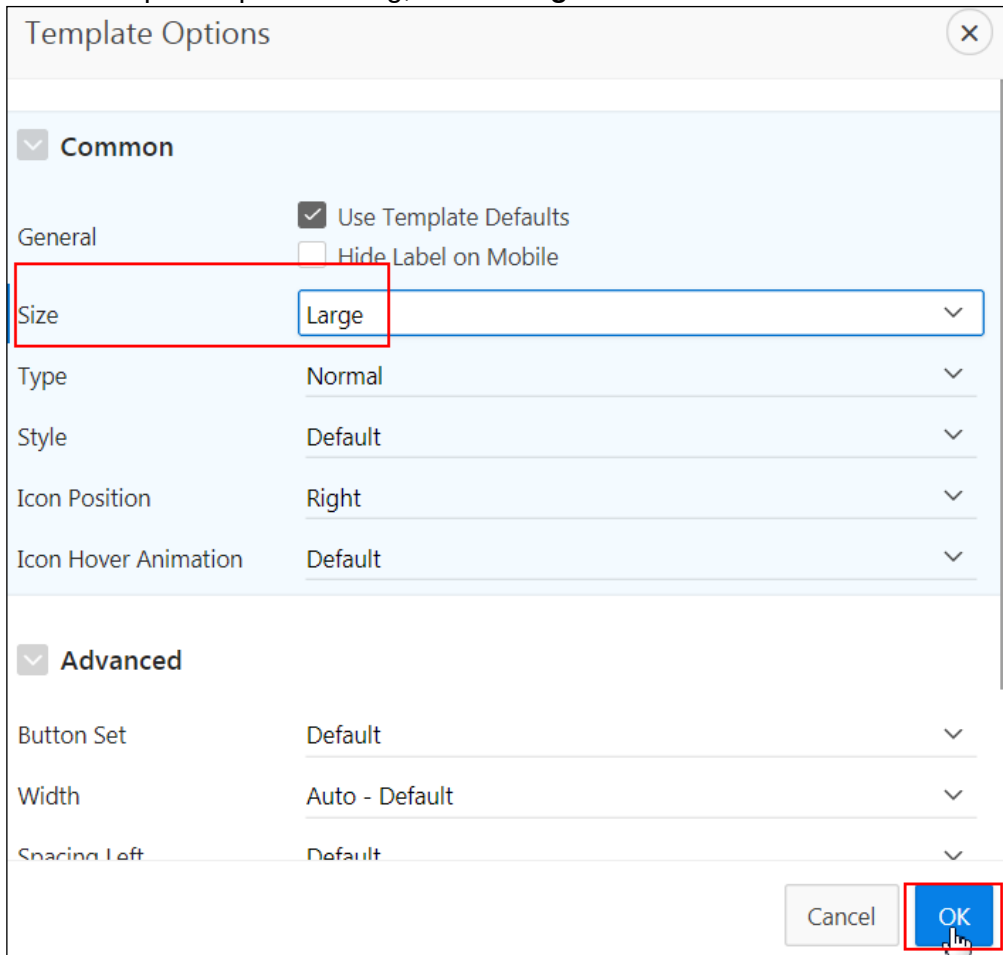
- p. Select **Yes** for **Appearance > Hot**.

The screenshot shows the 'Appearance' configuration panel for a button. It has a dropdown menu labeled 'Appearance' with a downward arrow. Below this, there are three sections: 'Button Template' with a dropdown menu showing 'Text with Icon' and a right arrow button; 'Hot' with a 'Yes' button (highlighted with a red box) and a 'No' button; and 'Template Options' with a button labeled 'Use Template Defaults, Right'. At the bottom, there is a 'CSS Classes' section with a menu icon.

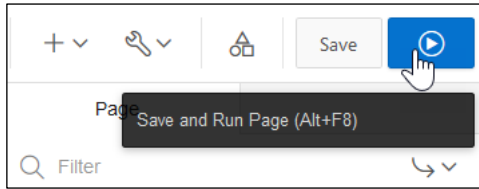
- q. Under **Appearance > Template Options**, click **Use Template Defaults, Right**. The Template Options dialog opens.



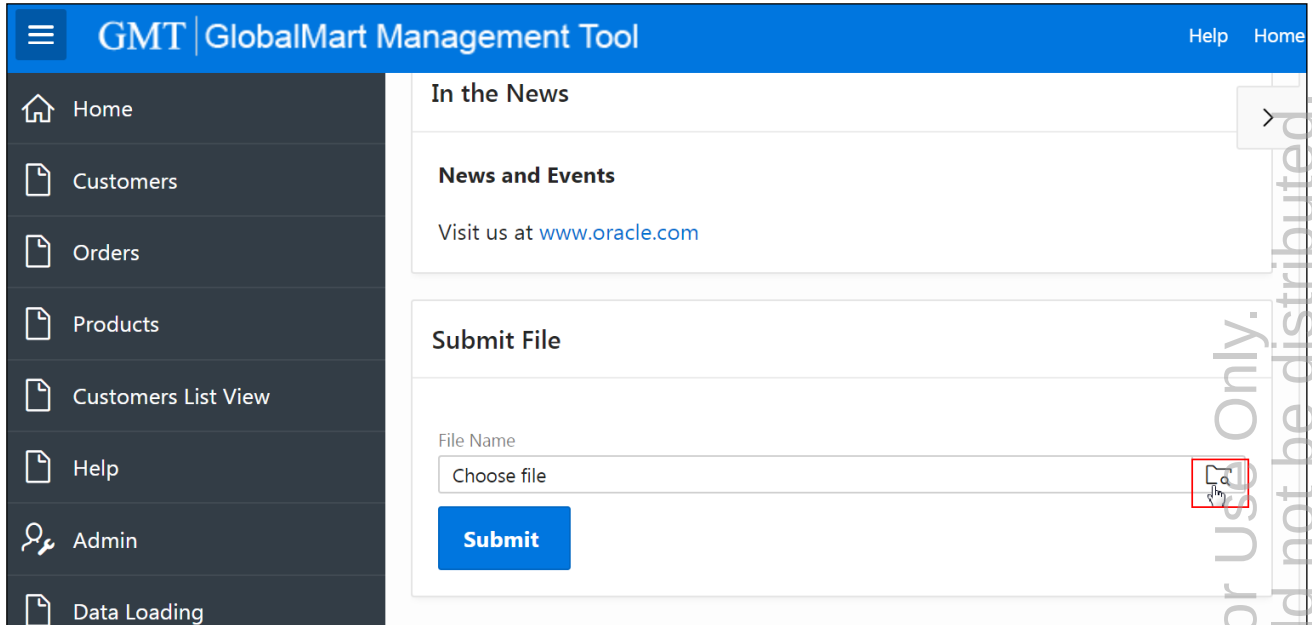
- r. In the Template Options dialog, select **Large** for **Size** and click **OK**.



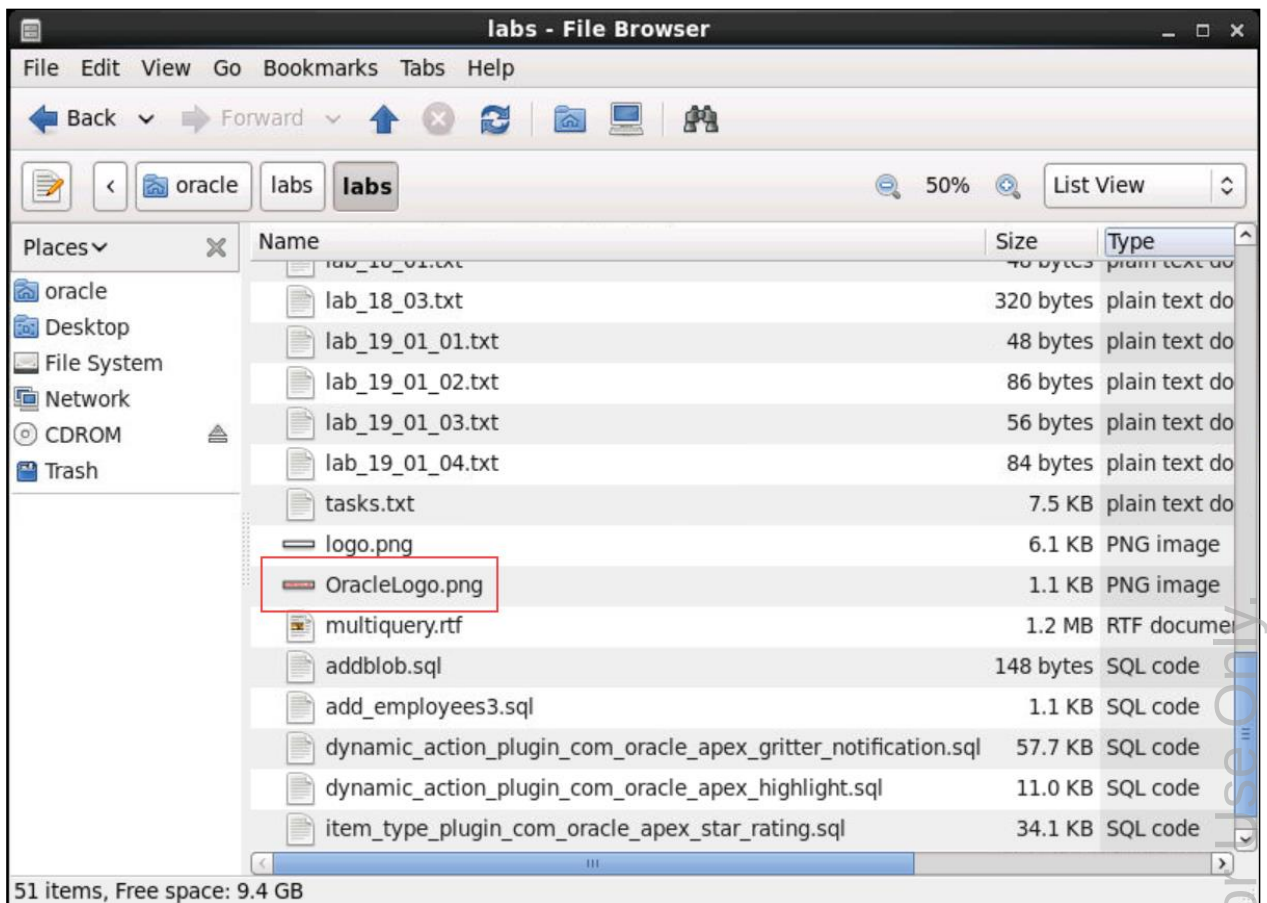
- s. Click the **Save and Run Page** icon.



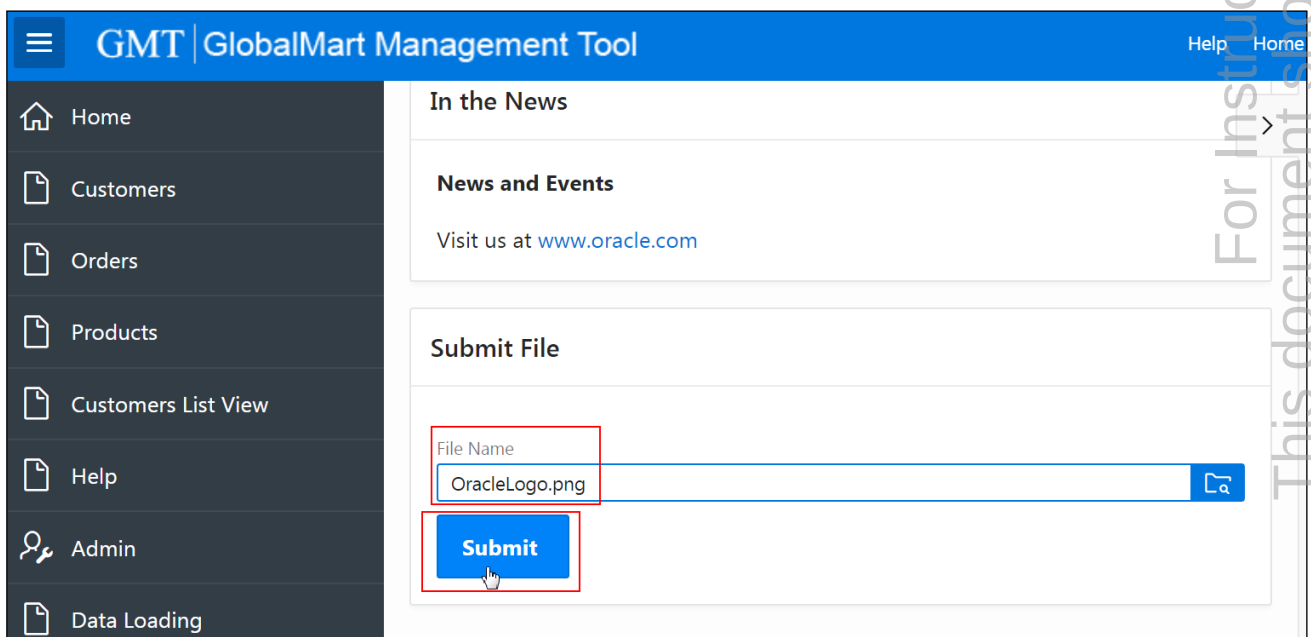
- t. The GMT app opens. Click the **Browse** icon in the **File Name** field.



- u. Select the `/home/oracle/labs/labs/OracleLogo.png` file and click **Open**.



- v. Click **Submit** to upload the file.



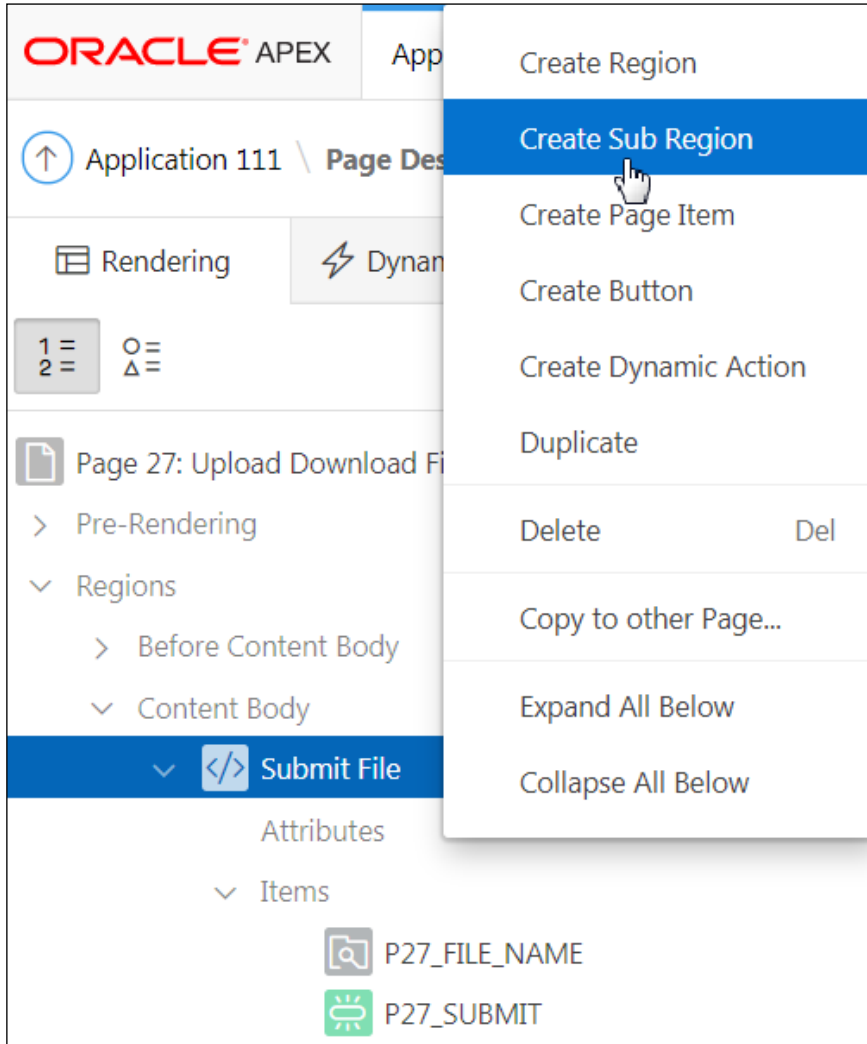
- w. The file is submitted. At this point, you want to create a report to show the list of files that are uploaded already. Click **Edit Page 27** on the Developer toolbar to open Page 27 in Page Designer.



2. Create an Interactive Report subregion called Uploaded Files with the following SQL query (located in /home/oracle/labs/labs/lab_18_02.txt):

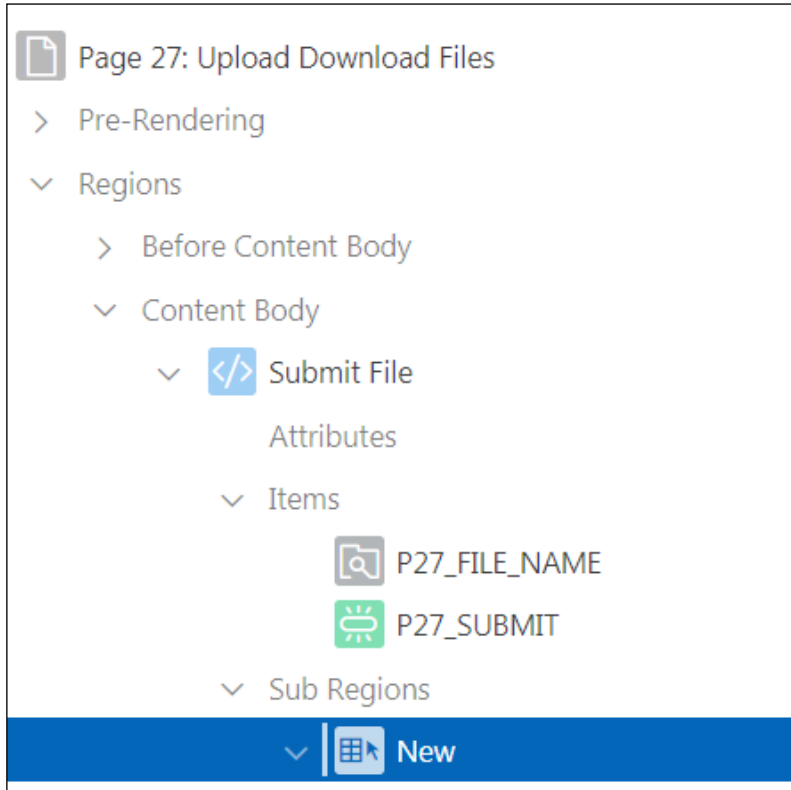
```
SELECT id, filename FROM APEX_APPLICATION_TEMP_FILES order by  
created_on desc
```

- a. You will create the report as a subregion. Right-click the **Submit File** region and select **Create Sub Region**.



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- b. Ensure that the new subregion is selected in the Rendering tab.



- c. In the Region tab, enter the following values for the respective fields:
- **Identification > Title:** Uploaded Files
 - **Identification > Type:** Select **Interactive Report**
 - **Source > Type:** Select **SQL Query**
 - **Source > SQL Query:** Copy the SQL query from `/home/oracle/labs/labs/lab_18_02.txt` and paste here.

▼

Identification

Title

Uploaded Files

Type

Interactive Report

▼

⋮

▼

Source

Location

Local Database

▼

Type

SQL Query

▼

SQL Query

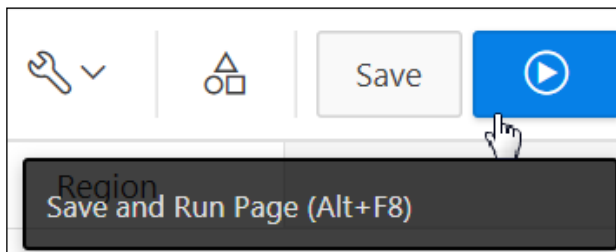
↗

SELECT id, filename FROM APEX_APPLICATION_TEMP_FILES order by created_on desc

Page Items to Submit

⋮

- d. Click the **Save and Run Page**.



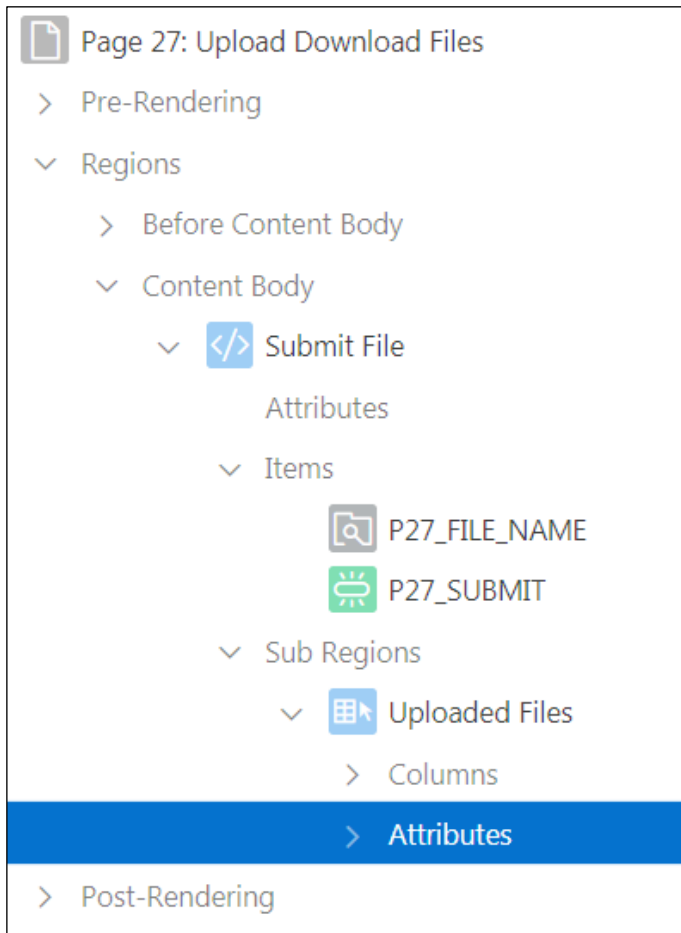
- e. The report that you just created shows all the documents that have been uploaded. Note that your list may be different depending on what you uploaded. Next, you provide a link to download the document. Click **Edit Page 27** on the Developer toolbar.

The screenshot displays the GMT GlobalMart Management Tool interface. On the left is a dark sidebar with navigation links: Home, Customers, Orders, Products, Customers List View, Help, Admin, and Data Loading. The main content area is titled 'Submit File' and contains a 'File Name' input field with a 'Choose file' button and a 'Submit' button. Below this is a search bar with a 'Go' button and an 'Actions' dropdown. A table lists uploaded files with columns 'Id' and 'Filename'. One file is listed with Id '20441864583259493' and Filename 'OracleLogo.png', which is highlighted with a red box. The bottom of the interface features a Developer toolbar with buttons for Home, Application 111, Edit Page 27 (highlighted with a red box), Session, View Debug, Debug, Page Info, Quick Edit, Theme Roller, and a settings icon.

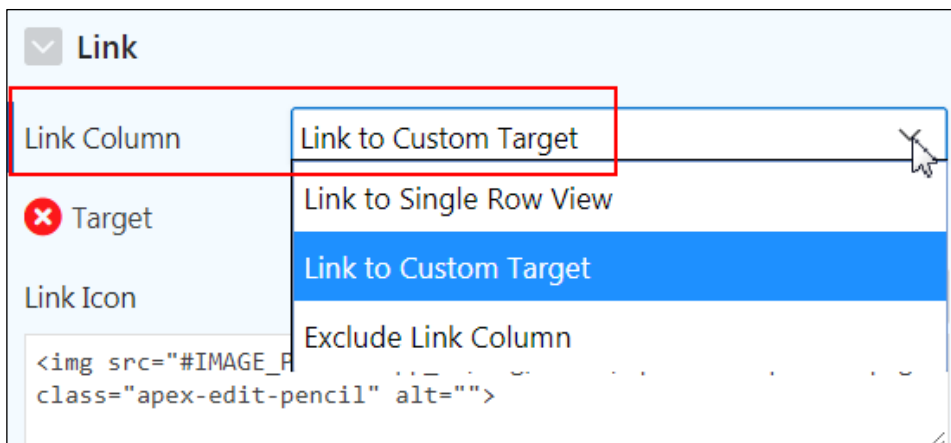
Id	Filename
20441864583259493	OracleLogo.png

3. Modify the report to create a link on the ID column to download the file.

a. In the Rendering tab, select **Uploaded Files > Attributes**.



b. In the Attributes tab, under **Link > Link Column**. Select **Link to Custom Target**.



- c. Under **Link > Target**, click **No Link Defined**. The Link Builder - Target dialog opens.

Link

Link Column: Link to Custom Target

Target: No Link Defined

Link Icon: `<img src=\"#IMAGE_PREFIX#app_ui/img/icons/apex-edit-pencil.png\" class=\"apex-edit-pencil\" alt=\"\">`

Link Attributes

Authorization Scheme: - Select -

- d. In the Link Builder - Target dialog, select **URL** for **Target > Type** and enter `p?n=#ID#` for **URL**. Click **OK**.

Link Builder - Target

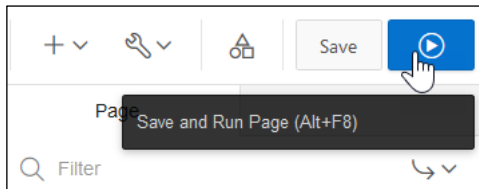
Target

Type: URL

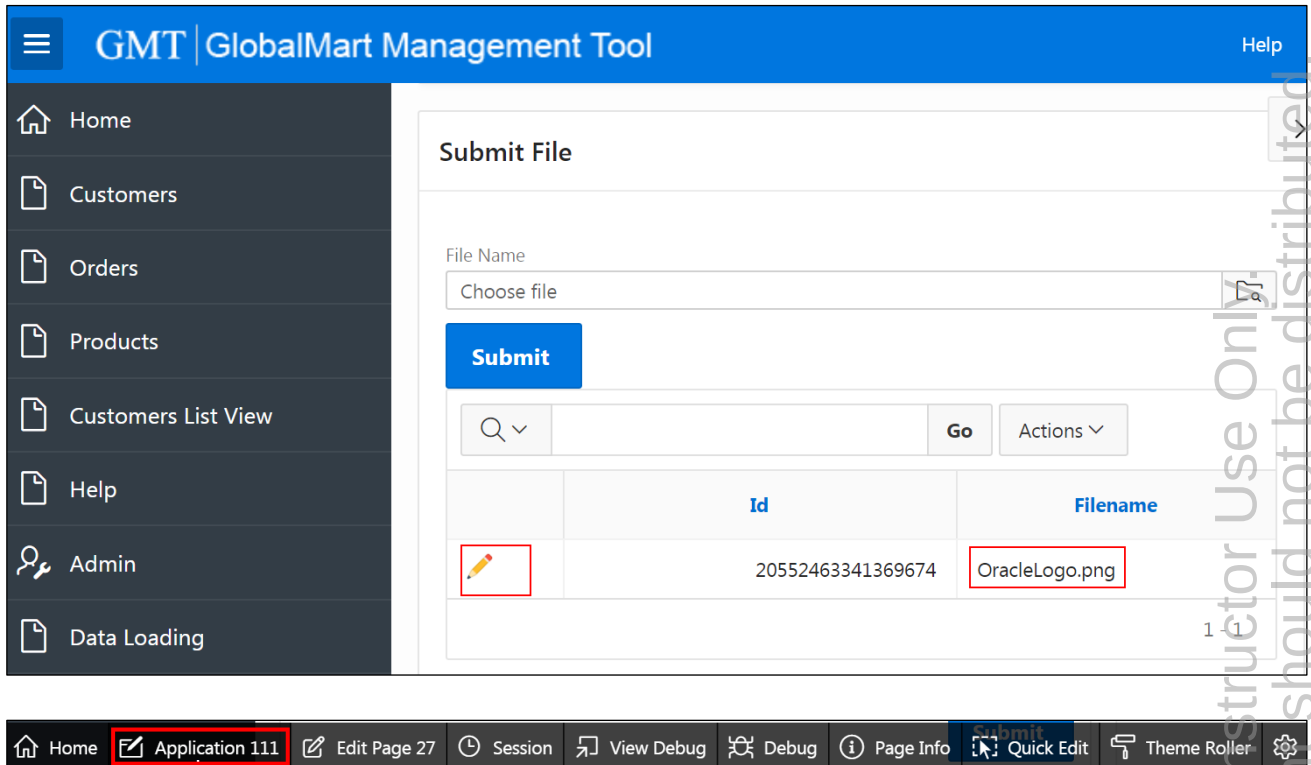
URL: p?n=#ID#

Cancel Clear OK

- e. Click the **Save and Run Page** icon.



- f. The report is displayed with a link indicated by the pencil icon to open and download the file. Click the **Application 111** link on the Developer toolbar.



	Id	Filename
	20552463341369674	OracleLogo.png

**Practices for Lesson 19:
Creating and Editing Charts**

Practices for Lesson 19: Overview

Practices Overview

In these practices, you will learn how to create and modify charts for your applications.

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Practice 19-1: Creating and Modifying Charts

Overview

In this practice, you create a new Horizontal Bar – 3D Bar chart. You then modify the chart and change it to a Vertical Bar chart and change some of the settings to see how they work.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a new chart page that includes a Horizontal Bar chart. Use `Employees by Department` for the **Title**, `Departments` for the **X-Axis**, and `Number of Employees` for the **Y-Axis**. Use the following query (located in `/home/oracle/labs/labs/lab_19_01.txt`). Test your page.

```
select null link,
department_name chart_label,
count(*) chart_value
from oehr.oehr_employees e,
oehr_departments d
where e.department_id = d.department_id
group by department_name
```

2. Modify your chart and change it to a Vertical Bar chart.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **apex_admin SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 19-1**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.

- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. On the Application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the `apex_admin` user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 18-3**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_dev` user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_18_03.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).

- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

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Practice 19-2: Creating Pie and Doughnut Charts

Overview

In this practice, you create a Pie chart and then change it to a Donut chart for the GMT application.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a Pie chart. Use **Departments Chart** for the **Title**. Use the following query (located in `/home/oracle/labs/labs/lab_19_02.txt`). Then change the Pie chart to a Donut chart.

```
select null link,
department_name chart_label,
count(*) chart_value
from oehr.oehr_employees e,
oehr_departments d
where e.department_id = d.department_id
group by department_name
```

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 19-2**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory.

- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. On the Application's home page click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 19-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_19_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Practice 19-3: Creating Combination Chart

Overview

In this practice, you create some additional charts, such as a combination chart of a Line and Bar chart for the GlobalMart Management Tool application.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. You build a Combination chart. First, you upload and run the `add_employees3.sql` script to create a table called `employees3`.
2. Create a Combination chart named `Salary Commission Combined Chart`. Use the following SQL Query (located in the `/home/oracle/labs/labs/lab_19_03_01.txt` file). Specify `Employee` for X-Axis Title and `Salary` for Y-Axis Title. Edit the chart by changing the Series attribute to Line.

```
SELECT NULL LINK,  
LAST_NAME LABEL,  
SALARY "Salary",  
COMMISSION_PCT "COMMISSION"  
FROM EMPLOYEES3  
WHERE MANAGER_ID = 100  
Order by LAST_NAME
```

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 19-3**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. On the Application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 19-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_19_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.

- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

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Solution 19-1: Creating and Modifying Bar Charts

Overview

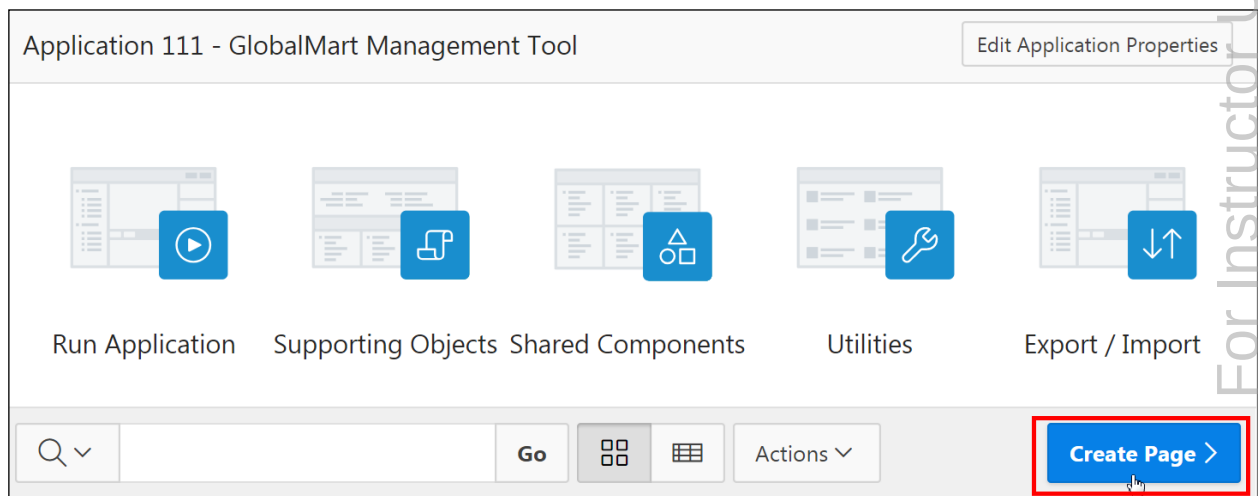
In this practice solution, steps are provided for creating and modifying charts.

Steps

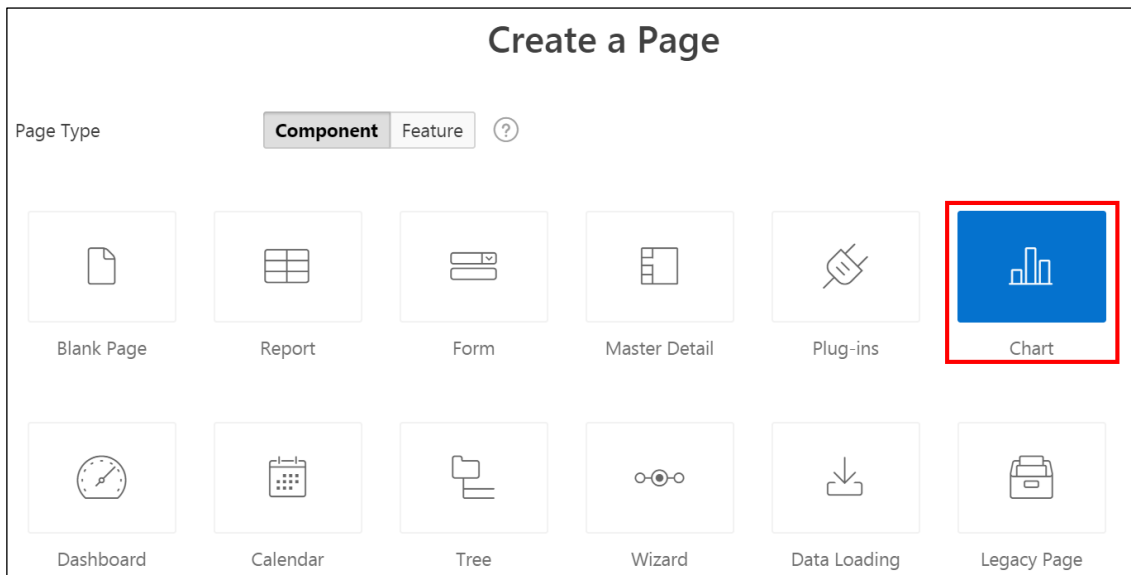
1. Create a new bar chart page that includes a Horizontal Bar. Use `Employees` by `Department` for the **Title**, `Departments` for the **X-Axis**, and `Number of Employees` for the **Y-Axis**. Use the following query (located in `/home/oracle/labs/labs/lab_19_01.txt`). Test your page.

```
select null link,
department_name chart_label,
count(*) chart_value
from oehr.oehr_employees e,
oehr_departments d
where e.department_id = d.department_id
group by department_name
```

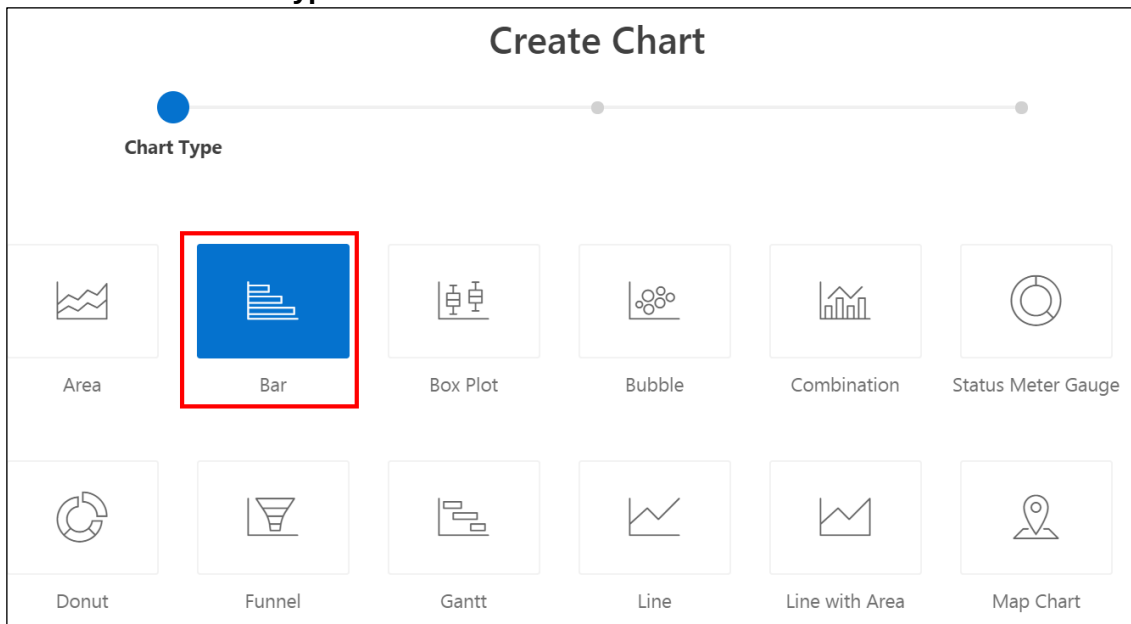
- a. In the GlobalMart Management Tool application home page, click **Create Page >**.



- b. Select **Chart**.



- c. Select Bar for **Chart Type**. Click **Next >**.



- d. For Page and Region Attributes, enter the Page Name `Employees by Department` and click **Next >**. The Page Number is 28.

The screenshot shows the 'Create Chart' wizard with a progress bar at the top. The first step, 'Page and Region Attributes', is highlighted with a blue dot. Below the title, there are four input fields: 'Page Number' (value: 28), 'Page Name' (value: Employees by Department), 'Page Mode' (value: Normal), and 'Breadcrumb' (value: - do not use breadcrumbs on page -). The 'Page Number' and 'Page Name' fields are highlighted with a red box. The 'Page Mode' field has a dropdown menu with 'Normal' and 'Modal Dialog' options. The 'Breadcrumb' field has a dropdown menu with '- do not use breadcrumbs on page -' as the selected option.

- e. Accept the defaults for **Navigation Preference** and click **Next >**.

The screenshot shows the 'Create Chart' wizard with a progress bar at the top. The second step, 'Navigation Menu', is highlighted with a blue dot. Below the title, there is a 'Navigation Preference' section with three radio button options: 'Do not associate this page with a navigation menu entry' (selected), 'Create a new navigation menu entry', and 'Identify an existing navigation menu entry for this page'. The first option is highlighted with a red box.

- f. For **Source Type**, select **SQL Query** and enter the following SQL Query.

```
select null link,  
department_name chart_label,  
count(*) chart_value  
from oehr.oehr_employees e,  
oehr_departments d  
where e.department_id = d.department_id  
group by department_name
```

After entering the SQL Query, validate the query by clicking the Validate icon. Click **Next>**.

Create Chart

Location: Local Database

Source Type: Table **SQL Query** (?)

* SQL Query (?)

Validation successful

```
1 select null link,  
2 department_name chart_label,  
3 count(*) chart_value  
4 from oehr.oehr_employees e,  
5 oehr_departments d  
6 where e.department_id = d.department_id  
7 group by department_name
```

Page Items to Submit: [] (?)

Maximum Rows: [] (?)

< Cancel **Next >**

g. For Column Mapping, enter the following and click **Create**.

- **Orientation:** Select `Horizontal`
- **Label Column:** Select `CHART_LABEL`
- **Value Column:** Select `CHART_VALUE`

Create Chart

Chart Type: **Bar** ?

Orientation: Horizontal ?

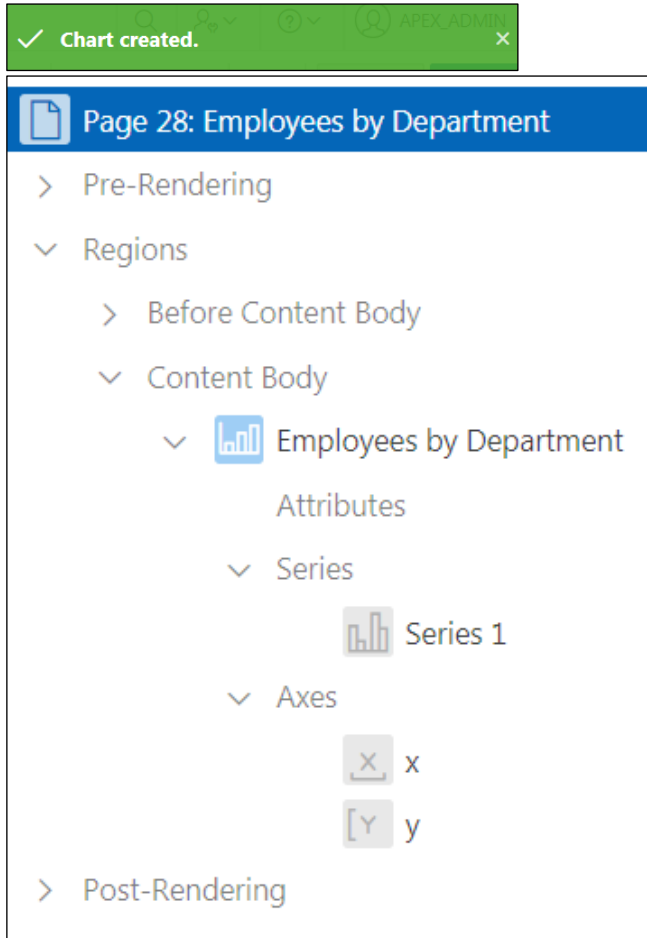
* Label Column: CHART_LABEL ?

* Value Column: CHART_VALUE ?

Column Mapping

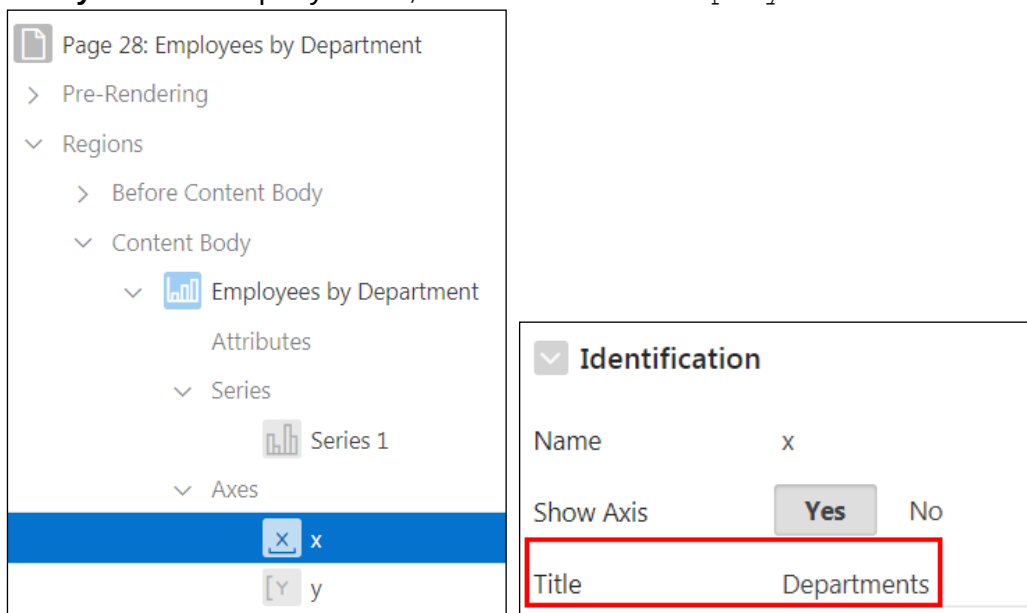
Cancel Create

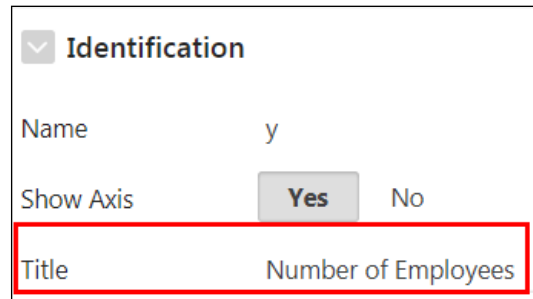
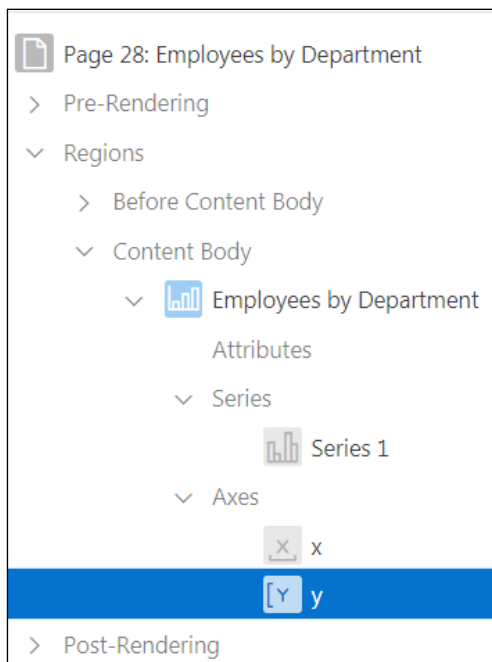
- h. The chart is created and opens in Page Designer view.



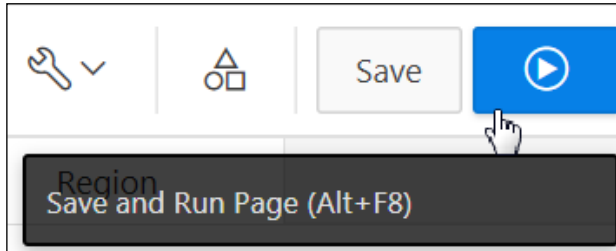
- i. Set the names for x axis and y axis as follows: In the Rendering tab, expand click **Axes** and define the following:

- Click **x**. On the Property Editor, enter `Departments` for **Title**.
- Click **y**. On the Property Editor, enter `Number of Employees` for **Title**.

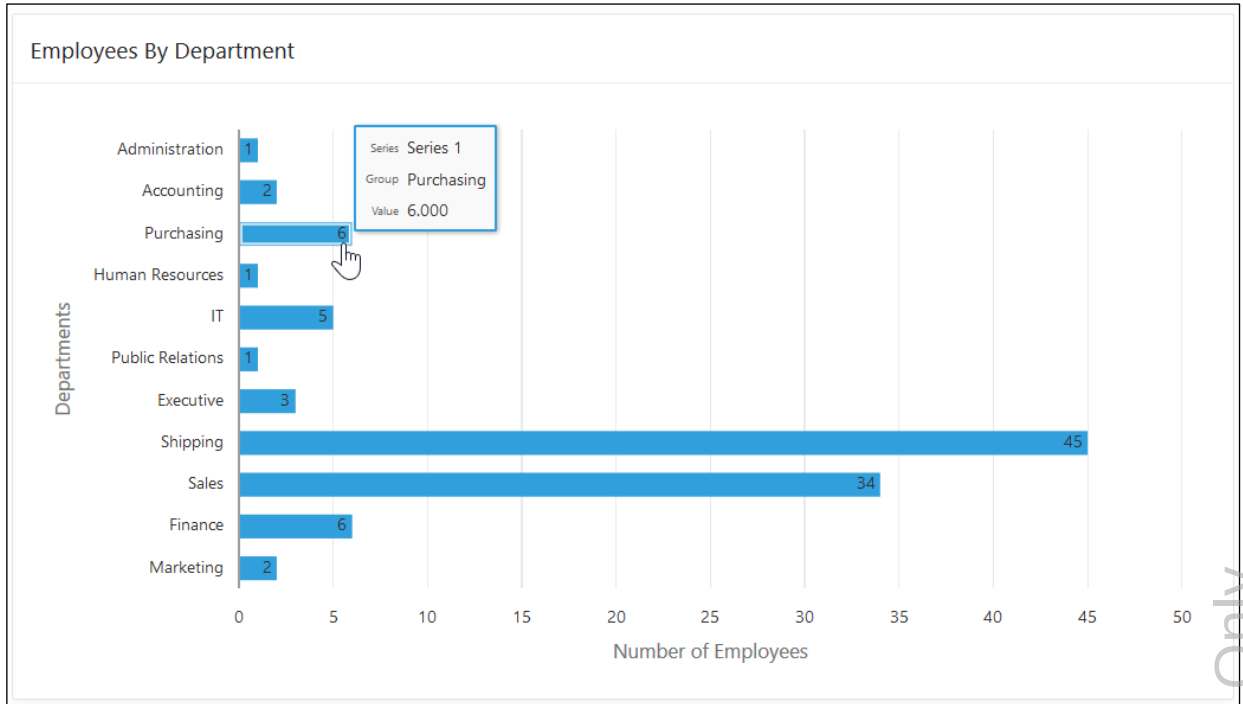




- j. Run the page by clicking the **Save and Run Page** icon.

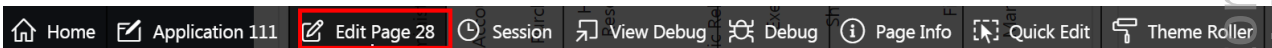


- k. Place your cursor over one of the bars to see the department and its value.



Note: You might need to activate the Adobe Flash plug-in.

- l. You now change the Horizontal Bar chart to a Vertical Bar chart and alter the way it looks. Click the **Edit Page 28** link on the Developer toolbar.



3. Modify your chart and change it to a **Vertical Bar Chart**.

- a. In the Rendering tab, select **Attributes** under **Regions > Content Body > Employees by Department**. On the Property Editor, under **Appearance**, select **Vertical** for **Orientation**.

Page 28: Employees by Department

> Pre-Rendering

> Regions

> Before Content Body

> Content Body

> Employees by Department

Attributes

> Series

Series 1

> Axes

x

y

> Appearance

Orientation

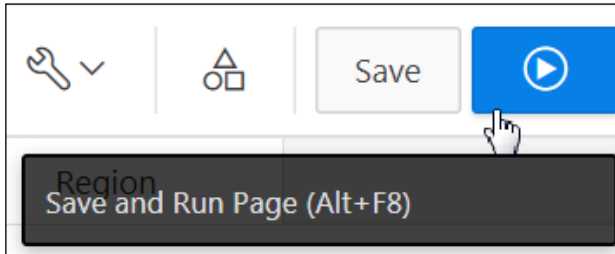
Vertical

Stack

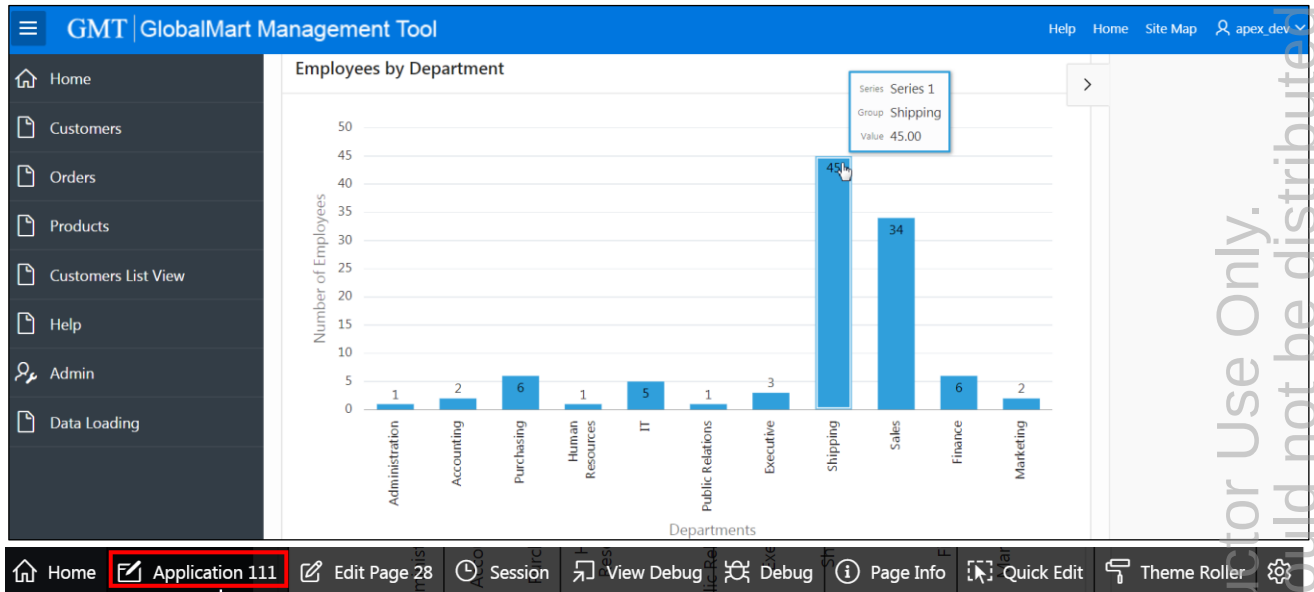
Vertical

Horizontal

- b. Click the **Save and Run Page** icon.



- c. Notice that the chart is now a Vertical Bar chart. Click the **Application 111** link on the Developer toolbar.



Solution 19-2: Creating a Pie Chart and Converting it to a Doughnut Chart

Overview

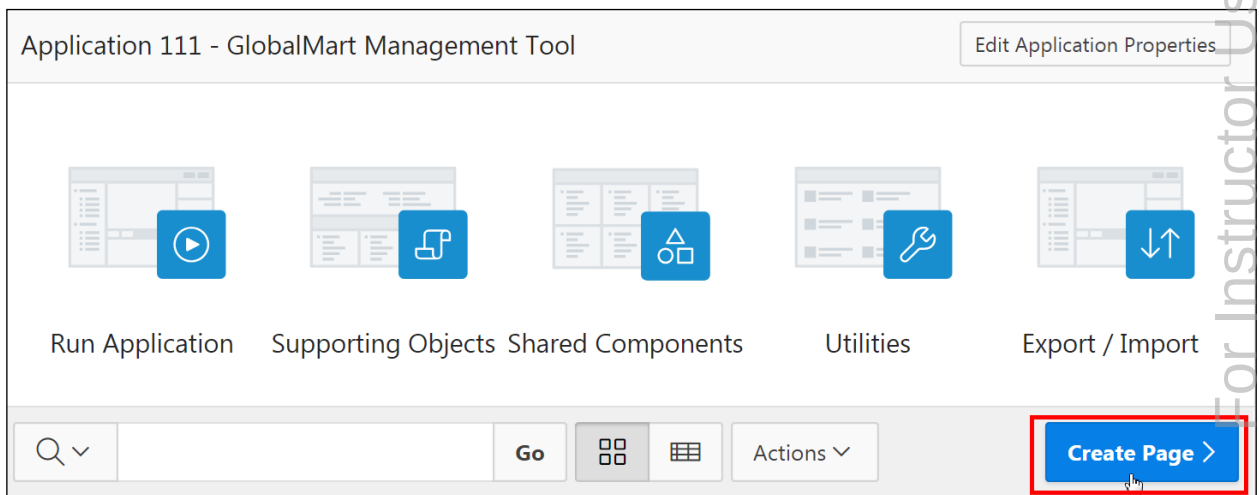
In this practice solution, steps are provided for creating a Pie chart for the GMT application.

Steps

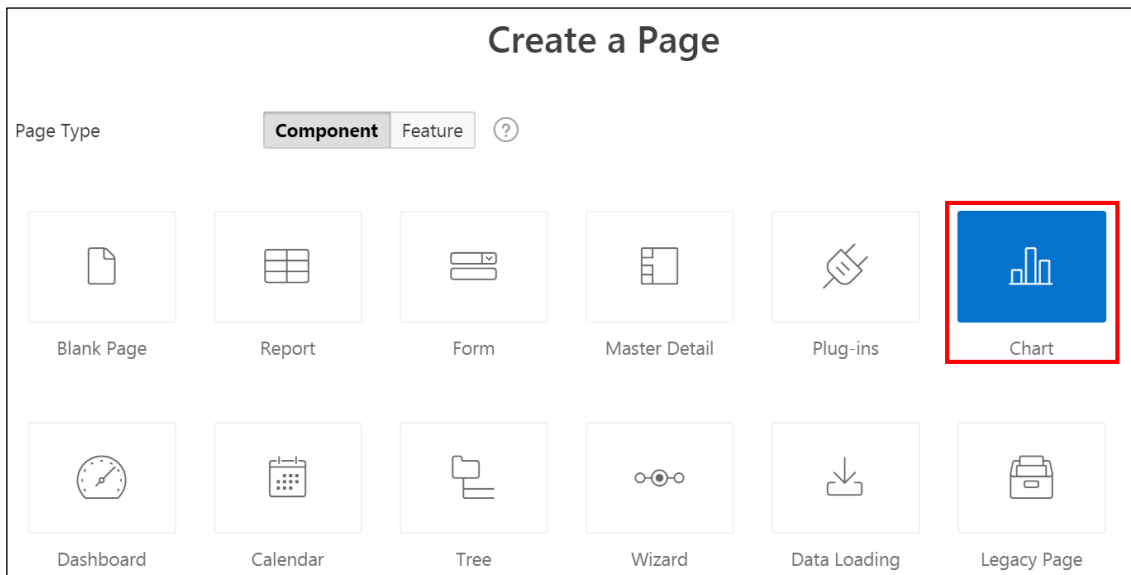
1. Create a Pie chart and then convert it to a Donut chart. Use **Departments Chart** for the **Title**. Use the following query (located in /home/oracle/labs/labs/lab_19_02.txt).

```
select null link,  
department_name chart_label,  
count(*) chart_value  
from oehr.oehr_employees e,  
oehr_departments d  
where e.department_id = d.department_id  
group by department_name
```

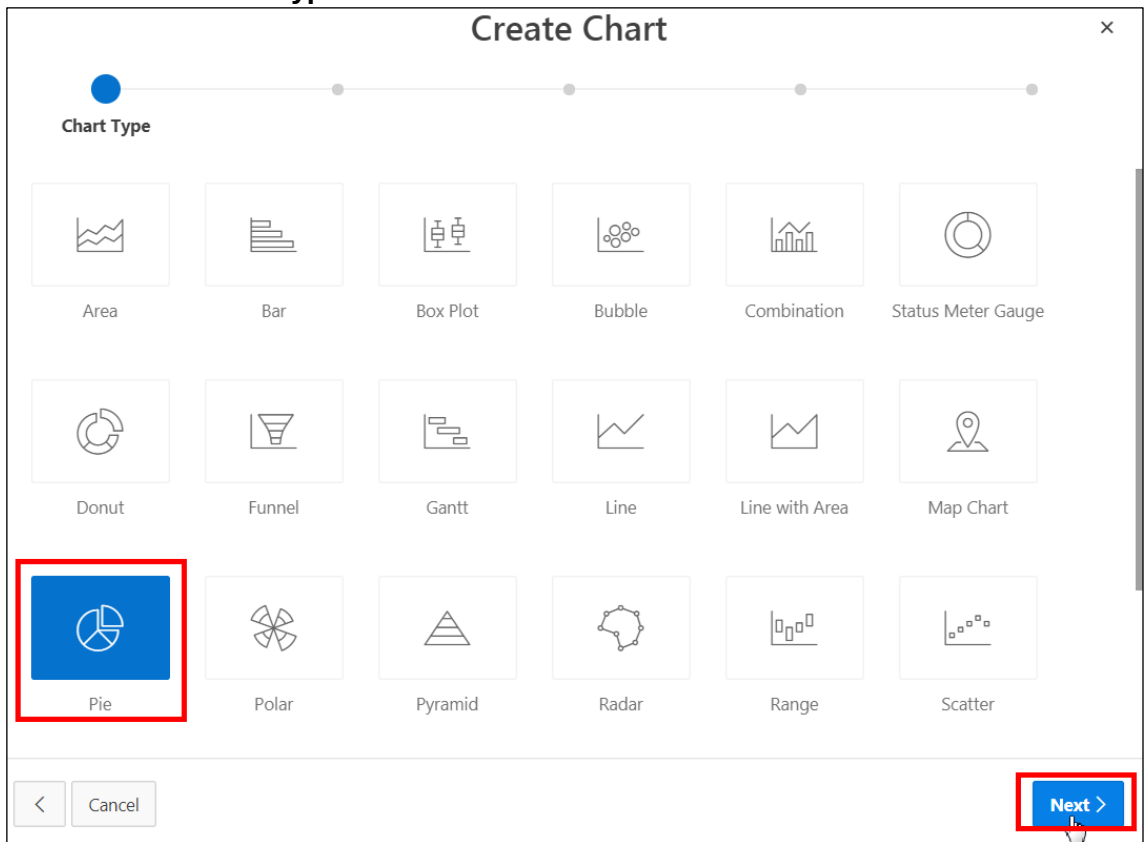
- a. In the GlobalMart Management Tool application home page, click **Create Page >**.



- b. For Page Type, select **Chart** and click **Next >**.



- c. Select **Pie** for **Chart Type** and click **Next >**.



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- d. For Page and Region Attributes, enter 29 for **Page Number** and Departments Chart for **Page Name**. Click **Next >**.

The screenshot shows the 'Create Chart' dialog box with a progress bar at the top. The first step, 'Page and Region Attributes', is highlighted with a green checkmark. The second step is highlighted with a blue circle. The 'Page Number' field is set to 29, and the 'Page Name' field is set to 'Departments Chart'. The 'Page Mode' is set to 'Normal', and the 'Breadcrumb' is set to '- do not use breadcrumbs on page -'. A red box highlights the 'Page Number' and 'Page Name' fields.

- e. Accept the default and click **Next >**.

The screenshot shows the 'Create Chart' dialog box with a progress bar at the top. The first two steps are highlighted with green checkmarks. The third step, 'Navigation Menu', is highlighted with a blue circle. The 'Navigation Preference' is set to 'Do not associate this page with a navigation menu entry'. Two other options are listed: 'Create a new navigation menu entry' and 'Identify an existing navigation menu entry for this page'. A red box highlights the 'Do not associate this page with a navigation menu entry' option.

- f. For **Source**, select SQL Query and enter the following SQL query (located in /home/oracle/labs/labs/lab_19_02.txt) and click **Next >**.

```
select null link,
department_name chart_label,
count(*) chart_value
from oehr.oehr_employees e,
oehr_departments d
where e.department_id = d.department_id
group by department_name
```


Create Chart

✓

✓

✓

Source

Location

Source Type **SQL Query** ⓘ

* SQL Query ⓘ

↶ ↷ 🔍 ↔ ⬆ A· ✓ ⚙

Validation successful ✕

```
1 select null link,
2 department_name chart_label,
3 count(*) chart_value
4 from oehr.oehr_employees e,
5 oehr_departments d
6 where e.department_id = d.department_id
7 group by department_name
```

Page Items to Submit ⓘ

Maximum Rows ⓘ

< Cancel Next >

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g. For Column Mapping, define the following values and click **Create**.

- **Label Column:** Select CHART_LABEL
- **Value Column:** Select CHART_VALUE

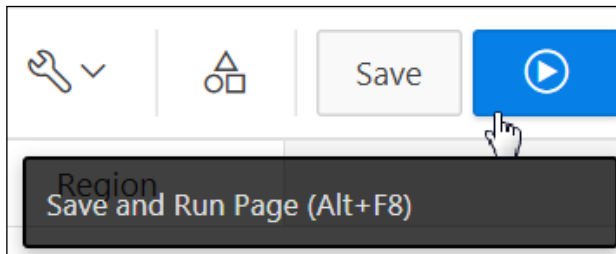
Chart Type: **Pie** ?

* Label Column CHART_LABEL ?

* Value Column CHART_VALUE ?

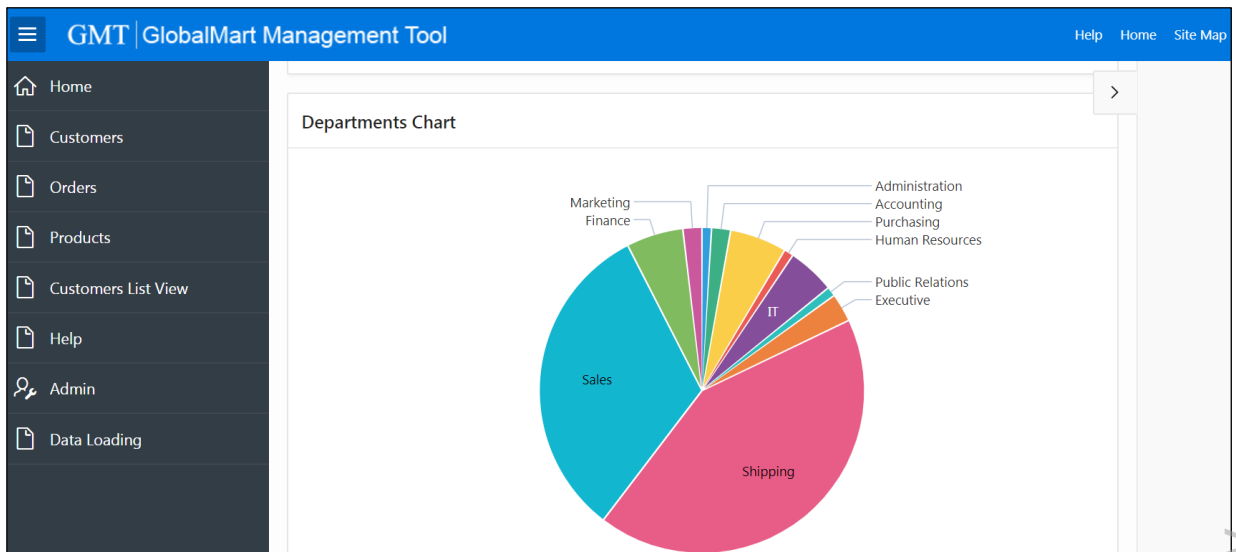
Cancel Create

h. The page now opens in Page Designer. Click the **Save and Run Page** icon.

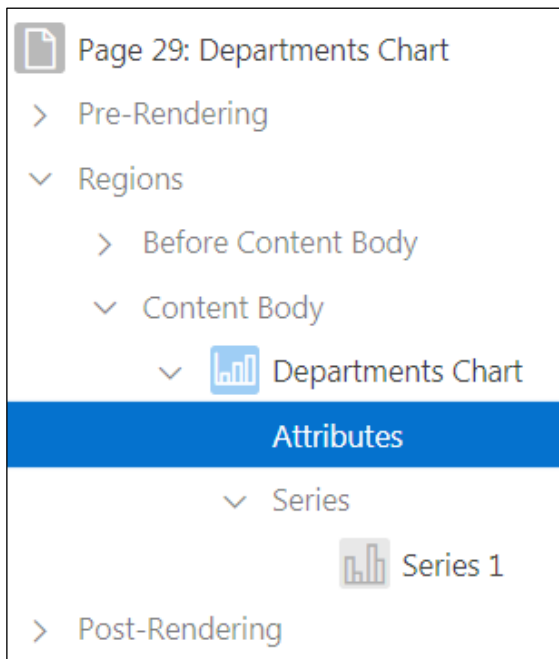


i. You may be asked to sign in. Enter your login credentials and click **Sign In**.

- j. The *Departments Chart* pie chart is displayed. Switch the window to the Page Designer page.

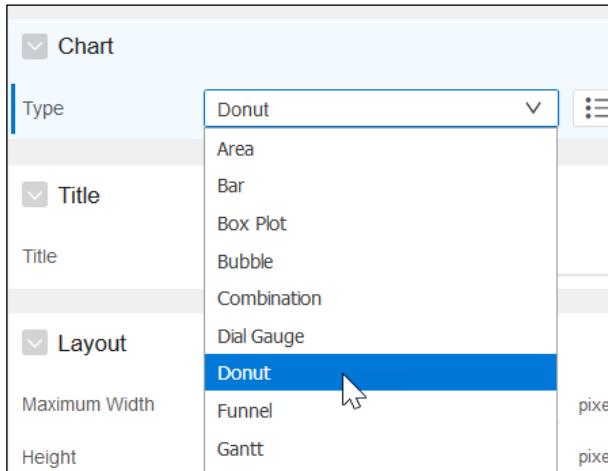


2. To convert the Pie chart to a Donut chart:
- a. Go to the Page Designer view of the Pie chart, and in the Rendering tab, click **Attributes**.

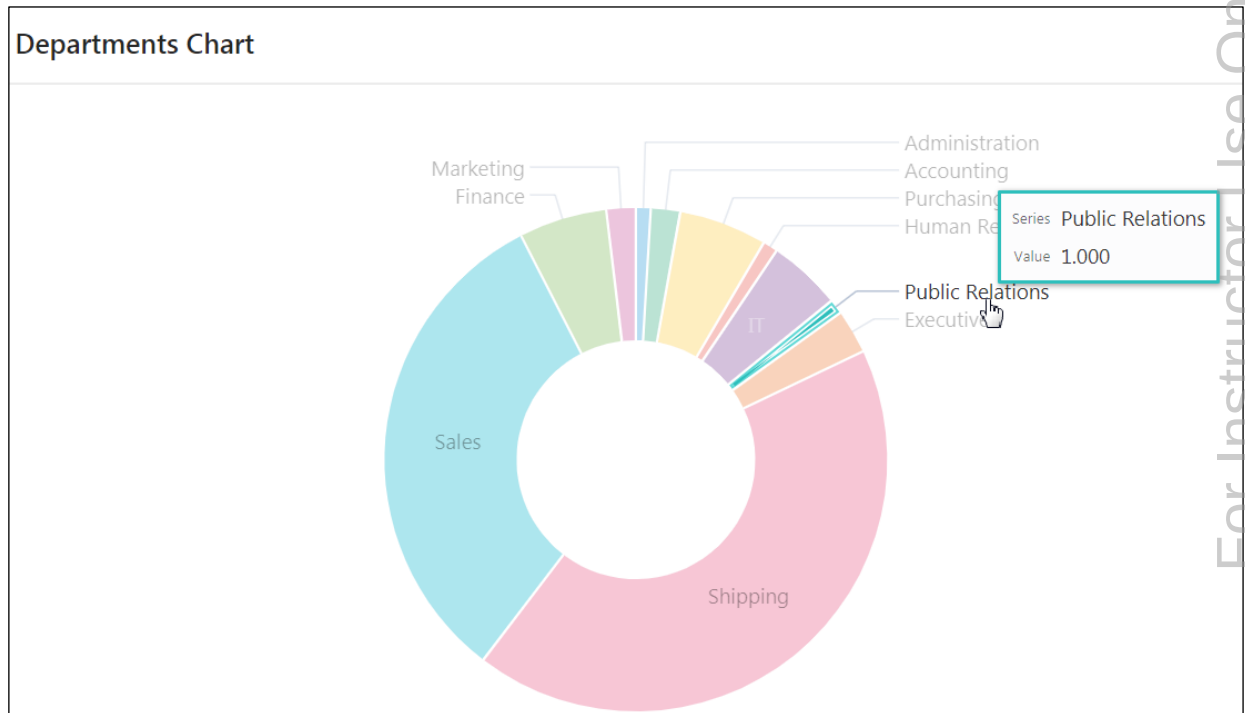
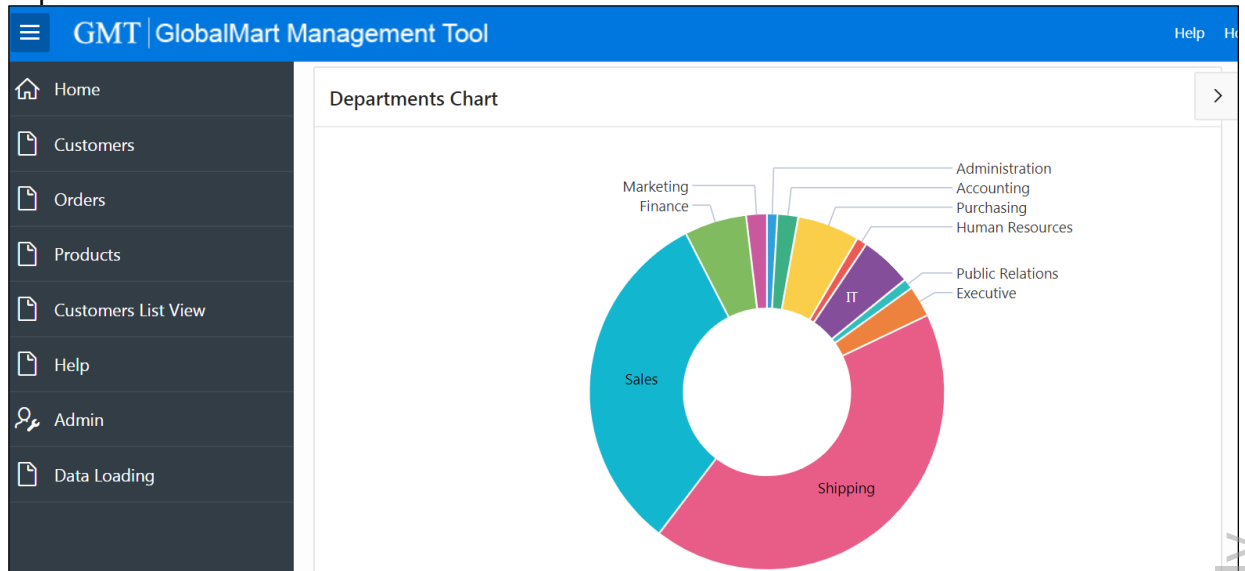


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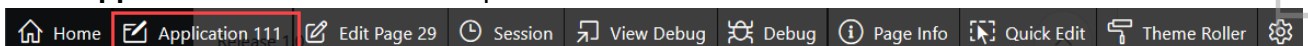
- b. On the Property Editor, change the Chart Type to **Donut**.



- c. Click **Save** and **Run**. The Donut chart is now created. Hover your mouse over the different departments in the chart.



- d. Click **Application 111** on the Developer toolbar.



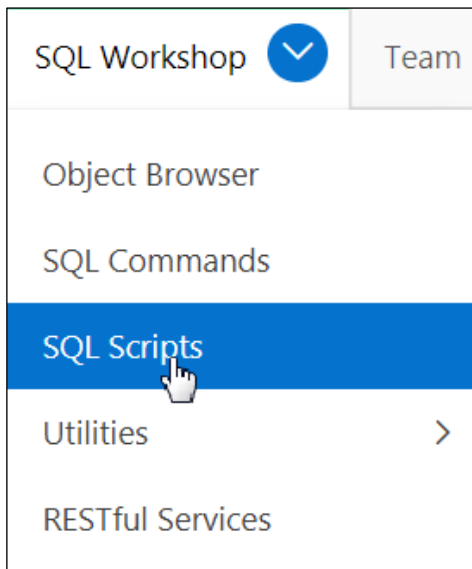
Solution 19-3: Creating Combination Chart

Overview

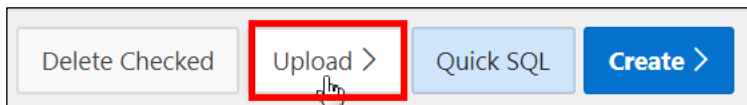
In this practice solution, steps are provided for creating a Combination chart for the GlobalMart Management Tool application.

Steps

1. You build a Combination chart. First, you upload and run the `add_employees3.sql` script to create a table called `employees3`.
 - a. Click the arrow in the **SQL Workshop** pull-down menu and select **SQL Scripts**.



- b. Click **Upload >**.



- c. Browse and select the `/home/oracle/labs/labs/add_employees3.sql` file and click **Upload**.

Upload Script

* File

Choose File

add_employees3.sql

Script Name

File Character Set

Unicode UTF-8

Upload

- d. Click the **Run** icon in the new row to create the `employees3` table.

Actions										Create
	Edit	Owner	Name	Created	Updated By	Updated	Bytes	Results	Run	
		APEX_DEV	add_employees3.sql	Now	APEX_DEV	Now	1,079	0		

- e. Click **Run Now**.



Run Script

You have requested to run the following script. Please confirm your request.

Script Name	add_employees3.sql
Created	on 04/24/2019 01:12:59 PM by APEX_DEV
Updated	on 04/24/2019 01:12:59 PM by APEX_DEV
Number of Statements	16
Script Size in Bytes	1,079



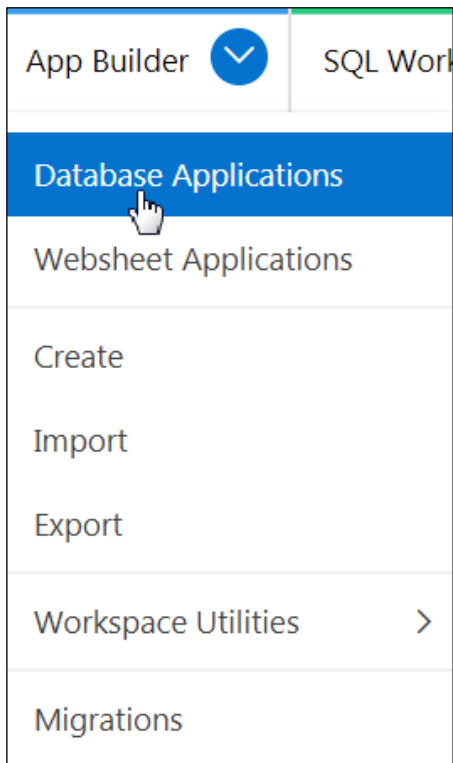
- f. Click the **View Results** icon. You can view the results, as shown in the screenshot here.

Script: **add_employees3.sql** ? Status: **Complete** ?

View: ☐ Detail ☒ Summary ? Rows: 15 ?

Number ↑	Elapsed	Statement	Feedback	Rows
1	0.72	CREATE TABLE OEHR.EMPLOYEES3 AS SELECT * FROM OEHR_EMPLOYEES	Table created.	0
2	0.11	alter table "EMPLOYEES3" modify("COMMISSION_PCT" NUMBER(7,2)	Table altered.	0
3	0.01	update employees3 set commission_pct=1400 where employee_id=	1 row(s) updated.	1
4	0.00	update employees3 set commission_pct=1500 where employee_id=	1 row(s) updated.	1
5	0.00	update employees3 set commission_pct=1200 where employee_id=	1 row(s) updated.	1
6	0.00	update employees3 set commission_pct=1500 where employee_id=	1 row(s) updated.	1
7	0.00	update employees3 set commission_pct=1700 where employee_id=	1 row(s) updated.	1
8	0.00	update employees3 set commission_pct=1600 where employee_id=	1 row(s) updated.	1
9	0.01	update employees3 set commission_pct=1000 where employee_id=	1 row(s) updated.	1

- g. Click the arrow in the **Application Builder** pull-down menu and select **Database Applications**.



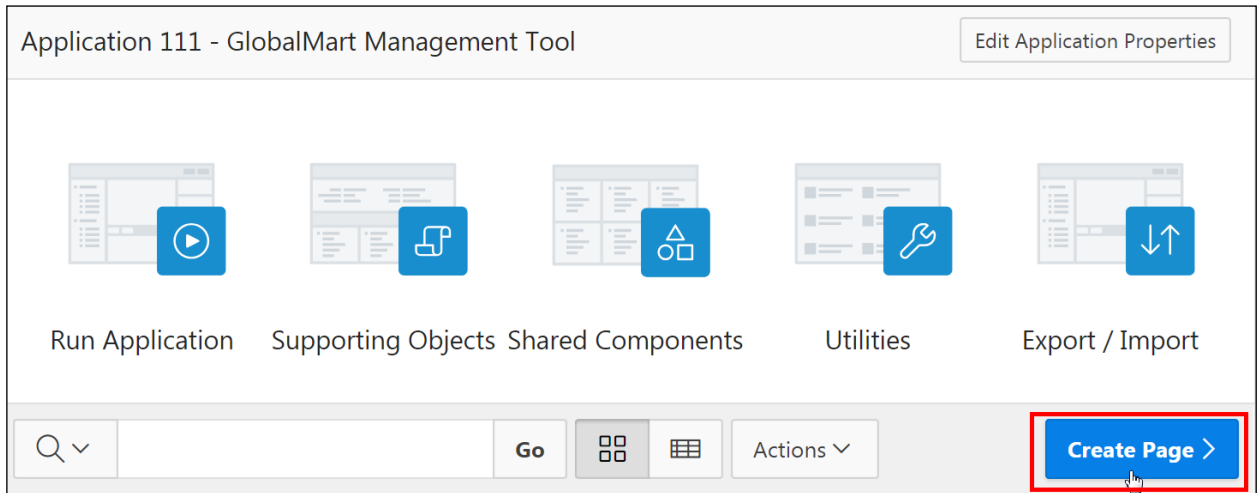
2. Create a Combination chart named **Salary Commission Combined Chart**. Use the following SQL Query (located in the `/home/oracle/labs/labs/lab_19_03_01.txt` file). Specify **Employee** for X-Axis Title and **Salary** for Y-Axis Title. Edit the chart by changing the Series attribute to Line.

```
SELECT NULL LINK,  
       LAST_NAME LABEL,  
       SALARY "Salary",  
       COMMISSION_PCT "COMMISSION"  
FROM EMPLOYEES3  
WHERE MANAGER_ID = 100  
Order by LAST_NAME
```

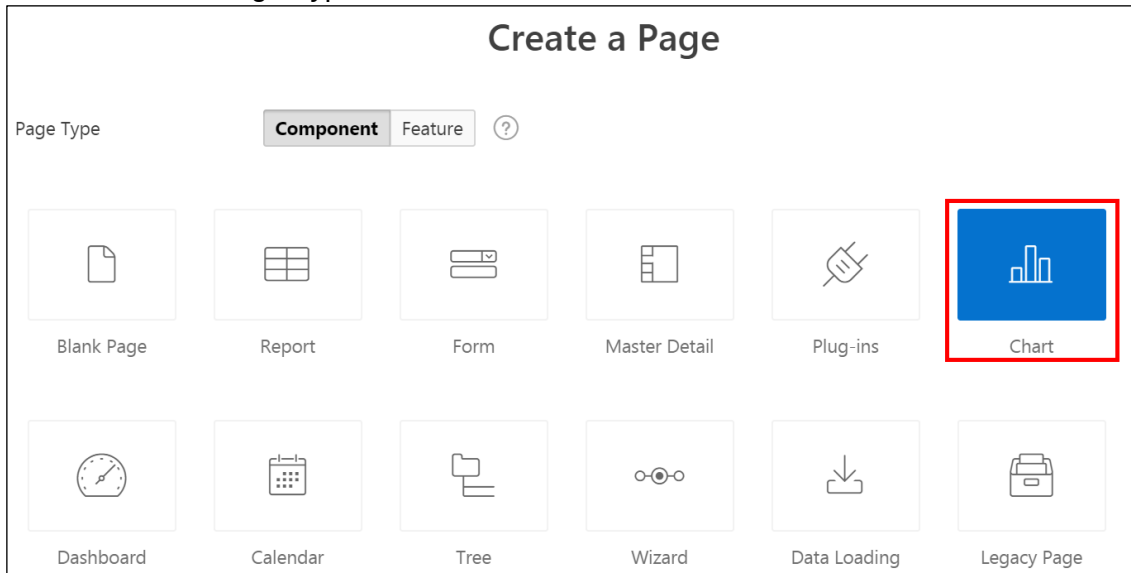
- a. Click the **GlobalMart Management Tool** application icon to navigate to the home page.



- b. In your application, click **Create Page >**.

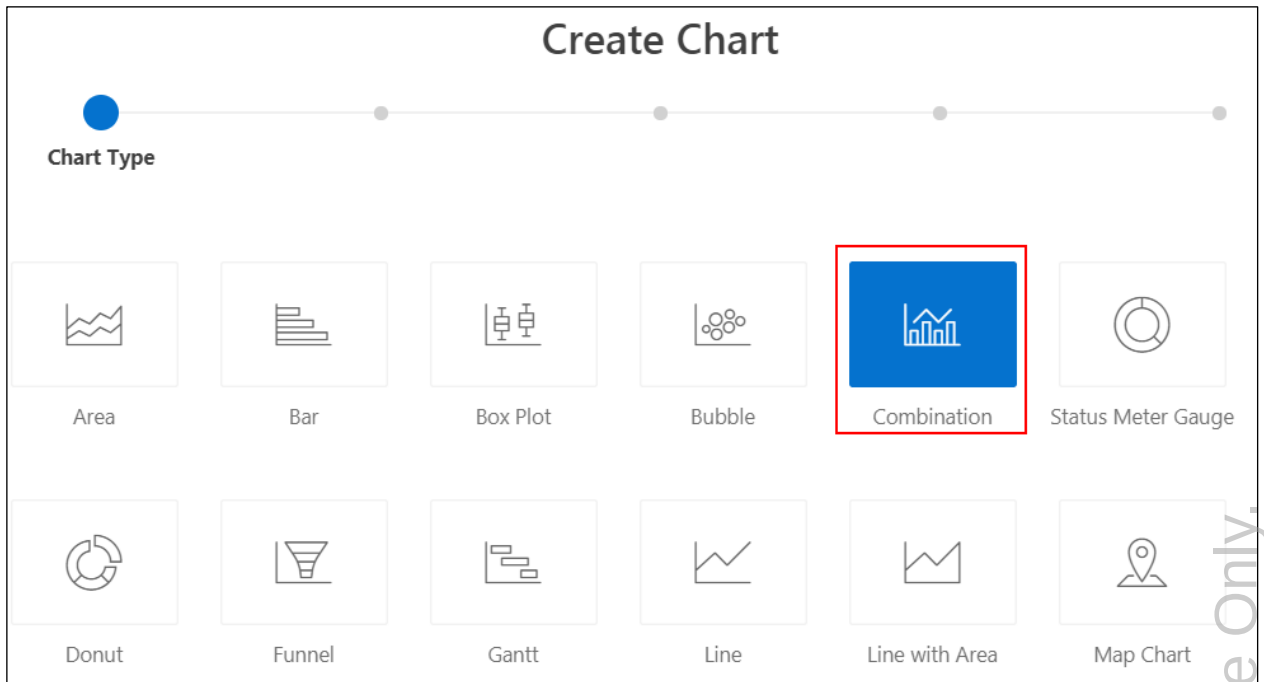


- c. Select Chart for Page Type.



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- d. For Chart Type, select **Combination** and click **Next >**.



- e. Enter 30 for **Page Number** and Salary Commission Combined Chart for **Page Name** and click **Next >**.

The screenshot shows the 'Create Chart' dialog with the 'Page and Region Attributes' step selected. The progress bar at the top now has a green checkmark at the first dot and a blue dot at the second dot (Page and Region Attributes). The 'Page Number' field is set to '30' and the 'Page Name' field contains 'Salary Commission Combined Chart'. Both fields are enclosed in a red rectangular box. The 'Page Mode' is set to 'Normal' and the 'Breadcrumb' is set to '- do not use breadcrumbs on page -'. Each field has a help icon (a question mark in a circle) to its right.

f. For Navigation Menu:

- **Navigation Preference:** Select **Create** a new navigation menu entry.
- **New Navigation Menu Entry**, by default **Salary Commission Combined Chart** is selected.
- **Parent Navigation Menu Entry:** Select **- No parent selected -**.
- Click **Next >**.

The screenshot shows the 'Create Chart' dialog box with the 'Navigation Menu' step selected in the progress bar. The 'Navigation Preference' section has three radio buttons: 'Do not associate this page with a navigation menu entry', 'Create a new navigation menu entry' (which is selected and highlighted with a red box), and 'Identify an existing navigation menu entry for this page'. Below this, the 'New Navigation Menu Entry' section has a text field containing 'Salary Commission Combined Chart' and a dropdown menu for 'Parent Navigation Menu Entry' with the option '- No parent selected -' selected. The 'Next >' button at the bottom right is highlighted with a red box.

g. For data source of this Combination chart, use the following SQL query (located in the /home/oracle/labs/labs/lab_19_03_01.txt) to enter in the SQL Query text area. Click **Next >**.

```
SELECT NULL LINK,  
       LAST_NAME LABEL,  
       SALARY "Salary",  
       COMMISSION_PCT "COMMISSION"  
FROM EMPLOYEES3  
WHERE MANAGER_ID = 100  
Order by LAST_NAME
```

Create Chart

✓

✓

✓

Source

Location Local Database

Source Type Table **SQL Query**

* SQL Query

↶ ↷ 🔍 ↔ ⬆ A· ✓ ⚙

Validation successful

```
1 SELECT NULL LINK,  
2 LAST_NAME LABEL,  
3 SALARY "Salary",  
4 COMMISSION_PCT "COMMISSION"  
5 FROM EMPLOYEES3  
6 WHERE MANAGER_ID = 100  
7 Order by LAST_NAME  
8
```

Page Items to Submit ⋮

Maximum Rows

< Cancel Next >

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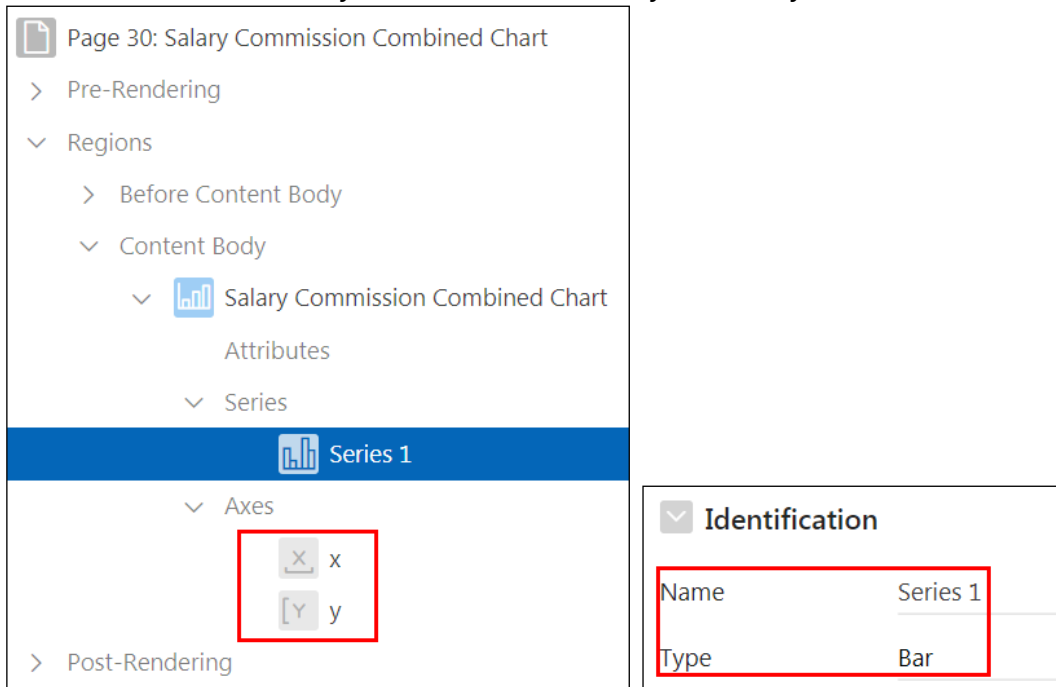
h. For Column Mapping, enter the following:

- **Orientation:** Select `Vertical`.
- **Label Column:** Select `LABEL`. This is the value for the x axis of the chart.
- **Value Column:** Select `Salary`. This is the value for the y axis of the chart.
- Click **Create**.

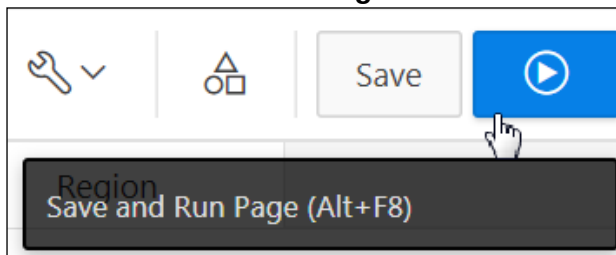
The screenshot shows a 'Create Chart' dialog box with a progress bar at the top. The progress bar has four green checkmarks and one blue circle labeled 'Column Mapping'. The 'Chart Type' is set to 'Combination'. The 'Orientation' is set to 'Vertical'. The 'Label Column' is set to 'LABEL'. The 'Value Column' is set to 'Salary'. The 'Create' button is highlighted with a red box. The 'Cancel' button is also visible.

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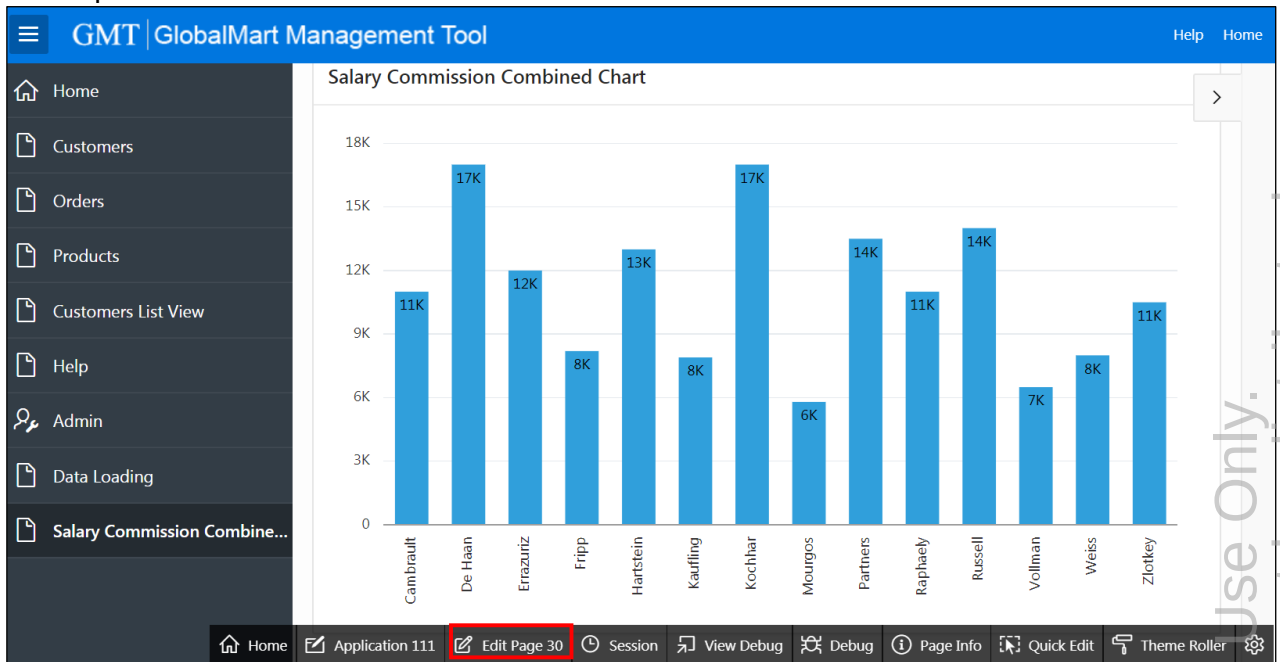
- i. The Combination chart is created, and the page opens in Page Designer. Go to the Rendering tab in Page Designer. Observe that only one series, that is **Series 1**, is created. Click **Series 1** to view its details in Property Editor. It shows that the chart for **Series 1** is a Bar chart. The x-axis and y-axis do not contain any label, as you can see in the chart.



- j. Click the **Save and Run Page** icon.

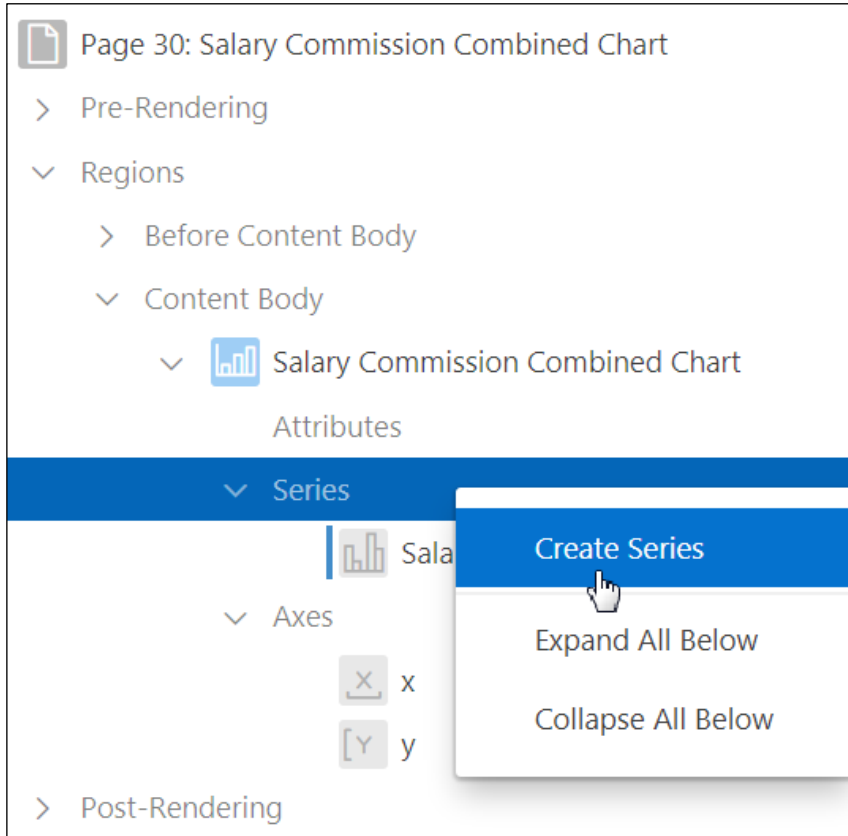


- k. Observe that although you have selected **Combination** chart for Chart Type, while creating, it has only generated a Vertical Bar chart depicting the salary of employees. To create the second chart to be included in the Combination chart, you have to create another series to depict the commission so that both the charts for the two series – salary and commission, are depicted in the same Combination chart. Click **Edit Page 30** on the Developer toolbar.



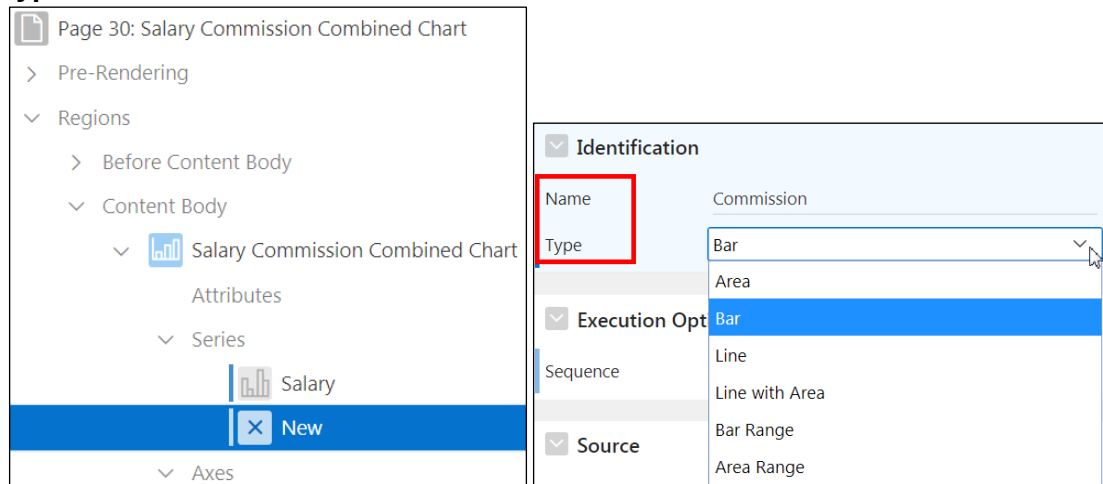
- l. Rename Series 1 to Salary and change the Bar chart to a Line chart. To do so, go to the Rendering tab and click **Series 1**. In the Property Editor, enter `Salary` in the **Name** field to rename the chart. Select **Line** for **Identification > Type**.

- m. Create the second series for a Bar chart to depict the commission. On the Rendering tab, right-click Series and select **Create Series**.



- n. On the Property Editor, under **Identification** enter the following:

- **Name:** Enter Commission
- **Type:** Select Bar



o. For **Source** of the Bar chart, define the following:

- **Type:** Select SQL Query
- **Source:** Enter the following SQL Query (located in the /home/oracle/labs/labs/lab_19_03_02.txt):

```
SELECT NULL LINK,  
       LAST_NAME LABEL,  
       SALARY "Salary",  
       COMMISSION_PCT "COMMISSION"  
FROM EMPLOYEES3  
WHERE MANAGER_ID = 100
```

Source

Location: Local Database

Type: SQL Query

SQL Query

```
SELECT NULL LINK,  
       LAST_NAME LABEL,  
       SALARY "Salary",  
       COMMISSION_PCT "COMMISSION"  
FROM EMPLOYEES3  
WHERE MANAGER_ID = 100
```

p. For **Column Mapping**, define the values for Label and Values as follows:

- **Label:** Select LABEL
- **Value:** Select Commission

Column Mapping

Series Name: - Select -

Label: LABEL

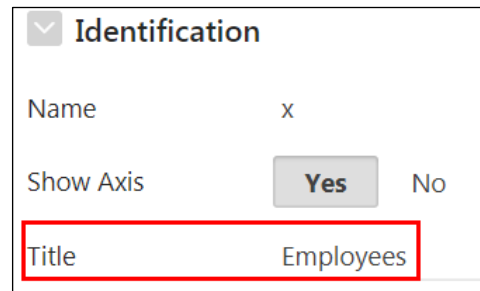
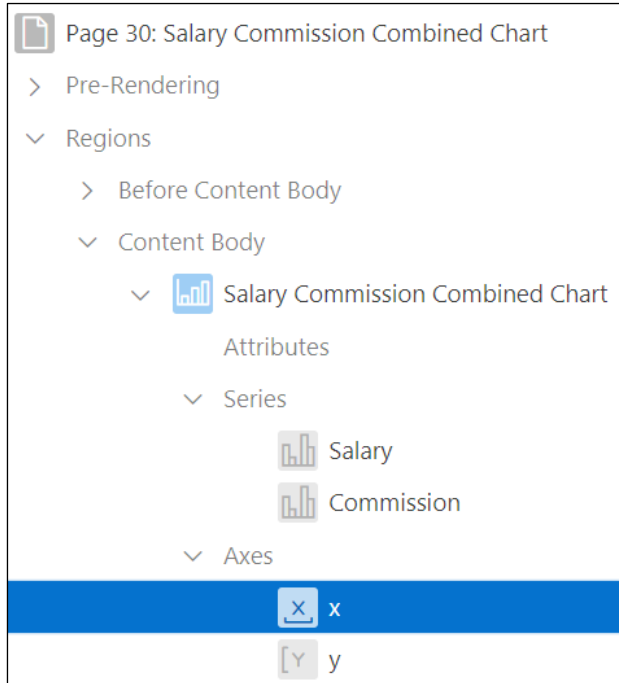
Value: COMMISSION

Z: - Select -

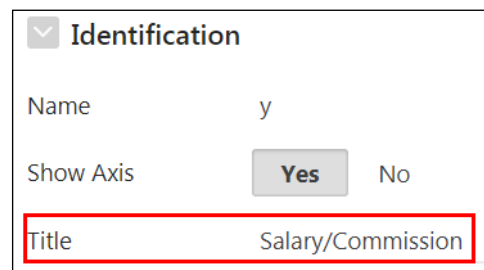
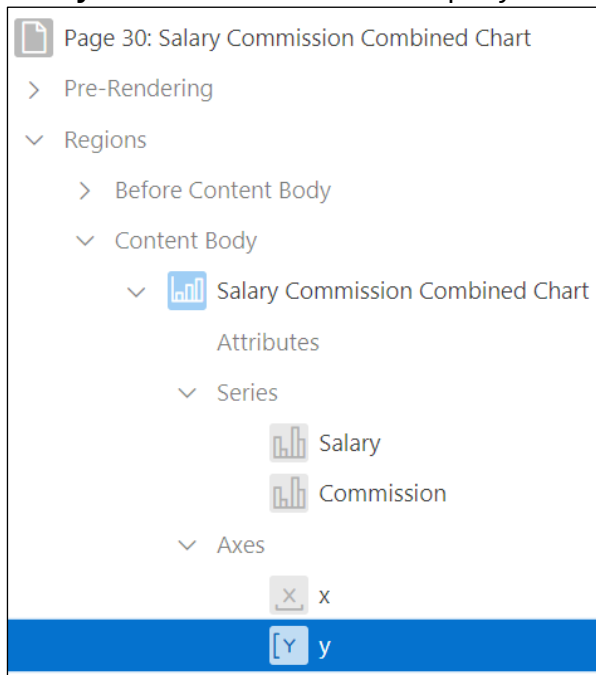
Custom Tooltip: - Select -

q. Now, go to the Rendering tab to name the x-axis and y-axis as **Employees** and **Salary/Commission**, respectively.

- Click **x**. Under Identification in Property Editor, enter **Employees** for Title.

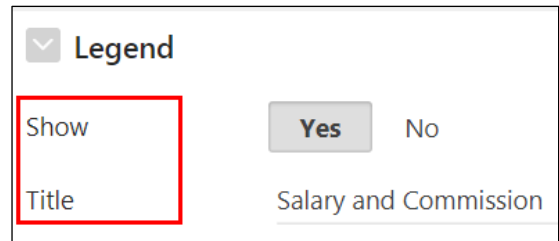
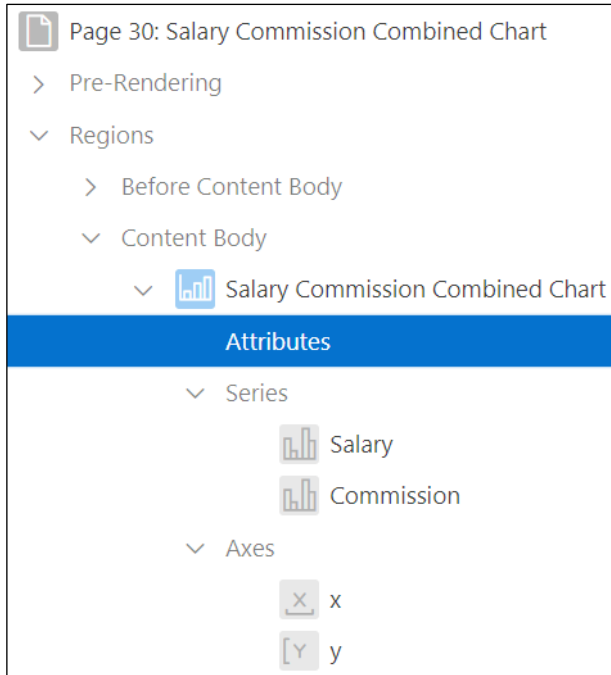


- Click **y**: Under Identification in Property Editor, enter **Salary/Commission** for Title.

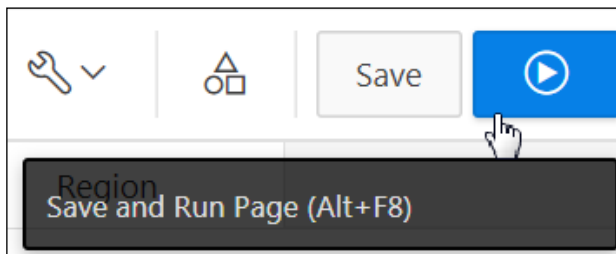


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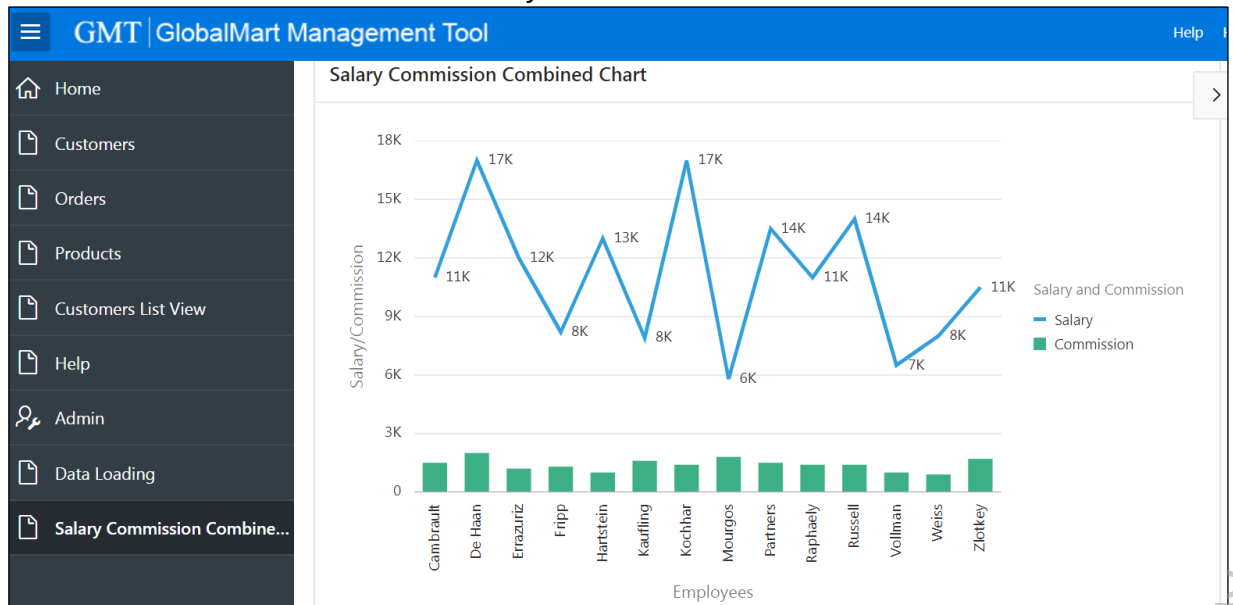
- r. In the Rendering tab, click **Attributes**. On the Property Editor, scroll down to **Legend** and click **Yes** for **Show**. For **Title**, enter **Salary and Commission**.



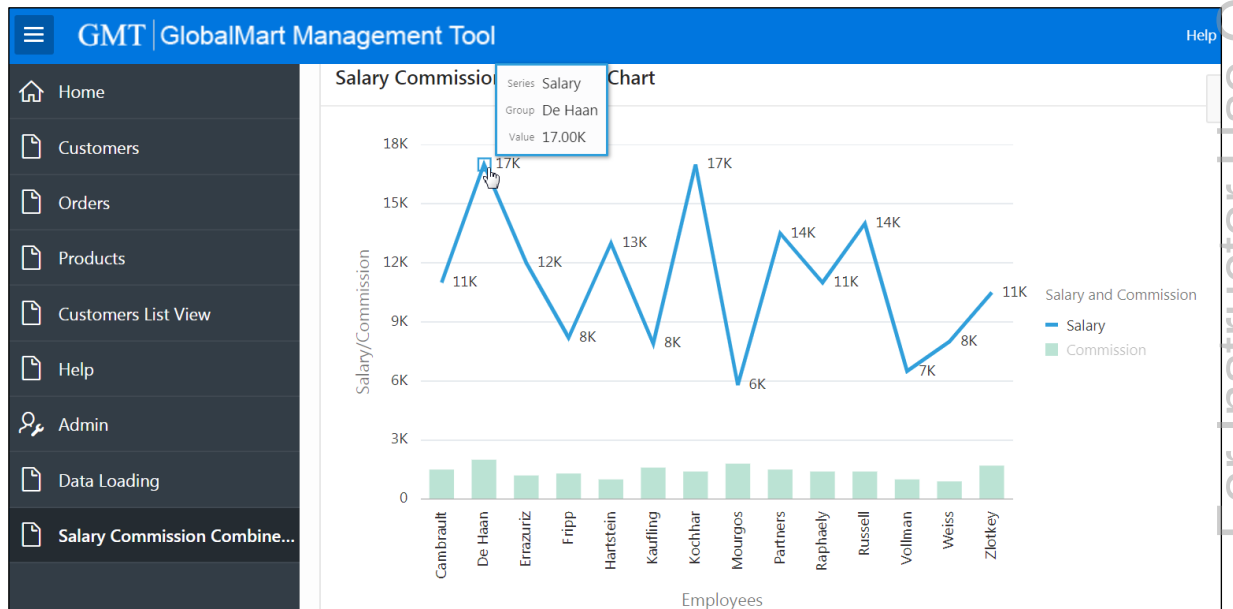
- s. Click the **Save and Run Page** icon.



- t. You can now see a combination of salary as Line and commission as Bar charts.



Place your cursor over one of the points in the Line chart to view the details.



Click **Application 111** on the Developer toolbar.



Practices for Lesson 20: Adding Calendars and Trees

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Practices for Lesson 20: Overview

Practices Overview

In these practices, you will create a calendar page that displays the order total per day, create a calendar for the GMT application, and create a new page with a tree region and link to the Dynamic Employee Details page.

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Practice 20-1: Creating a Calendar

Overview

In this practice, you create a calendar page that displays the order total per day. You also add a link to the page to be able to edit the details of an order.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a new Calendar page called **Order Calendar**. Base the calendar on the `ORDERS` table and set the **Date Column** to `ORDER_DATE` and **Display Column** to `ORDER_TOTAL`. Allow for dragging order total to different dates. Create links to Orders form so that you can create and edit the columns in the `ORDERS` table. Test your pages.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 20-1**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_dev` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).

- h. Click **Install**.
- i. Click **Edit Application**.
- j. On the Application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 19-3**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_19_03.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Note:

- You must also upload and run the `add_employees3.sql` script (browse and select the `/home/oracle/labs/labs/add_employees3.sql` file) to create a table called `employees3` (for running the Combination chart).

Practice 20-2: Creating a Tree

Overview

In this practice, you create a new page with a tree region and link it to the Dynamic Employee Details page. Add a button on the Dynamic Employee Details page to link back to the tree page.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a tree page called **Employees by Manager** that uses the default tree template. Select the appropriate options in the wizard so that the current query (located in `/home/oracle/labs/labs/lab_20_02.txt`) looks as follows:

```
select case when connect_by_isleaf = 1 then 0
         when level = 1 then 1
         else -1
         end as status,
         level,
         "LAST_NAME" as title,
         null as icon,
         "EMPLOYEE_ID" as value,
         null as tooltip,
         null as link
from "OEHR"."EMPLOYEES"
start with "MANAGER_ID" is null
connect by prior "EMPLOYEE_ID" = "MANAGER_ID"
order siblings by "LAST_NAME"
```

2. Add a button to the **Employee by Department** page called **Employee Hierarchy** that navigates to the **Employees by Manager** tree page.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.

- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 20-2**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. On the Application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.
- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 20-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.

- c. Click the **Browse** button.
- d. Browse and open the `sol_20_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Note:

- You must also upload and run the `add_employees3.sql` script (browse and select the `/home/oracle/labs/labs/add_employees3.sql` file) to create a table called `employees3` (for running the Combination chart).

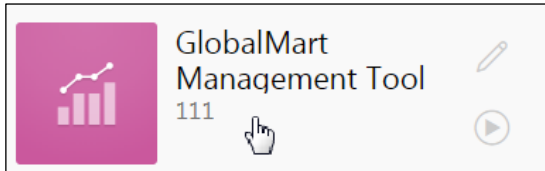
Solution 20-1: Creating a Calendar

Overview

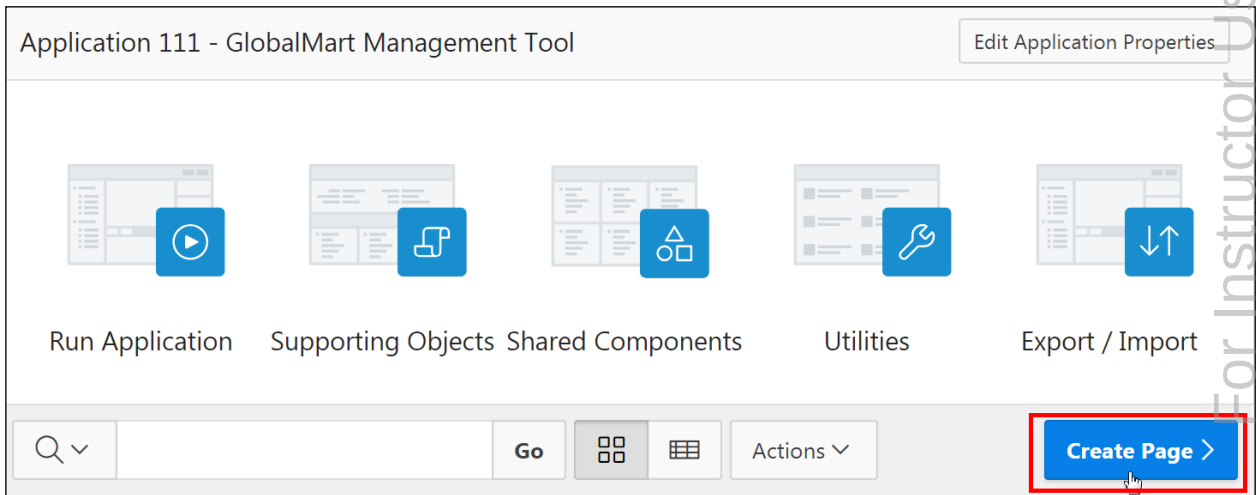
In this practice solution, steps are provided for creating a calendar for the GlobalMart Management Tool application.

Steps

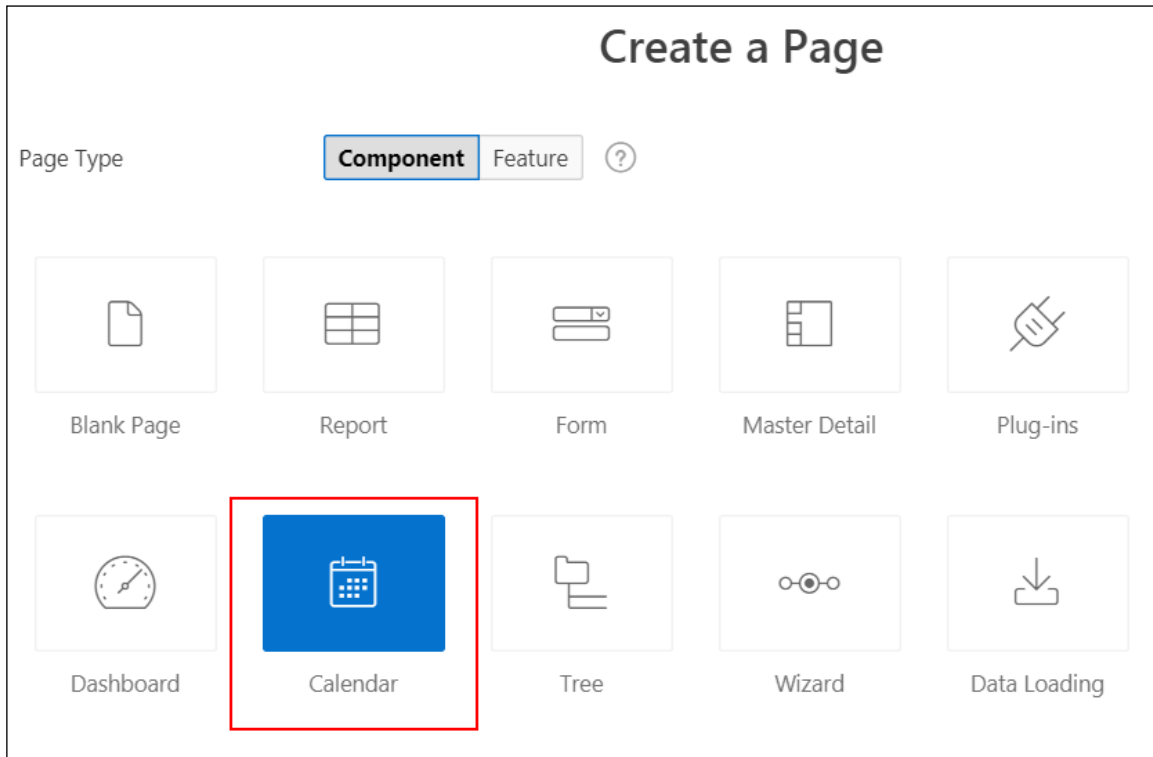
1. Create a Calendar page called **Order Calendar**. Create the calendar based on the `OEHR_ORDERS` table and set the **Date Column** to `ORDER_DATE` and **Display Column** to `ORDER_TOTAL`. Allow for dragging order total to different dates. Create links to the Orders form page so that you can create new entries and edit the columns in the `OEHR_ORDERS` table. Test your pages.
 - a. Select the **GlobalMart Management Tool** application.



- b. Click **Create Page >**.



- c. Select **Calendar** for page type. Click **Next >**.



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d. In the Page Attributes dialog, define the following and click **Next >**:

- **Page Number:** Enter 31
- **Page Name:** Enter Orders Calendar
- **Breadcrumb:** Select **Breadcrumb**
- **Parent Entry:** Select **Home (page 1)**
- **Entry Name:** **Orders Calendar** (selected by default)

The screenshot shows the 'Create Page' dialog with the 'Page Attributes' tab selected. The 'Region Type' is set to 'Calendar'. The 'Page Number' is 31, 'Page Name' is 'Orders Calendar', 'Page Mode' is 'Normal', 'Page Group' is '- Select Page Group -', 'Breadcrumb' is 'Breadcrumb', 'Parent Entry' is 'Home (Page 1)', and 'Entry Name' is 'Orders Calendar'. Red boxes highlight the 'Page Number', 'Page Name', 'Breadcrumb', 'Parent Entry', and 'Entry Name' fields.

e. For Navigation Preference, define the following and click **Next >**.

- **Navigation Preference:** Click **Create a new navigation menu entry**
- **New Navigation Menu Entry:** **Orders Calendar** (selected by default)
- **Parent Navigation Menu Entry:** Select **Home**

Create Page

✓ ✓ Navigation Menu

Navigation Preference ?

☐ Do not associate this page with a navigation menu entry
☒ **Create a new navigation menu entry**
☐ Identify an existing navigation menu entry for this page

* New Navigation Menu Entry ?

Orders Calendar

Parent Navigation Menu Entry ?

- No parent selected -
Home
 Customers
 Orders
 Products
 Customers List View
 Help

- f. Select **Source Type** as **Table**. From the **Table / View Name** drop-down list, select **OEHR_ORDERS (table)**. By default, all the columns are selected. Click **Next >**.

Create Page

✓ ✓ ✓ Source

Data Source ?

Local Database

Source Type ?

☒ **Table** ☐ SQL Query

* Table / View Owner ?

OEHR

* Table / View Name ?

OEHR_ORDERS (table)

* Select Columns ?

⇐
ORDER_ID (Number)

>>
ORDER_DATE (Timestamp(6) With Local Time Zone)

>
ORDER_MODE (Varchar2)

>
CUSTOMER_ID (Number)

<
ORDER_STATUS (Number)

<
ORDER_TOTAL (Number)

<<
SALES_REP_ID (Number)

PROMOTION_ID (Number)

↑
↑
↓
↓

> **SQL Examples**

< Cancel
Next >

- g. Select `ORDER_TOTAL` for **Display Column** and ensure `ORDER_DATE` is selected for **Start Date Column**. Click **Create**.

Create Page

Display Column: ORDER_TOTAL

Start Date Column: ORDER_DATE

End Date Column: - Select -

Show Time: No

- h. The Order Calendar is created, and it opens in Page Designer. This completes the task of creating the Order Calendar based on the `OEHR_ORDERS` table and setting the **Display Column** and the **Start Date Column** to `ORDER_TOTAL` and `ORDER_DATE`, respectively.
- i. Now, in Page Designer, define the attributes for Primary Key: Click **Attributes** under Order Calendar on the Rendering tab. In the Property Editor on the right pane, under **Settings**, set the **Primary Key Column** to `ORDER_ID`.

Page 31: Orders Calendar

Attributes

Settings

Display Column: ORDER_TOTAL

Start Date Column: ORDER_DATE

End Date Column: - Select -

Primary Key Column: ORDER_ID

Show Time: Yes No

- j. To enable drag and drop of calendar entries, select **Attributes** on Rendering tab. On the right pane under **Settings**, click **Yes** for Drag and Drop and enter the following code, (located in `/home/oracle/labs/labs/lab_20_01.txt`):

```
BEGIN
  UPDATE "OEHR_ORDERS"
    SET "ORDER_DATE" = to_date(:APEX$NEW_START_DATE,
'YYYYMMDDHH24MISS')
    where "ORDER_ID" = :APEX$PK_VALUE;
END;
```

Additional Calendar Views

☒ List

☒ Navigation

Drag and Drop **Yes** No

Drag and Drop PL/SQL Code

```

BEGIN
  UPDATE "OEHR_ORDERS"
  SET "ORDER_DATE" = to_date(:APEX$NEW_START_DATE, 'YYYYMMDDHH24MISS')
  where "ORDER_ID" = :APEX$PK_VALUE;
END;

```

Create Link No Link Defined

- k. To define links to Orders form page in your calendar, under Settings, click **No Link Defined under Create Link.**

Create Link No Link Defined

View / Edit Link No Link Defined

- l. In the Link Builder - Create Link dialog, define the following:
- **Type:** Select **Page in this application**
 - **Page:** Select page 13 (Oehr Orders page)
 - Under **Set Items**, select P13_ORDER_ID for **Name**.
 - Click **OK**.

Link Builder - Create Link

Target

Type: Page in this application

Page: 13

Set Items

Name	Value
P13_ORDER_ID	&ORDER_ID

Clear Session State

Clear Cache

Reset Pagination: Yes No

Cancel Clear **OK**

- m. To define links to Orders form page in your calendar, under Settings, click **No Link Defined** under **View / Edit Link**.

Create Link

Page 13

View / Edit Link

No Link Defined

Maximum Events / Day

No Link Defined

- n. In the Link Builder – View/Edit Link dialog, define the following:
- **Type:** Select **Page in this application**
 - **Page:** Select page 13 (Oehr Orders page)
 - Under Set Items, select P13_ORDER_ID for **Name** and select &ORDER_ID for **Value**.

- Click **OK**.

Link Builder - View / Edit Link

☒ **Target**

Type Page in this application

Page 13

☒ **Set Items**

Name	Value
P13_ORDER_ID	&ORDER_ID.

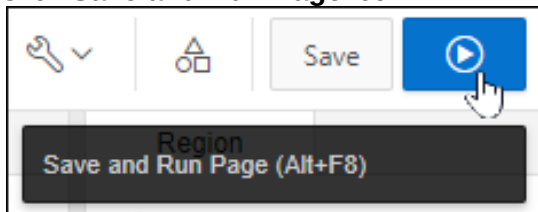
☒ **Clear Session State**

Clear Cache

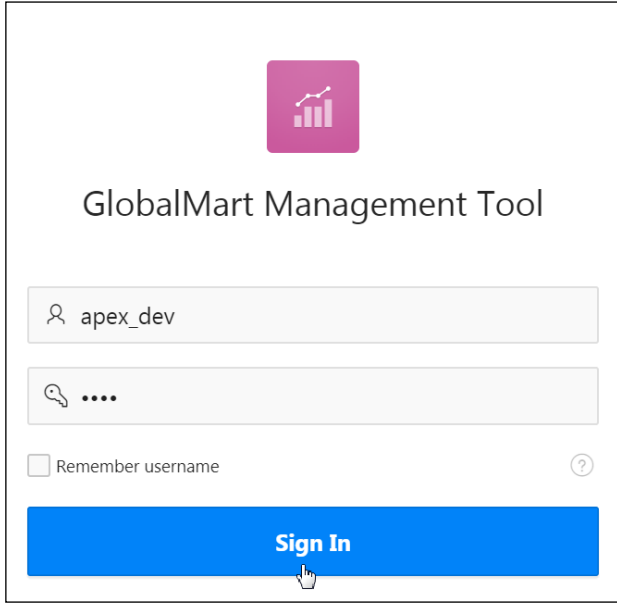
Reset Pagination Yes No

Cancel Clear **OK**

- o. Click **Save and Run Page** icon.

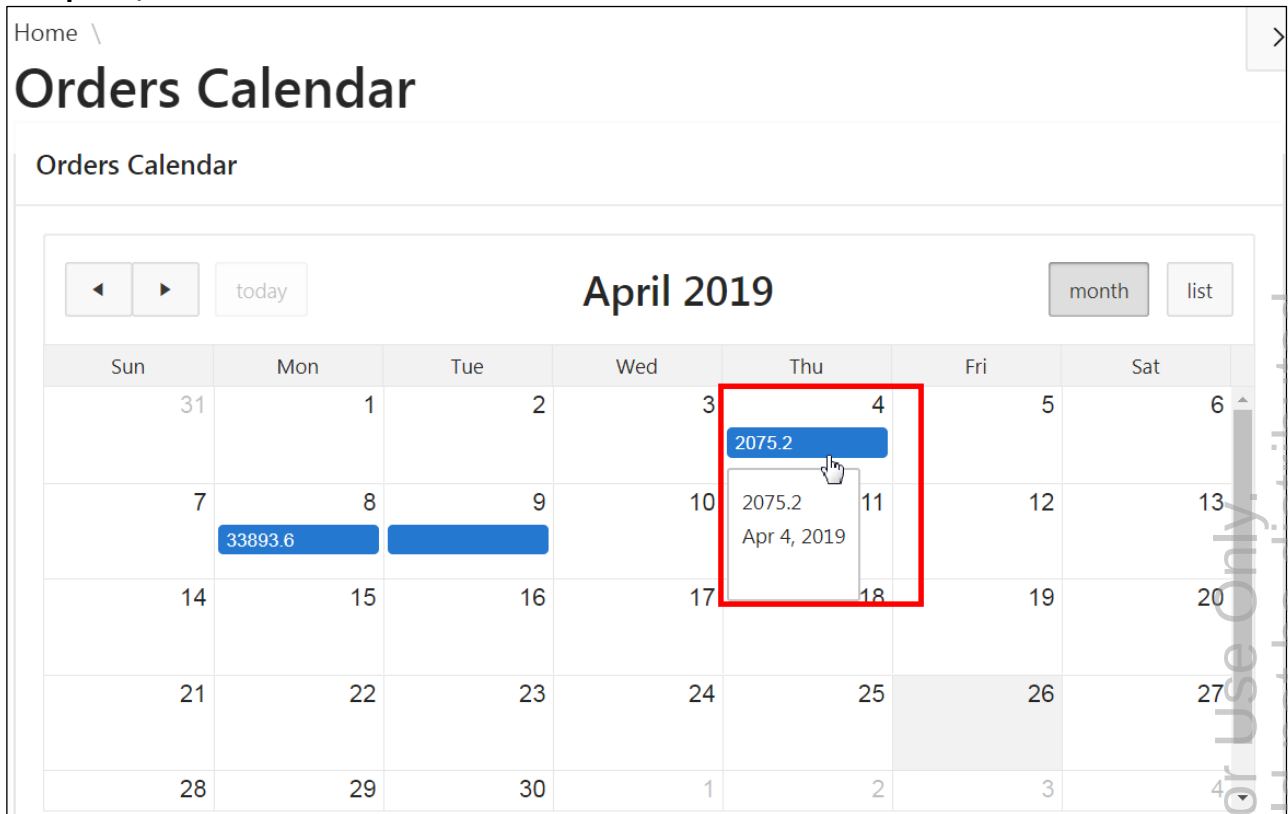


- p. If prompted, enter your developer credentials.



The image shows a login interface for the 'GlobalMart Management Tool'. At the top is a purple square icon with a white bar chart. Below the icon is the title 'GlobalMart Management Tool'. There are two input fields: the first contains the username 'apex_dev' and has a magnifying glass icon; the second contains masked characters '....' and has a key icon. Below these fields is a checkbox labeled 'Remember username' and a help icon (a circle with a question mark). At the bottom is a large blue button with the text 'Sign In' in white. A mouse cursor is pointing at the 'Sign In' button.

- q. The Orders Calendar that you created now opens. Find a month where there is an order total listed for a particular day (for example: April 2019). Click the amount link for the entry on **April 4, 2019**.



- r. Change the Order Date to a different date in the same month by using the **Date Picker**. For example, change the order date to April 10, 2019, and click **Apply Changes**.

Home \

Oehr Orders

Oehr Orders

Order Date
10-Apr-19

Order Mode
direct

Customer Id
106

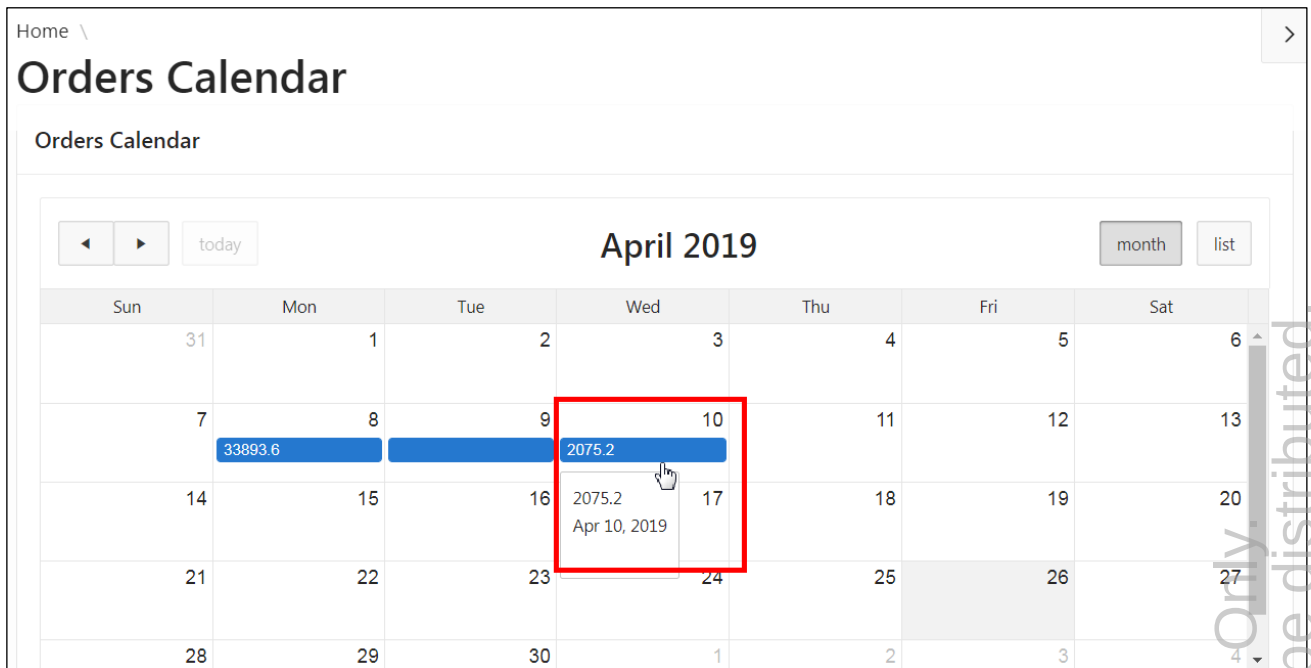
Order Status
5

Order Total
2075.2

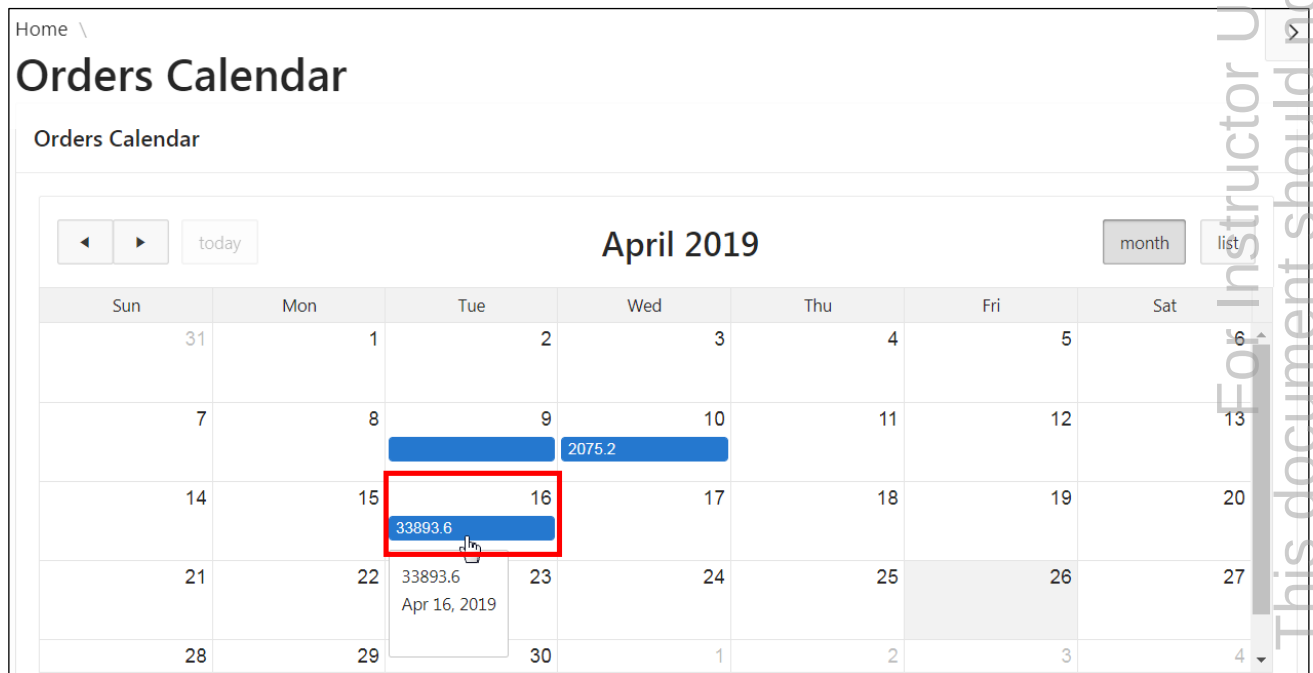
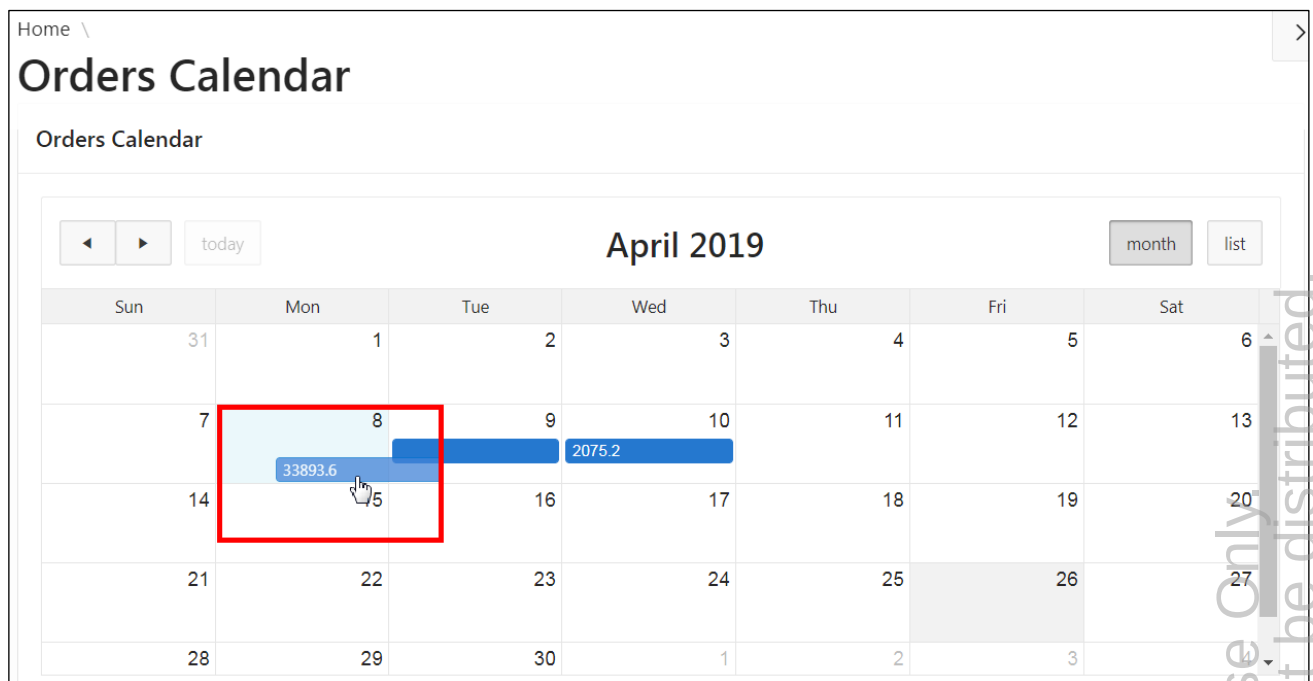
Sales Rep Id
160

Cancel Delete Apply Changes

- s. The change is applied. Note that the Order Total has moved to the date you changed it to, that is, to **April 10, 2019**.



- t. You can also drag an order total to a different day. Drag the order total from April 8, 2019, to April 16, 2019. Note that after you release your mouse, the order total moves to the new location.



- u. Click the **Application 111** link on the Developer toolbar.



To learn more about what you can do with calendaring, install the Sample Calendars packaged application.

Solution 20-2: Creating a Tree

Overview

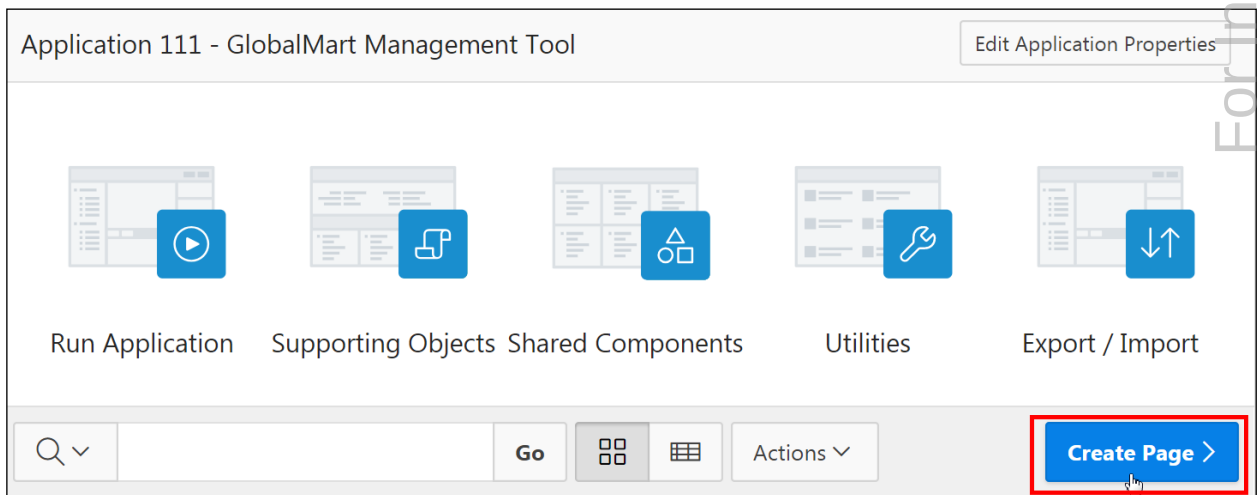
In this practice solution, steps are provided for creating a new page with a tree region and linking it to the Dynamic Employee Details page in the GMT application.

Steps

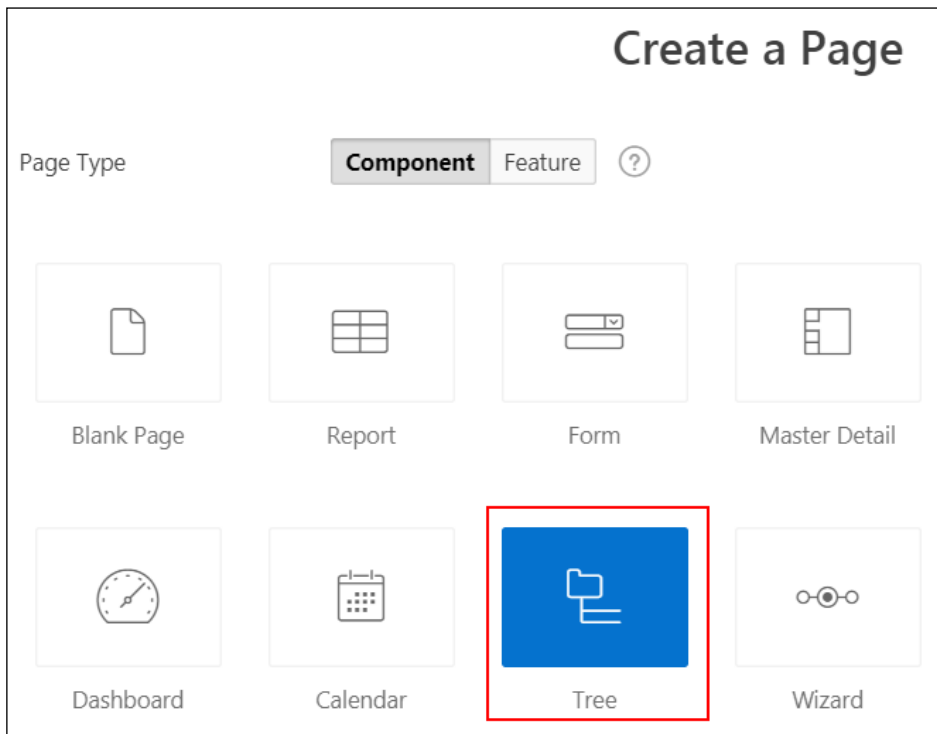
1. Create a tree page called **Employees by Manager** that uses the default tree template. Select the appropriate options in the wizard so that the current query (located in /home/oracle/labs/labs/lab_20_02.txt) looks as follows:

```
select case when connect_by_isleaf = 1 then 0
      when level = 1          then 1
      else                    -1
end as status,
level,
"LAST_NAME" as title,
null as icon,
"EMPLOYEE_ID" as value,
"FIRST_NAME" as tooltip,
null as link
from "OEHR"."OEHR_EMPLOYEES"
start with "MANAGER_ID" is null
connect by prior "EMPLOYEE_ID" = "MANAGER_ID"
order siblings by "LAST_NAME"
```

- a. In the GMT application home page, click **Create Page >**.



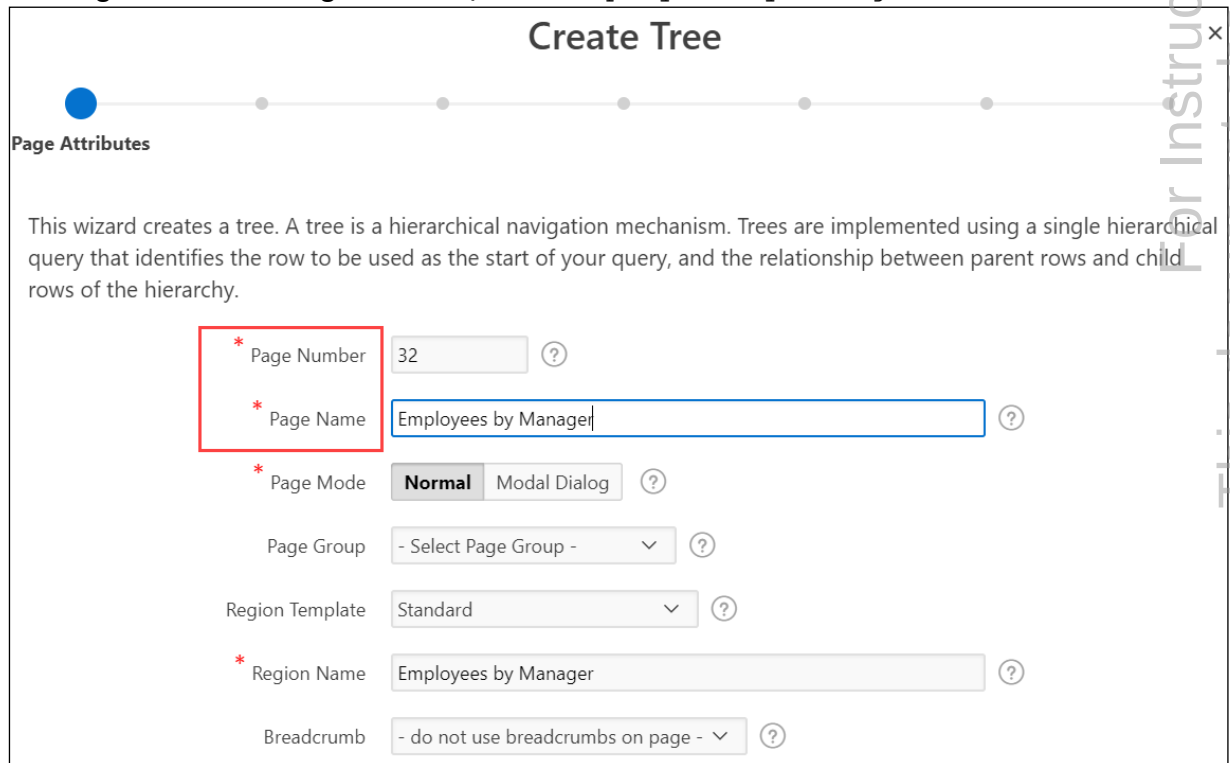
- b. Select the **Tree** option and click **Next>**.



The "Create a Page" wizard screen shows a "Page Type" section with two tabs: "Component" (selected) and "Feature". Below the tabs are eight icons representing different page types: Blank Page, Report, Form, Master Detail, Dashboard, Calendar, Tree, and Wizard. The "Tree" icon, which depicts a folder structure, is highlighted with a red rectangular box.

- c. In the Page Attribute section of the wizard, enter the following:

- In the **Page Number** field, the applicable page number is allotted as applicable. In this example, it is 32.
- For **Page Name** and **Region Name**, enter `Employees by Manager`.



The "Create Tree" wizard screen shows a progress bar at the top with five steps. The first step, "Page Attributes", is active. Below the progress bar is a descriptive paragraph: "This wizard creates a tree. A tree is a hierarchical navigation mechanism. Trees are implemented using a single hierarchical query that identifies the row to be used as the start of your query, and the relationship between parent rows and child rows of the hierarchy." Below this text are several input fields, each with a red asterisk indicating it is required. The "Page Number" field contains the value "32". The "Page Name" field contains the text "Employees by Manager". The "Page Mode" field has two options: "Normal" (selected) and "Modal Dialog". The "Page Group" field is a dropdown menu with the text "- Select Page Group -". The "Region Template" field is a dropdown menu with the text "Standard". The "Region Name" field contains the text "Employees by Manager". The "Breadcrumb" field is a dropdown menu with the text "- do not use breadcrumbs on page -".

- d. Click **Next >**.

e. For navigation preference:

- For **Navigation Preference**, select **Create a new navigation menu entry**.
- For **New Navigation Menu Entry**, ensure that the entry `Employees by Manager` is selected.
- For **Parent Navigation Menu Entry**, select **Home**.

Create Tree

Navigation Menu

Navigation Preference

- ☐ Do not associate this page with a navigation menu entry
- ☒ **Create a new navigation menu entry**
- ☐ Identify an existing navigation menu entry for this page

* New Navigation Menu Entry: Employees by Manager

Parent Navigation Menu Entry: - No parent selected -
Home
... (Orders Calendar)
Customers
Orders
Products
Customers List View

- Click **Next >**.

f. For Table/View Owner Name, select `OEHR_EMPLOYEES` (table) and click **Next >**.

Create Tree

Table / View Owner and Name

Select the owner of the table or view from which you want to draw the tree query.

* Table / View Owner: OEHR

* Table / View Name: OEHR_EMPLOYEES (table)

- g. For Query, select the following values for the respective fields (some are populated by default) and click **Next >**:
- **ID:** Select `EMPLOYEE_ID` (Number)
 - **Parent ID:** Select `MANAGER_ID` (Number)
 - **Node Text:** Select `LAST_NAME` (Varchar2)
 - **Start With:** Select `MANAGER_ID` (Number)
 - **Start Tree:** Value is NULL

Create Tree

✓

✓

✓

●

Query

A tree is based on a query and returns data that can be represented in a hierarchy. A **start with .. connect by** clause will be used to generate the hierarchical query for your tree. Use this page to identify the column you want to use as the ID, the Parent ID, and text that should appear on the nodes. The Start With column will be used to specify the root of the hierarchical query, and its value can be based on an existing item, static value or SQL query returning a single value.

*

ID

EMPLOYEE_ID (Number)

▼

?

*

Parent ID

MANAGER_ID (Number)

▼

?

*

Node Text

LAST_NAME (Varchar2)

▼

?

*

Start With

MANAGER_ID (Number)

▼

?

*

Start Tree

Value is NULL

▼

?

> Example Start With Query

- h. You can see the query that is generated by expanding the **Current Query** tab in the Where and Order by section of the wizard. Note that in the **Order Siblings By** field, `LAST_NAME` (Varchar2) is selected. Click **Next >**.

Create Tree

Order Siblings By (for example ENAME) `LAST_NAME (Varchar2)` ?

Current Query

```
select case when connect_by_isleaf = 1 then 0
           when level = 1 then 1
           else -1
end as status,
level,
"LAST_NAME" as title,
null as icon,
"EMPLOYEE_ID" as value,
null as tooltip,
null as link
from "OEHR"."OEHR_EMPLOYEES"
start with "MANAGER_ID" is null
connect by prior "EMPLOYEE_ID" = "MANAGER_ID"
order siblings by "LAST_NAME"
```

< Cancel **Next >**

- i. In the Tree Attributes section, select Database Column for **Tooltip** and FIRST_NAME (Varchar2) for **Tooltip Column**.

Create Tree

Identify the button, tooltip and link attributes you want to define on your tree. To make leaf node text a link, select **Existing Application Item**.

Include Buttons: ☒ **Collapse All** ☒ **Expand All** (?)

Selected Node Page Item: (?)

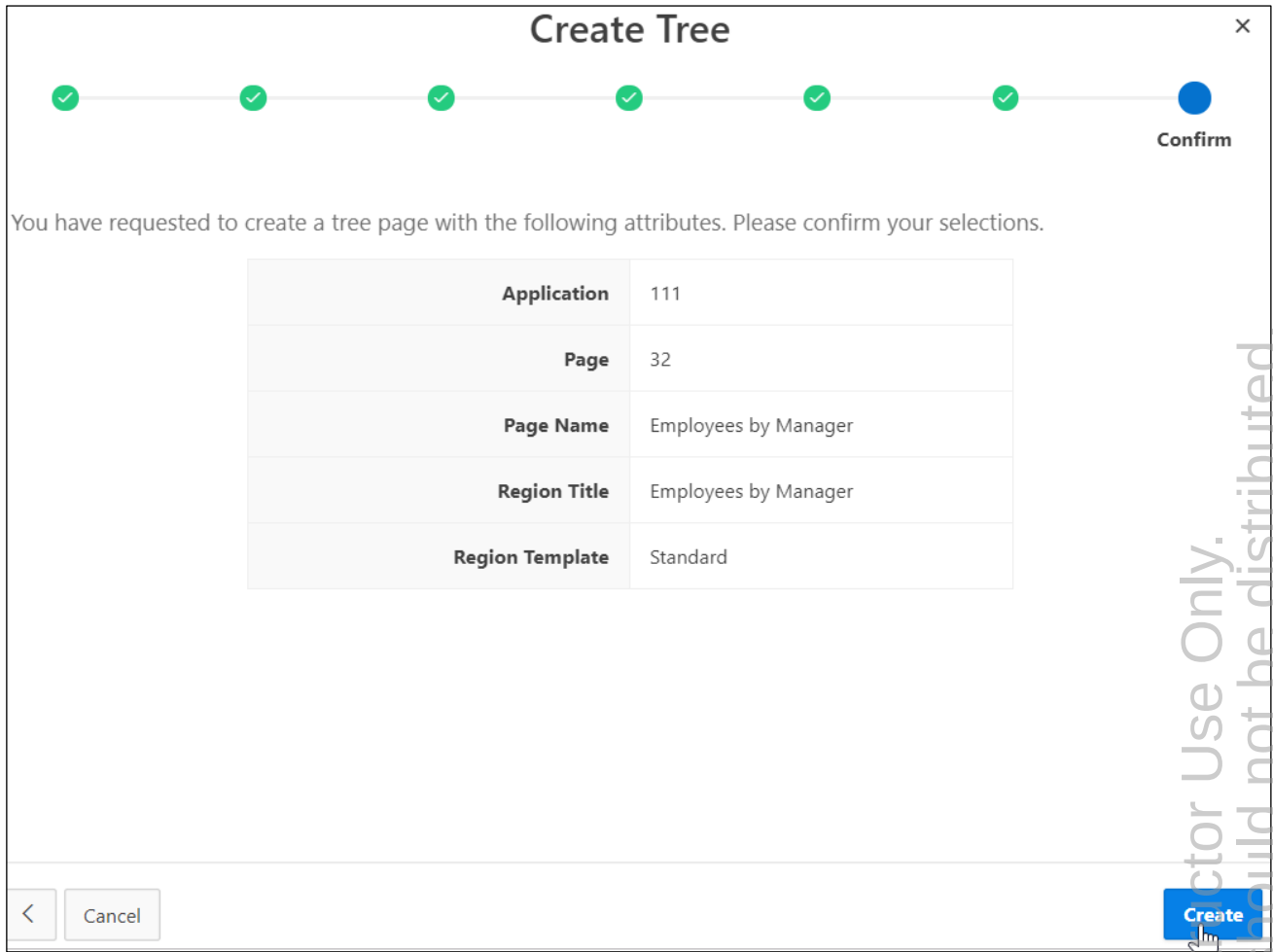
Tooltip: Database Column (?)

* Tooltip Column: FIRST_NAME (Varchar2) (?)

Link Option: ☒ **Nothing** (?) ☐ Existing Application Item

- j. Click **Next >**.

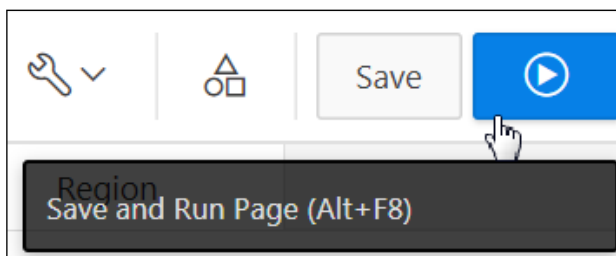
- k. In the Confirm section of the wizard, click **Create**.



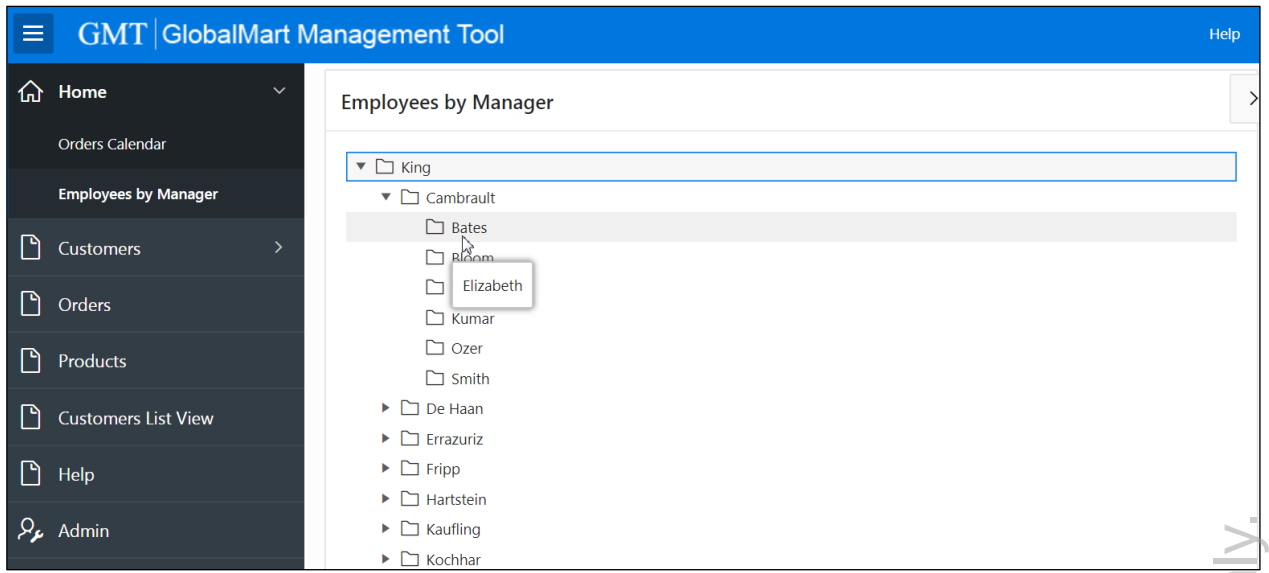
The 'Create Tree' wizard is shown at the 'Confirm' step. A progress bar at the top has six green checkmarks followed by a blue circle labeled 'Confirm'. Below the progress bar, a message states: 'You have requested to create a tree page with the following attributes. Please confirm your selections.' A table lists the attributes: Application (111), Page (32), Page Name (Employees by Manager), Region Title (Employees by Manager), and Region Template (Standard). At the bottom, there are buttons for '< Cancel' and 'Create'.

Application	111
Page	32
Page Name	Employees by Manager
Region Title	Employees by Manager
Region Template	Standard

- l. The page for the Employees by Manager tree opens in Page Designer. Click the **Save and Run Page** icon.



- m. Expand one of the manager nodes. Notice that when you place your cursor over one of the names, you see the `FIRST_NAME` of the person as a tooltip.

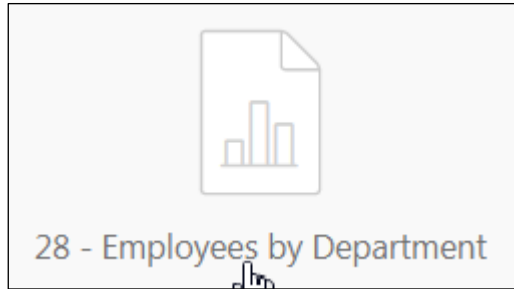


- n. Click the **Application 111** link on the Developer toolbar.

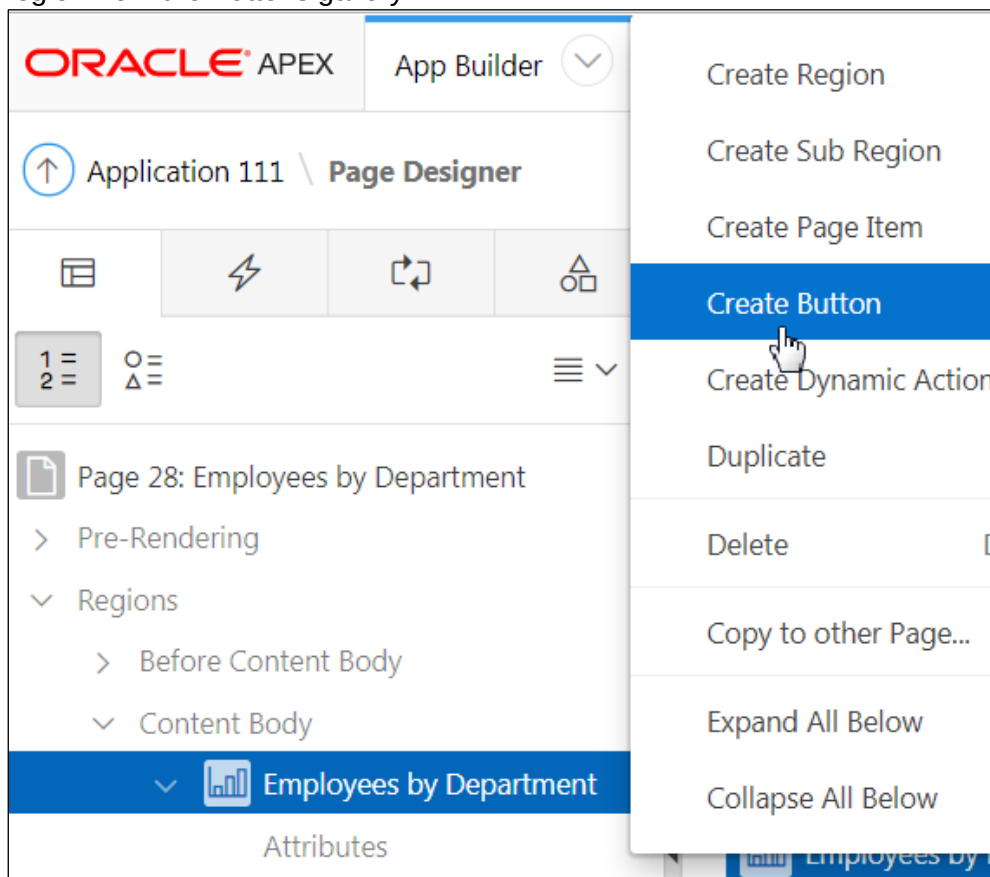


2. Add a button to the **Employee by Department** page called **Employee Hierarchy** that navigates to the **Employees by Manager** tree page.

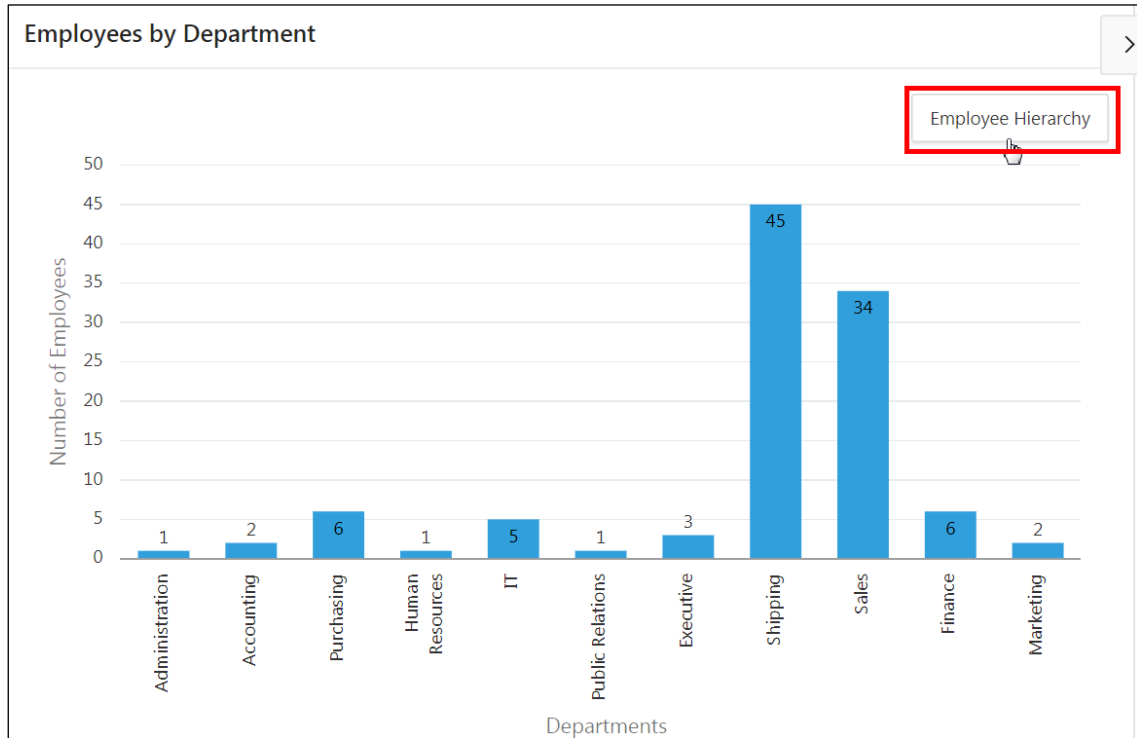
- a. Click the Chart page **28- Employees by Department**.



- b. In the Rendering tab, right-click **Employees by Department** region and click **Create Button**. Alternatively, you can drag a HTML button into the Employees by Department region from the Buttons gallery.



- e. The **Employees by Department** page now has the **Employee Hierarchy** button. Click **Employee Hierarchy** in the **Employees by Department** chart.



- f. It opens the **Employees By Manager** tree under **Home**. Click the **Application 111** link on the **Developer** toolbar.

GMT | GlobalMart Management Tool

Home (selected)

- Orders Calendar
- Employees by Manager**
- Customers
- Orders
- Products
- Customers List View
- Help

Employees by Manager

- King
 - Cambraut
 - Bates
 - Bloom
 - Fox
 - Kumar
 - Ozer
 - Smith
 - De Haan
 - Errazuriz
 - Fripp
 - Hartstein

Developer Toolbar: Home, **Application 111** (highlighted), Edit Page 32, Session, View Debug, Debug, Page Info, Quick Edit, Theme Roller, Settings.

**Practices for Lesson 21:
Managing Application Feedback**

Practices for Lesson 21: Overview

Practices Overview

There is one practice in this lesson. In this practice, you will create a feedback form in your application, submit feedback, and review and categorize the feedback.

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Practice 21-1: Adding and Monitoring Feedback in Your Application

Overview

In this practice, you create a feedback form in your application, submit feedback, and review and categorize the feedback.

Note: The solution script (`sol_21_01.sql`) generated at the end of this practice is just a starting point for this practice and not the solution.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a feedback form for your GMT application.
2. Create the following feedback from your application:

Feedback #	Feedback Text	Experience	Feedback Type
1.	The date format on the Orders form needs to be changed from <code>dd-mm-yy</code> to <code>mm-dd-yyyy</code>	Neutral	General Comment
2.	Excellent application. Very user-friendly and intuitive UI.	Positive	General comment
3.	Error when deleting a line item on an order	Negative	Bug
4.	Add label help text on Orders page	Neutral	General comment

3. Review your feedback on the GMT application home page.
4. Review your feedback on the Team Development page.
5. Log feedback 1 ("Date format needs to be changed on Master detail") as a Feature with a status of *"Functionally complete - 80%."*
6. Acknowledge feedback 2 ("Excellent application. Very user-friendly and intuitive UI.")
7. Log feedback 3 ("Error when deleting a line item on an order") as a bug. Set the severity to *"3. Significant Impact"* and assign it to `brad.knight`.
8. Edit the bug and change the status to *"80. Fixed in development"* and set the priority to *"1. As soon as possible."*
9. Log feedback 4 (Add label help text on Orders page) as a **To Do**, assign it to `susie.parker`, and change the status to *"Assigned - 30%."* Delete the feedback entry after logging it as a To Do.

10. Edit the To Do, changing the status to “*Work Progressing - 80%*”, specify the category to *Documentation*, and assign it to susie.parker.
11. Review the Team Development dashboard.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 21-1**.

If you haven't completed **Practice 16-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_16_02.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Click **Install**.
- i. Click **Edit Application**.
- j. On the Application's home page, click **Shared Components**.
- k. On the Shared Components page, click **Application Access Control**.
- l. Click on the **Roles** and **User Role Assignments** tab. You see that no user or user roles are imported when you imported the application.
- m. Sign out of Application Express.
- n. Sign in to the Application Express workspace as the **apex_admin** user.
- o. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- p. Click the **Upload >** button.
- q. Browse and upload the `createusers_addroles.sql` file in the `/home/oracle/labs/labs` directory.
- r. Note that you must change the application ID in the script to reflect your new application ID. Click the edit icon and make the necessary change in the script and click **Save**.

- s. Click **Upload**.
- t. Run the script by clicking the **Run** icon. Ensure that you have selected OEHR schema on the top left corner.
- u. Click **Run Now**.
- v. Make sure that the script executed successfully and three users (`brad.knight`, `susie.parker`, and `john.bell`) are created with their respective roles as ADMINISTRATOR, CONTRIBUTOR, READER. Now, if you go to the Application Access Control link and click on the **Roles** and **User Role Assignments** tab, you see that the above users are created with their respective roles.

If you haven't completed **Practice 20-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_dev` user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_20_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Make sure that **Yes** is selected for the **Install Supporting Objects** option and click **Next >**.
- i. Click **Install**.
- j. Verify that the application and the supporting objects are successfully installed.

Solution 21-1: Adding and Monitoring Feedback in Your Application

Overview

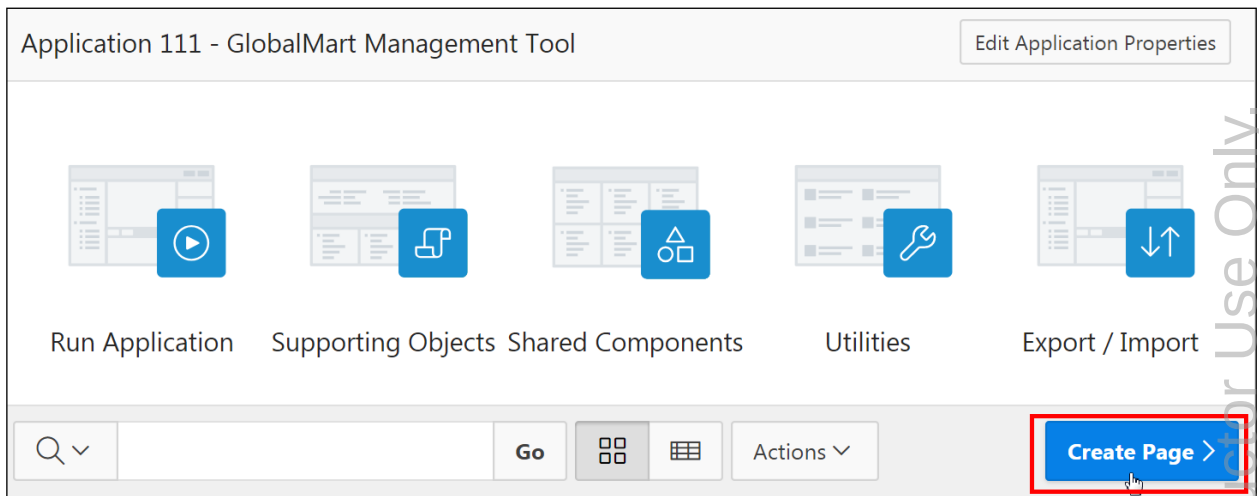
In this practice solution, steps are provided for adding and monitoring feedback in the GMT application.

Note: The solution script (`sol_21_01.sql`) generated at the end of this practice is just a starting point for this practice and not the solution.

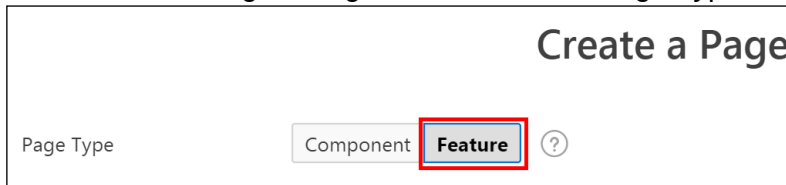
Steps

1. Create a feedback form for your GMT application.

- a. In the GlobalMart Management Tool application home page, click **Create Page >**.



- b. In the Create a Page dialog, click **Feature** for Page Type.

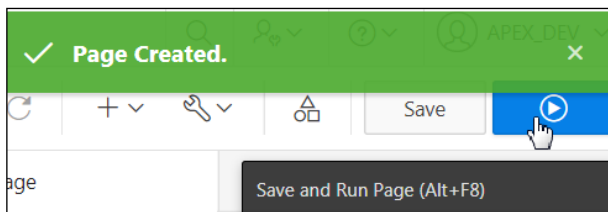


- d. In the Create Feedback Pages dialog, select the following (some are selected by default) and click **Create**.

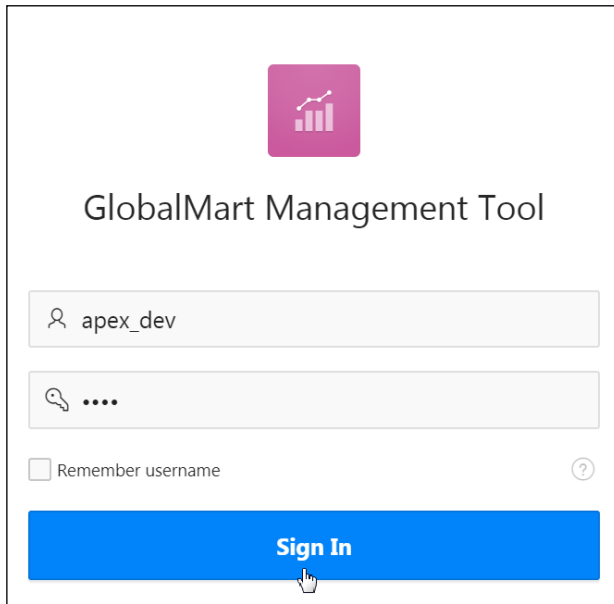
- **Starting Page Number:** 10020
- **Include in Navigation Bar?:** Yes.
- **Administration Page Preference:** Click **Identify an existing page**
- **Existing Administration Page:** Select **1**. (This is the home page.)

Note: When you create the feedback page, you will notice that the navigation bar will have an entry for feedback, and the application feedback will be automatically enabled.

- e. The feedback page now opens in Page Designer. The message “Page Created” is displayed. Note that Home Page opens in Page Designer with the Feedback region (you had selected Page 1: Home) as the Administration page in the above step.
- f. Click the **Save and Run Page** icon.



- g. You are redirected to the **Sign In** page of the GMT application. Enter your username and password and click **Sign In**.



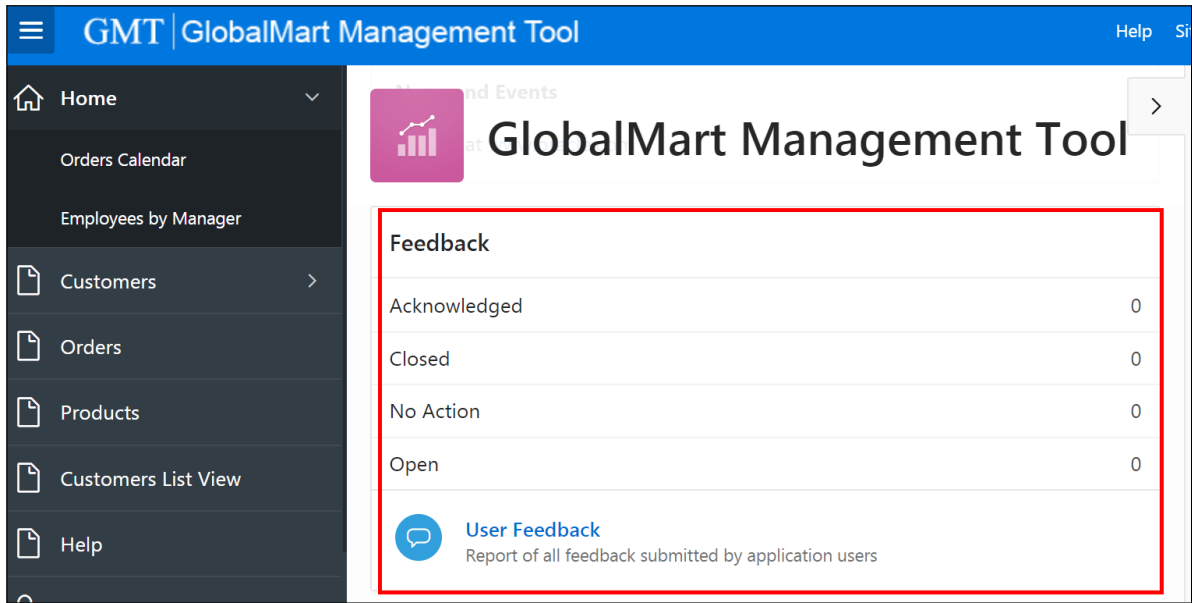
Also, note that 10020 (Feedback), 10021 (Feedback Submitted), 10023 (Manage Feedback), and 10024 (Feedback) get created. However, these are all dialog pages, and you cannot run them directly. You will get an error message if you try to run them.

✓ Changes saved. Dialog pages can not be run directly from Page Designer. Review this page by running a page that launches this dialog. ✕

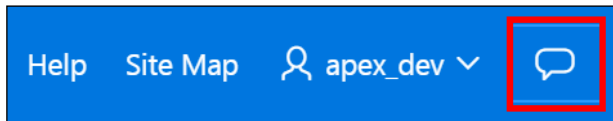
2. Create the following feedback in your GMT application:

Feedback #	Feedback Text	Experience	Feedback Type
1.	The date format on the Orders form needs to be changed from dd-mm-yy to mm-dd-yyyy	Neutral	General Comment
2.	Excellent tool. Very user-friendly and intuitive UI.	Positive	General comment
3.	Error when deleting a line item on an order	Negative	Bug
4.	Add label help text on Orders page.	Neutral	General Comment

- a. After signing in to the GMT application, note that the home page now contains a region on Feedback.

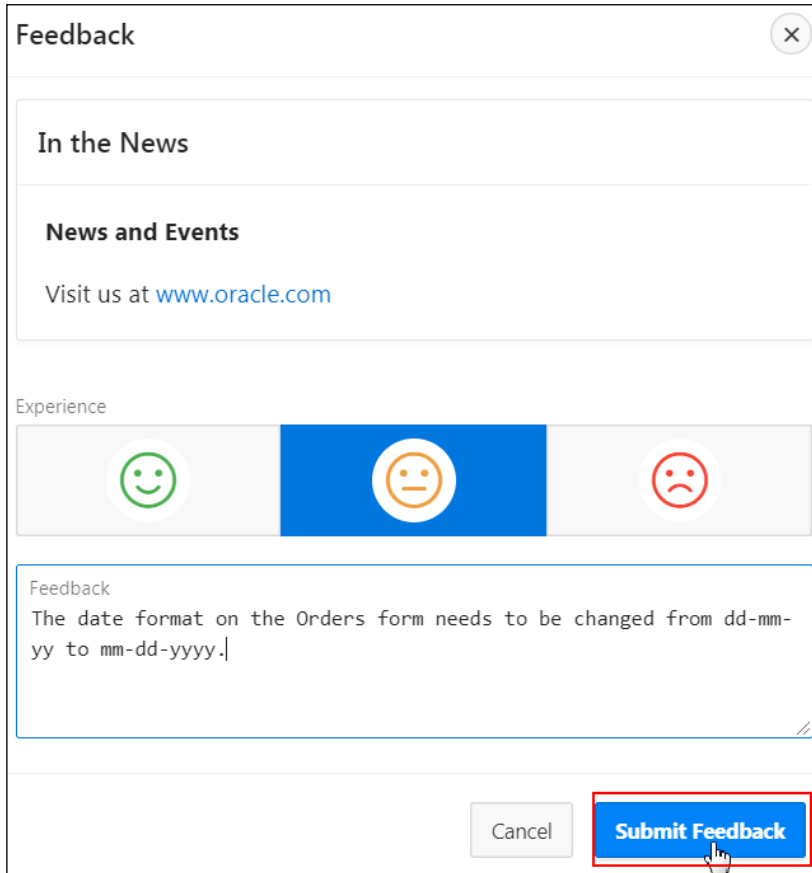


- b. Click the feedback icon  on the navigation bar.



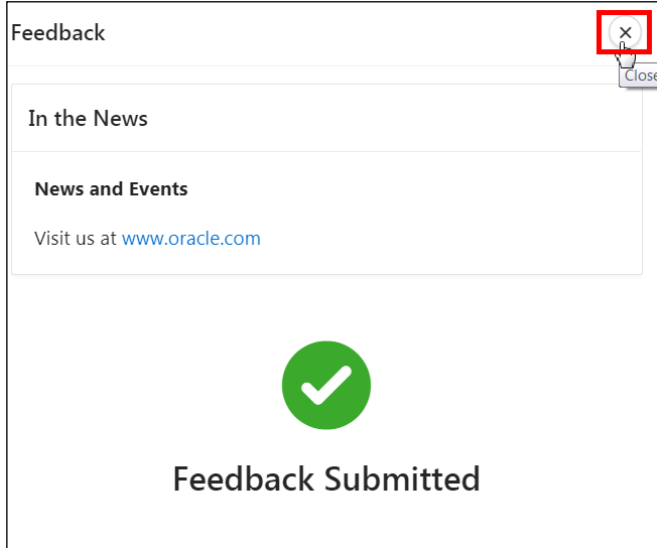
- c. The Feedback dialog opens. To create the first feedback, enter the following in the Feedback dialog:

- **Experience:** Click 
- **Feedback:** Enter The date format on the Orders form needs to be changed from dd-mm-yy to mm-dd-yyyy.
- Click **Submit Feedback**

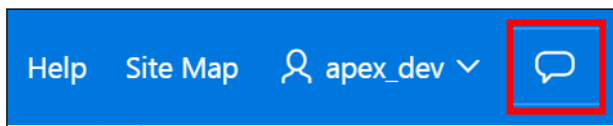


The image shows a 'Feedback' dialog box with a close button (X) in the top right corner. The dialog is divided into several sections. The first section is titled 'In the News' and contains a sub-section 'News and Events' with the text 'Visit us at www.oracle.com'. Below this is an 'Experience' section with three icons: a green smiley face, a blue neutral face (which is highlighted with a blue background), and a red frowny face. At the bottom of the dialog is a text area labeled 'Feedback' containing the text 'The date format on the Orders form needs to be changed from dd-mm-yy to mm-dd-yyyy.' Below the text area are two buttons: 'Cancel' and 'Submit Feedback'. The 'Submit Feedback' button is highlighted with a red border and a mouse cursor is pointing at it.

- d. After submitting the feedback, you get the “Feedback Submitted” message. Close the dialog to go back to the GMT application home page.



- e. To create the second feedback item, click the feedback icon  once again on the navigation bar.



- f. In the Feedback dialog, enter the following:

- **Experience:** Click  to indicate “positive”.
- **Feedback:** Enter `Excellent application. The UI is user friendly and quite intuitive.`
- Click **Submit Feedback**.

Feedback
✕

In the News

News and Events
Visit us at www.oracle.com

Experience

😊

😐

😞

Feedback
Excellent application. The UI is user friendly and quite intuitive.

Cancel
Submit Feedback

- g. After submitting the feedback, you get the “Feedback Submitted” message. Close the dialog to go back to the GMT application home page.

Feedback
✕
Close

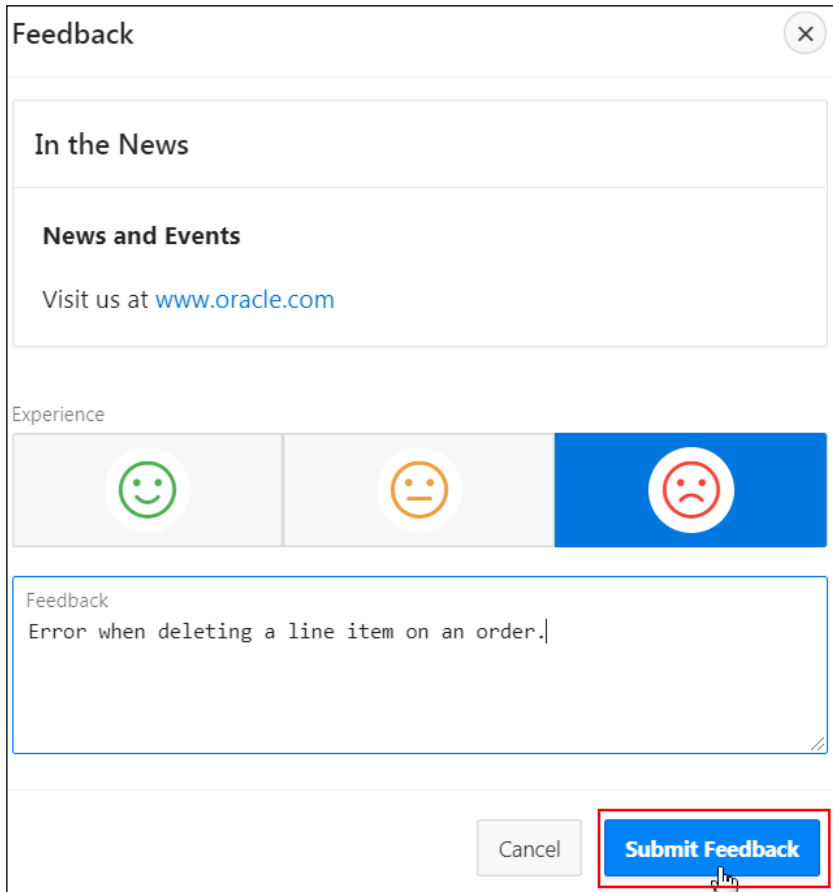
In the News

News and Events
Visit us at www.oracle.com

✅

Feedback Submitted

- h. To create the third feedback, click the feedback icon  on the navigation bar.
- i. In the Feedback dialog, enter the following:
- **Experience:** Click .
 - **Feedback:** Enter `Error when deleting a line item on an order.`
 - Click **Submit Feedback**.



The image shows a 'Feedback' dialog box with a close button (X) in the top right corner. It contains three sections: 'In the News', 'News and Events', and 'Experience'. The 'Experience' section has three buttons with smiley face icons: a green happy face, an orange neutral face, and a red sad face. The red sad face button is selected. Below the 'Experience' section is a text area labeled 'Feedback' containing the text 'Error when deleting a line item on an order.'. At the bottom right, there are two buttons: 'Cancel' and 'Submit Feedback'. The 'Submit Feedback' button is highlighted with a red rectangle and a mouse cursor is pointing at it.

Feedback

In the News

News and Events

Visit us at www.oracle.com

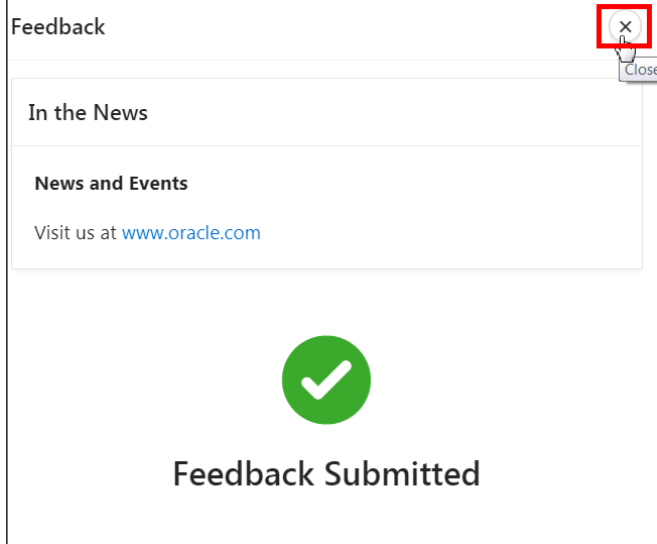
Experience

Feedback

Error when deleting a line item on an order.

Cancel Submit Feedback

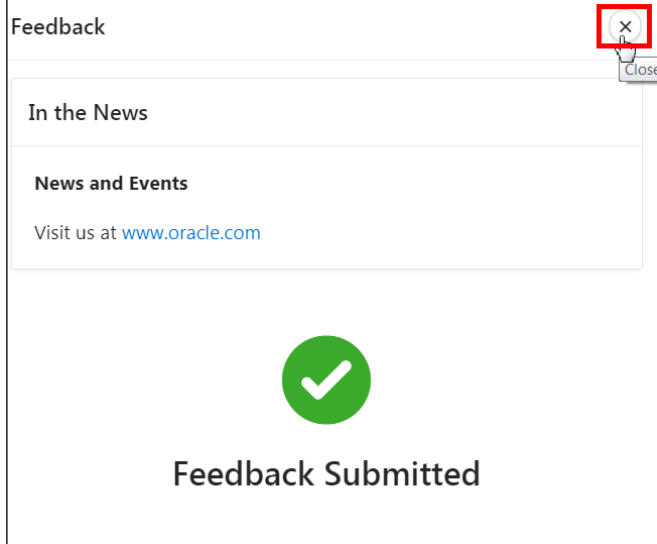
- j. Close the Feedback Submitted dialog by clicking close.



- k. To create the fourth and the last feedback, click the feedback icon  on the navigation bar.
- l. In the Feedback dialog, enter the following:
- **Experience:** Click .
 - **Feedback:** Enter Add label help text on Orders page.
 - Click **Submit Feedback**.

A screenshot of the 'Feedback' dialog box. The dialog has a title bar with a close button (X). Below the title bar, there is a section titled 'In the News' with a sub-section 'News and Events' and a link 'Visit us at www.oracle.com'. Below this, there is a section titled 'Experience' with three icons: a green smiley face, a blue neutral face (highlighted with a red square), and a red frowny face. Below the 'Experience' section, there is a text area labeled 'Feedback' containing the text 'Add label help text on Orders page.' At the bottom, there are two buttons: 'Cancel' and 'Submit Feedback' (highlighted with a red square).

- m. Close the Feedback Submitted dialog by clicking close.



- n. You have entered all your feedbacks. Now you want to view it in Team Development. Click **Home** on the Developer toolbar.

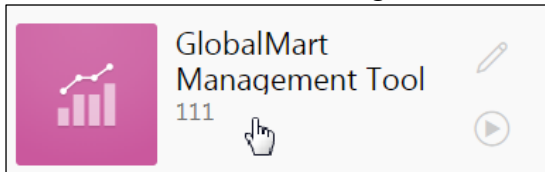


3. Review your feedback in the GMT application home page.

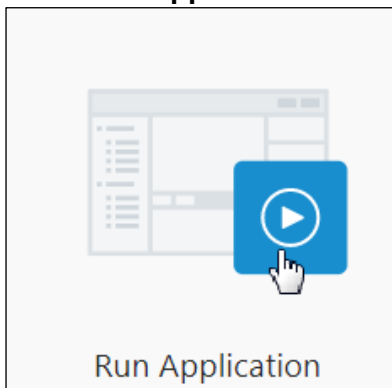
You can review the feedback in two ways – from the application home page and also from the **Oracle Application Express > Team Development** page.

You review and manage feedback from the GMT application as an end user:

- a. Select the **GlobalMart Management Tool** application.



- b. Select **Run Application**.



- c. The Home Page opens. You can see that the Feedback section displays the count of the feedbacks submitted. You can view the feedback details by clicking **User Feedback**.

A screenshot of the 'GMT | GlobalMart Management Tool' home page. The page has a blue header with the title and a 'Help' link. A dark sidebar on the left contains navigation links: Home, Orders Calendar, Employees by Manager, Customers, Orders, Products, Customers List View, and Help. The main content area has a pink icon and the title 'GlobalMart Management Tool'. Below this is a 'Feedback' section with a table showing the status of feedbacks. The table has two columns: the status and a count. The 'No Action' row has a count of 4, which is highlighted with a red box. Below the table is a 'User Feedback' link, also highlighted with a red box, with the text 'Report of all feedback submitted by application users' below it.

Feedback	
No Action	4
Acknowledged	0
Closed	0
Open	0

[User Feedback](#)
Report of all feedback submitted by application users

- d. When you click User Feedback, the Manage Feedback page opens and displays feedback details.

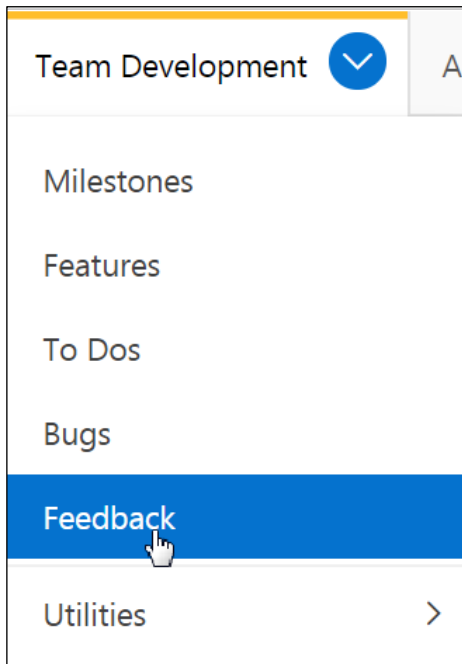
Manage Feedback ✕									
	Application Page	Filed	Filed By	Feedback	Rating	Status	Response	Updated ↓	Updated By
	31. Credit Limit	18 minutes ago	apex_dev	Add label help text on Orders page.		No Action	-	18 minutes ago	apex_dev
	31. Credit Limit	22 minutes ago	apex_dev	Error when deleting a line item on an order.		No Action	-	22 minutes ago	apex_dev
	31. Credit Limit	25 minutes ago	apex_dev	Excellent application. The UI is user friendly and quite intuitive.		No Action	-	25 minutes ago	apex_dev
	31. Credit Limit	38 minutes ago	apex_dev	The date format on the Orders form needs to be changed from dd-mm-yy to mm-dd-yyyy		No Action	-	38 minutes ago	apex_dev

- e. Click the edit icon  for any feedback that you want to update. The feedback form for the specific feedback opens where you can enter your (end user) response and change the status. Your (end user) response is visible as “Public Response” in the Developer view. Click **Apply Changes**. Note that in this activity we have not made any changes. Close the **Manage Feedback** window.
- f. Click **Edit Page 1** link on the Developer toolbar.

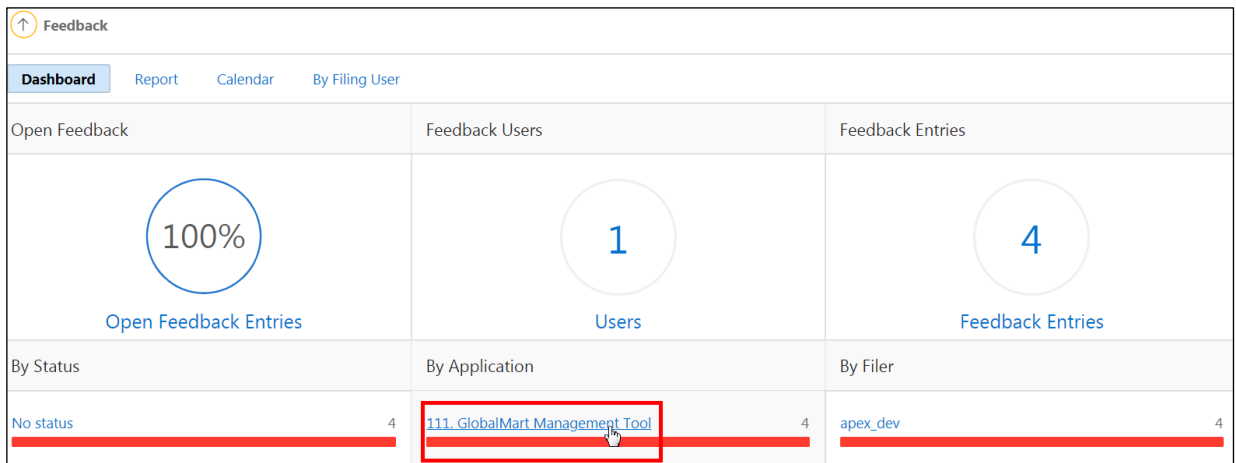


4. Review your feedback in Team Development.

- a. Page 1: Home opens in Page Designer view. Go to **Team Development** and click **Feedback**. The Feedback Dashboard opens.



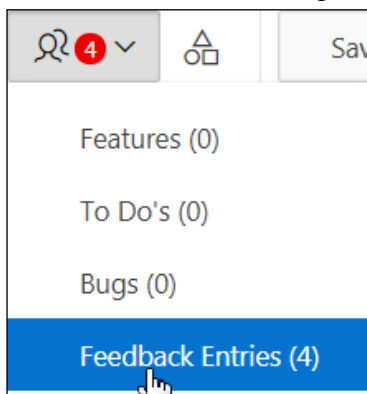
- b. On the Feedback Dashboard, click **111. GlobalMart Management Tool** to view the feedbacks grouped by application.



The Feedback Report is displayed. Currently the Status column displays **0. No Status**.

	Number	Feedback	Type	Status
☒	4	Add label help text on Orders page.	General Comment	0. No status
☐	3	Error when deleting a line item on an order.	General Comment	0. No status
☐	2	Excellent application. The UI is user friendly and quite intuitive.	General Comment	0. No status
☐	1	The date format on the Orders form needs to be changed from dd-mm-yy to mm-dd-yyyy	General Comment	0. No status

Note that you can also see your feedbacks by clicking the Team Development icon  > **Feedback Entries** in Page Designer view.



5. Log feedback **1** (“The Date format on the Orders form”) as a feature with a status of “Functionally complete - 80%.”
 - a. Click the Feedback Number **1** link to edit it.

Feedback \ Report			
Dashboard Report Calendar By Filing User			
Search: All Text Columns Go Open Feedback Actions Edit Save			
☐		Number	Feedback
☒		4	Add label help text on Orders page.
☐		3	Error when deleting a line item on an order.
☐		2	Excellent application. The UI is user friendly and quite intuitive.
☐		1	The date format on the Orders form needs to be changed from dd-mm-yy to mm-dd-yyyy

- b. The View Your Feedback dialog displays the feedback details. Click **Log as Feature**.

View / Edit Feedback

Number: 1

Application: 111 - GlobalMart Management Tool

User: APEX_DEV

Created: 94 minutes ago

Type: General Comment

Page: 31 - Credit Limit

Email: apexdev@oracle.com

Rating: 😊

User Provided Comment: ?

The date format on the Orders form needs to be changed from dd-mm-yy to mm-dd-yyyy

Disposition

- c. Specify the following and click **Create Feature**:
 - **Feature Name:** Enter Date format change
 - **New Release:** Enter 19.2
 - **Assigned To:** Select John.Bell
 - **Start Date:** <Select today's date>
 - **Due Date:** <Select the date one month from now>
 - **Desirability:** Select 3. Desirable (if not selected by default)

- **Development Priority:** Select **3. Normal Priority** (if not selected by default)
- **Feature Status:** Select **Functionally complete - 80%**

Log as Feature

Feature Details

Feedback: The date format on the Orders form needs to be changed from dd-mm-yy to mm-dd-yyyy.

Type: General Comment

Created By: APEX_DEV

Logged: 20 minutes ago

Application:

Page:

* Feature Name: Date format change

Release: - Select Release -

New Release: 19.2

Assigned To: john.bell

Start Date: 05/10/2019

Due Date: 06/10/2019

Desirability: 3. Desirable

Development Priority: 3. Normal priority

Approval Status: - Select Approval Status -

Feature Status: Functionally complete - 80%

Feature Description

Cancel

Create Feature

Notice that assigning a feedback as Feature Bug changes the Status to **4.Closed**.

		Num...	Feedback	Type	Status	Developer C...	Public Response
☒		4	Add label help text on Orders page.	General Comment	0. No status	-	-
☐		3	Error when deleting a line item on an order.	General Comment	0. No status	-	-
☐		2	Excellent application. The UI is user friendly and quite intuitive.	General Comment	0. No status	-	-
☐		1	The date format on the Orders form needs to be changed from dd-mm...	General Comment	4. Closed	-	Thank you for you...

6. Acknowledge feedback 2.

- a. Click the feedback 2 link. The View/Feedback dialog opens.

Feedback \ Report					
Dashboard Report Calendar By Filing User					
Search: All Text Columns Go Open Feedback Actions Edit Save Reset					
☐		Num...	Feedback	Type	Status
☒		4	Add label help text on Orders page.	General Comment	0. No status
☐		3	Error when deleting a line item on an order.	General Comment	0. No status
☐		2	Excellent application. The UI is user friendly and quite intuitive.	General Comment	0. No status
☐		1	The date format on the Orders form needs to be changed from dd-mm...	General Comment	4. Closed

In the View/Edit Feedback dialog, change the Status to **1 – Acknowledged** and click **Apply Changes**.

View / Edit Feedback

Feedback

Number: **2**

Type: **General Comment**

Application: **111 - GlobalMart Management Tool**

Page: **1 - Home**

User: **APEX_DEV**

Email: **apexdev@oracle.com**

Created: **25 minutes ago**

Rating:

User Provided Comment:

Excellent application. The UI is user friendly and quite intuitive.

Disposition

Status: **1. Acknowledged**

Notice the change in the Status column. It displays now as **1. Acknowledged**.

☐		Num...	Feedback	Type	Status
☒		4	Add label help text on Orders page.	General Comment	0. No status
☐		3	Error when deleting a line item on an order.	General Comment	0. No status
☐		2	Excellent application. The UI is user friendly and quite intuitive.	General Comment	1. Acknowledged
☐		1	The date format on the Orders form needs to be changed from dd-mm...	General Comment	4. Closed

7. Log feedback 4 (“Add label help text on Orders page”) as a To Do, assign it to `susie.parker`, and change the status to “Assigned - 30%.”
- a. Click the Feedback Number 4 link to edit it.

Dashboard

Report

Calendar

By Filing User

Q

Search: All Text Columns

Go

Open Feedback

Actions

Edit

Save

Reset

☐		Num...	Feedback	Type	Status
☒		4	Add label help text on Orders page.	General Comment	0. No status
☐		3	Error when deleting a line item on an order.	General Comment	0. No status
☐		2	Excellent application. The UI is user friendly and quite intuitive.	General Comment	1. Acknowledged
☐		1	The date format on the Orders form needs to be changed from dd-mm...	General Comment	4. Closed

- b. You want to change feedback 4 to an **Enhancement Request** and a **To Do**. Enter the following:
- **Type:** Select Enhancement Request
 - Click **Log as To Do**

View / Edit Feedback

Feedback

Number: 4

Application: 111 - GlobalMart Management Tool

User: APEX_DEV

Created: 24 minutes ago

* Type: Enhancement Request

Page: 1 - Home

Email: apexdev@oracle.com

Rating: ☹

User Provided Comment: ?

Add label help text on Orders page.

Disposition

Log as Bug

Log as To Do

Log as Feature

Status

0. No status

- c. In the Convert Feedback to a To Do dialog, specify the following and click **Create To do**:
- **Release:** Select 19.2
 - **Assigned To:** Select `susie.parker`
 - **To do status:** Select Assigned - 30%

x

Convert Feedback as To Do

Feedback

Converting feedback to a to do will set the feedback status to closed. Please enter a public response for the feedback.

Public Response

Thank you for your feedback, a task has been created to address this issue.

To Do Details

Feedback: Add label help text on Orders page.	Type: Enhancement Request
Created By: APEX_DEV	Feedback Logged: Friday May 10, 2019 11:34
Application: 111. GlobalMart Management Tool	Page: 1. Home
To Do Name: Add label help text on Orders page. ?	Release: 19.2 ?
Assigned To: susie.parker ?	
To do status: Assigned - 30% ?	Category: - No Category - ?

8. Edit the To Do, changing the status to "Work Progressing - 80%" and specifying Documentation for the category.
 - a. Click the arrow next to the **Team Development** pull-down menu and select **To Dos**.

Team Development
▼

Milestones

Features

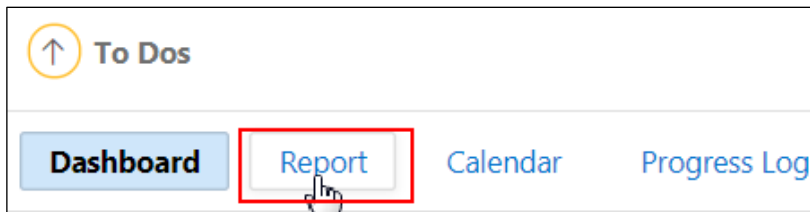
To Dos

Bugs

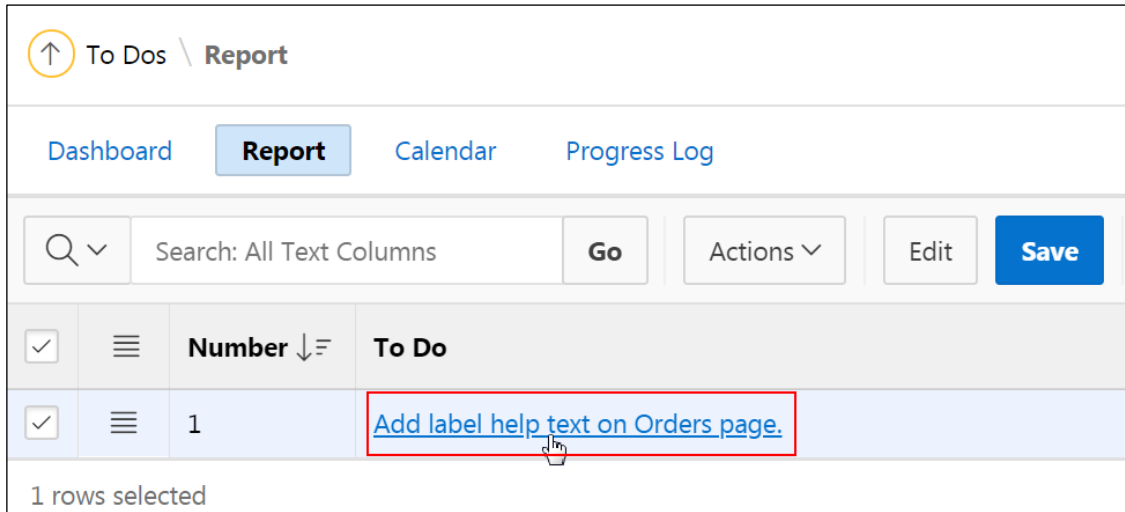
Feedback

Utilities >

- b. Click **Report**.



- c. In the To Do Report, click the **Add label help text on Orders page** link.



- d. Change the **To Do > Status** to **Work Progressing - 80%**, enter **Documentation** for **Details > New Category**, and click **Apply Changes**.

The screenshot shows a 'To Do' form with the following details:

- Status:** Work Progressing - 80% (highlighted with a red box)
- New Category:** Documentation (highlighted with a red box)
- Apply Changes:** Button highlighted with a red box in the top right corner.

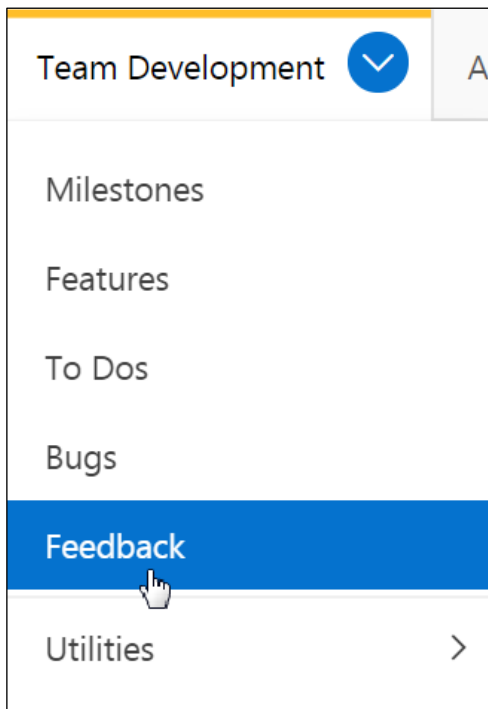
You get the message **Action Processed**. Notice that the **Percent Complete** column is displayed as 80.

The screenshot shows a 'To Dos' report with the following table:

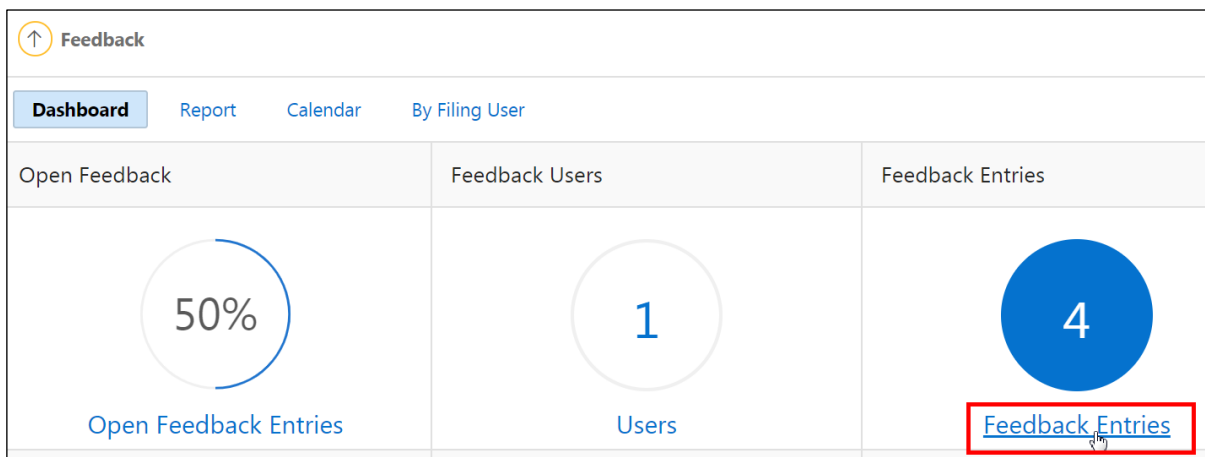
	Number	To Do	Feature	Percent Complete
☒	1	Add label help text on Orders page.		80

1 rows selected

9. Log feedback 3 (“Error when deleting a line item on an order”) as a bug. Set the severity to “4. Moderate Impact” and assign it to `brad.knight`.
- a. Click the arrow next to the **Team Development** pull-down menu and select **Feedback**.



- b. Click the **Feedback Entries** link.



- c. Click the Feedback number **3** link to edit it.

Feedback \ Report

Dashboard **Report** Calendar By Filing User

Search: All Text Columns Go Open Feedback Actions Edit Save Reset

		Num...	Feedback	Type	Status
☒		4	Add label help text on Orders page.	Enhancement Request	4. Closed
☐		3	Error when deleting a line item on an order.	General Comment	0. No status
☐		2	Excellent application. The UI is user friendly and quite intuitive.	General Comment	1. Acknowledged
☐		1	The date format on the Orders form needs to be changed from dd-mm...	General Comment	4. Closed

- d. In the View/Edit Feedback dialog that opens, click **Log as Bug** to mark Feedback 3 as a bug. The Convert Feedback to Bug dialog opens.

View / Edit Feedback

Feedback

Number: **3** * Type: General Comment

Application: **111 - GlobalMart Management Tool** Page: **1 - Home**

User: **APEX_DEV** Email: **apexdev@oracle.com**

Created: **36 minutes ago** Rating: 😞

User Provided Comment: ?
Error when deleting a line item on an order.

Disposition

Log as Bug Log as To Do Log as Feature

- e. In the Convert Feedback to Bug dialog, specify the following and click **Create Bug**.
- **Severity:** Select 3. Significant Impact
 - **Release:** Select 19.2
 - **Assign To:** Select brad.knight

Convert Feedback to Bug

Feedback

Converting feedback to a bug will set the feedback status to closed. Please enter a public response for the feedback.

Public Response ?
Thank you for your feedback, a bug has been created for this issue.

Bug Details

Feedback: Error when deleting a line item on an order.

Type: General Comment

Created By: APEX_DEV

Feedback Logged: Friday May 10, 2019 11:33

Application: 111. GlobalMart Management Tool

Page: 1. Home

Bug Title
Error when deleting a line item on an order.

Severity
3. Significant Impact ?

Release
19.2 ?

Assign To
brad.knight ?

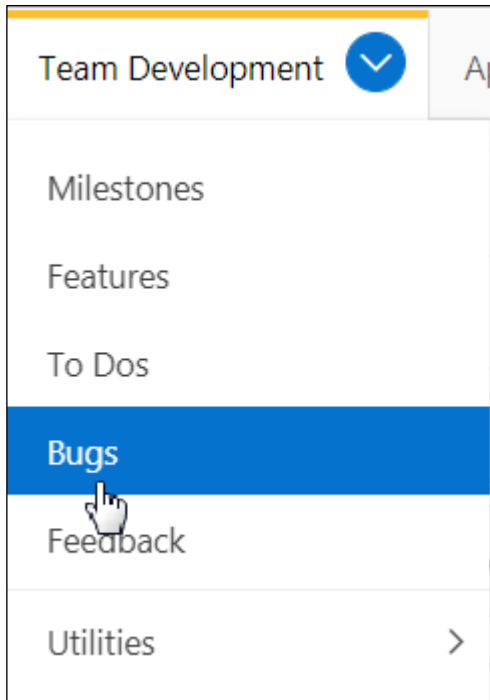
New Assignee ?

Cancel

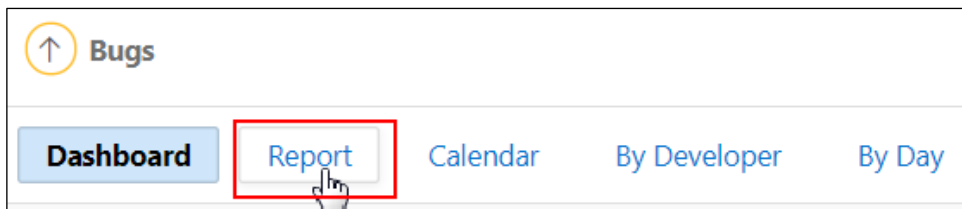
Create Bug

10. Edit the bug and change the status to **80. Fixed in development** and set the priority to **1. As soon as possible**.

a. Click the arrow next to the **Team Development** pull-down menu and select **Bugs**.



b. Click **Report**.



c. Click the **Error when deleting a line item on an order** link.

Bugs \ Report								
Dashboard Report Calendar By Developer By Day								
Q v Go Actions v								
Bug Title	Bug	Assignee	Release	Updated By	Updated ↓	Severity	Priority	Status
Error when deleting a line item on an order.	1	brad.knight	19.2	apex_dev	2 minutes ago	3. Significant Impact	-	0. No Status

- d. On the Bug: 1 details page, enter the following:
- **Status:** Select 80. Fixed in development
 - **Priority:** Select 1. As soon as possible
 - Click **Apply Changes**

Bug: 1

Cancel Delete **Apply Changes**

Show All Bug Resolution Description Application Con... Context Impact Customer

Bug

* Bug Title Error when deleting a line item on an order. ?

Status 80. Fixed in development ?

Severity 3. Significant Impact ?

Priority 1. As soon as possible ?

Created By: APEX_DEV , 4 minutes ago ?

Notice the changes displayed in Status and Priority columns.

↑ Bugs \ Report

✓ Action Processed.

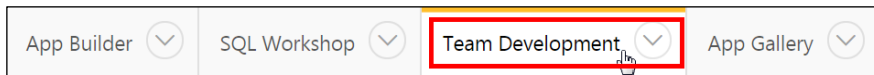
Dashboard **Report** Calendar By Developer By Day

Q Go Actions

Bug Title	Bug	Assignee	Release	Updated By	Updated ↓	Severity	Priority	Status
Error when deleting a line item on an order.	1	brad.knight	19.2	apex_dev	Now	3. Significant Impact	1. As soon as possible	080. Fixed in development

11. Review the Team Development dashboard.

- a. Click **Team Development** in the navigation bar.



- b. An overview of all your features, milestones, to dos, and bugs is displayed.

